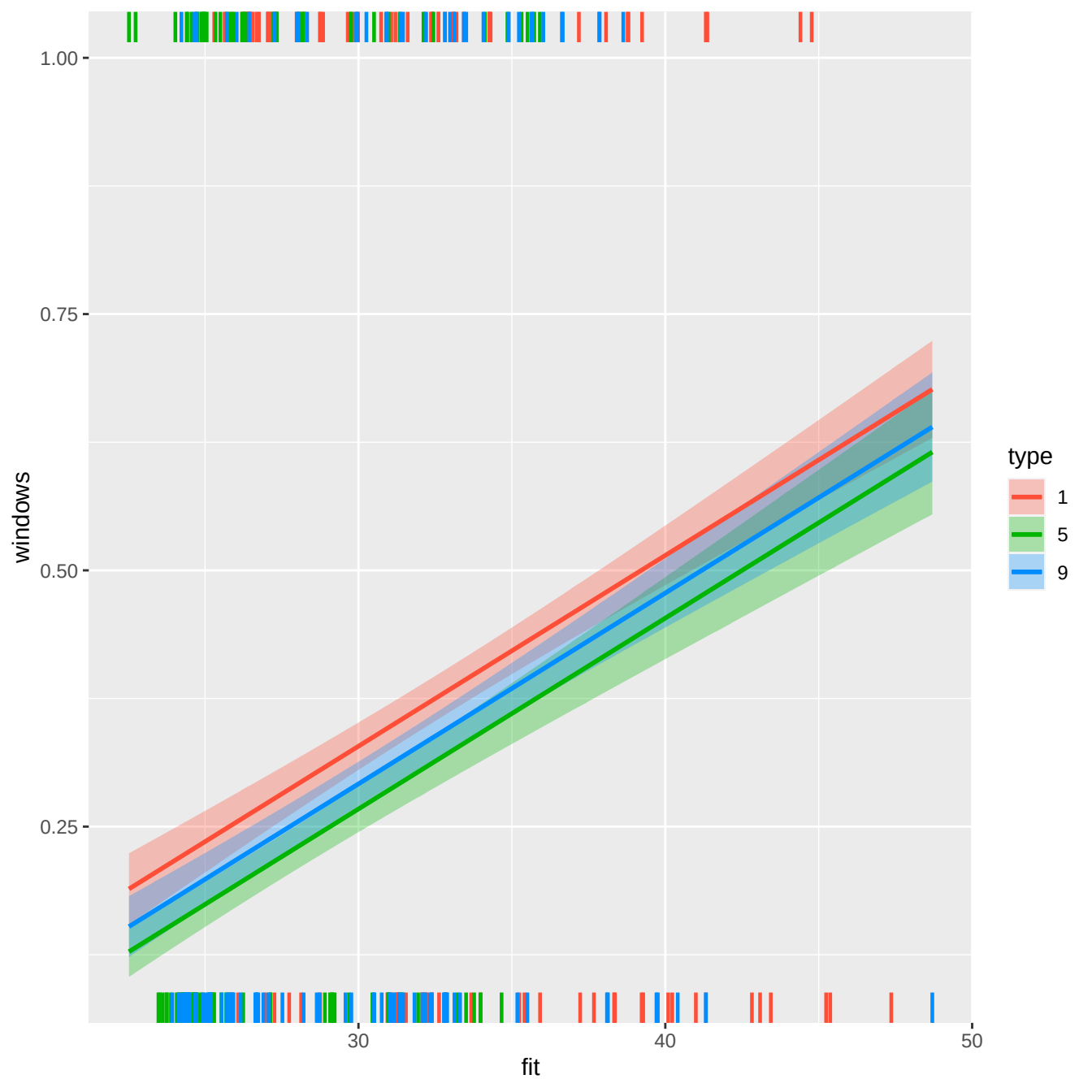
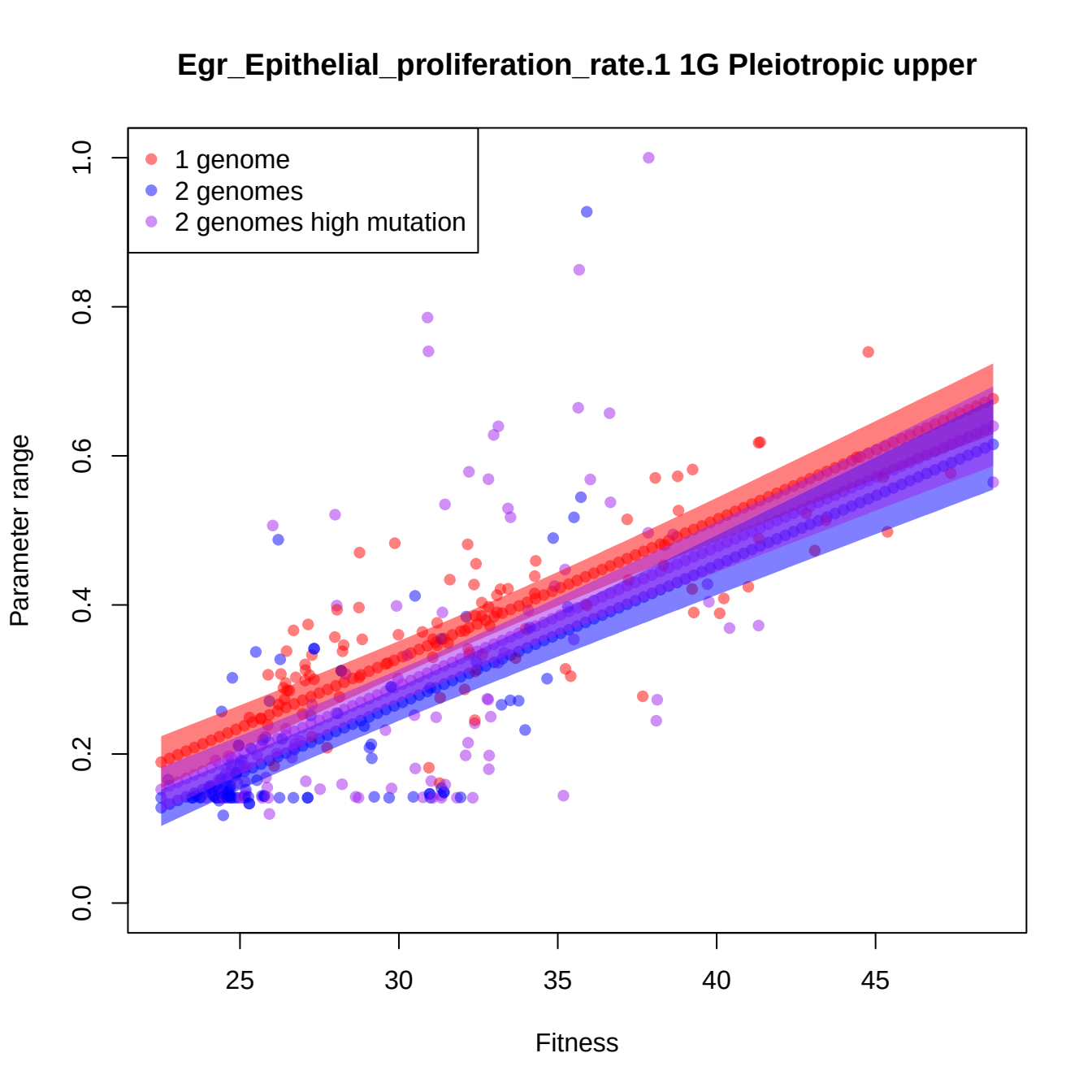
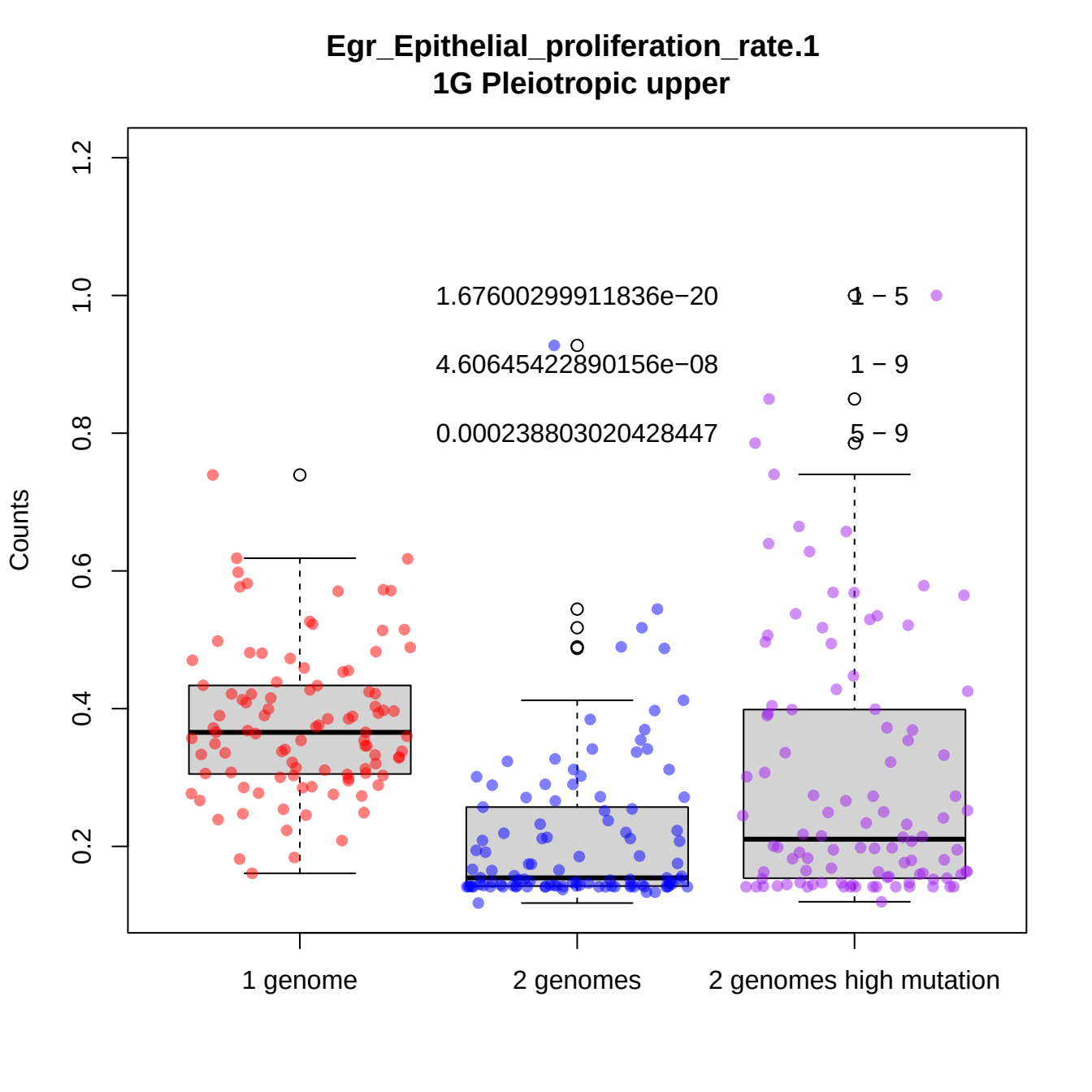
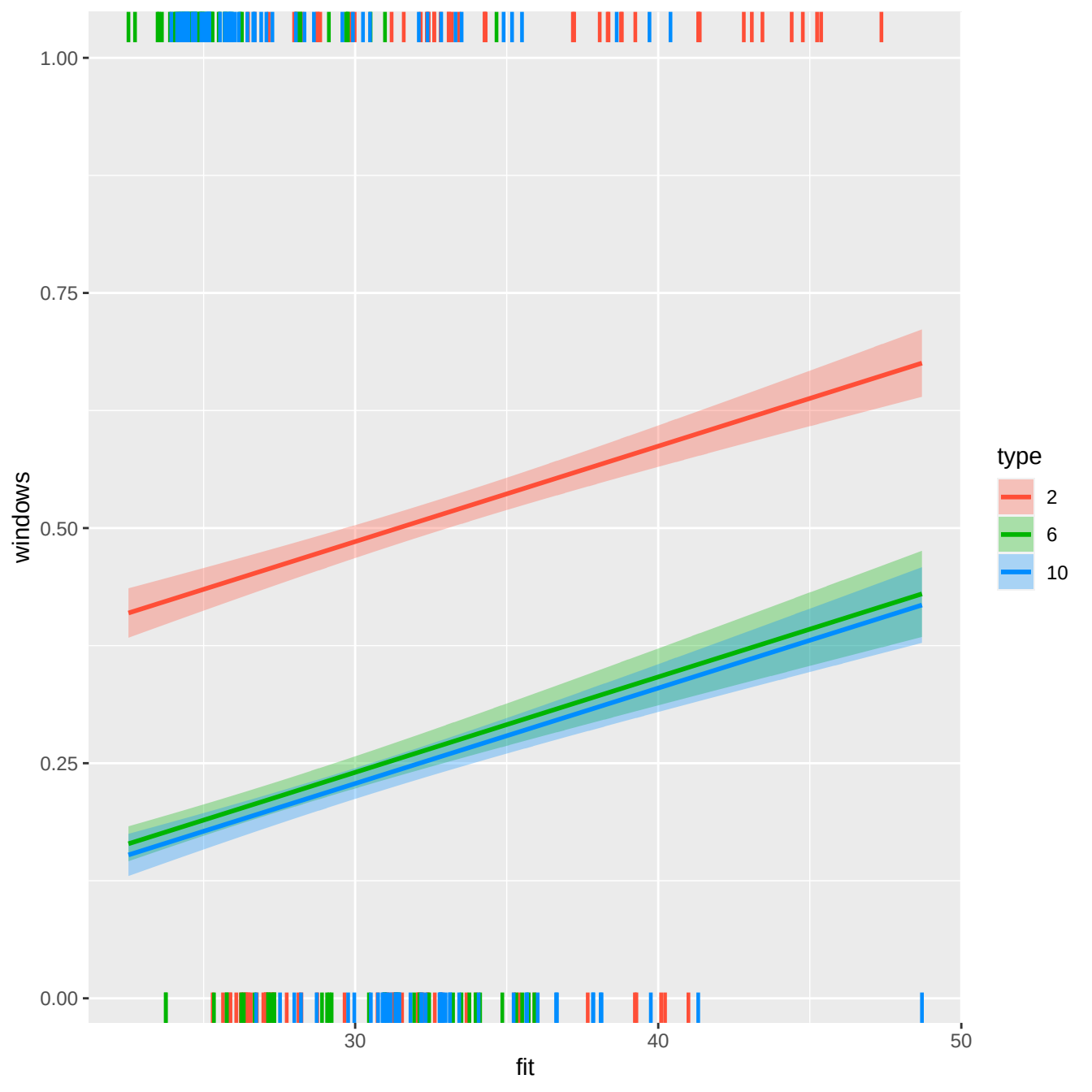
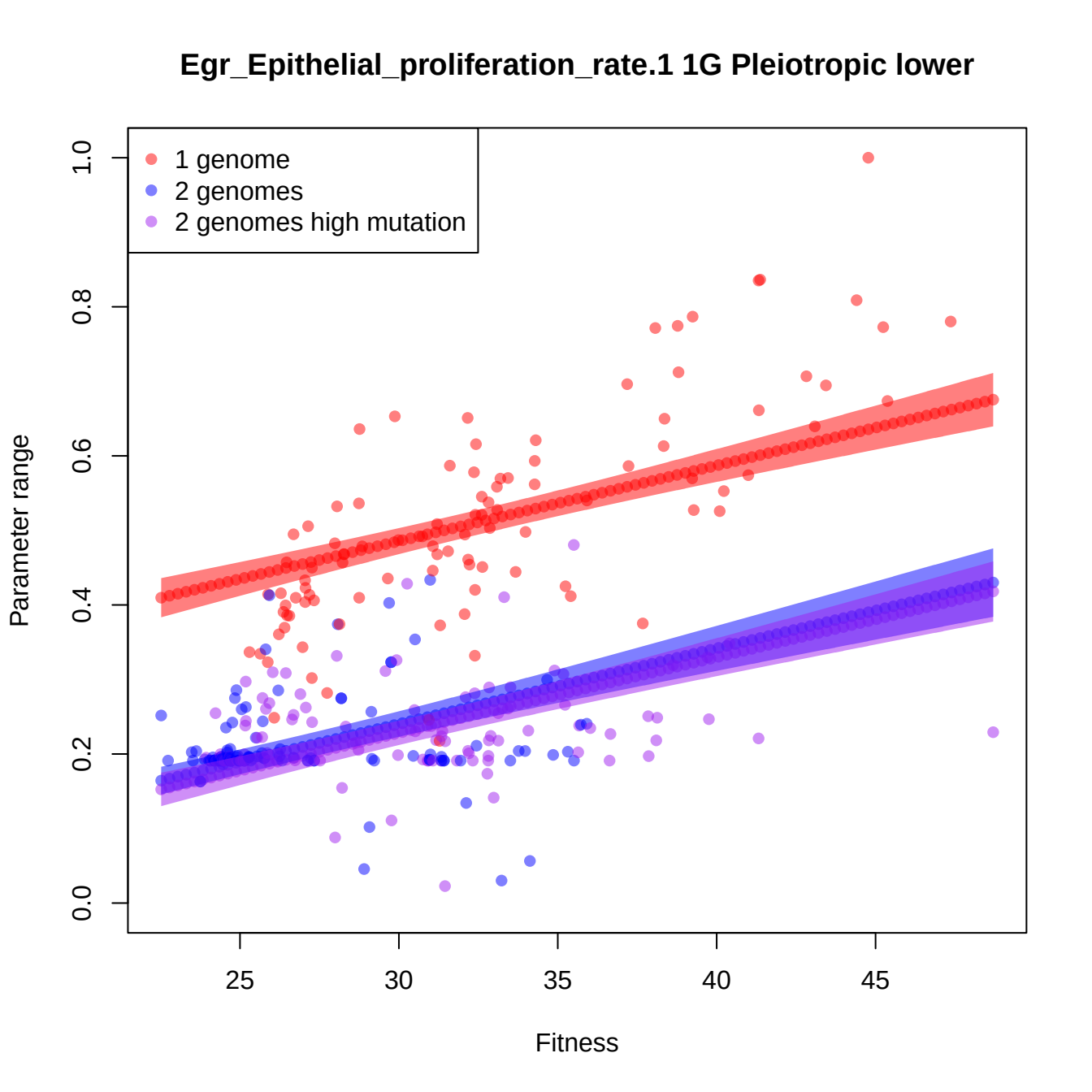
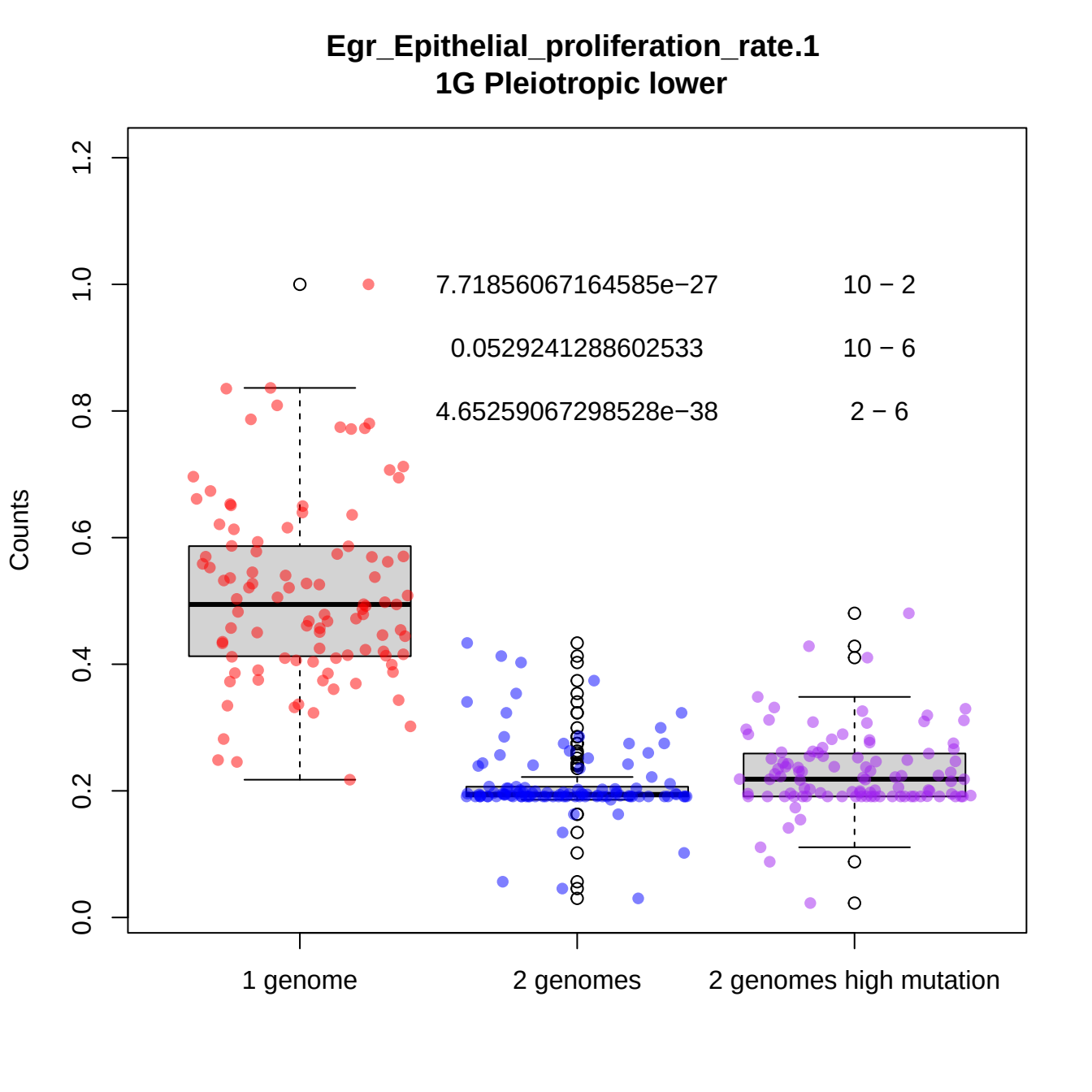

1.00
0.75
0.50
0.25
windows
30
40
50
fit
type
1
5
9

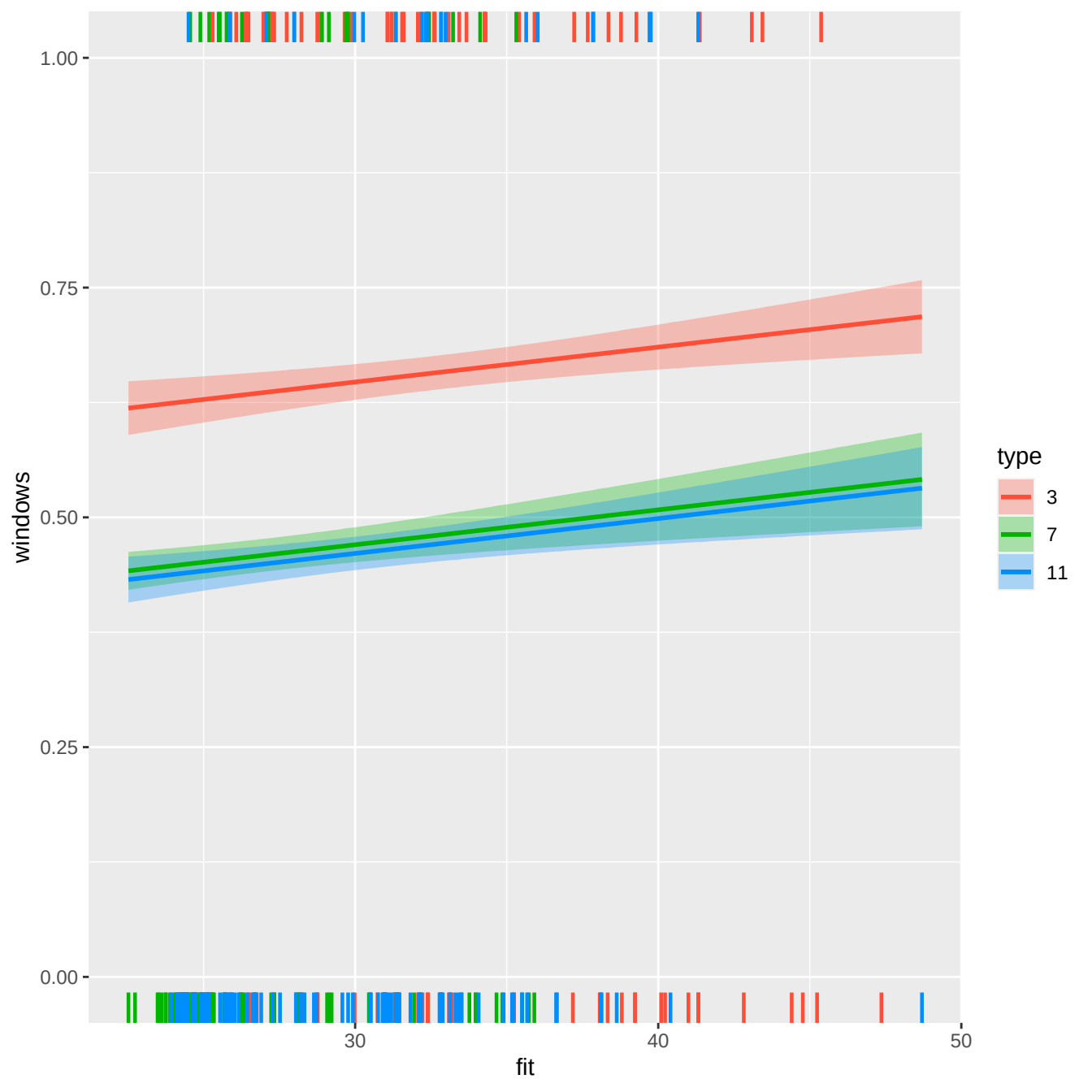

Egr_Epithelial_proliferation_rate.1 1G Pleiotropic upper
1 genome
2 genomes
2 genomes high mutation
Parameter range
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

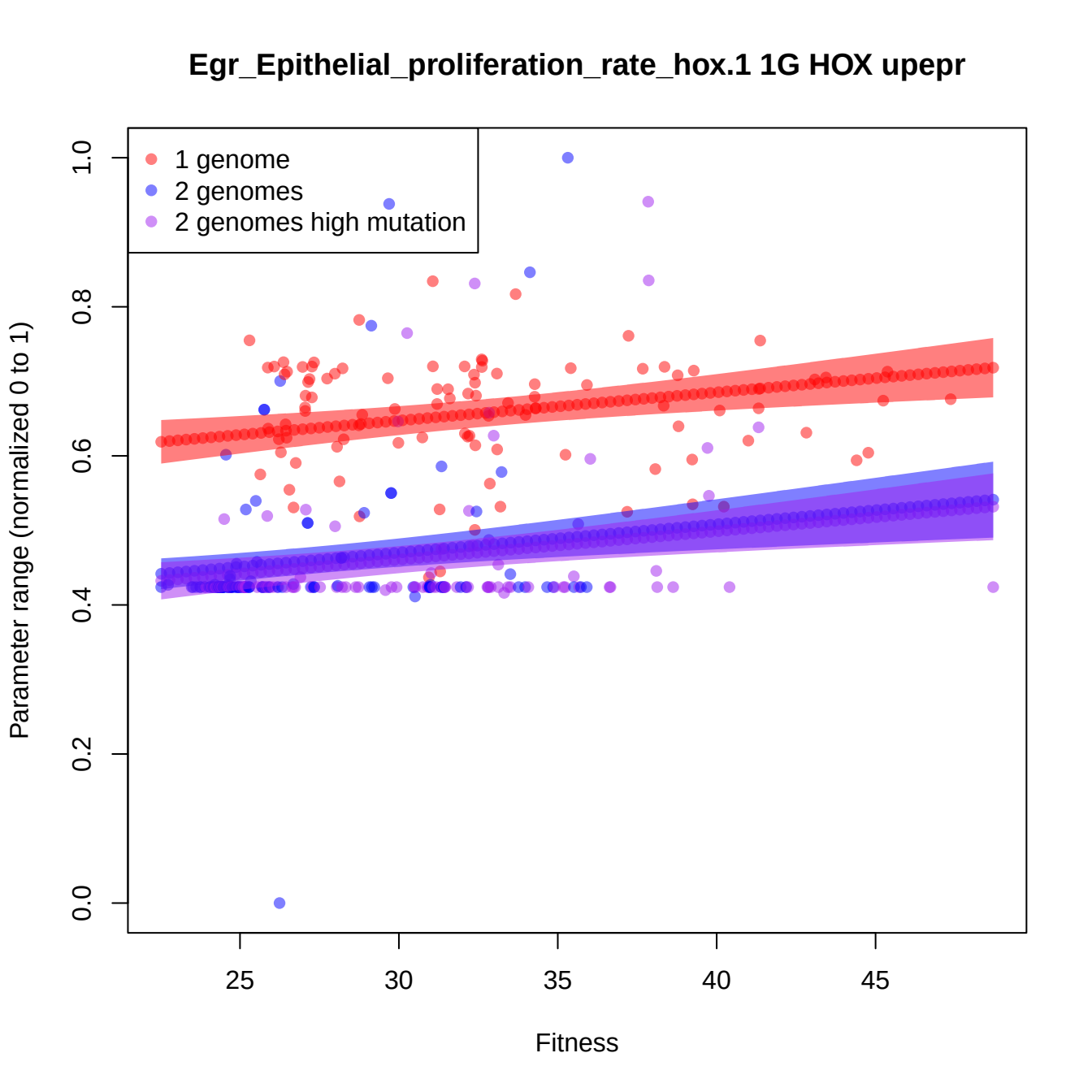

Egr_Epithelial_proliferation_rate.1
1G Pleiotropic upper
Counts
1.2
1.0
0.8
0.6
0.4
0.2
1.67600299911836e−20
4.60645422890156e−08
0.000238803020428447
1 − 5
1 − 9
5 − 9
1 genome
2 genomes
2 genomes high mutation

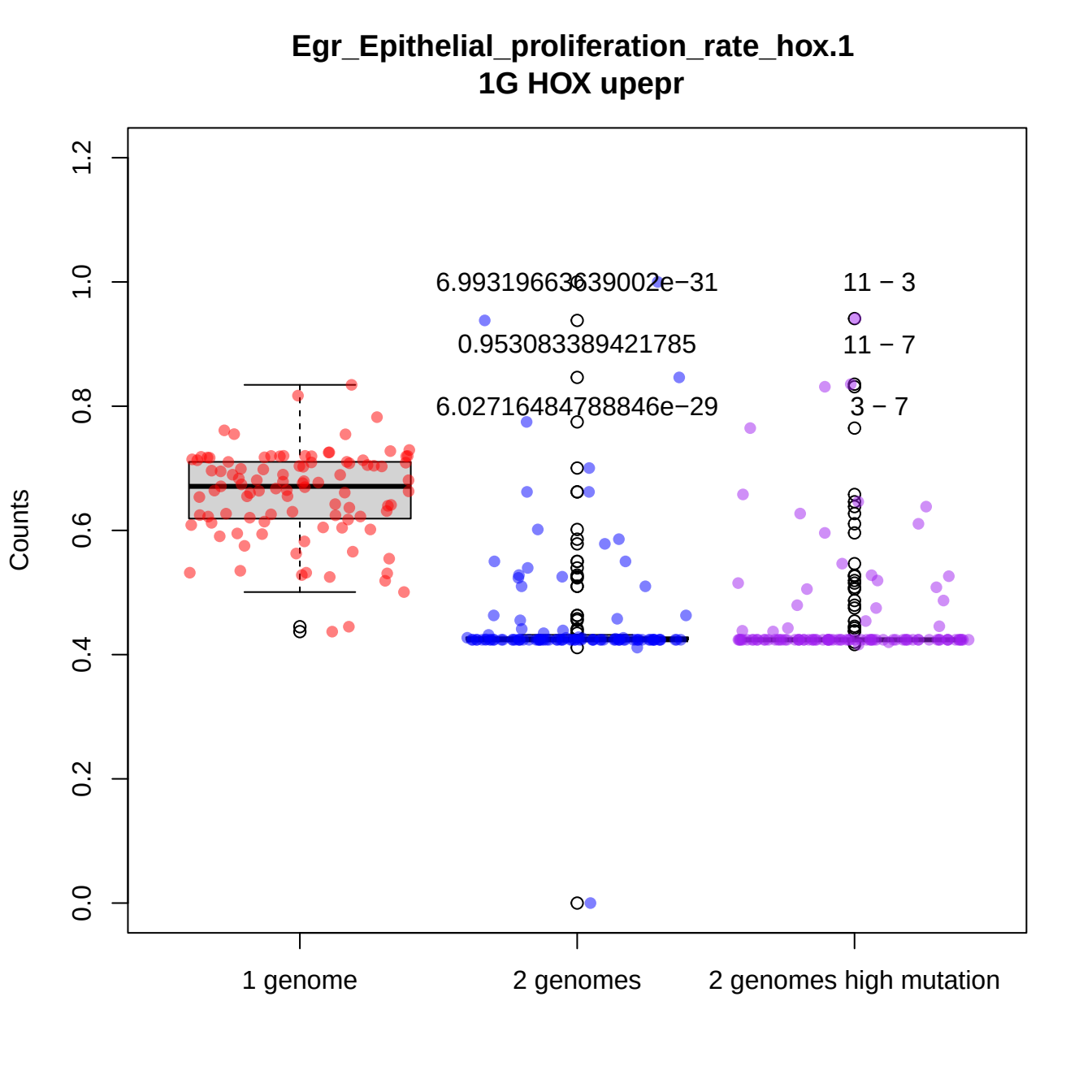

1.00
0.75
0.50
0.25
0.00
windows
30
40
50
fit
type
2
6
10

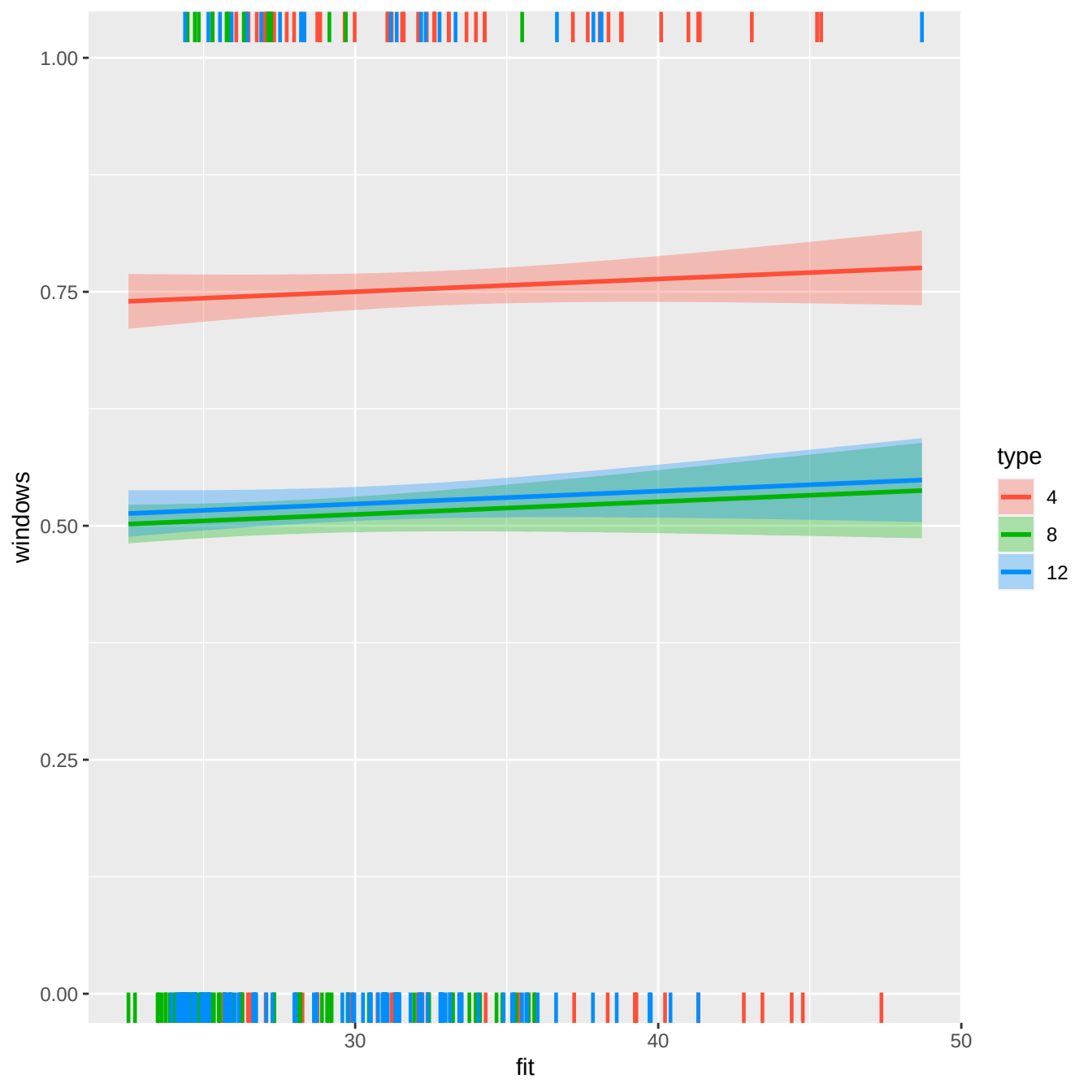

Egr_Epithelial_proliferation_rate.1 1G Pleiotropic lower
1 genome
2 genomes
2 genomes high mutation
Parameter range
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

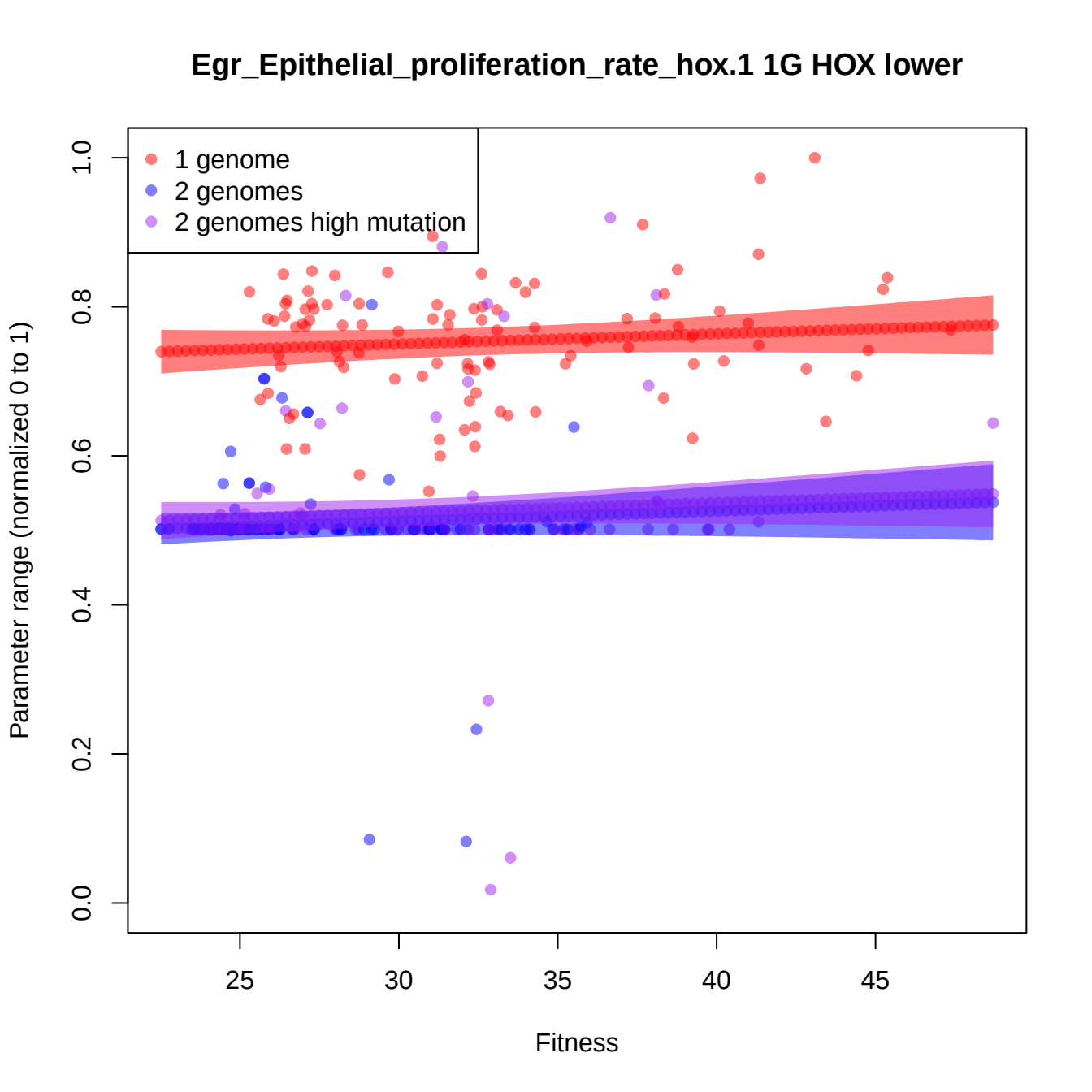

Egr_Epithelial_proliferation_rate.1
1G Pleiotropic lower
7.71856067164585e−27
10 − 2
0.0529241288602533
10 − 6
4.65259067298528e−38
2 − 6
Counts
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1 genome
2 genomes
2 genomes high mutation

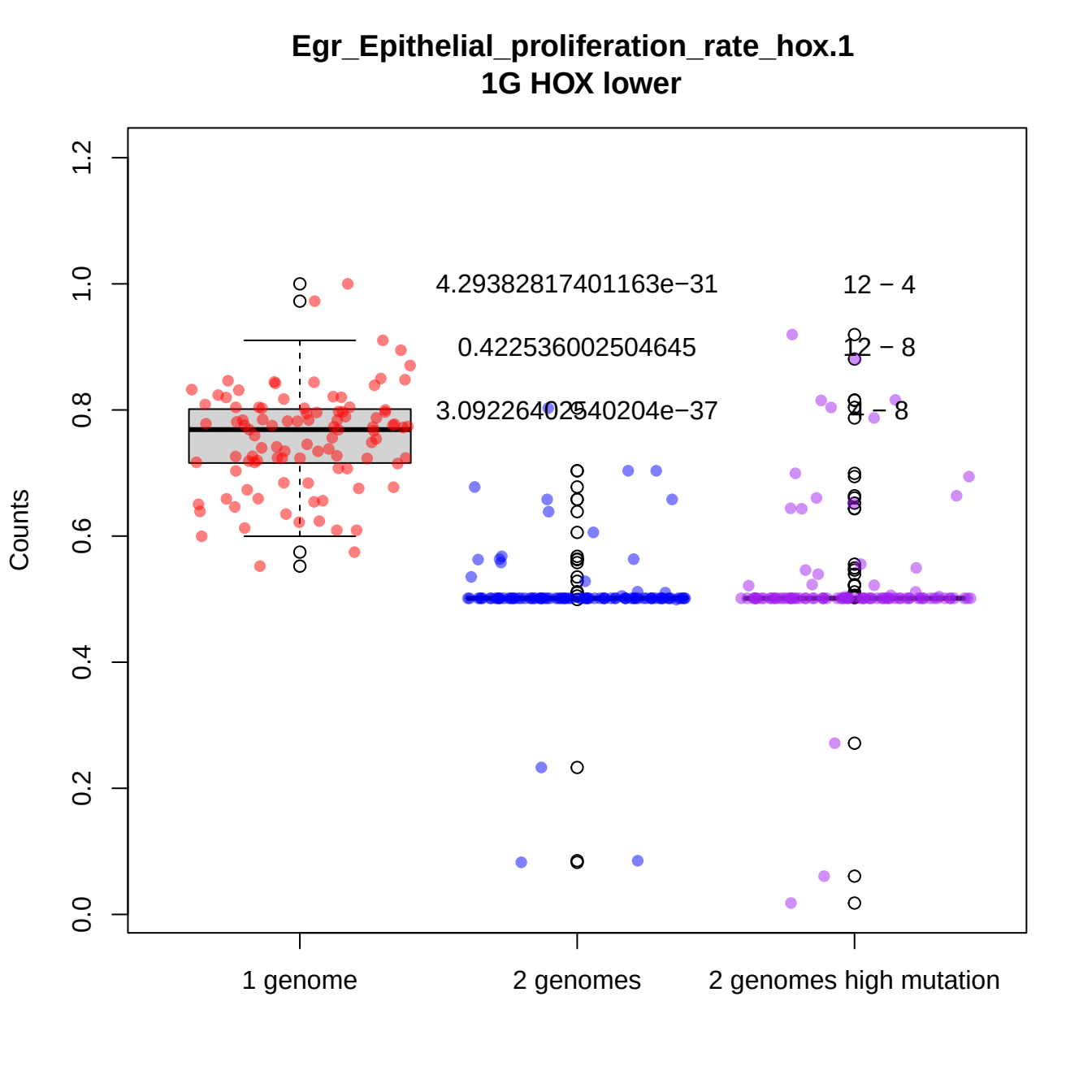

1.00
0.75
0.50
0.25
0.00
windows
30
40
50
fit
type
3
7
11

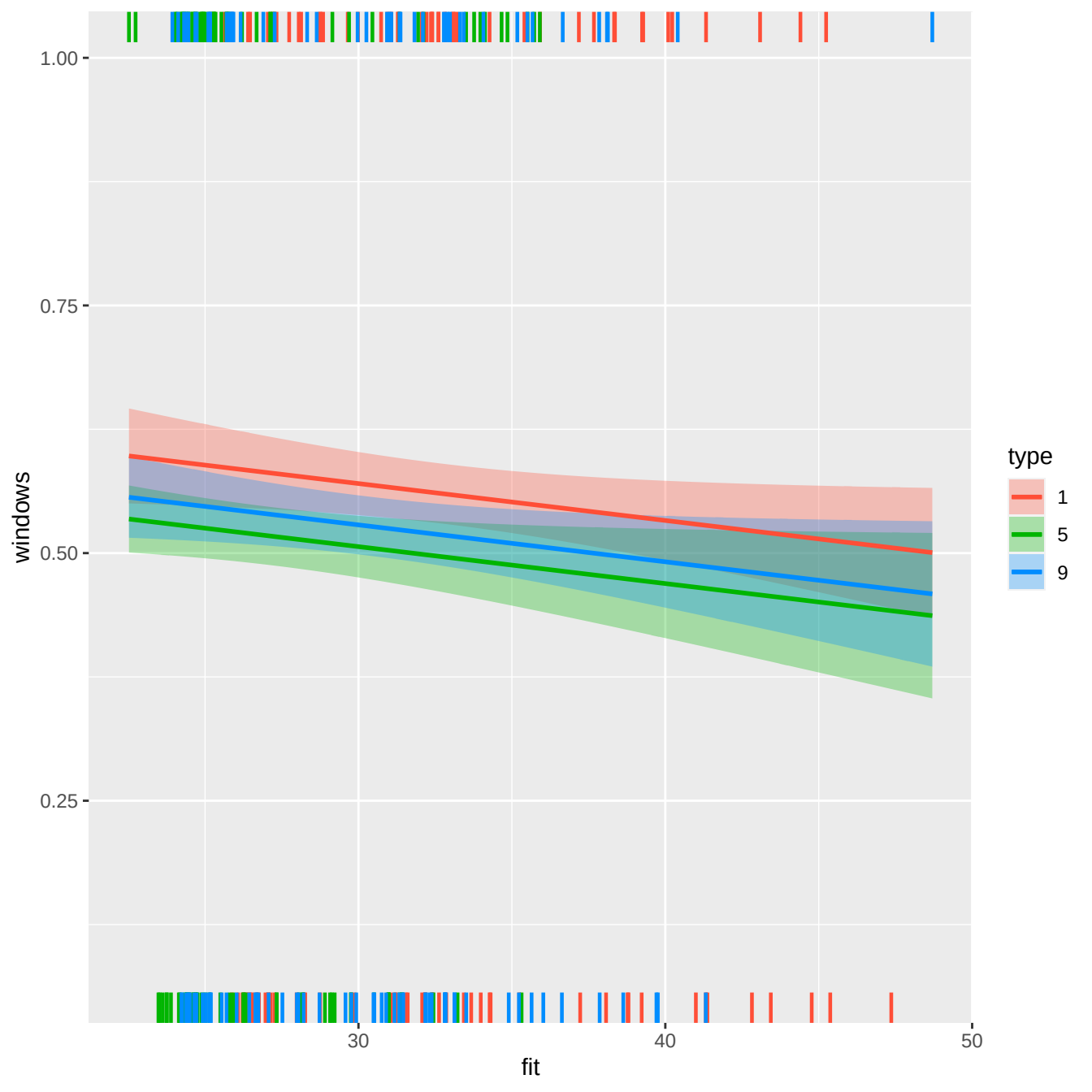

Egr_Epithelial_proliferation_rate_hox.1 1G HOX upepr
1 genome
2 genomes
2 genomes high mutation
Parameter range (normalized 0 to 1)
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

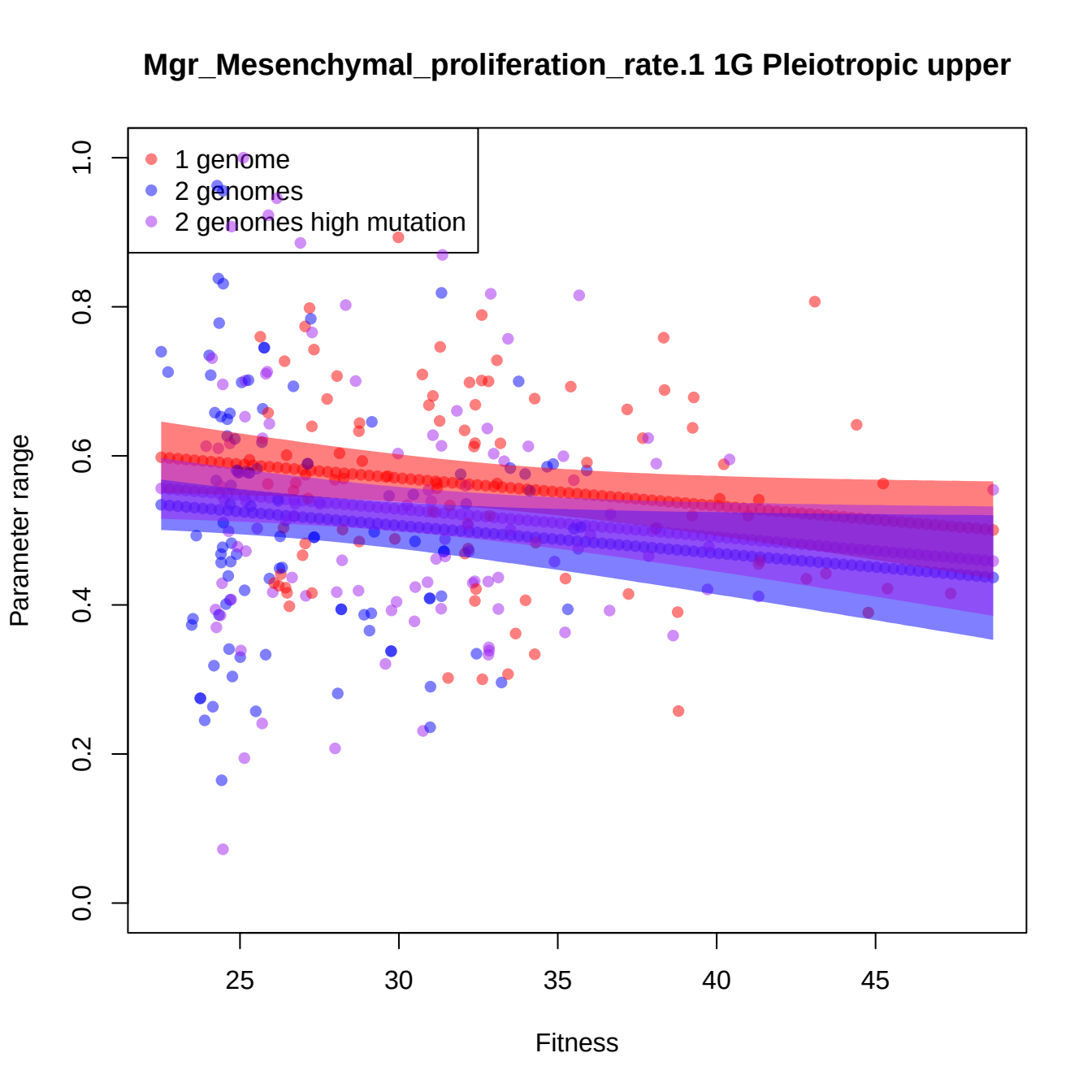

Egr_Epithelial_proliferation_rate_hox.1
1G HOX upepr
Counts
0.0
0.2
0.4
0.6
0.8
1.0
1.2
6.99319663639002e−31
0.953083389421785
6.02716484788846e−29
11 − 3
11 − 7
3 − 7
1 genome
2 genomes
2 genomes high mutation

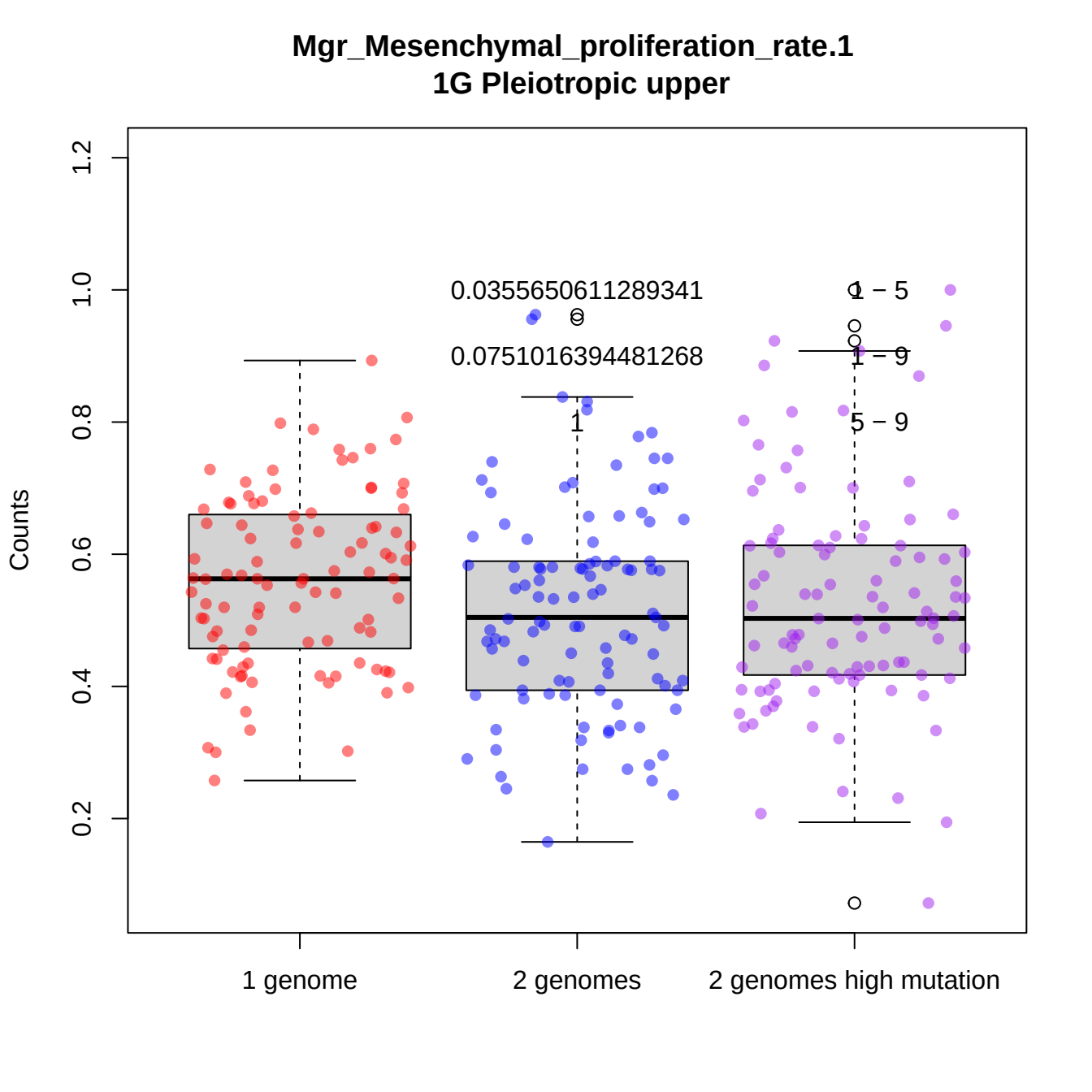

1.00
0.75
0.50
0.25
0.00
windows
30
40
50
fit
type
4
8
12

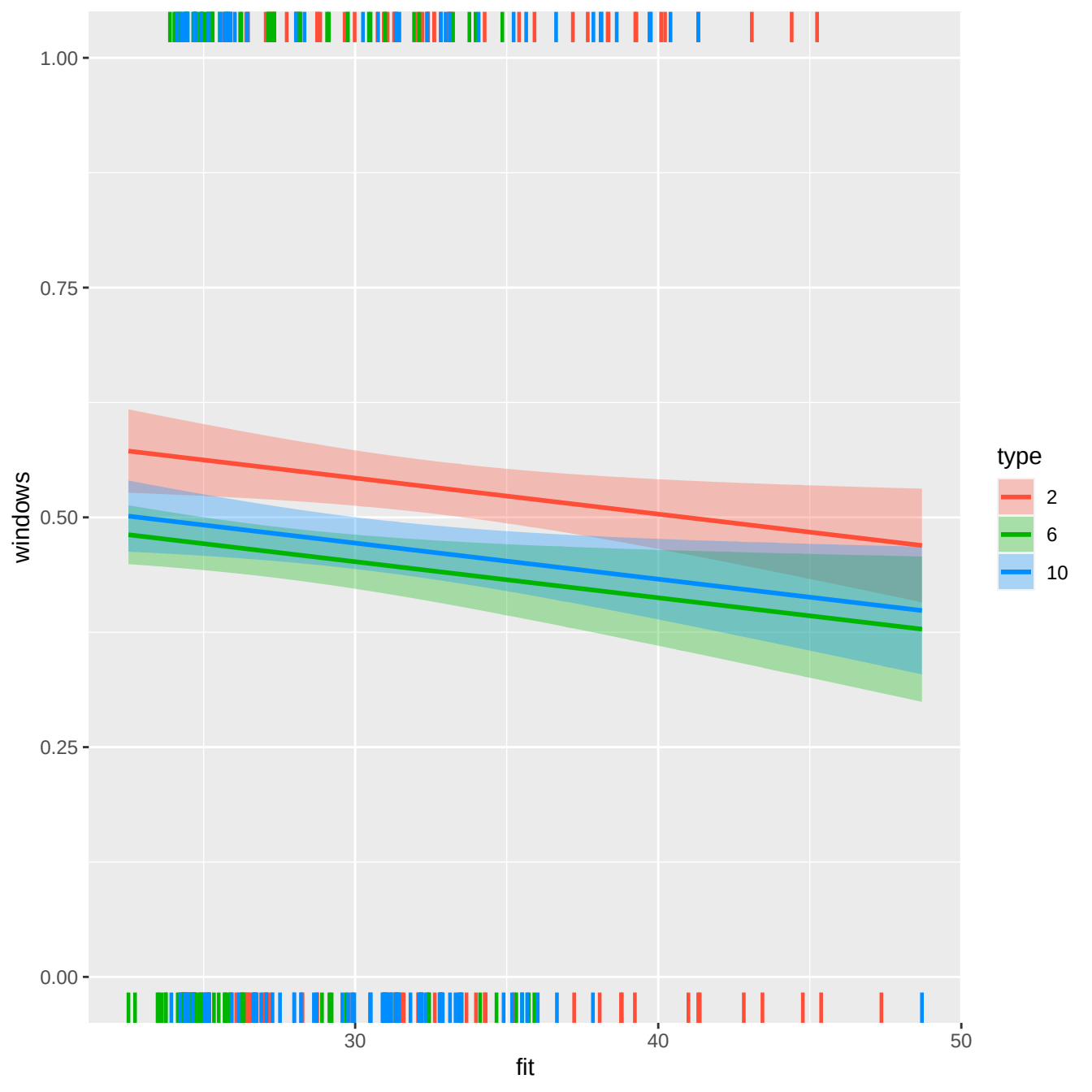

Egr_Epithelial_proliferation_rate_hox.1 1G HOX lower
1 genome
2 genomes
2 genomes high mutation
Parameter range (normalized 0 to 1)
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

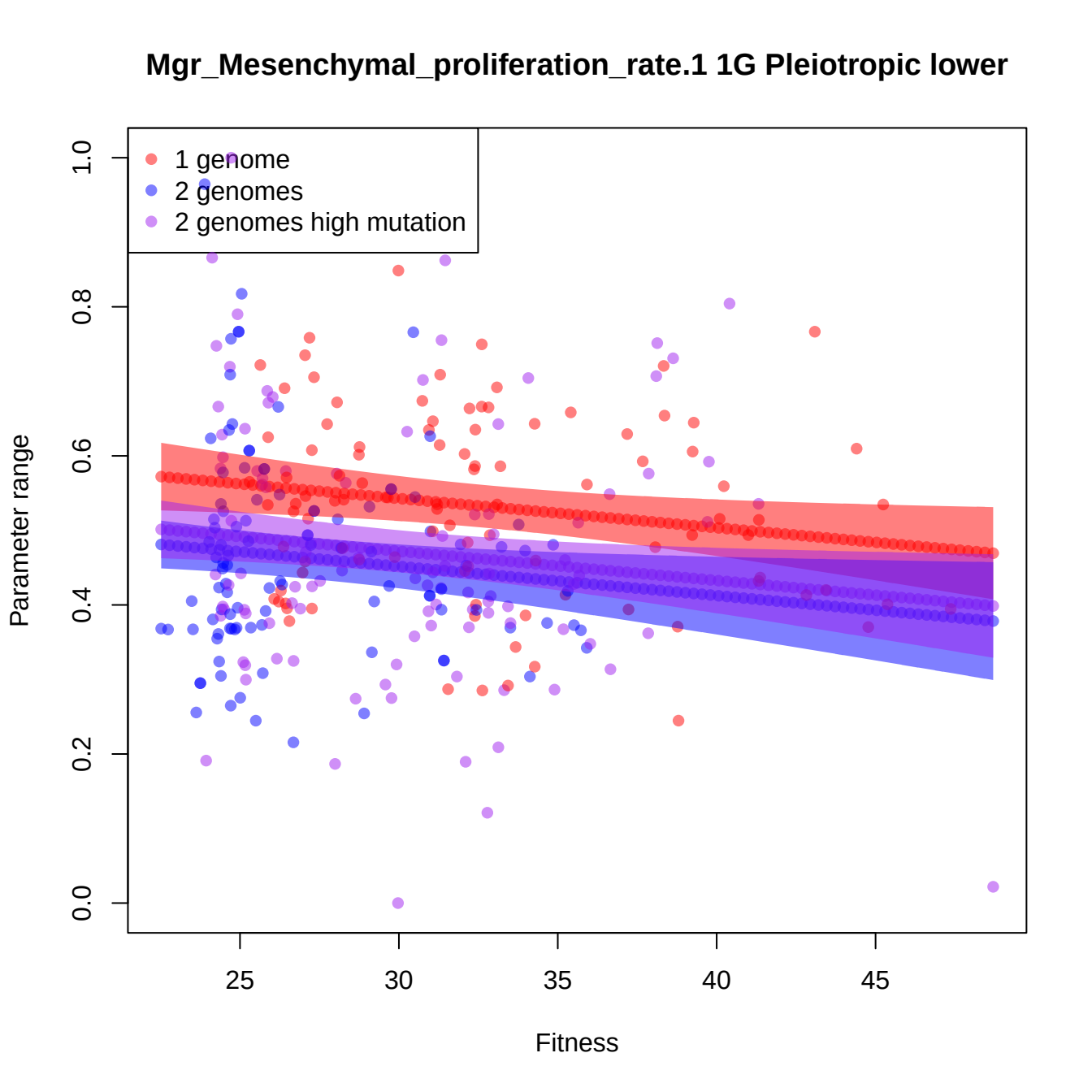

Egr_Epithelial_proliferation_rate_hox.1
1G HOX lower
Counts
0.0
0.2
0.4
0.6
0.8
1.0
1.2
4.29382817401163e−31
0.422536002504645
3.09226402540204e−37
12 − 4
12 − 8
4 − 8
1 genome
2 genomes
2 genomes high mutation

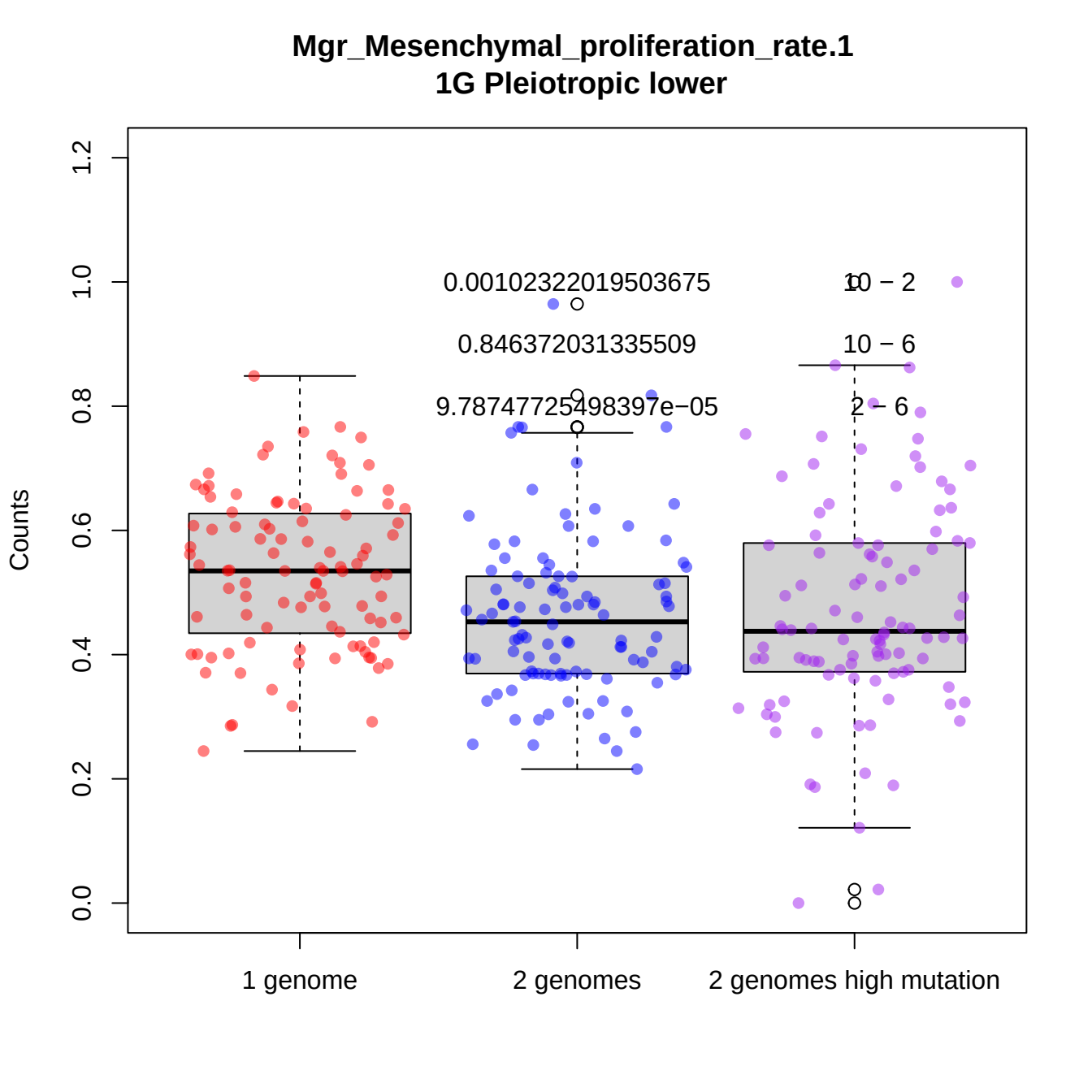

1.00
0.75
0.50
0.25
windows
30
40
50
fit
type
1
5
9

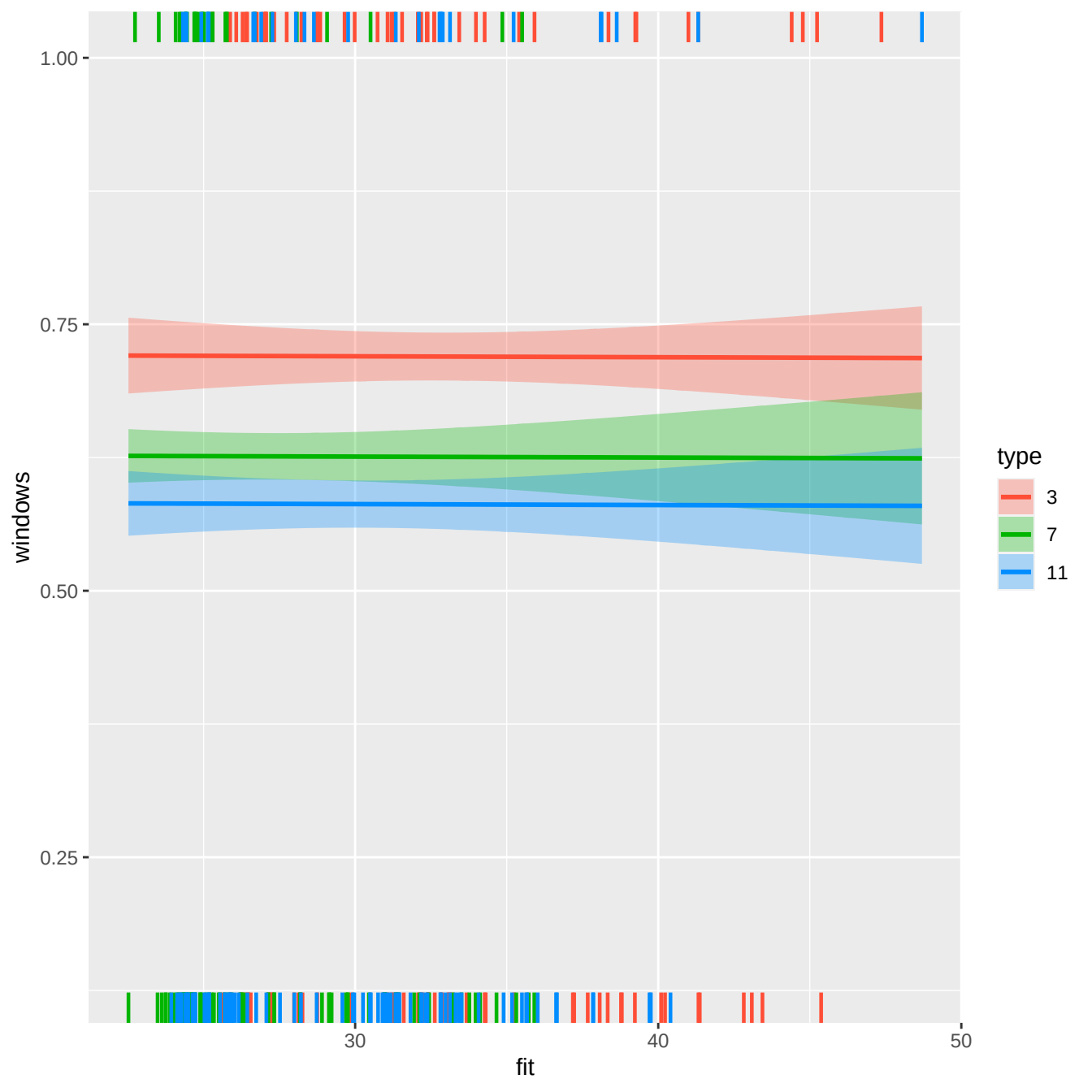

Mgr_Mesenchymal_proliferation_rate.1 1G Pleiotropic upper
1 genome
2 genomes
2 genomes high mutation
Parameter range
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

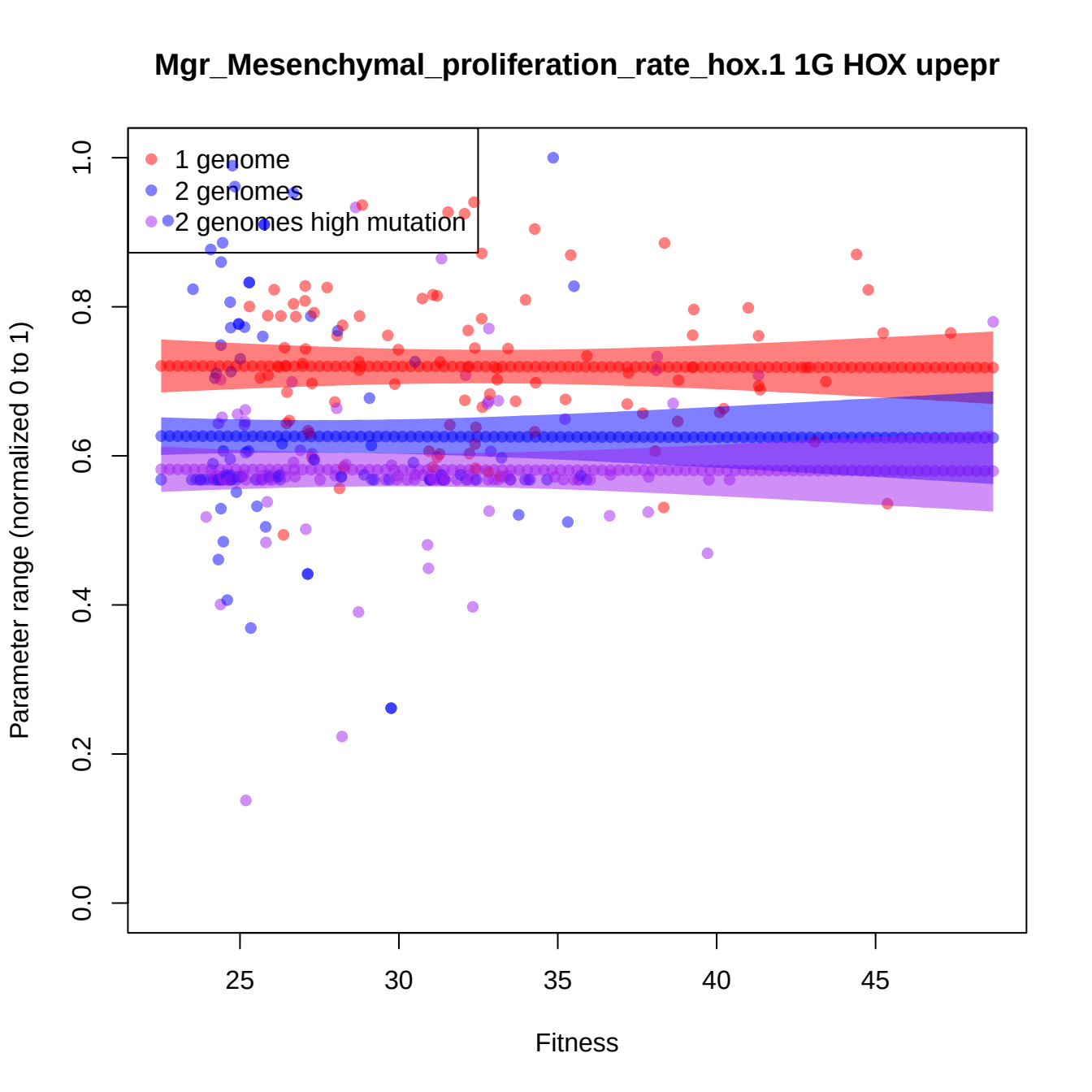

Mgr_Mesenchymal_proliferation_rate.1
1G Pleiotropic upper
Counts
1.2
1.0
0.8
0.6
0.4
0.2
0.0355650611289341
0.0751016394481268
1 − 5
1 − 9
5 − 9
1 genome
2 genomes
2 genomes high mutation

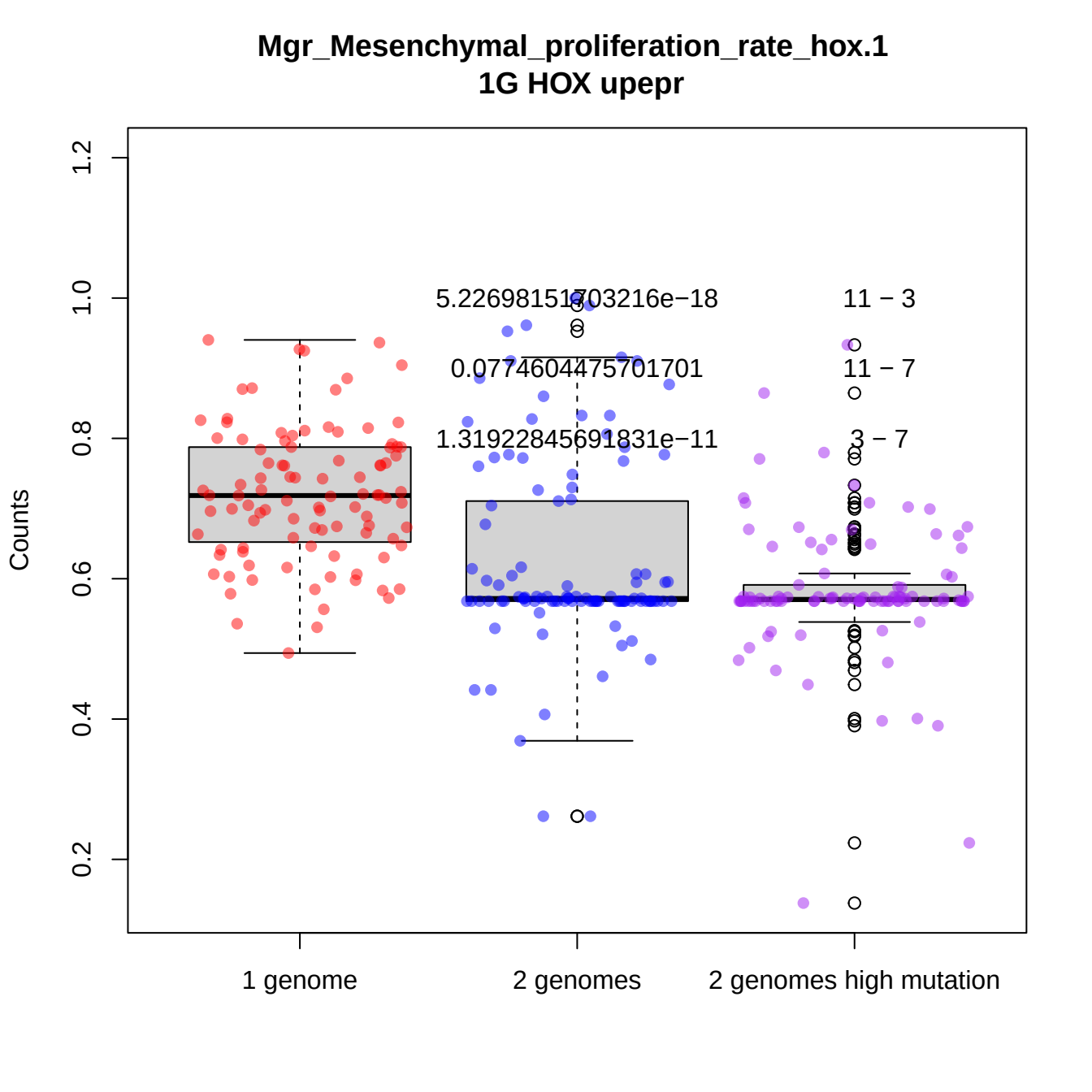

1.00
0.75
0.50
0.25
0.00
windows
30
40
50
fit
type
2
6
10

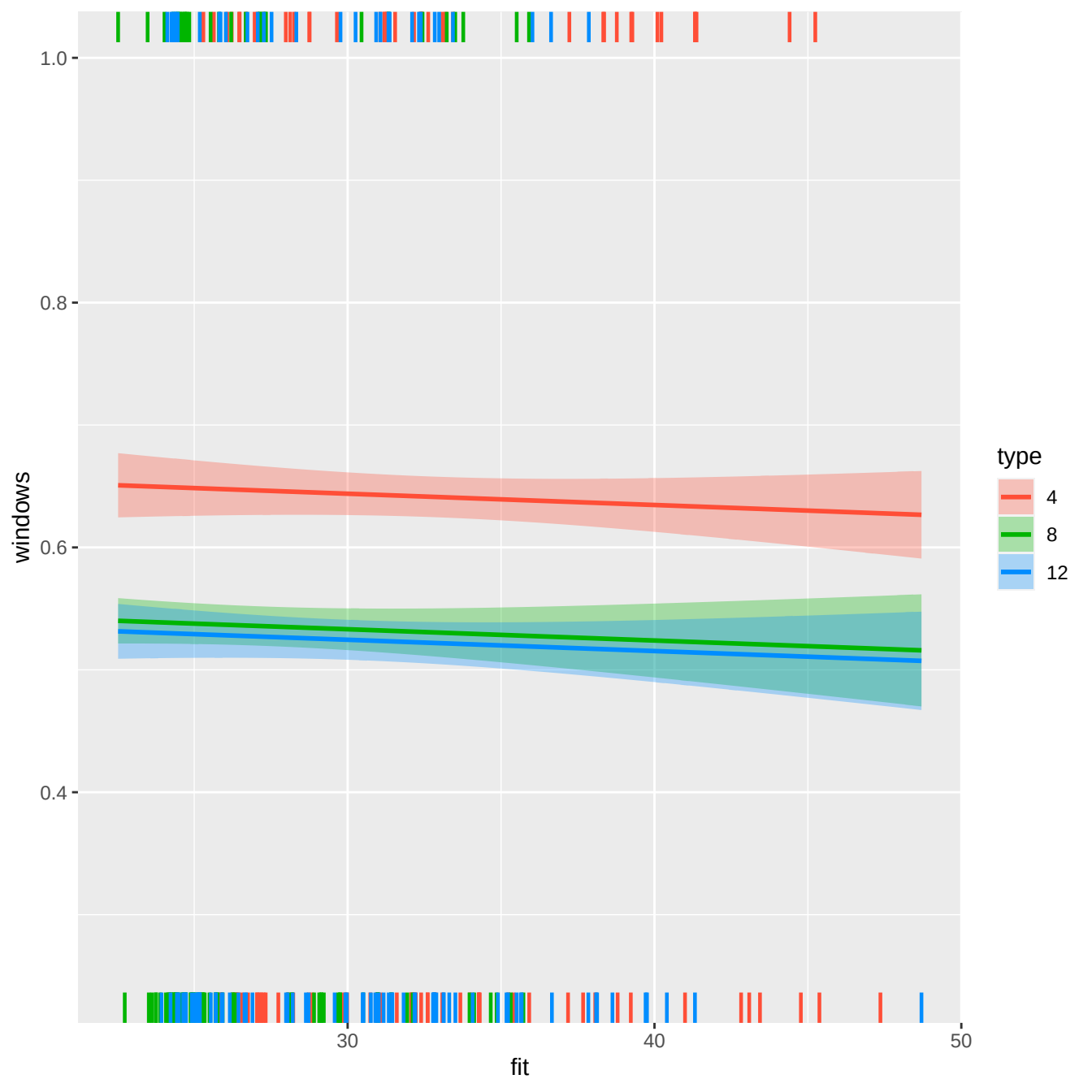

Mgr_Mesenchymal_proliferation_rate.1 1G Pleiotropic lower
1 genome
2 genomes
2 genomes high mutation
Parameter range
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

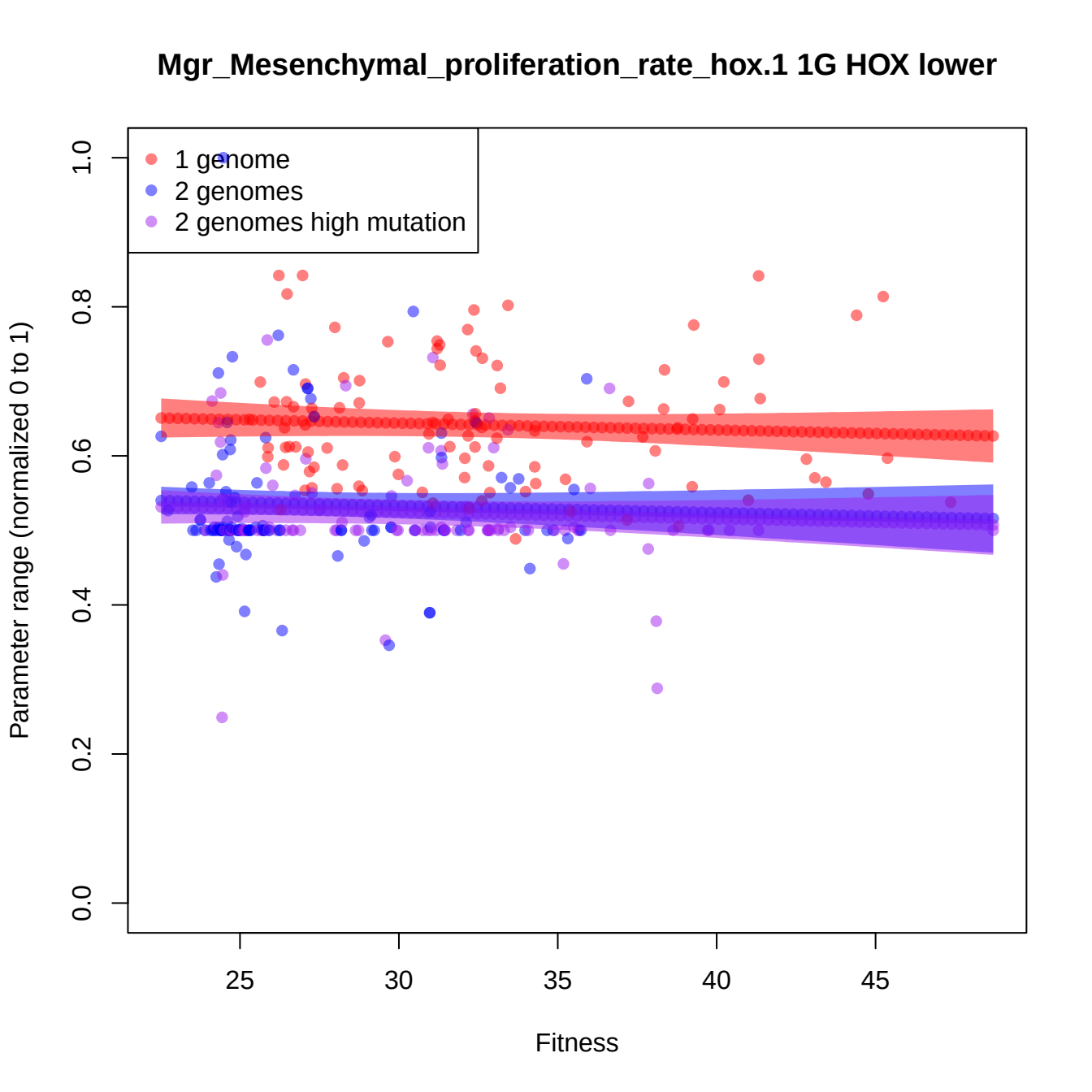

Mgr_Mesenchymal_proliferation_rate.1
1G Pleiotropic lower
Counts
0.0
0.2
0.4
0.6
0.8
1.0
1.2
0.00102322019503675
0.846372031335509
9.78747725498397e−05
10 − 2
10 − 6
2 − 6
1 genome
2 genomes
2 genomes high mutation

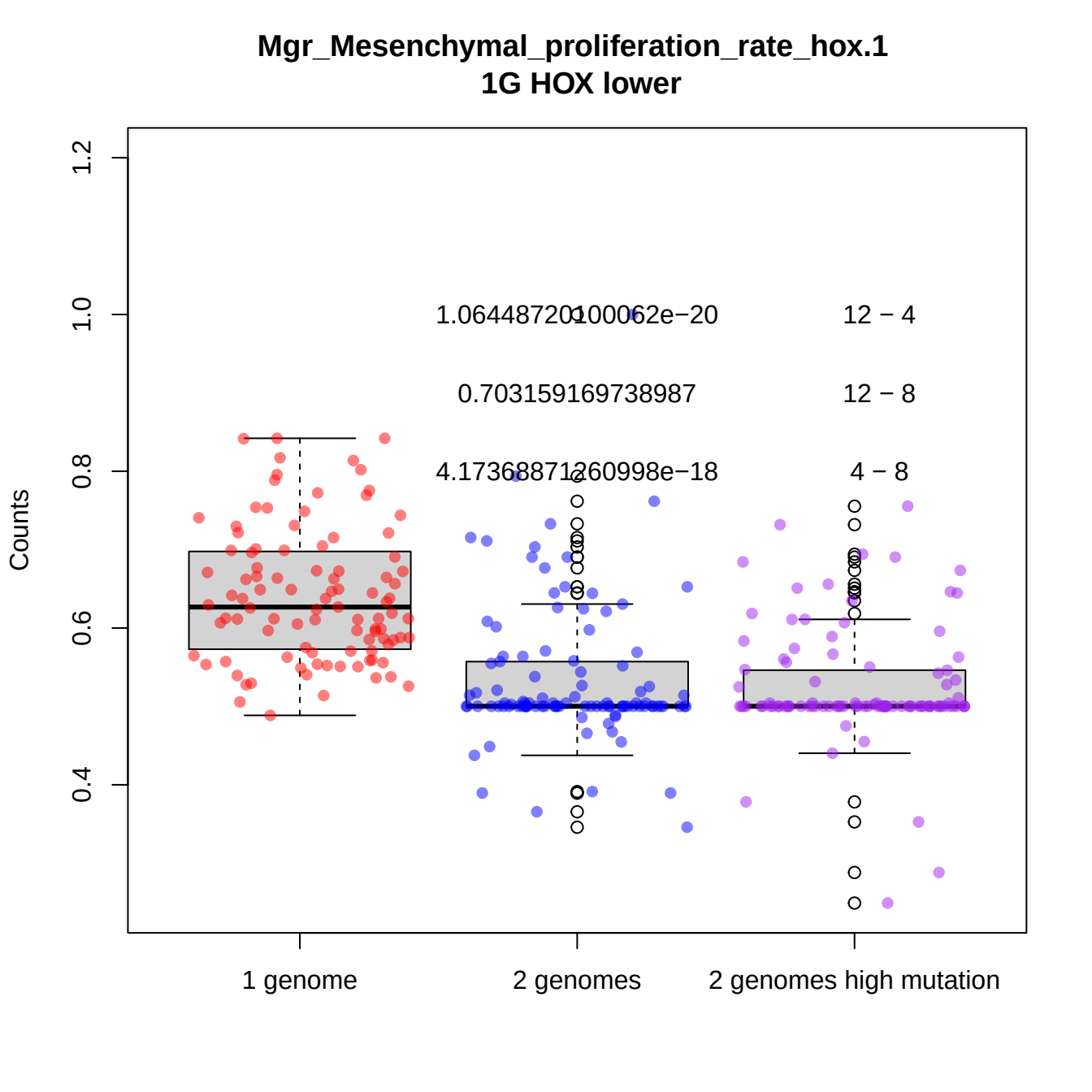

1.00
0.75
0.50
0.25
windows
30
40
50
fit
type
3
7
11

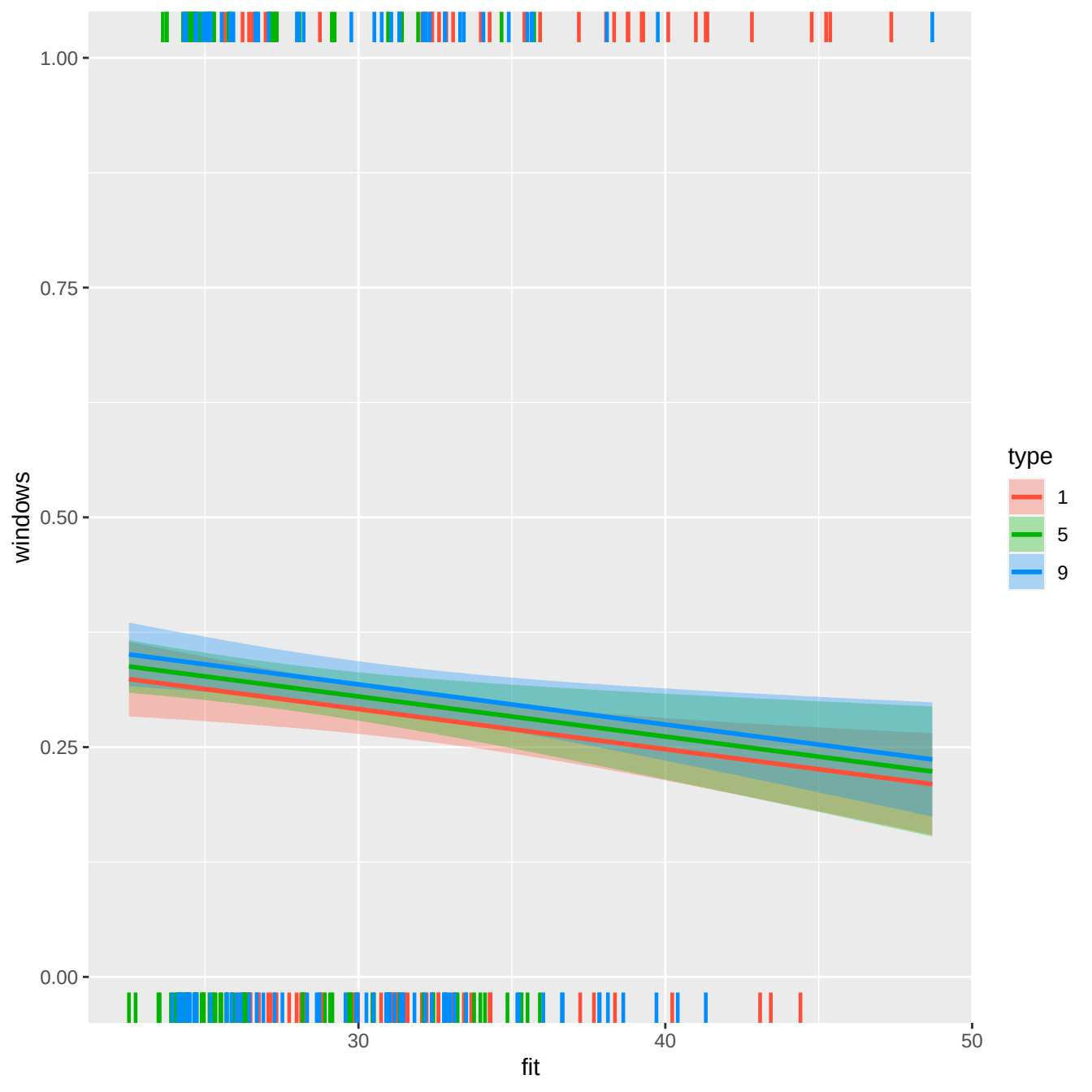

Mgr_Mesenchymal_proliferation_rate_hox.1 1G HOX upepr
1 genome
2 genomes
2 genomes high mutation
Parameter range (normalized 0 to 1)
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

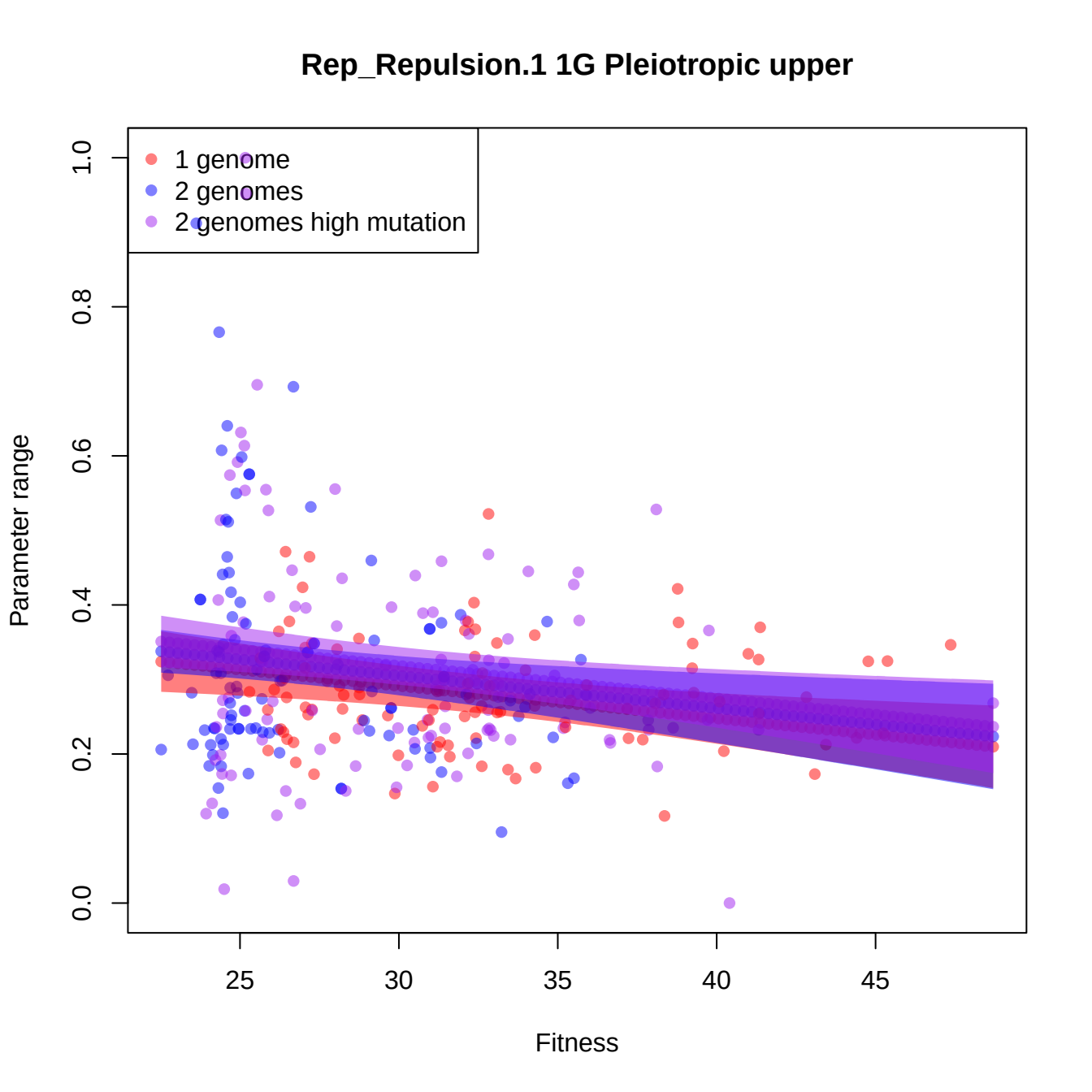

Mgr_Mesenchymal_proliferation_rate_hox.1
1G HOX upepr
Counts
1.2
1.0
0.8
0.6
0.4
0.2
5.226981517703216e−18
0.07746044757017 01
1.319228456918 31e−11
11 − 3
11 − 7
3 − 7
1 genome
2 genomes
2 genomes high mutation

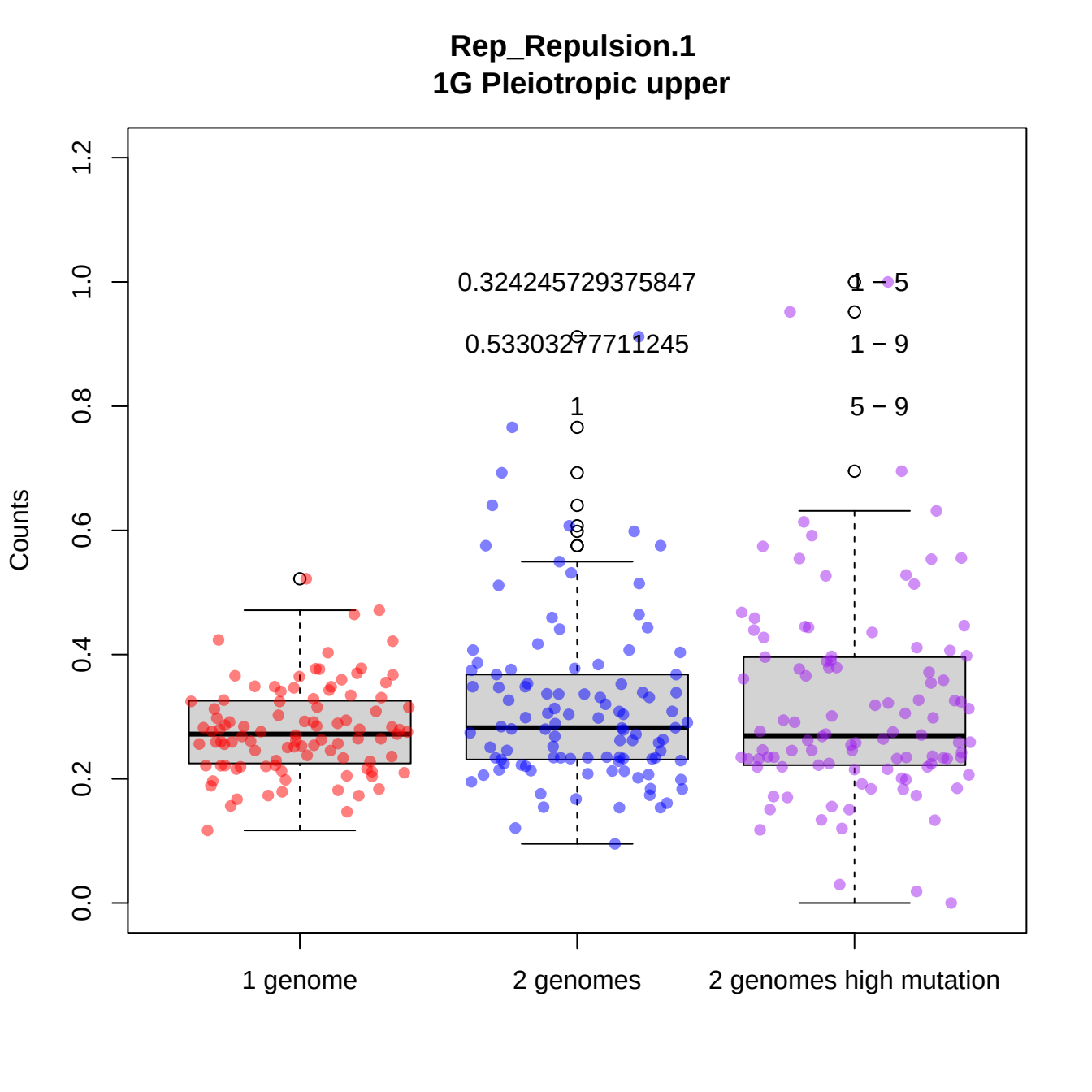

1.0
0.8
0.6
0.4
windows
30
40
50
fit
type
4
8
12

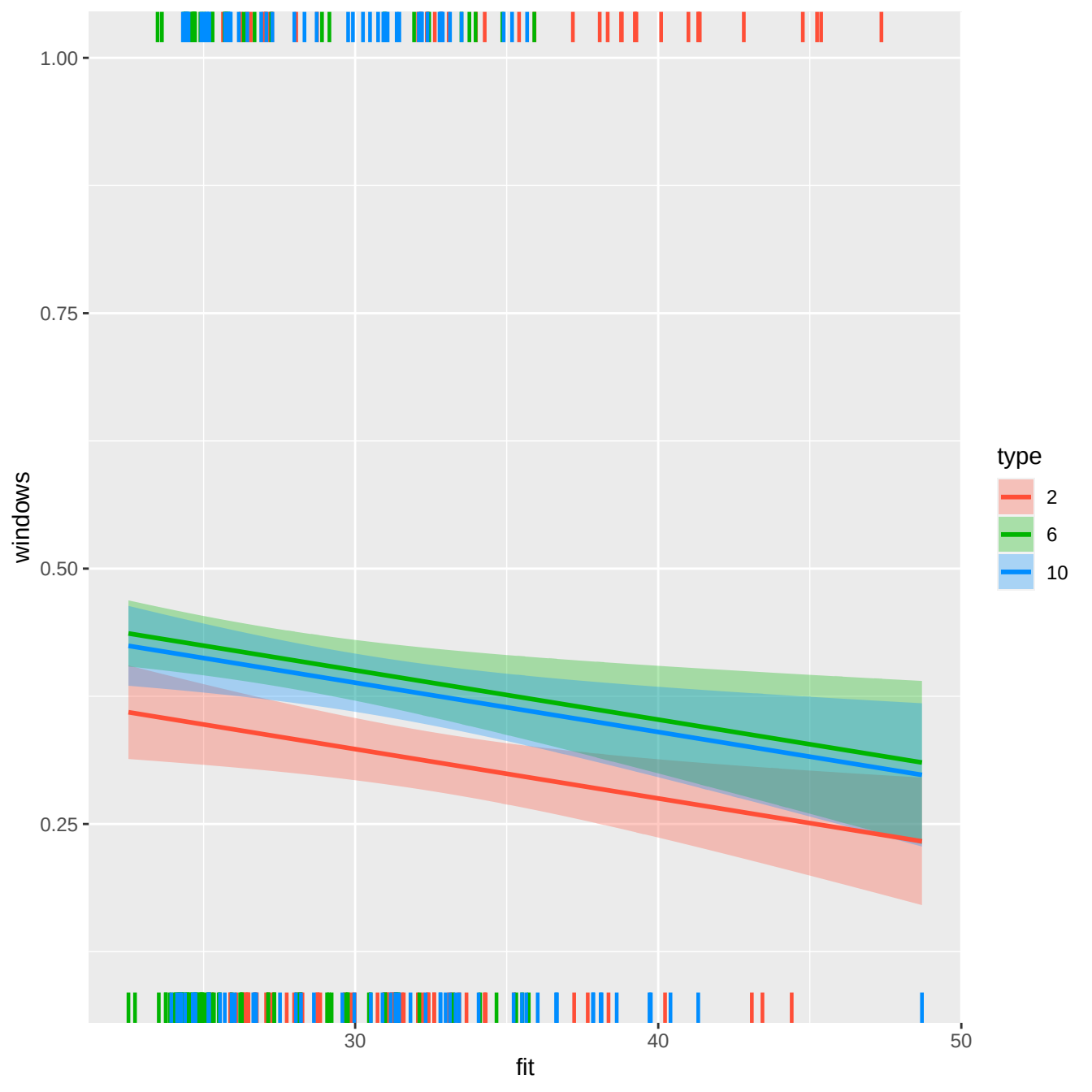

Mgr_Mesenchymal_proliferation_rate_hox.1 1G HOX lower
1 genome
2 genomes
2 genomes high mutation
Parameter range (normalized 0 to 1)
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

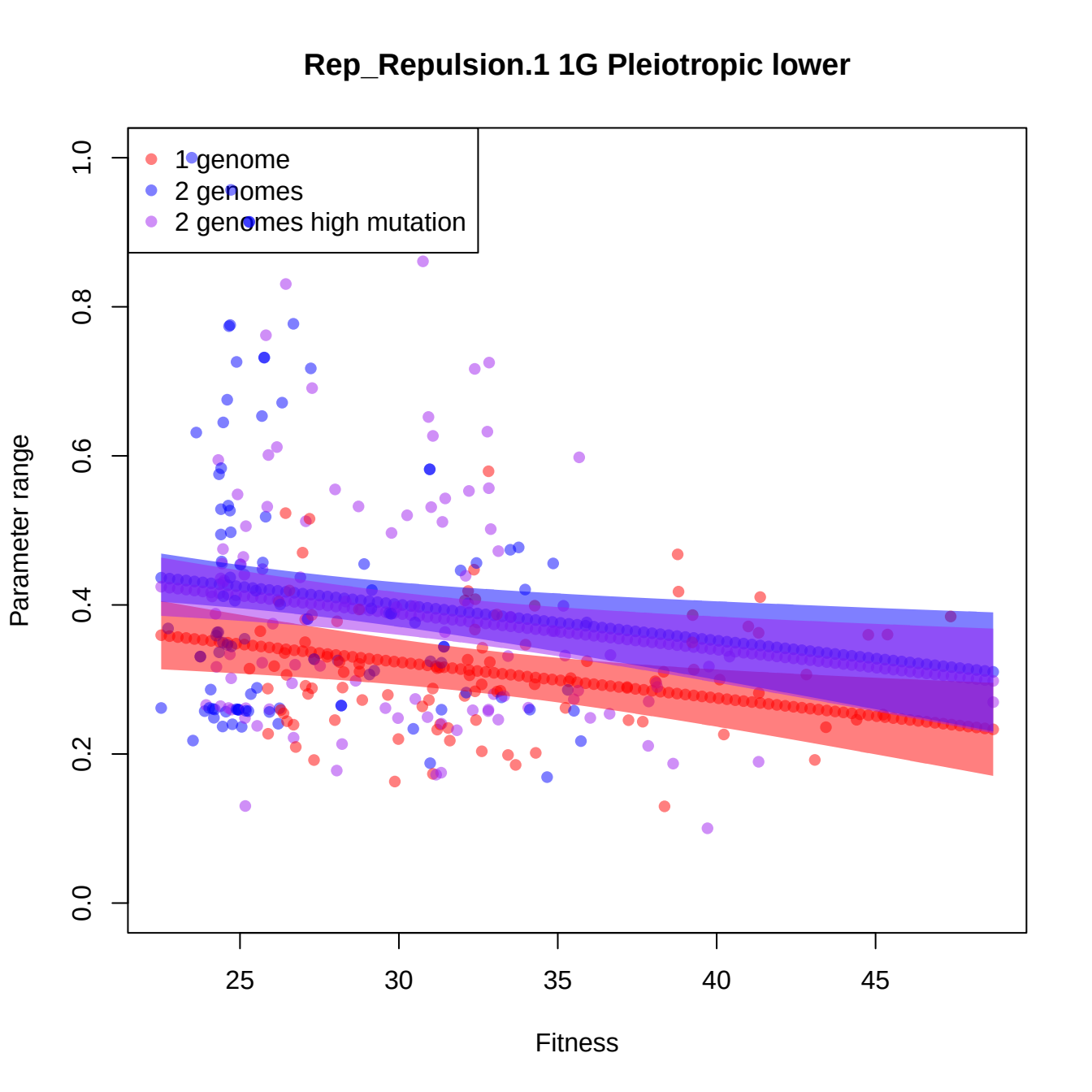

Mgr_Mesenchymal_proliferation_rate_hox.1
1G HOX lower
Counts
1.2
1.0
0.8
0.6
0.4
1.0644872001000062e−20
0.703159169738987
4.1736887126098e−18
12 − 4
12 − 8
4 − 8
1 genome
2 genomes
2 genomes high mutation

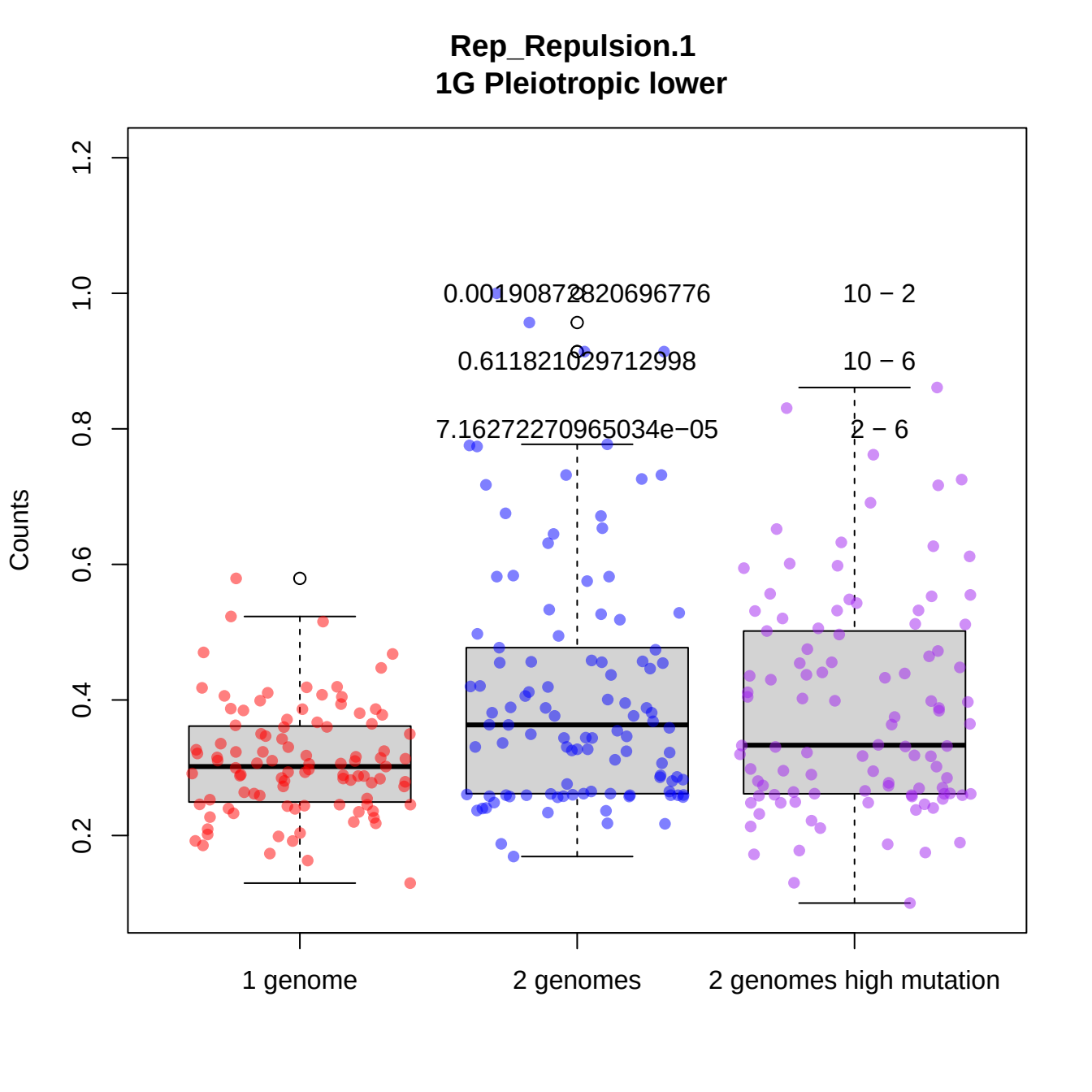

1.00
0.75
0.50
0.25
0.00
windows
30
40
50
fit
type
1
5
9

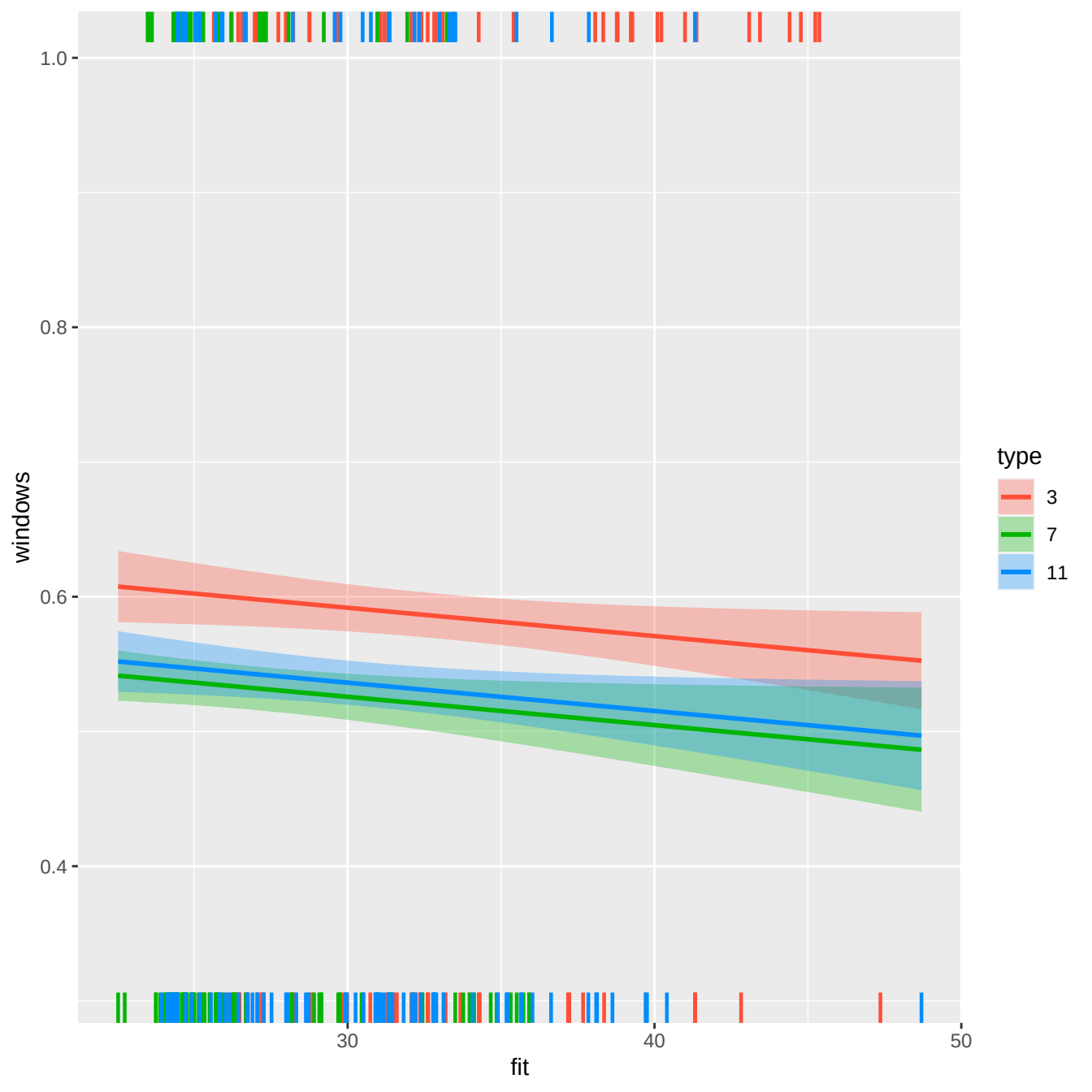

Rep_Repulsion.1 1G Pleiotropic upper
1 genome
2 genomes
2 genomes high mutation
Parameter range
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

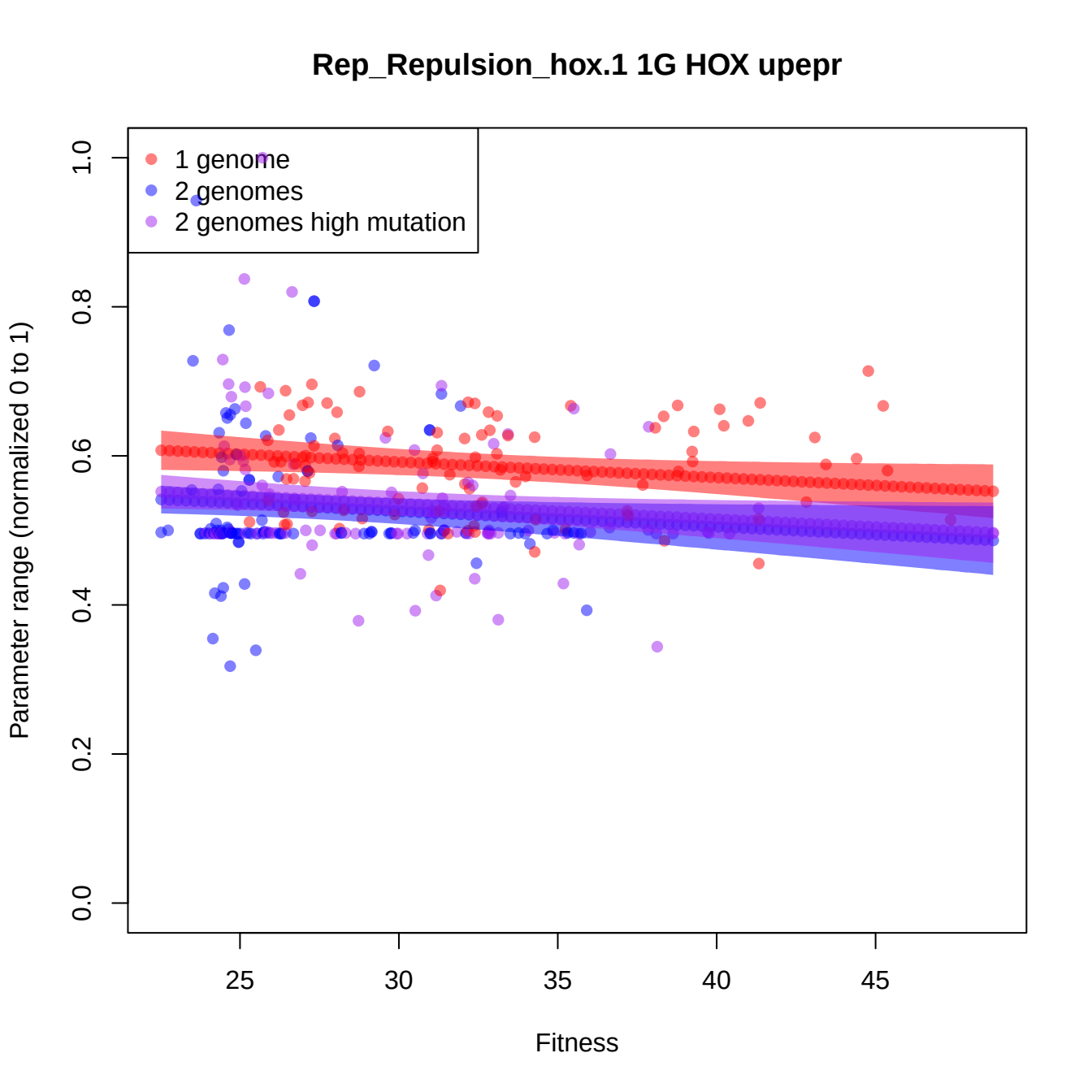

Rep_Repulsion.1
1G Pleiotropic upper
Counts
0.0
0.2
0.4
0.6
0.8
1.0
1.2
0.324245729375847
0.533032777111245
1
1 − 5
1 − 9
5 − 9
1 genome
2 genomes
2 genomes high mutation

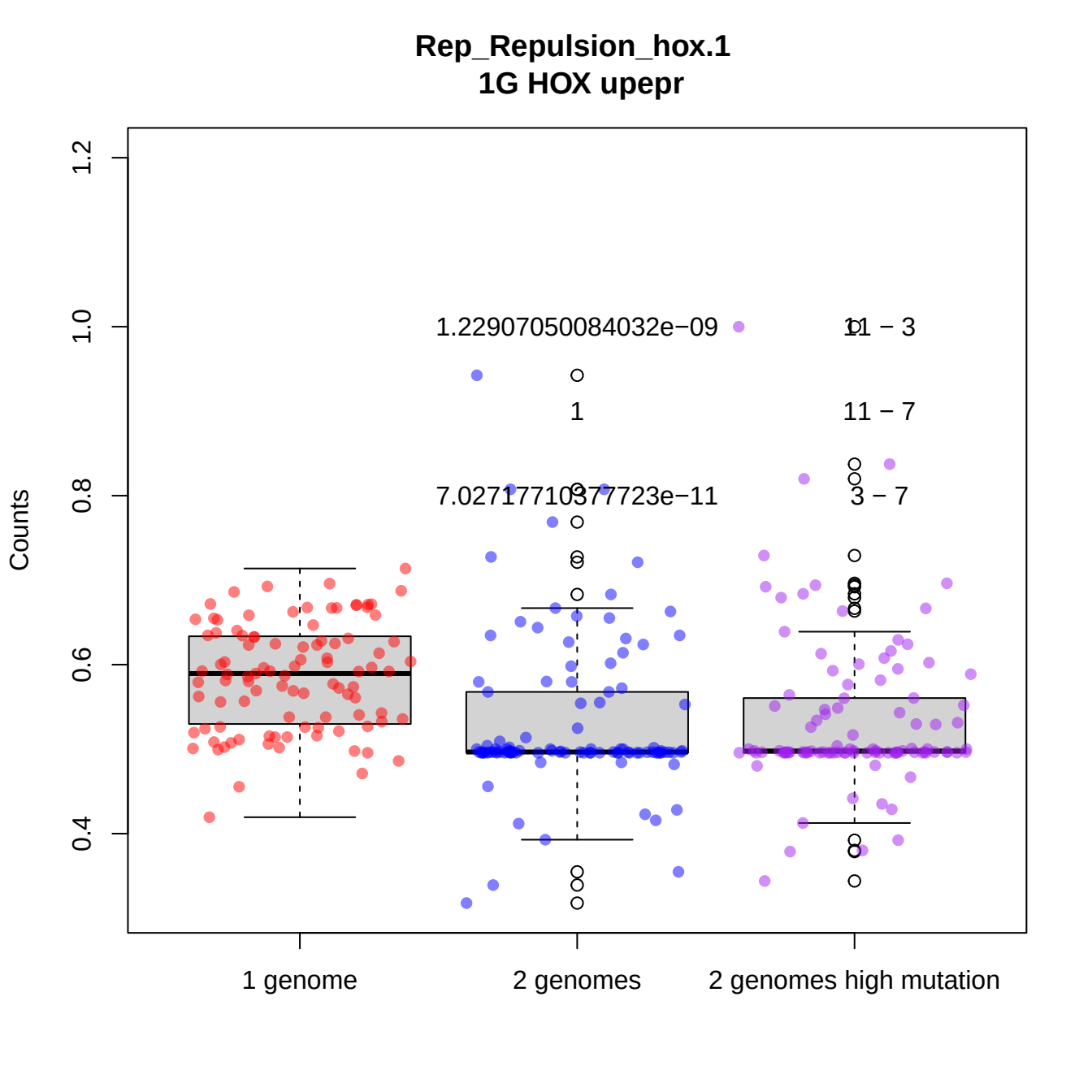

1.00
0.75
0.50
0.25
windows
30
40
50
fit
type
2
6
10

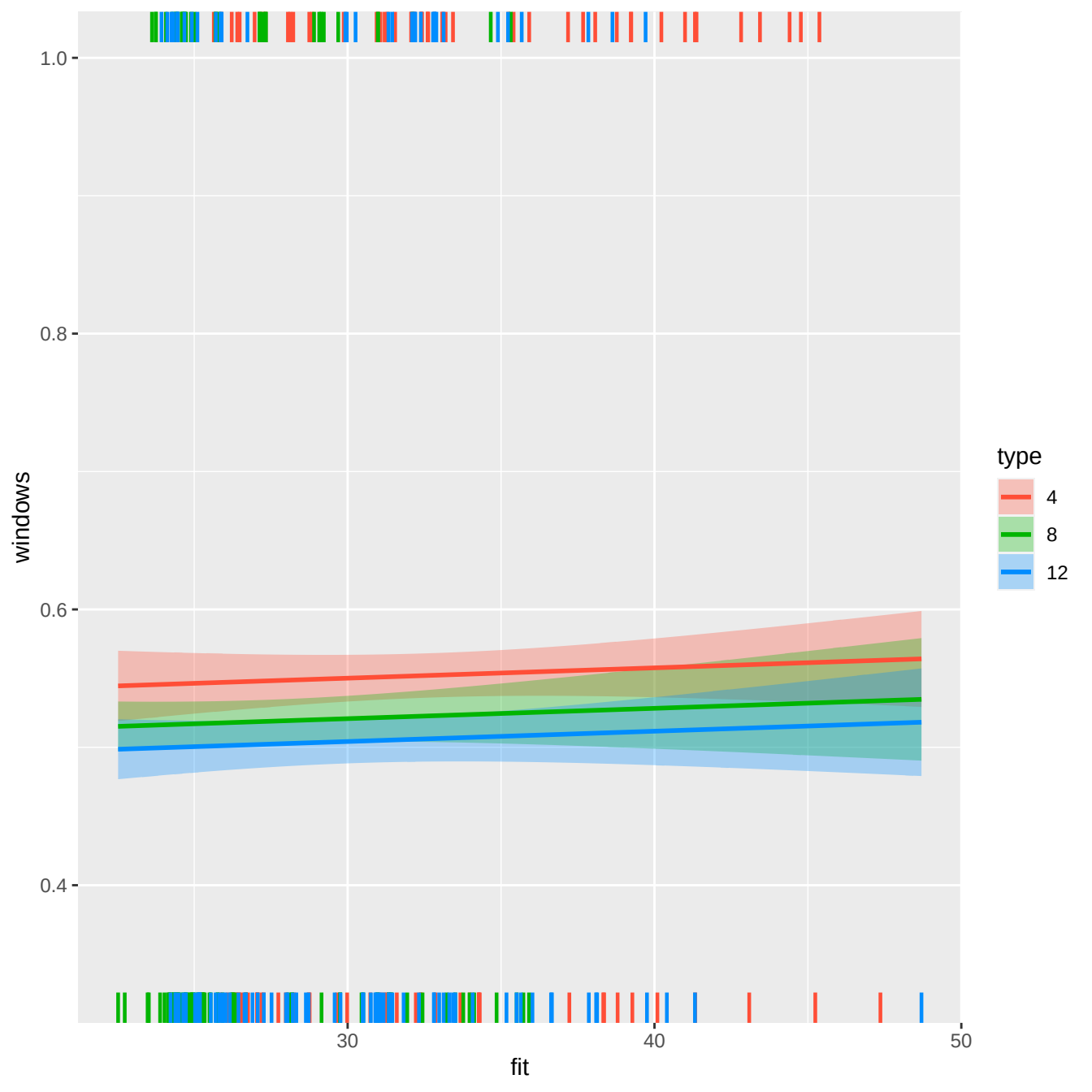

Rep_Repulsion.1 1G Pleiotropic lower
1 genome
2 genomes
2 genomes high mutation
Parameter range
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

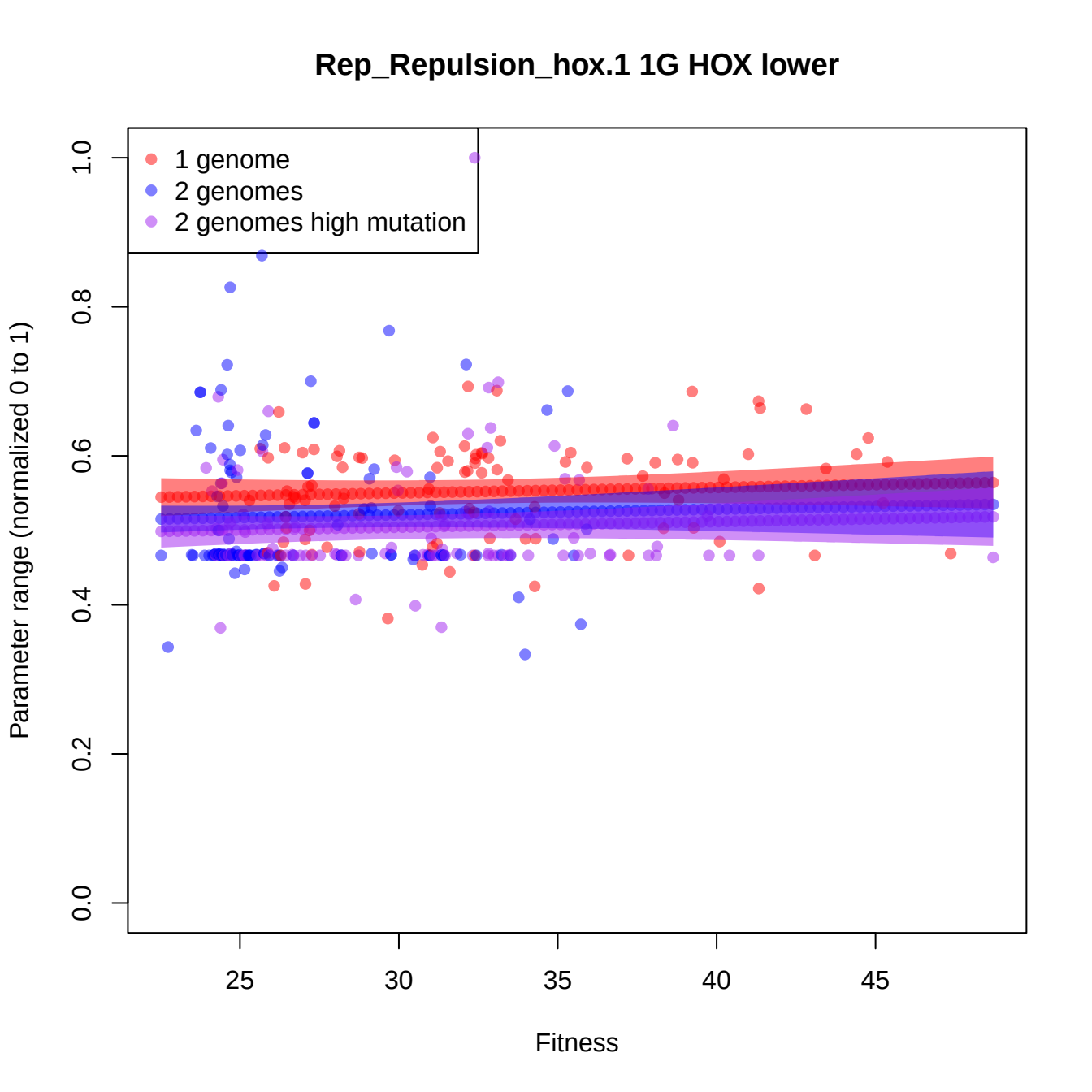

Rep_Repulsion.1
1G Pleiotropic lower
Counts
1.2
1.0
0.8
0.6
0.4
0.2
0.0019087282069677 6
0.6118210297129 98
7.16272270965034e−05
10 − 2
10 − 6
2 − 6
1 genome
2 genomes
2 genomes high mutation

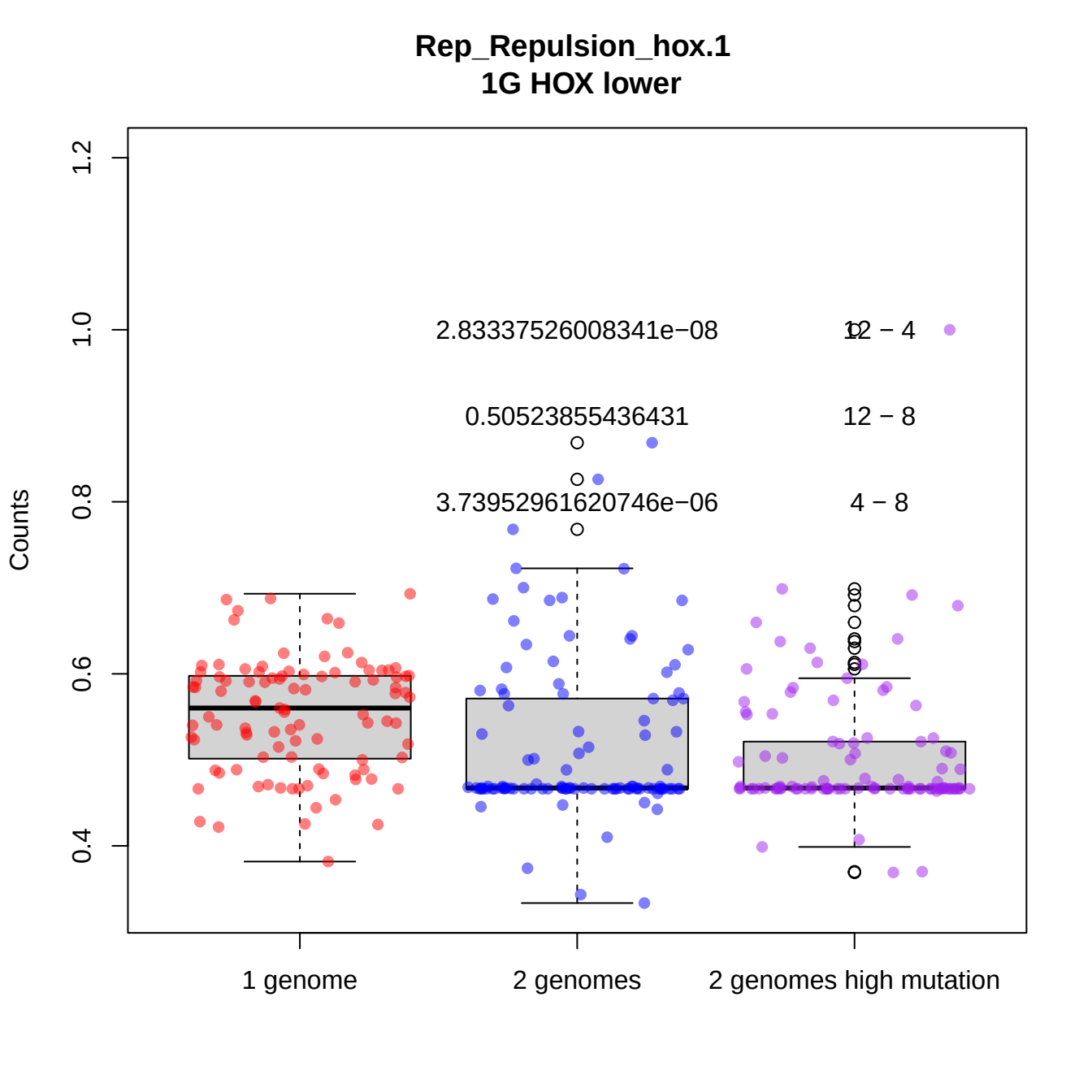

1.0
0.8
0.6
0.4
windows
30
40
50
fit
type
3
7
11

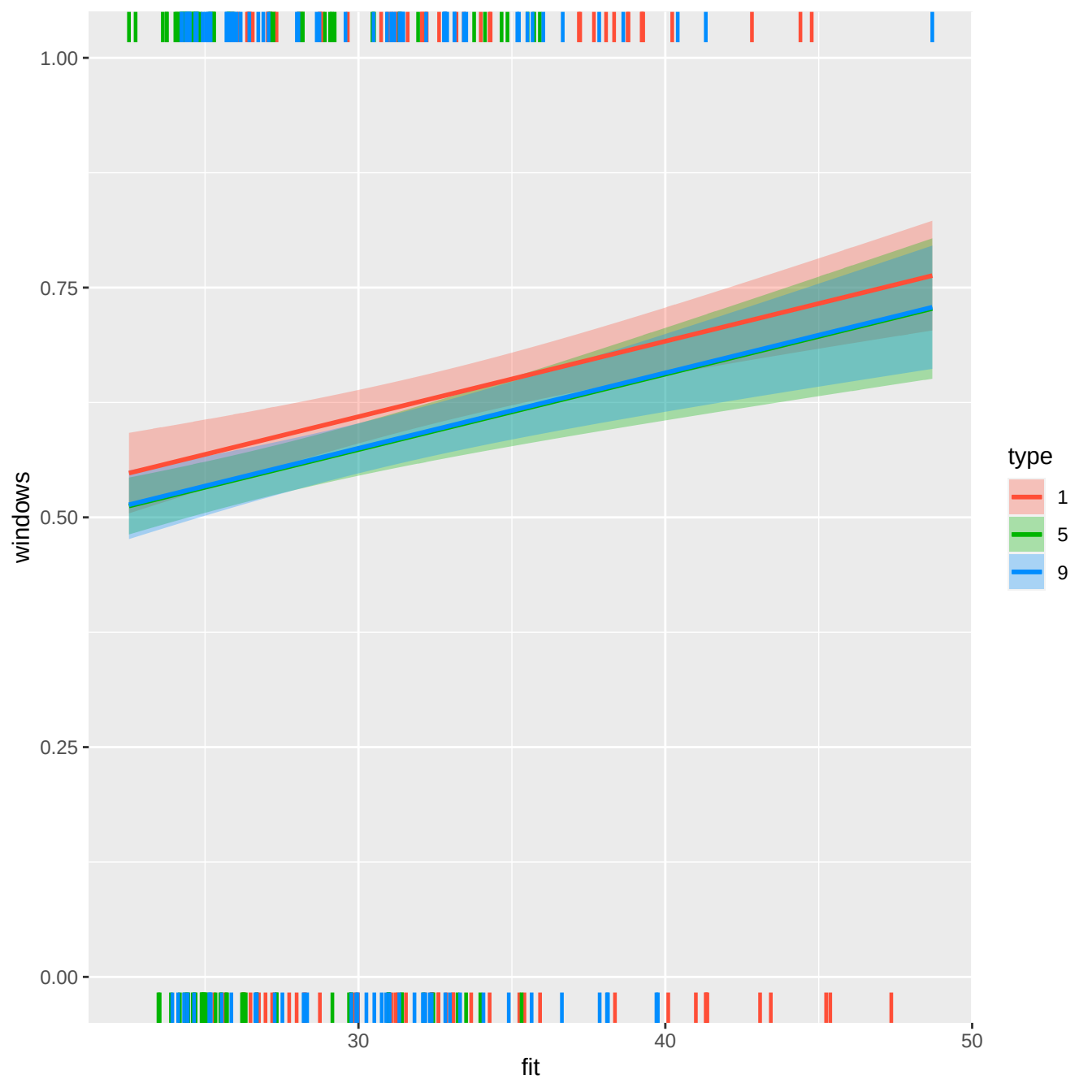

Rep_Repulsion_hox.1 1G HOX upepr
1 genome
2 genomes
2 genomes high mutation
Parameter range (normalized 0 to 1)
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

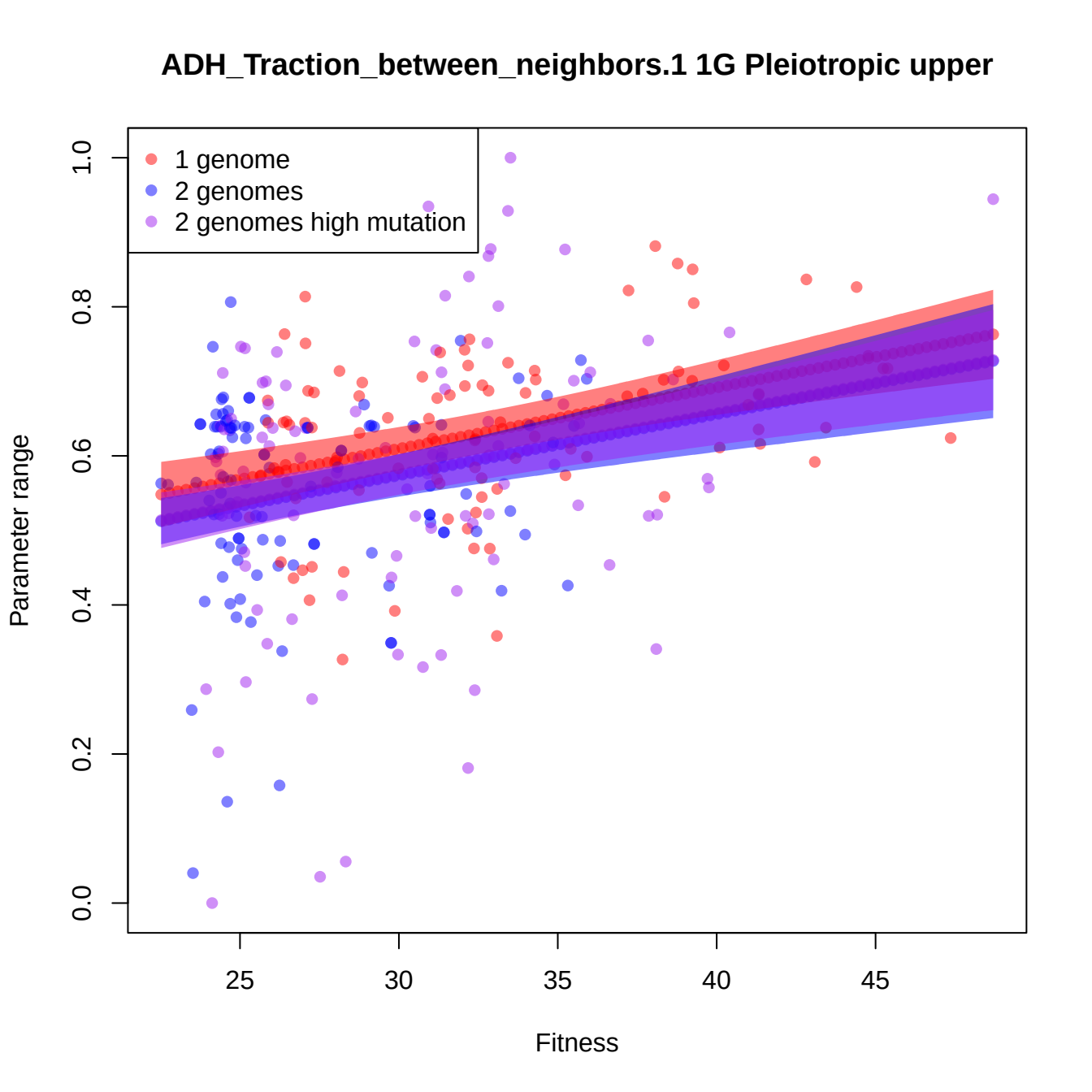

Rep_Repulsion_hox.1
1G HOX upepr
Counts
1.2
1.0
0.8
0.6
0.4
1.22907050084032e−09
1
7.02717710377723e−11
11 − 3
11 − 7
3 − 7
1 genome
2 genomes
2 genomes high mutation

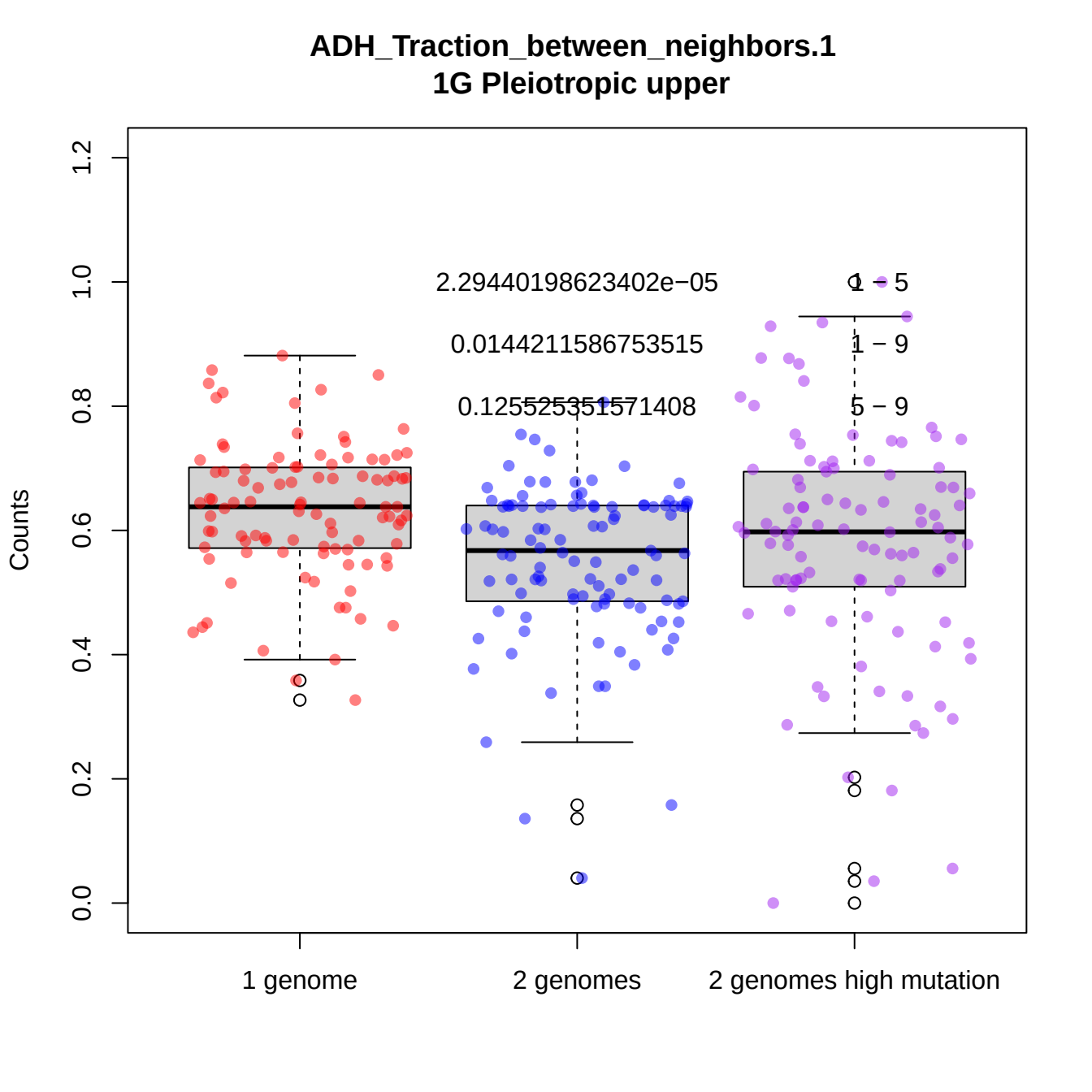

1.0
0.8
0.6
0.4
windows
30
40
50
fit
type
4
8
12

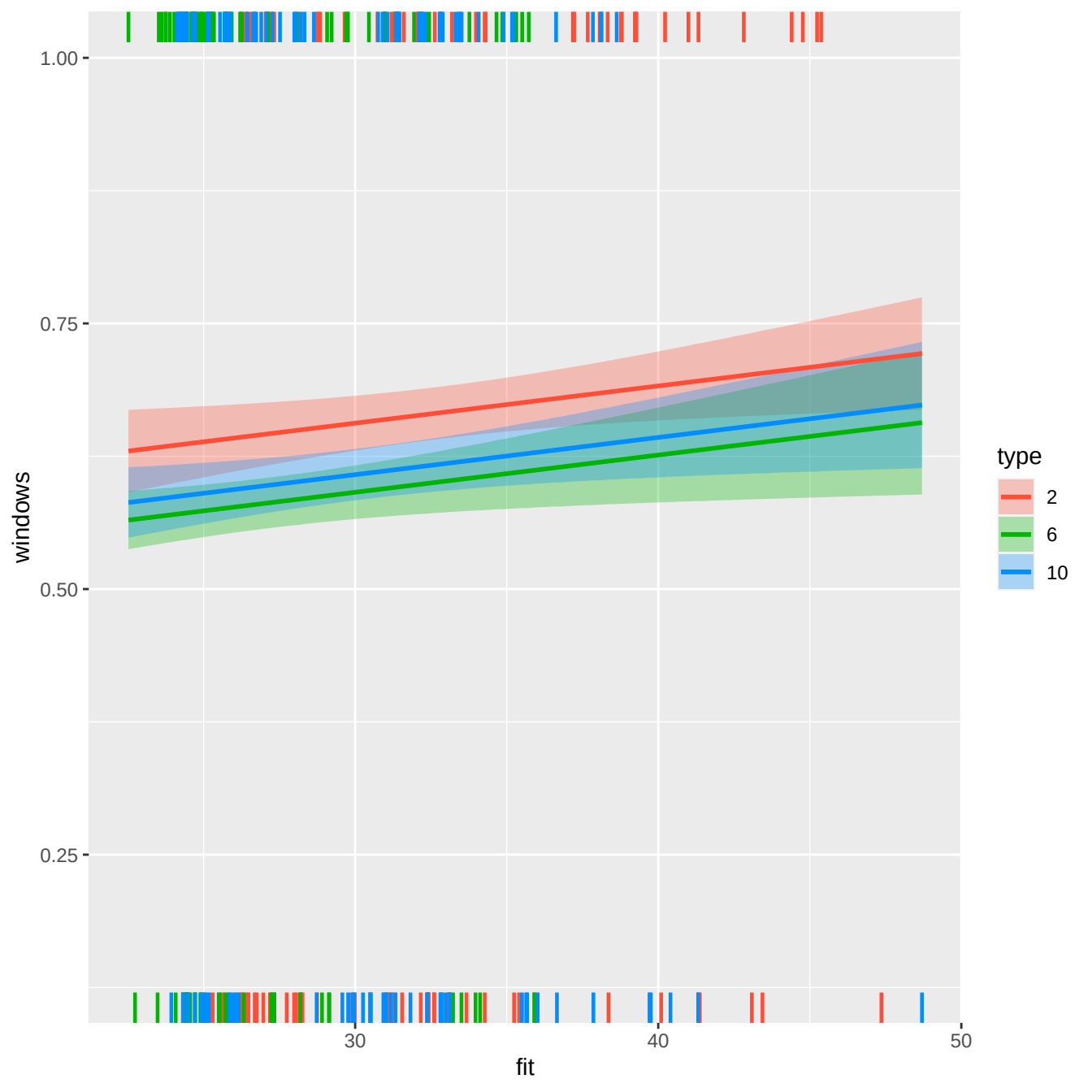

Rep_Repulsion_hox.1 1G HOX lower
1 genome
2 genomes
2 genomes high mutation
Parameter range (normalized 0 to 1)
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

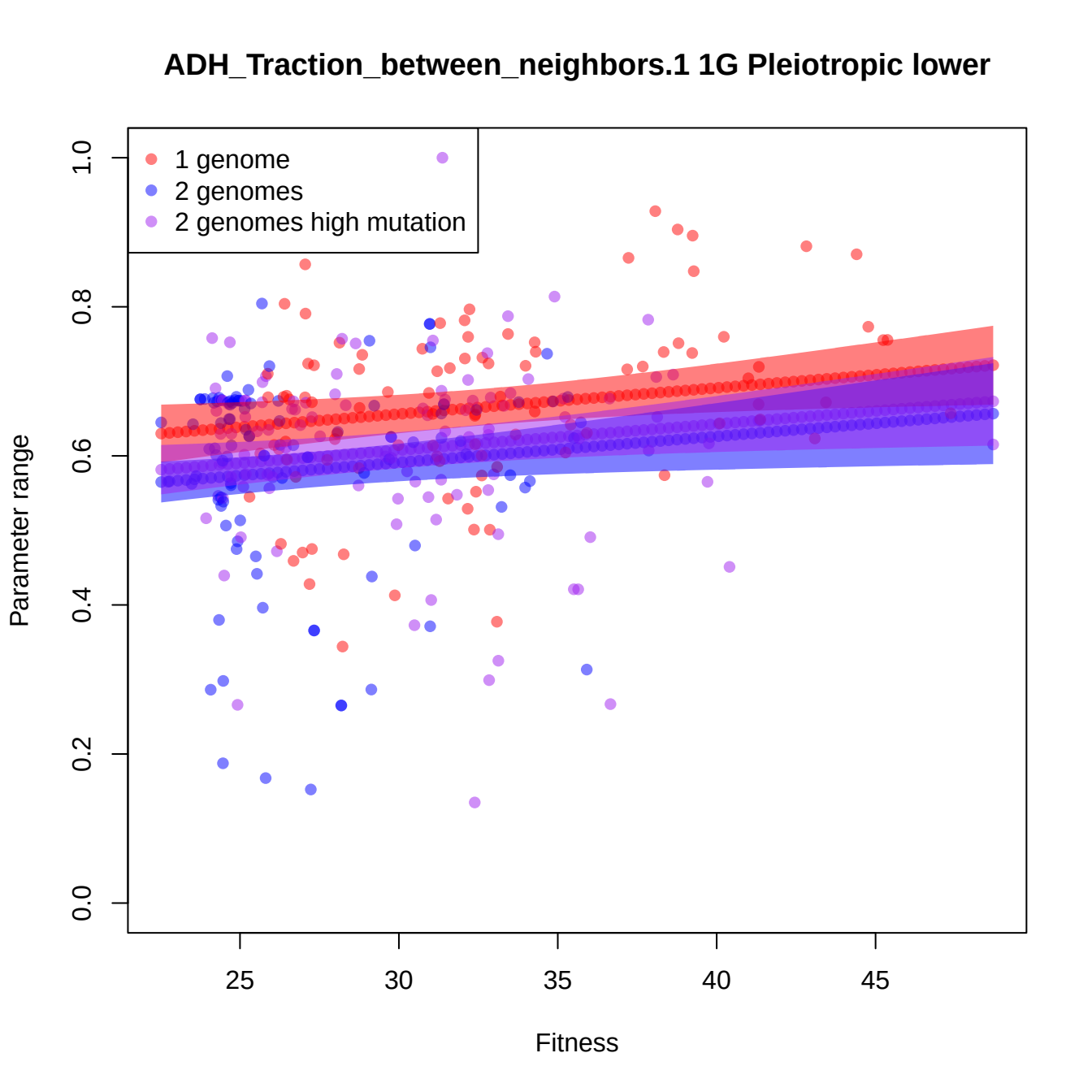

Rep_Repulsion_hox.1
1G HOX lower
Counts
1.2
1.0
0.8
0.6
0.4
2.83337526008341e−08
0.50523855436431
3.73952961620746e−06
12 − 4
12 − 8
4 − 8
1 genome
2 genomes
2 genomes high mutation

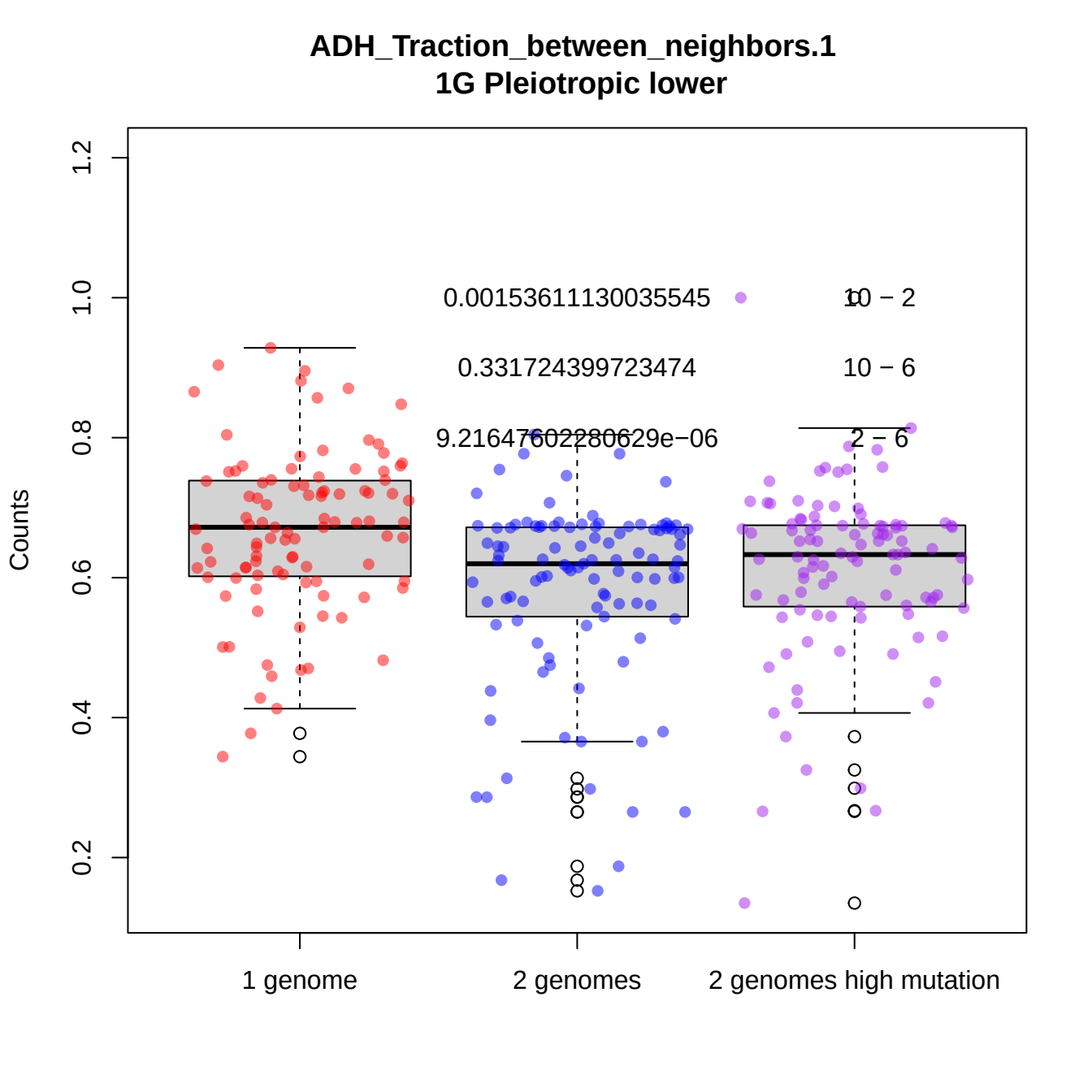

1.00
0.75
0.50
0.25
0.00
windows
30
40
50
fit
type
1
5
9

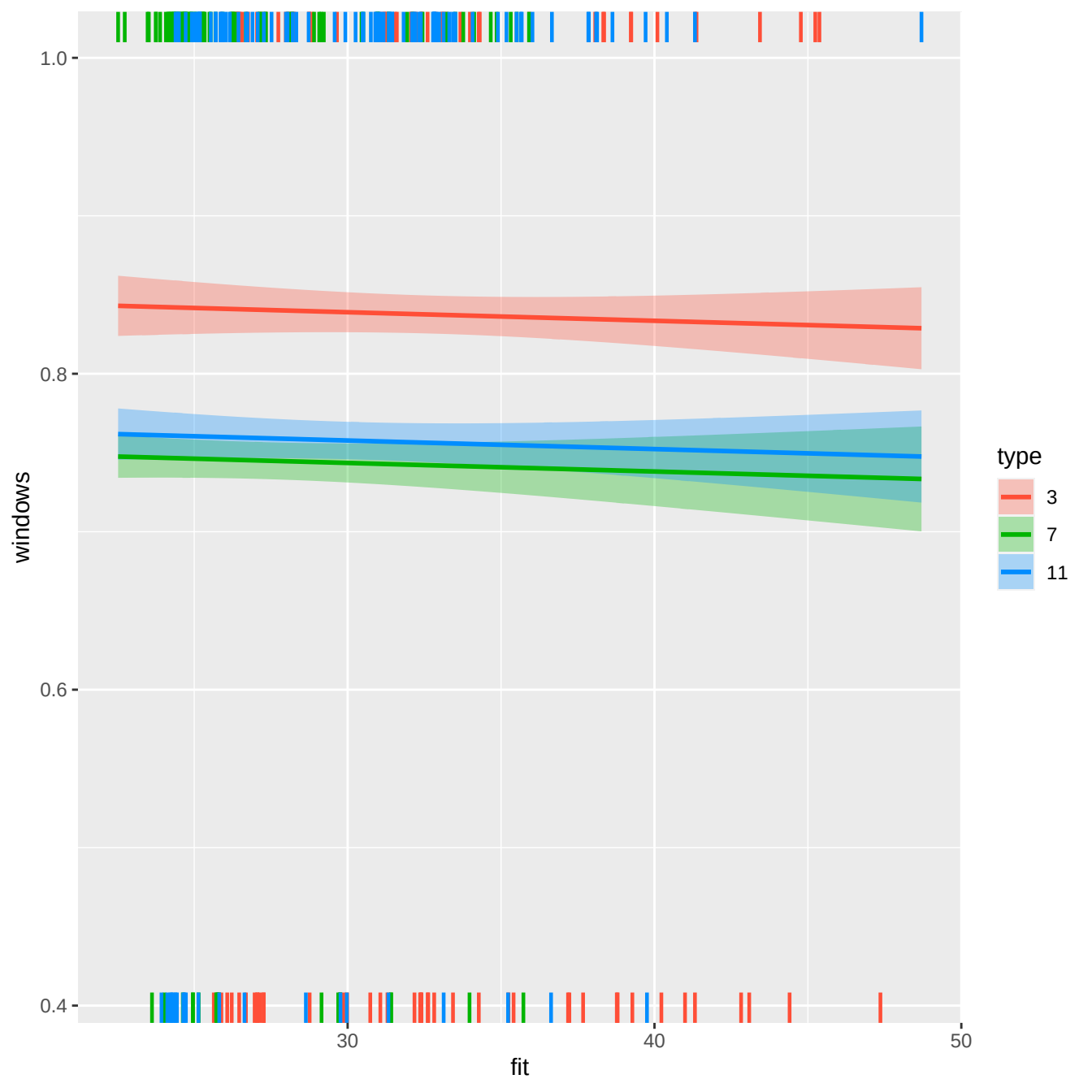

ADH_Traction_between_neighbors.1 1G Pleiotropic upper
1 genome
2 genomes
2 genomes high mutation
Parameter range
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

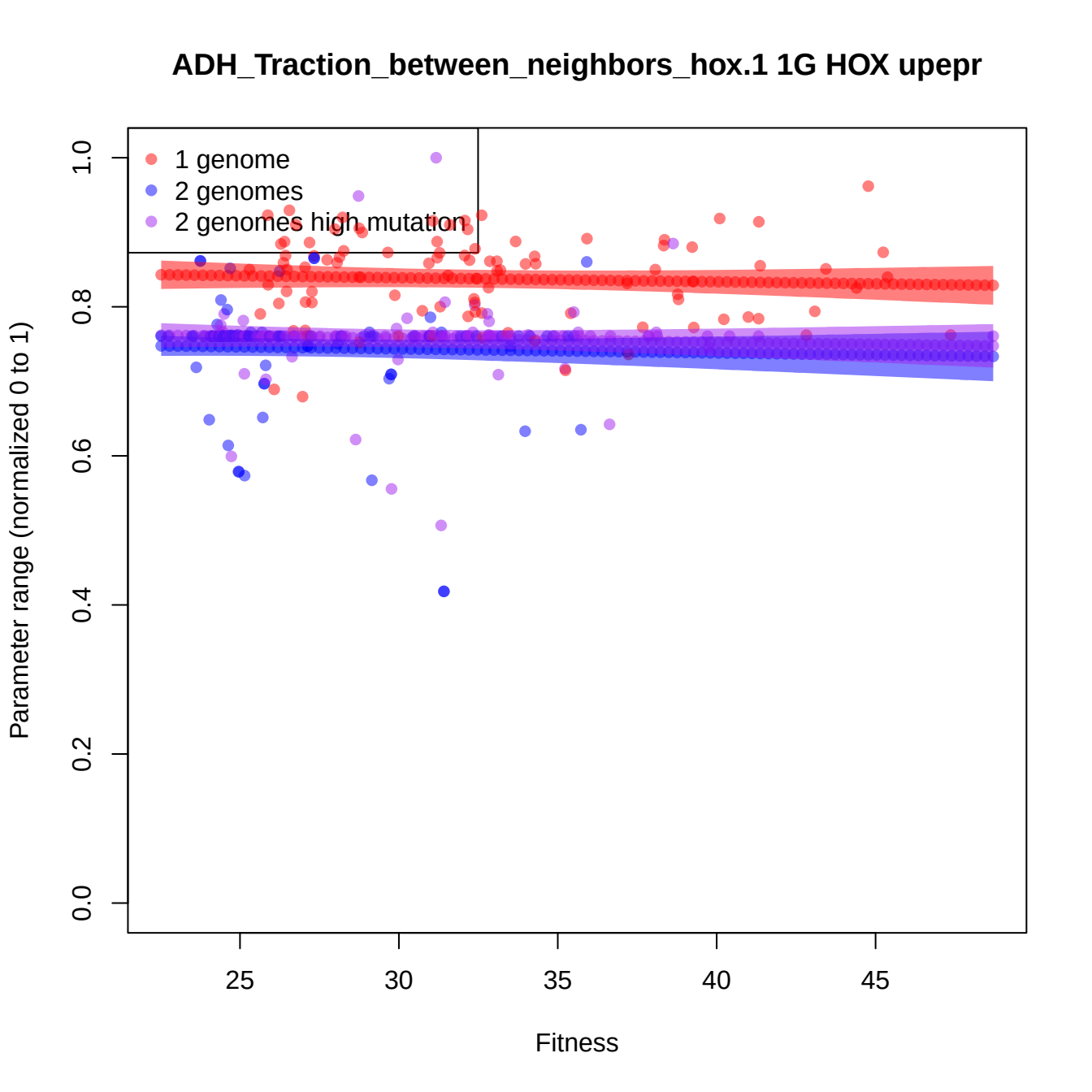

ADH_Traction_between_neighbors.1
1G Pleiotropic upper
Counts
0.0
0.2
0.4
0.6
0.8
1.0
1.2
2.29440198623402e−05
0.0144211586753515
0.125525351571408
1 − 5
1 − 9
5 − 9
1 genome
2 genomes
2 genomes high mutation


1.00
0.75
0.50
0.25
windows
30
40
50
fit
type
2
6
10

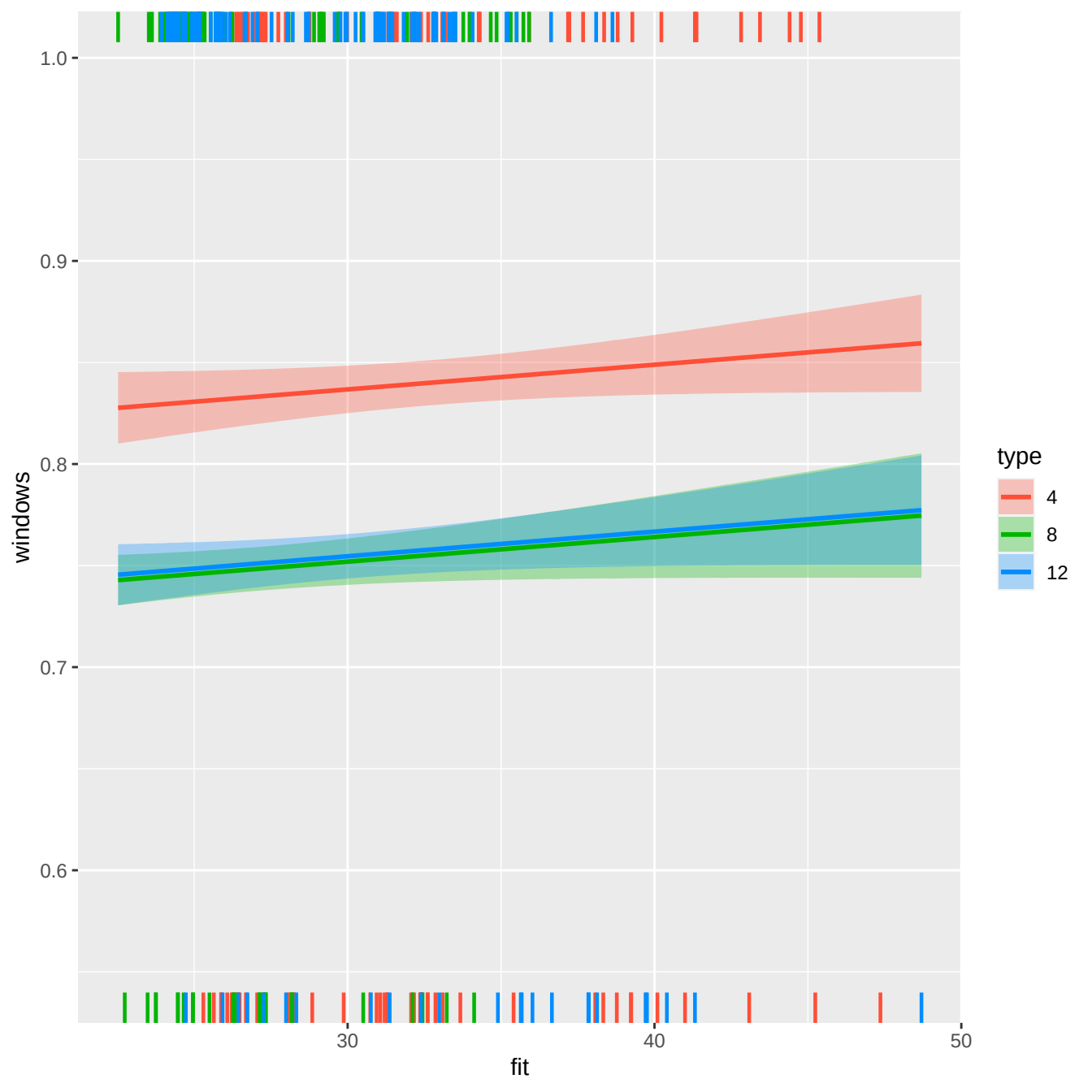

ADH_Traction_between_neighbors.1 1G Pleiotropic lower
1 genome
2 genomes
2 genomes high mutation
Parameter range
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

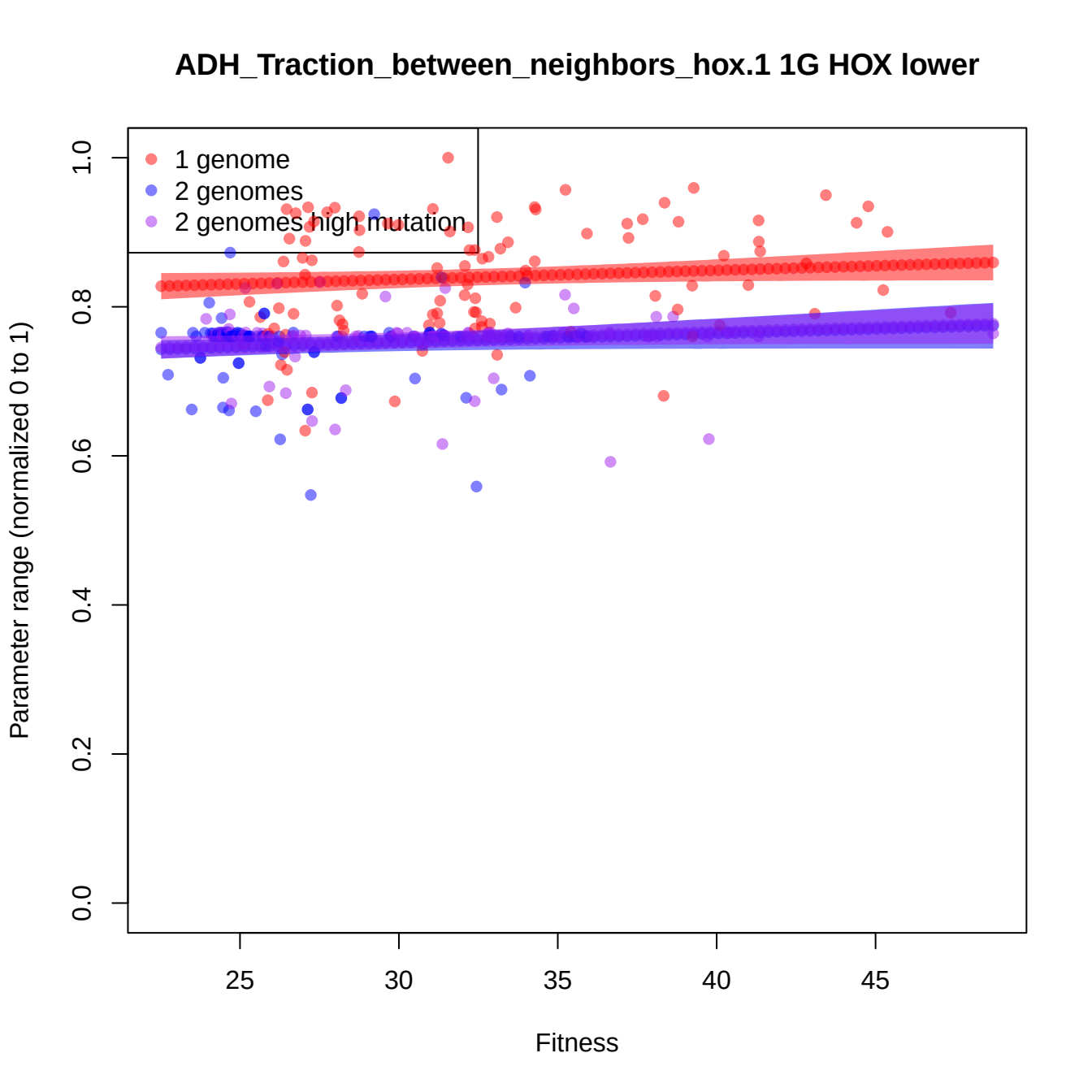

ADH_Traction_between_neighbors.1
1G Pleiotropic lower
Counts
1.2
1.0
0.8
0.6
0.4
0.2
0.00153611130035545
0.331724399723474
9.216476022806 29e−06
10 − 2
10 − 6
2 − 6
1 genome
2 genomes
2 genomes high mutation

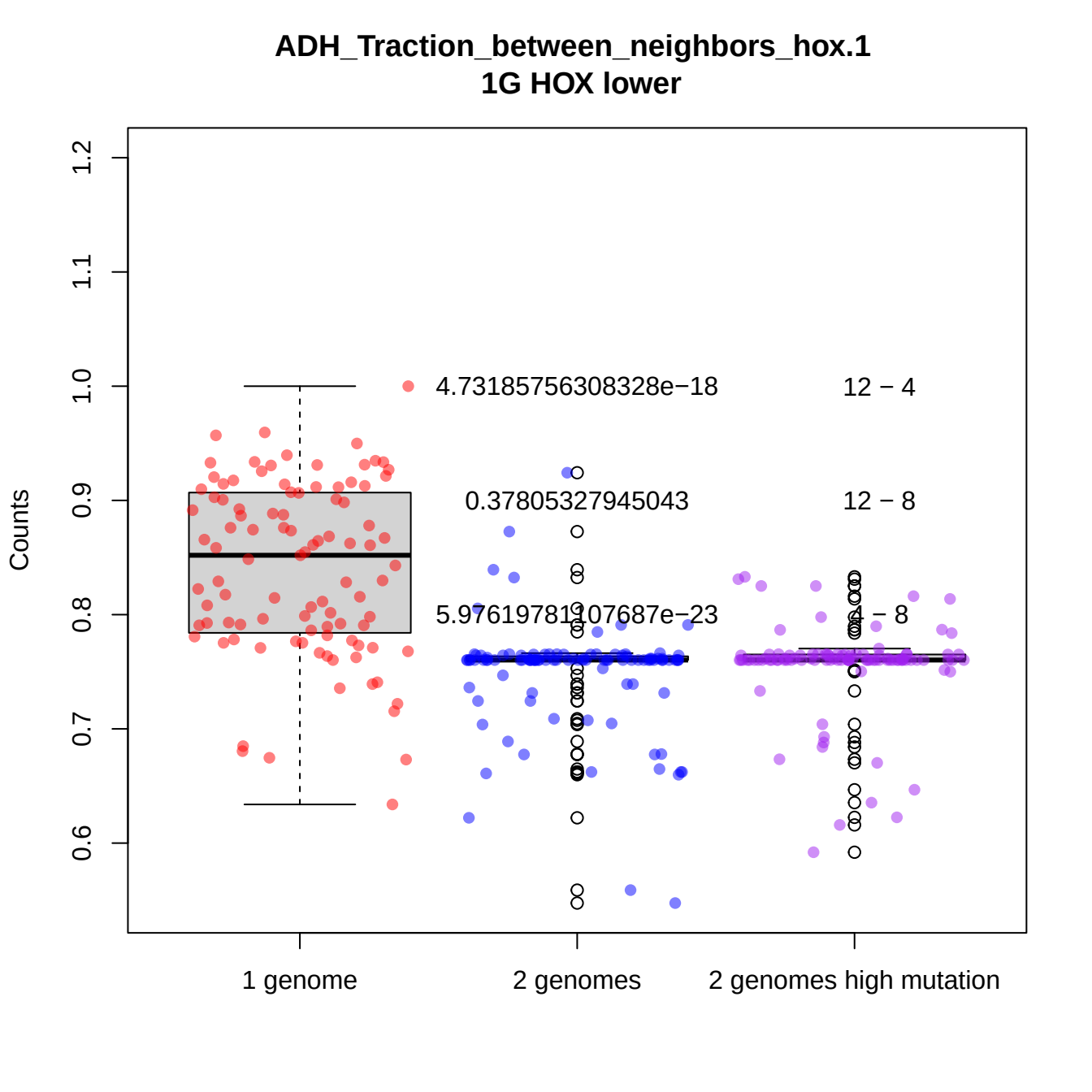

1.0
0.8
0.6
0.4
windows
30
40
50
fit
type
3
7
11

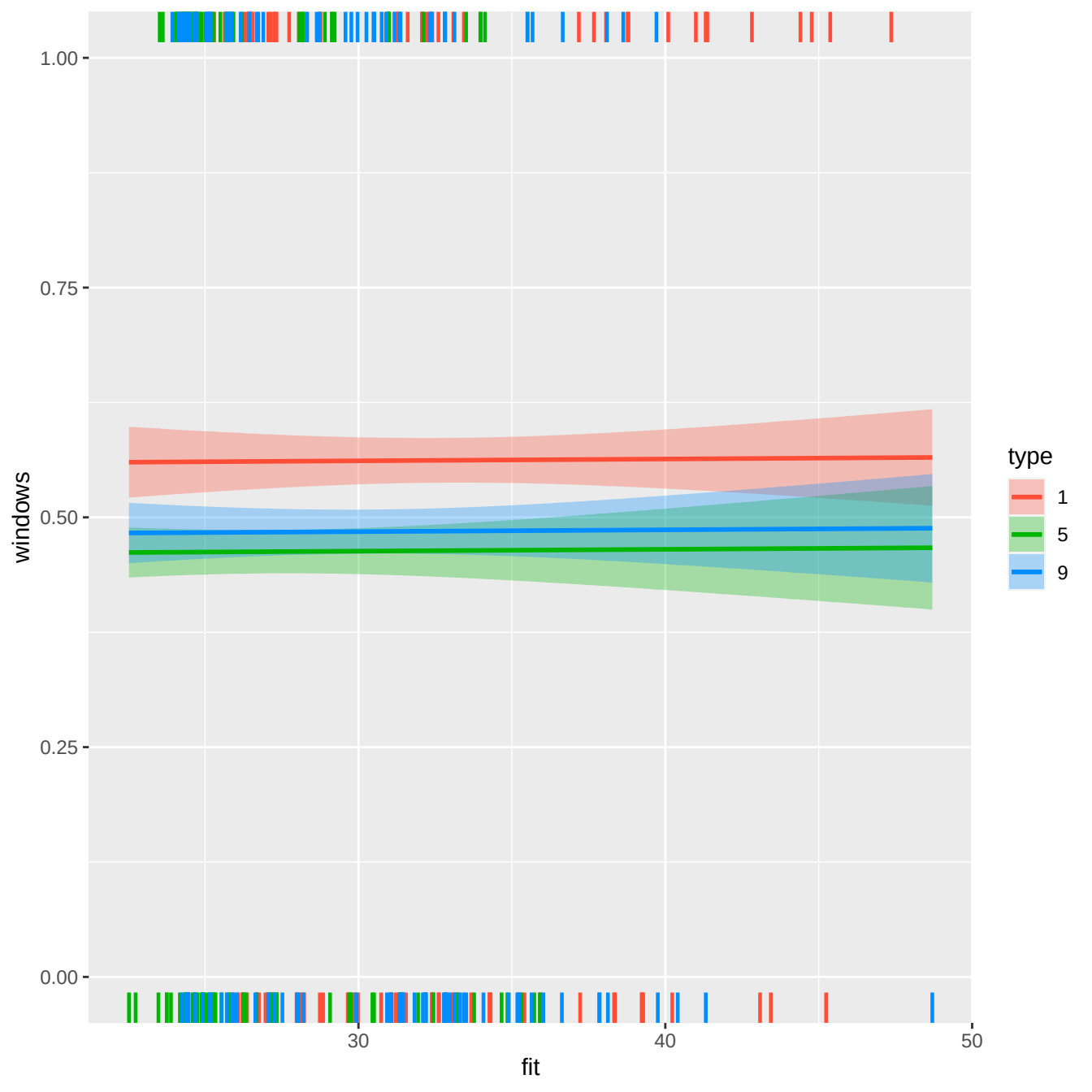

ADH_Traction_between_neighbors_hox.1 1G HOX upepr
1 genome
2 genomes
2 genomes high mutation
Parameter range (normalized 0 to 1)
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

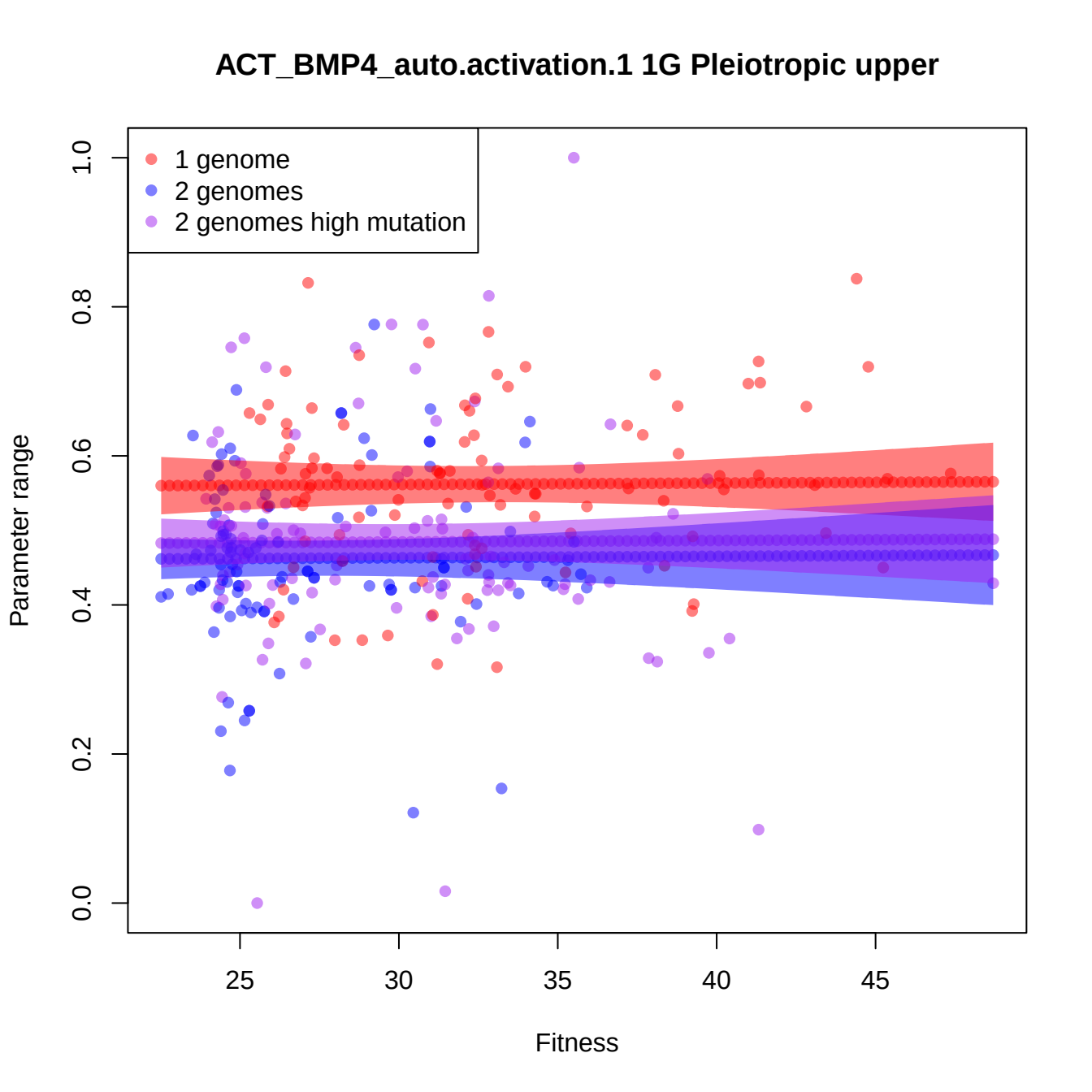

ADH_Traction_between_neighbors_hox.1
1G HOX upepr
Counts
1.2
1.0
0.8
0.6
0.4
5.151458321675e−21
0.481753014499117
1.29155664265988e−25
11 − 3
11 − 7
− 7
1 genome
2 genomes
2 genomes high mutation

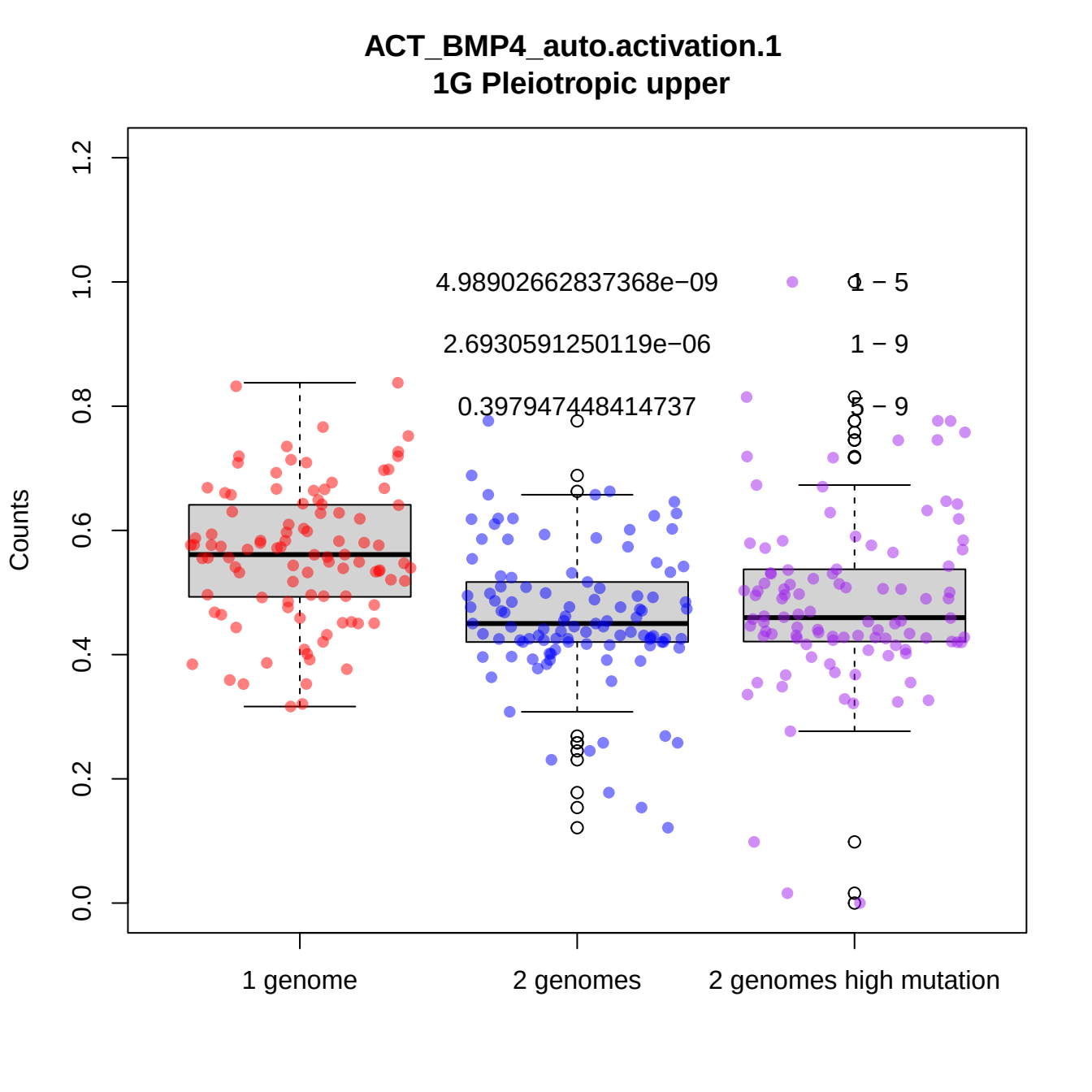

1.0
0.9
0.8
0.7
0.6
windows
30
40
50
fit
type
4
8
12

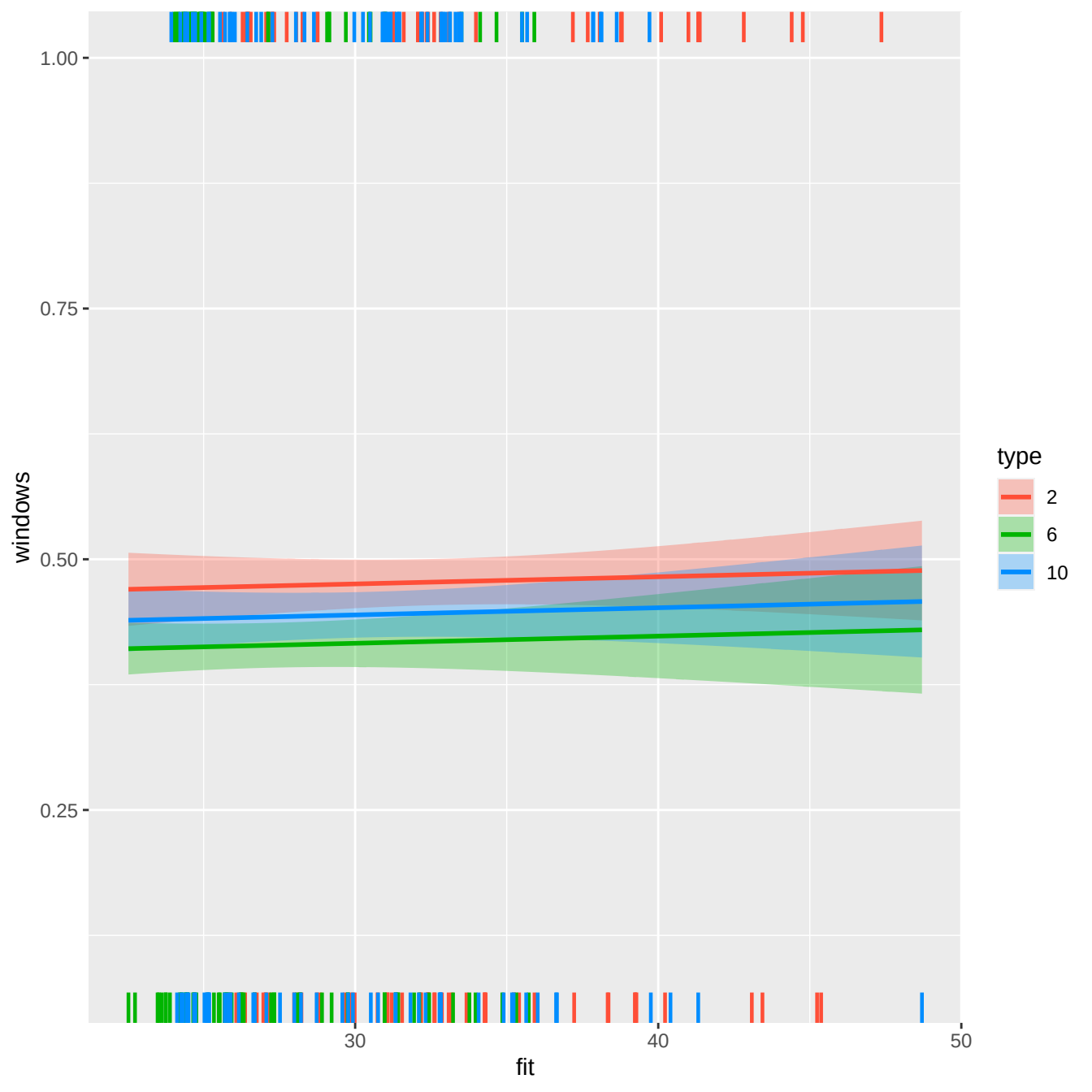

ADH_Traction_between_neighbors_hox.1 1G HOX lower
1 genome
2 genomes
2 genomes high mutation
Parameter range (normalized 0 to 1)
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

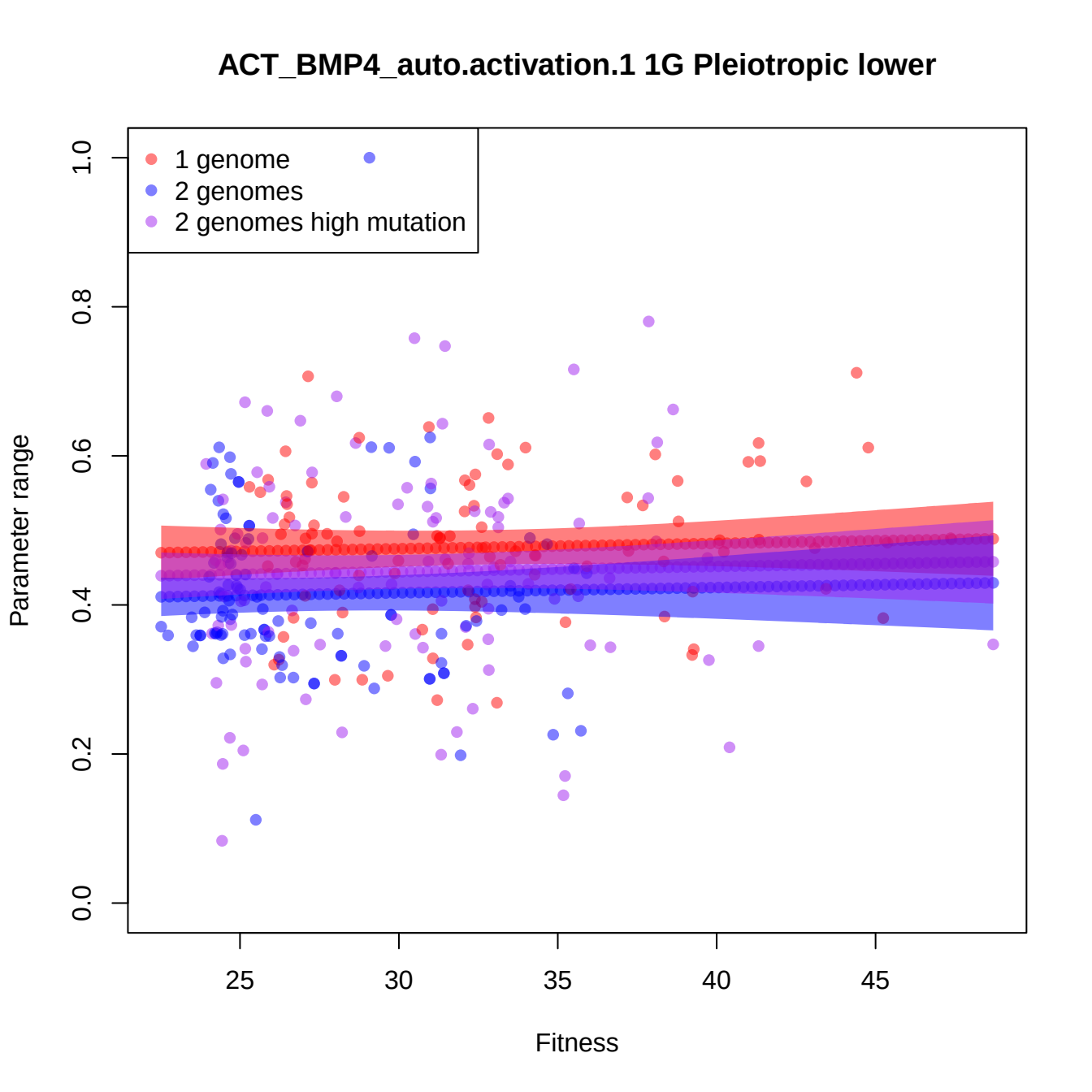

ADH_Traction_between_neighbors_hox.1
1G HOX lower
Counts
1.2
1.1
1.0
0.9
0.8
0.7
0.6
4.73185756308328e−18
12 − 4
0.37805327945043
12 − 8
5.97619781107687e−23
4 − 8
1 genome
2 genomes
2 genomes high mutation

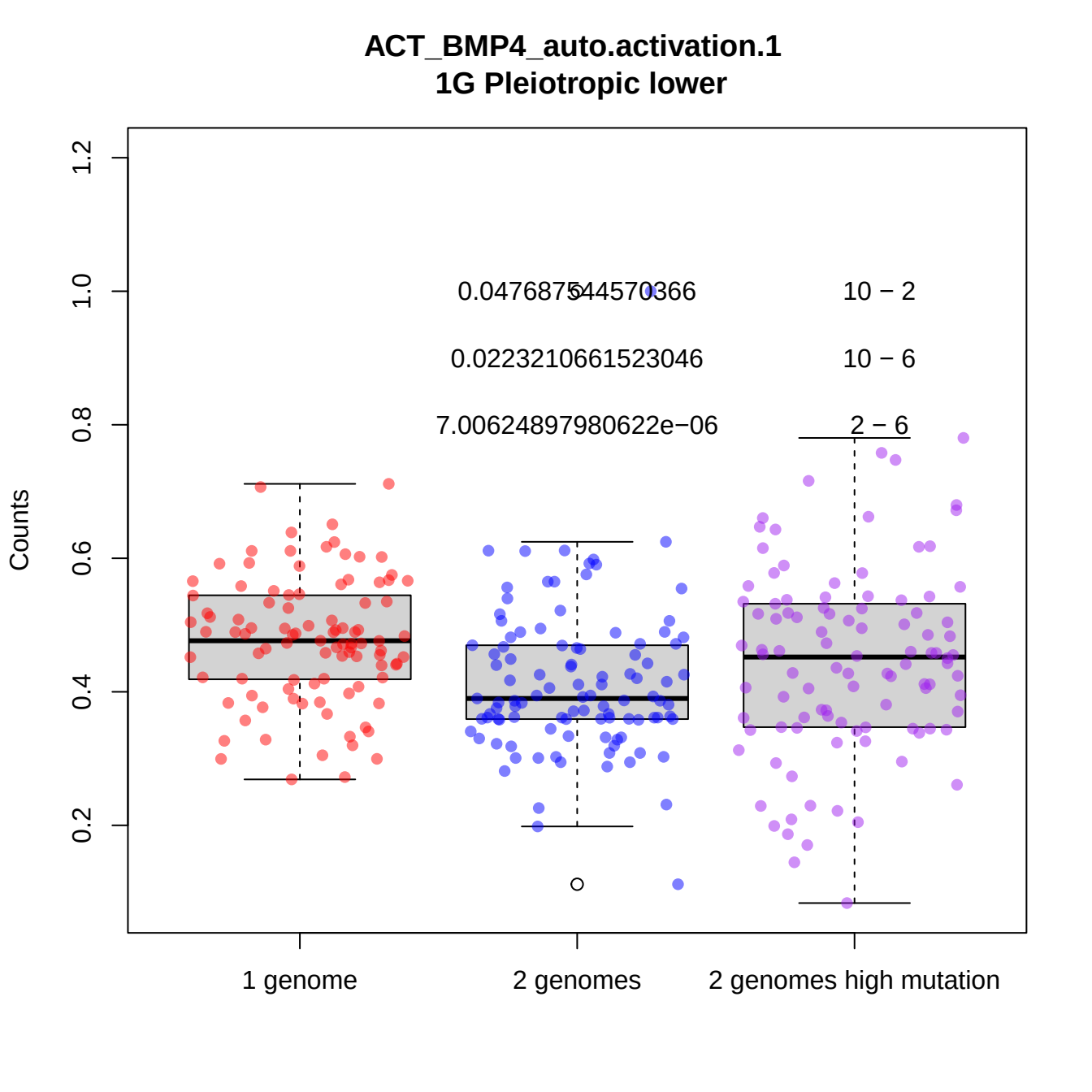

1.00
0.75
0.50
0.25
0.00
windows
30
40
50
fit
type
1
5
9

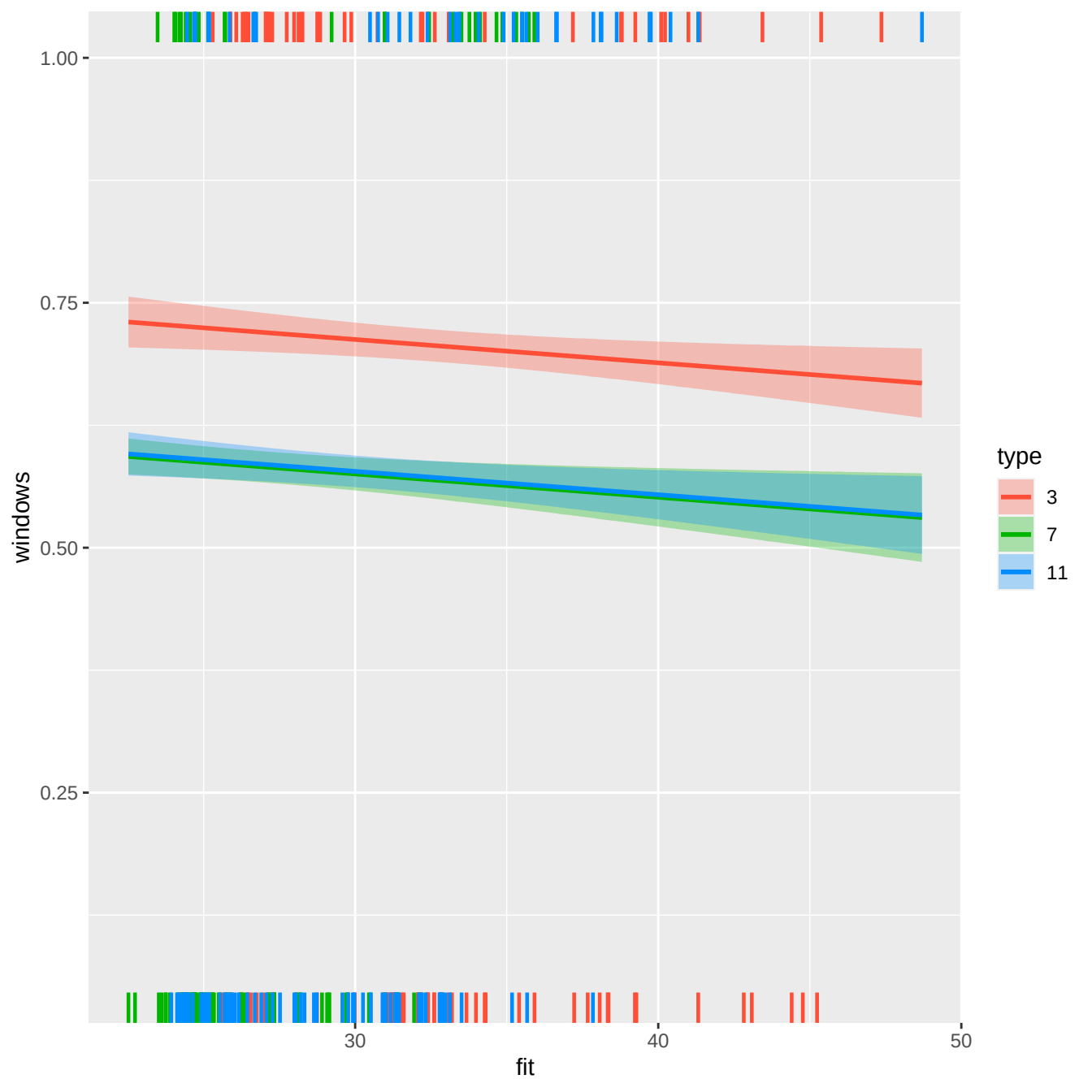

ACT_BMP4_auto.activation.1 1G Pleiotropic upper
1 genome
2 genomes
2 genomes high mutation
Parameter range
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

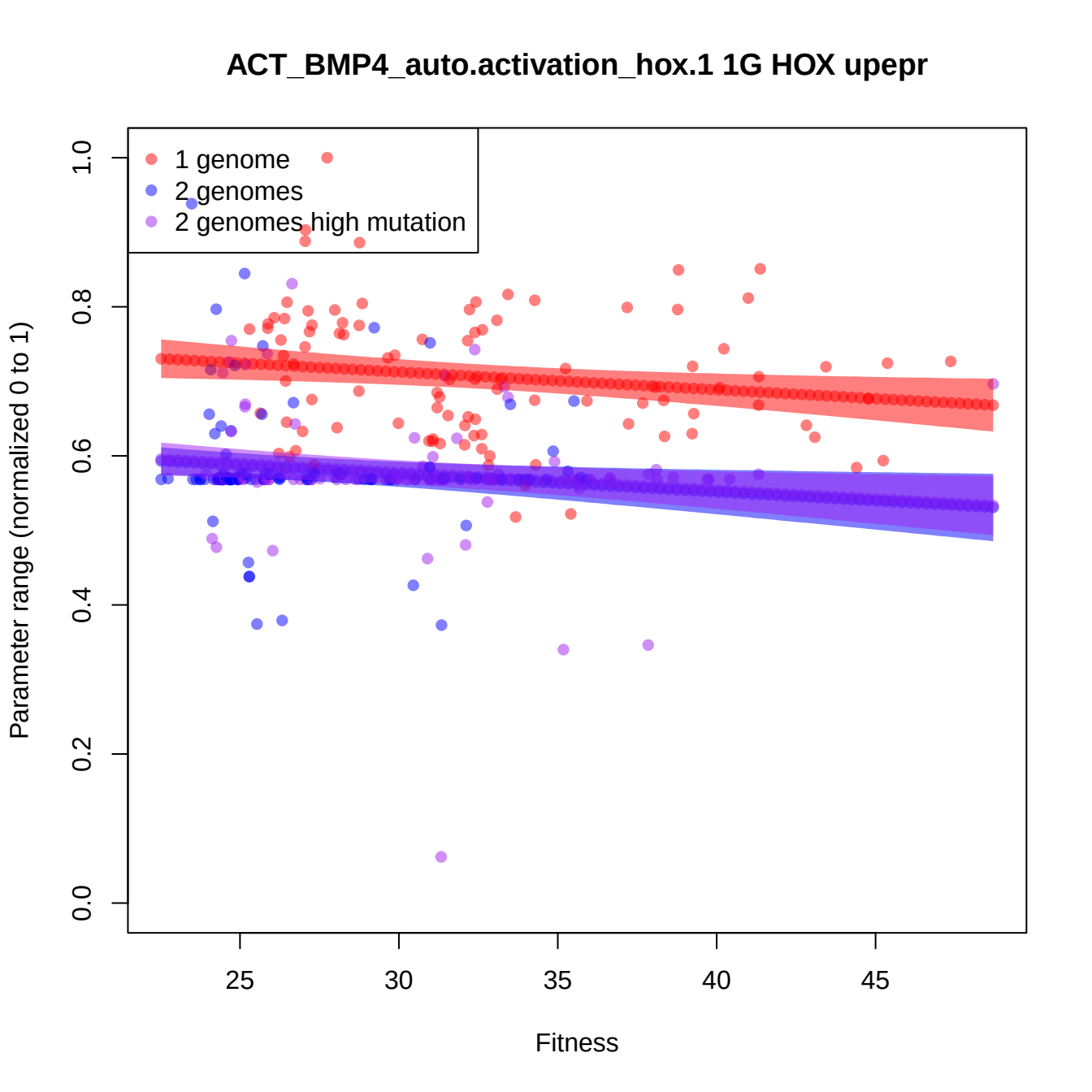

ACT_BMP4_auto.activation.1
1G Pleiotropic upper
4.98902662837368e−09
2.6930591250119e−06
0.397947448414737
1 − 5
1 − 9
5 − 9
Counts
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1 genome
2 genomes
2 genomes high mutation

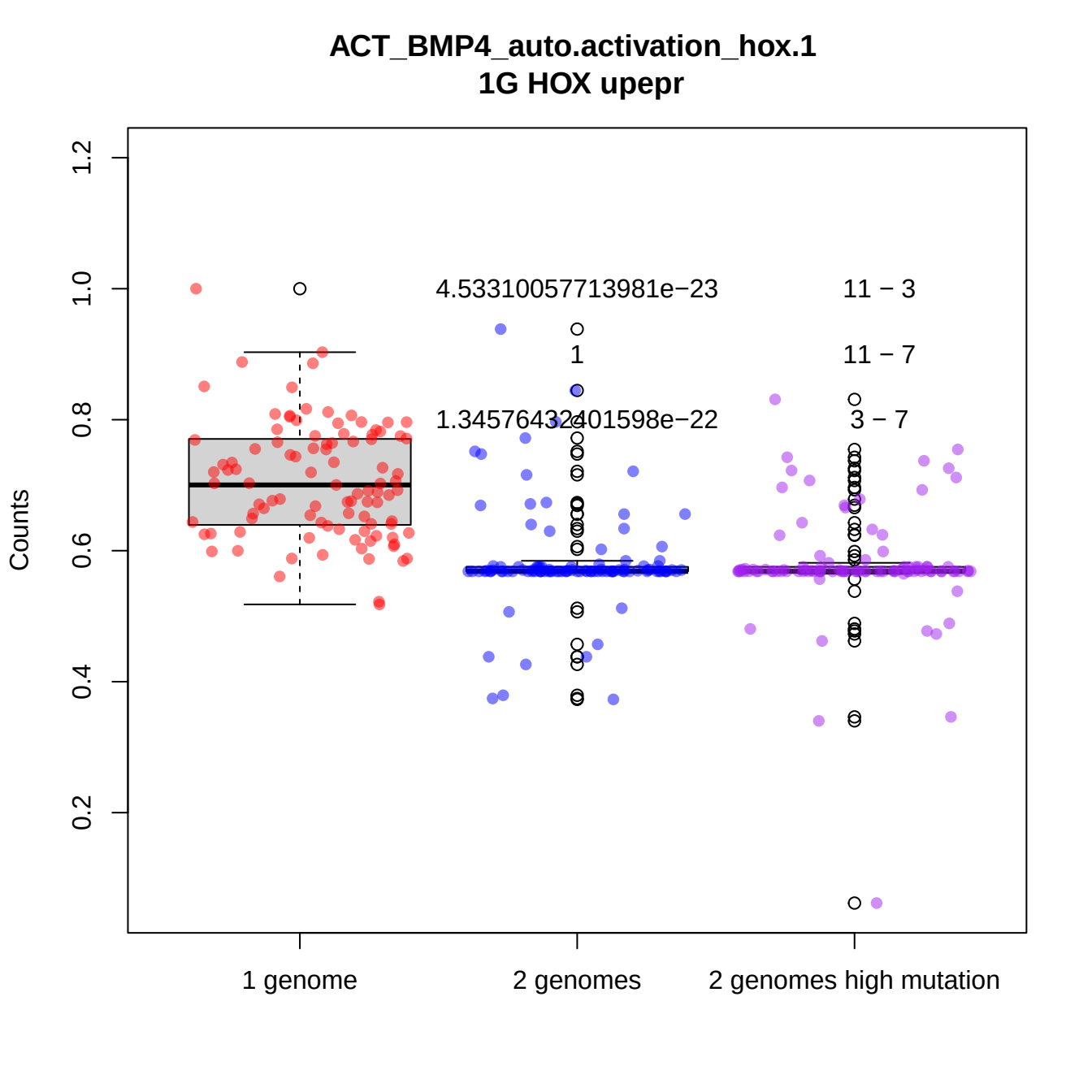

1.00
0.75
0.50
0.25
windows
30
40
50
fit
type
2
6
10

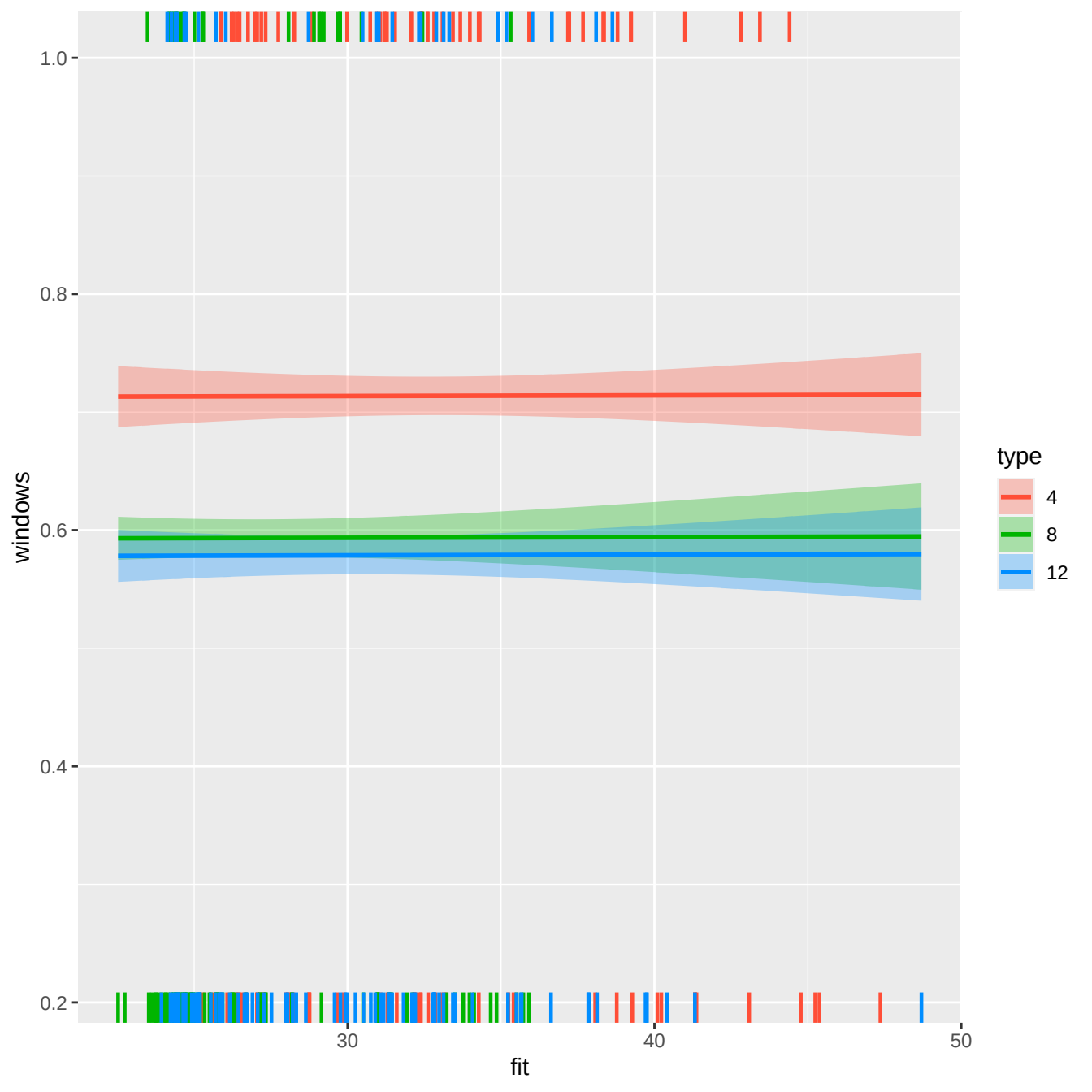

ACT_BMP4_auto.activation.1 1G Pleiotropic lower
1 genome
2 genomes
2 genomes high mutation
Parameter range
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

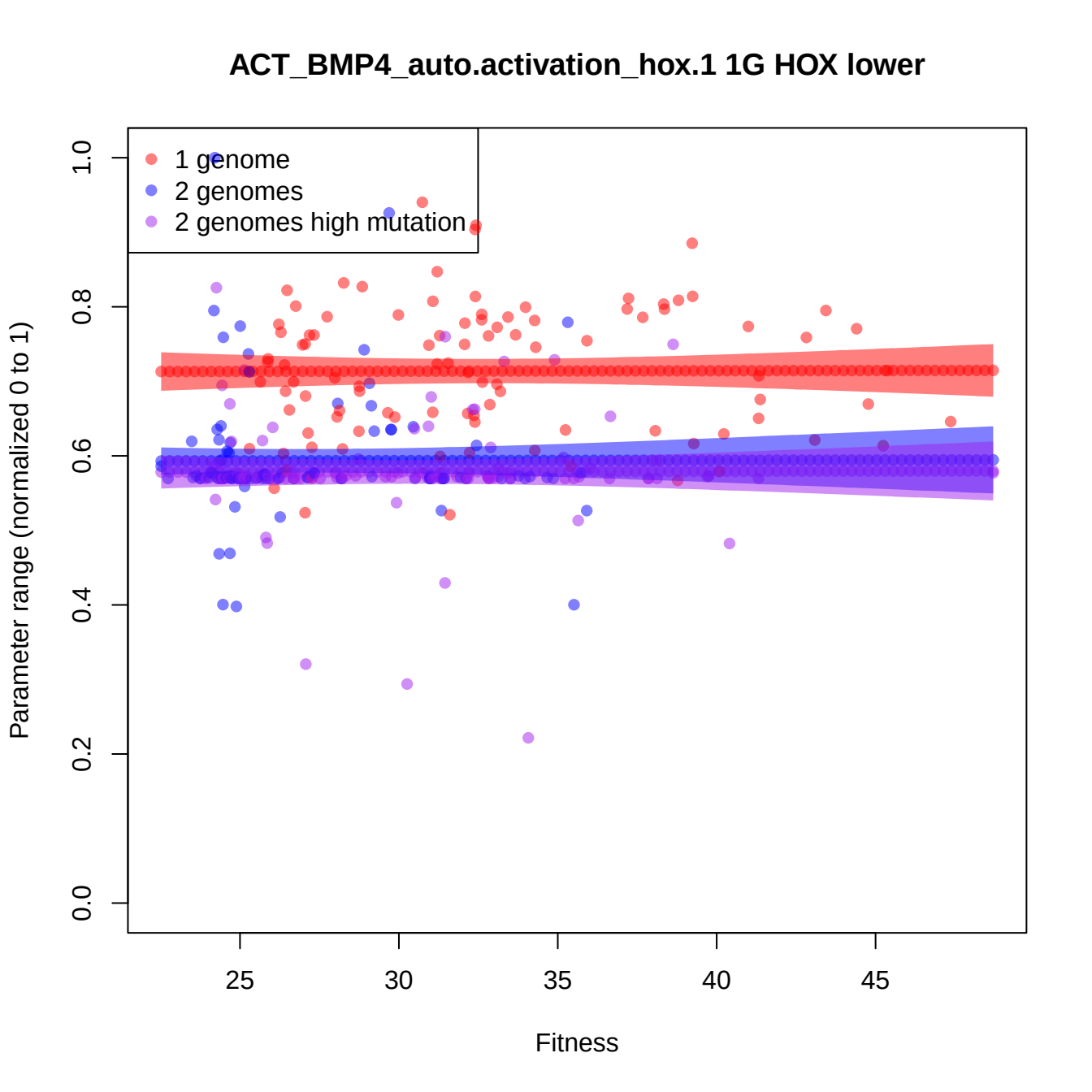

ACT_BMP4_auto.activation.1
1G Pleiotropic lower
0.047687544570366
10 − 2
0.0223210661523046
10 − 6
7.00624897980622e−06
2 − 6
Counts
0.2
0.4
0.6
0.8
1.0
1.2
1 genome
2 genomes
2 genomes high mutation

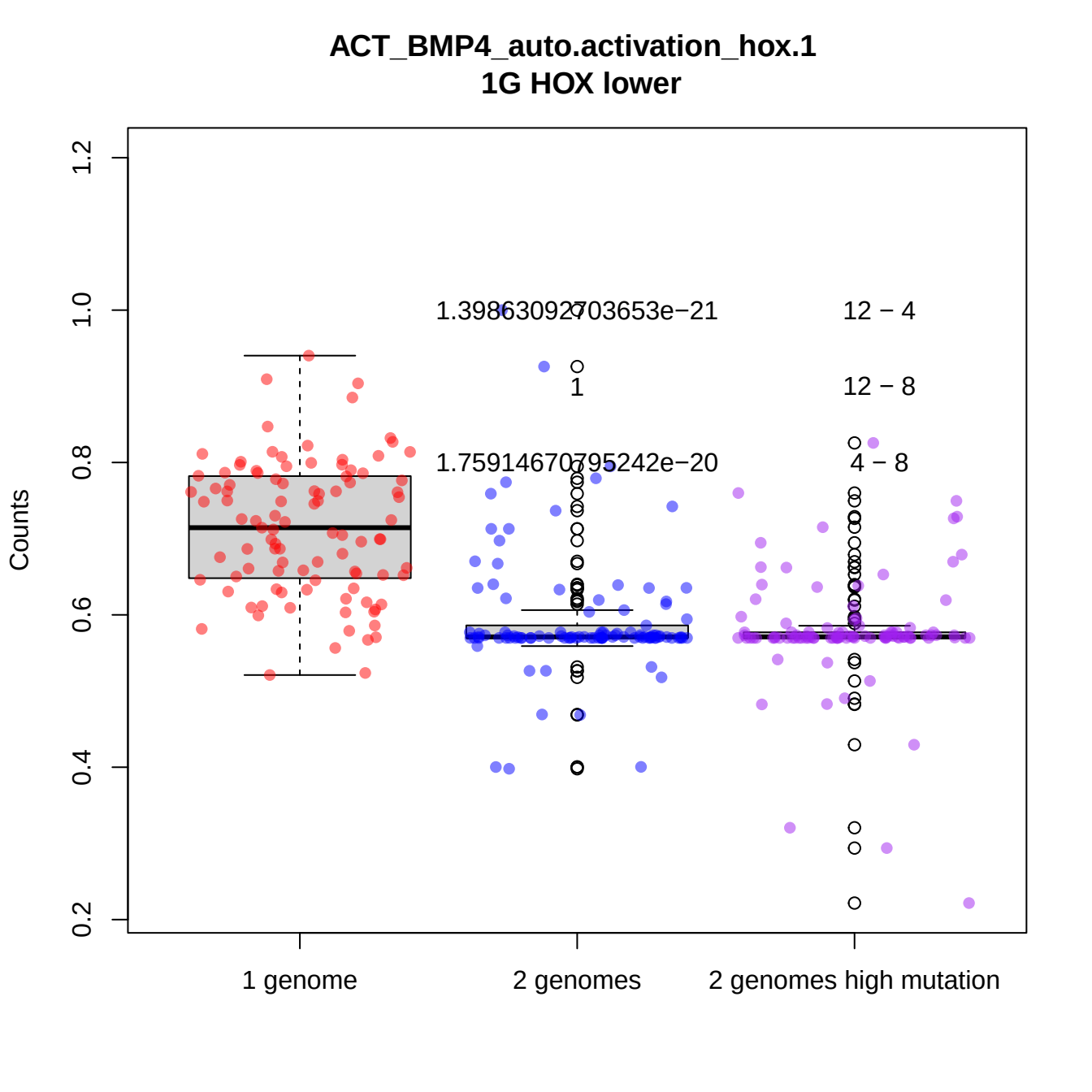

1.00
0.75
0.50
0.25
windows
30
40
50
fit
type
3
7
11

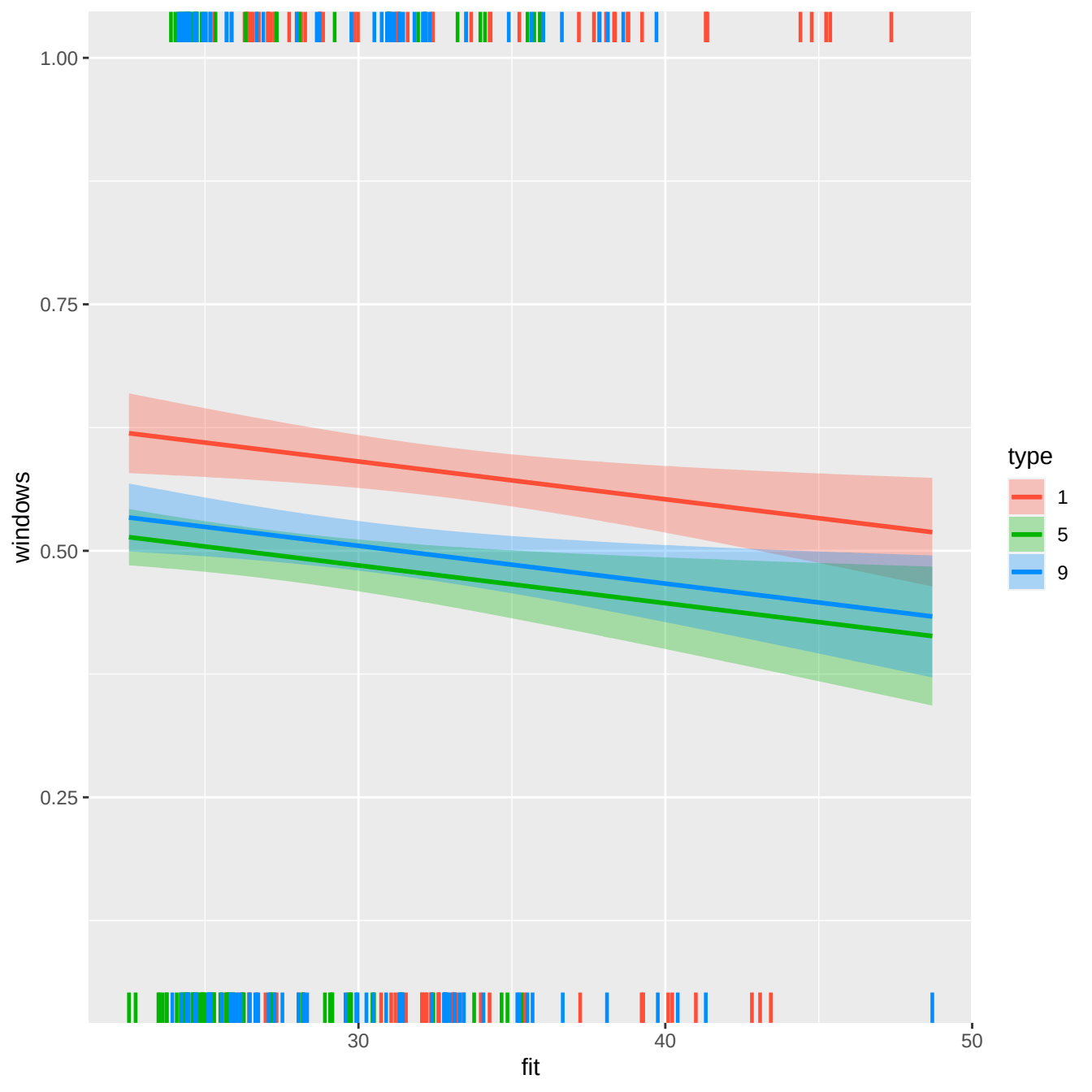

ACT_BMP4_auto.activation_hox.1 1G HOX upepr
1 genome
2 genomes
2 genomes high mutation
Parameter range (normalized 0 to 1)
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

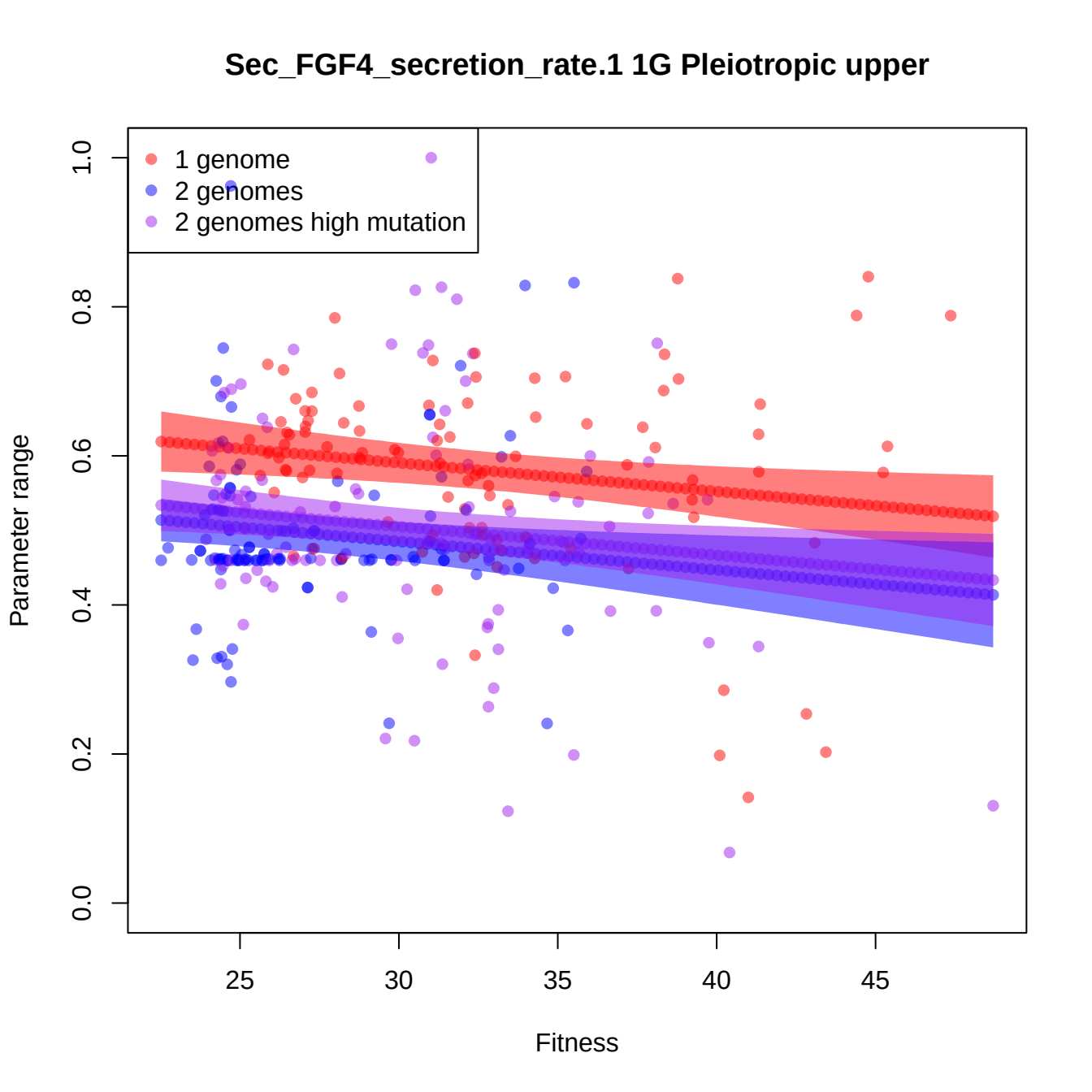

ACT_BMP4_auto.activation_hox.1
1G HOX upepr
Counts
1.2
1.0
0.8
0.6
0.4
0.2
4.53310057713981e−23
1
1.34576432401598e−22
11 − 3
11 − 7
3 − 7
1 genome
2 genomes
2 genomes high mutation

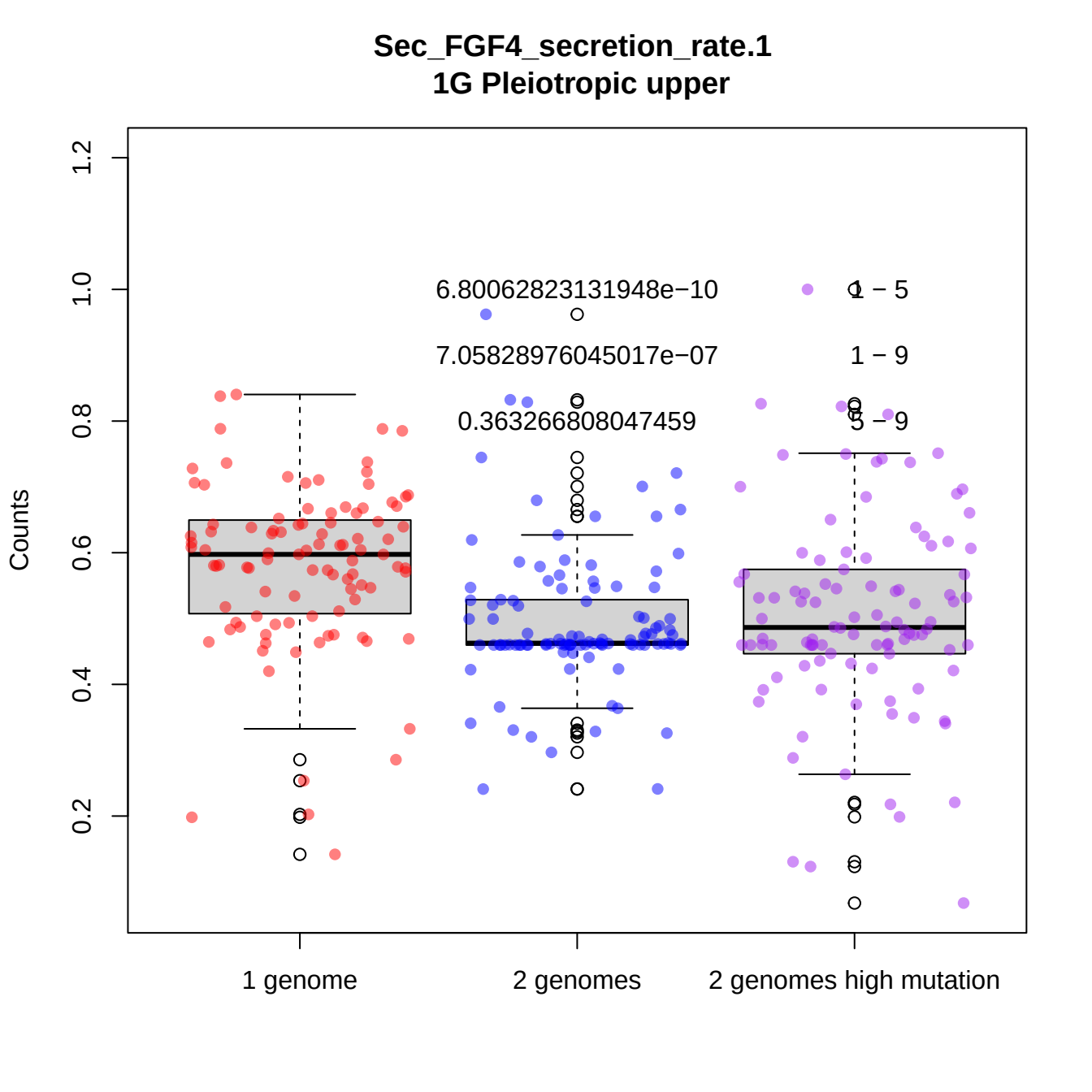

1.0
0.8
0.6
0.4
0.2
windows
30
40
50
fit
type
4
8
12

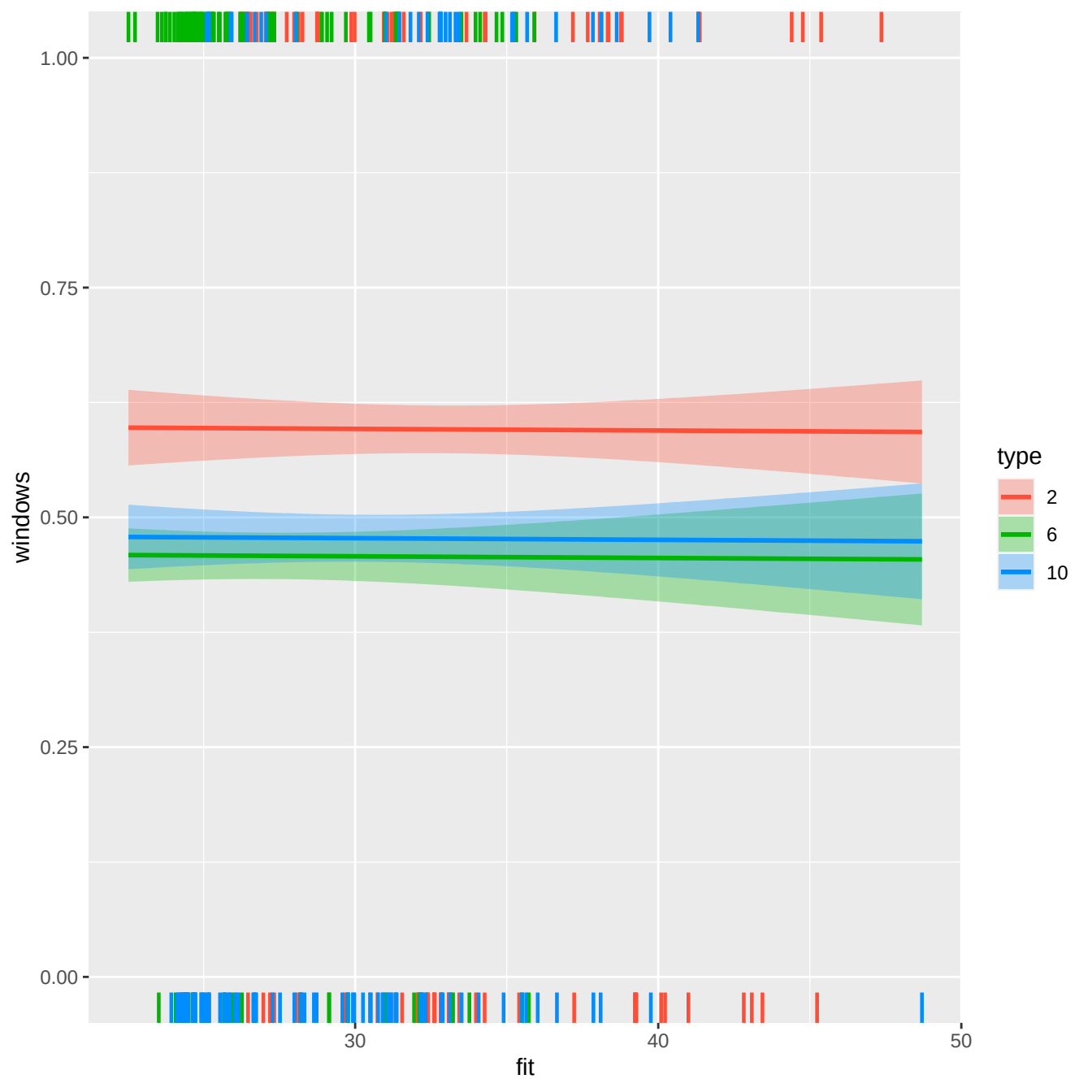

ACT_BMP4_auto.activation_hox.1 1G HOX lower
1 genome
2 genomes
2 genomes high mutation
Parameter range (normalized 0 to 1)
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

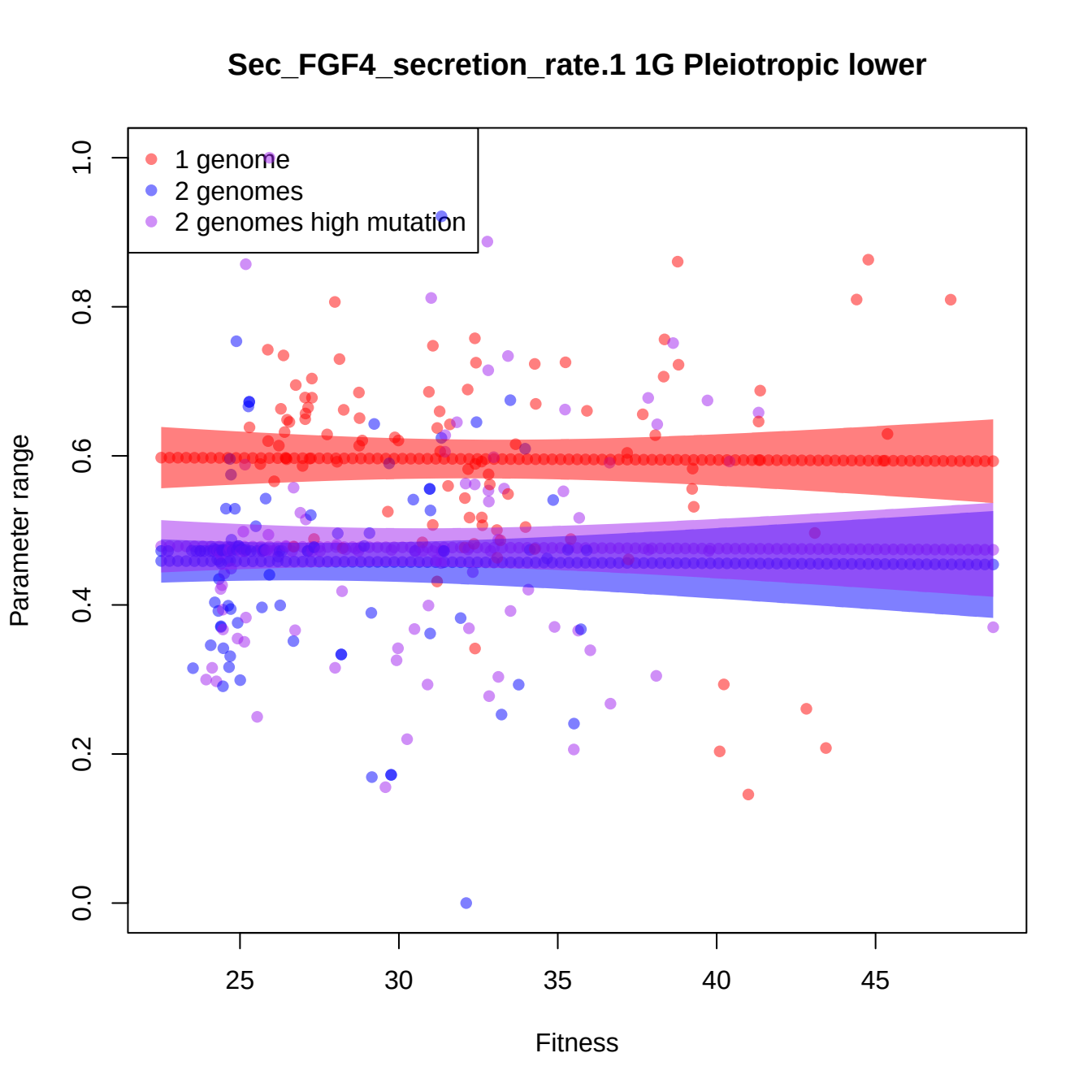

ACT_BMP4_auto.activation_hox.1
1G HOX lower
Counts
1.2
1.0
0.8
0.6
0.4
0.2
1.398630927036 53e−21
1
1.759146707952 42e−20
12 − 4
12 − 8
4 − 8
1 genome
2 genomes
2 genomes high mutation

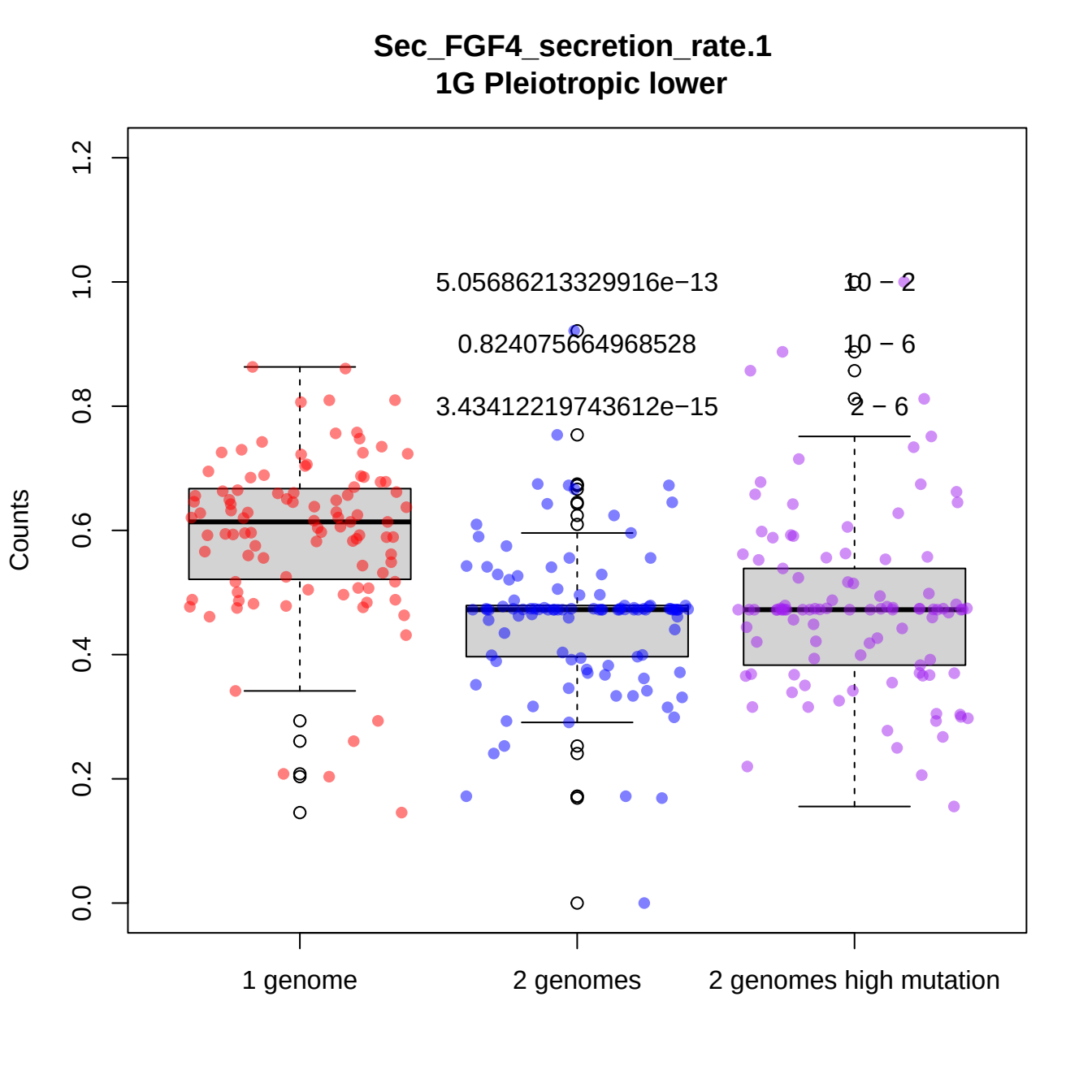

1.00
0.75
0.50
0.25
windows
30
40
50
fit
type
1
5
9

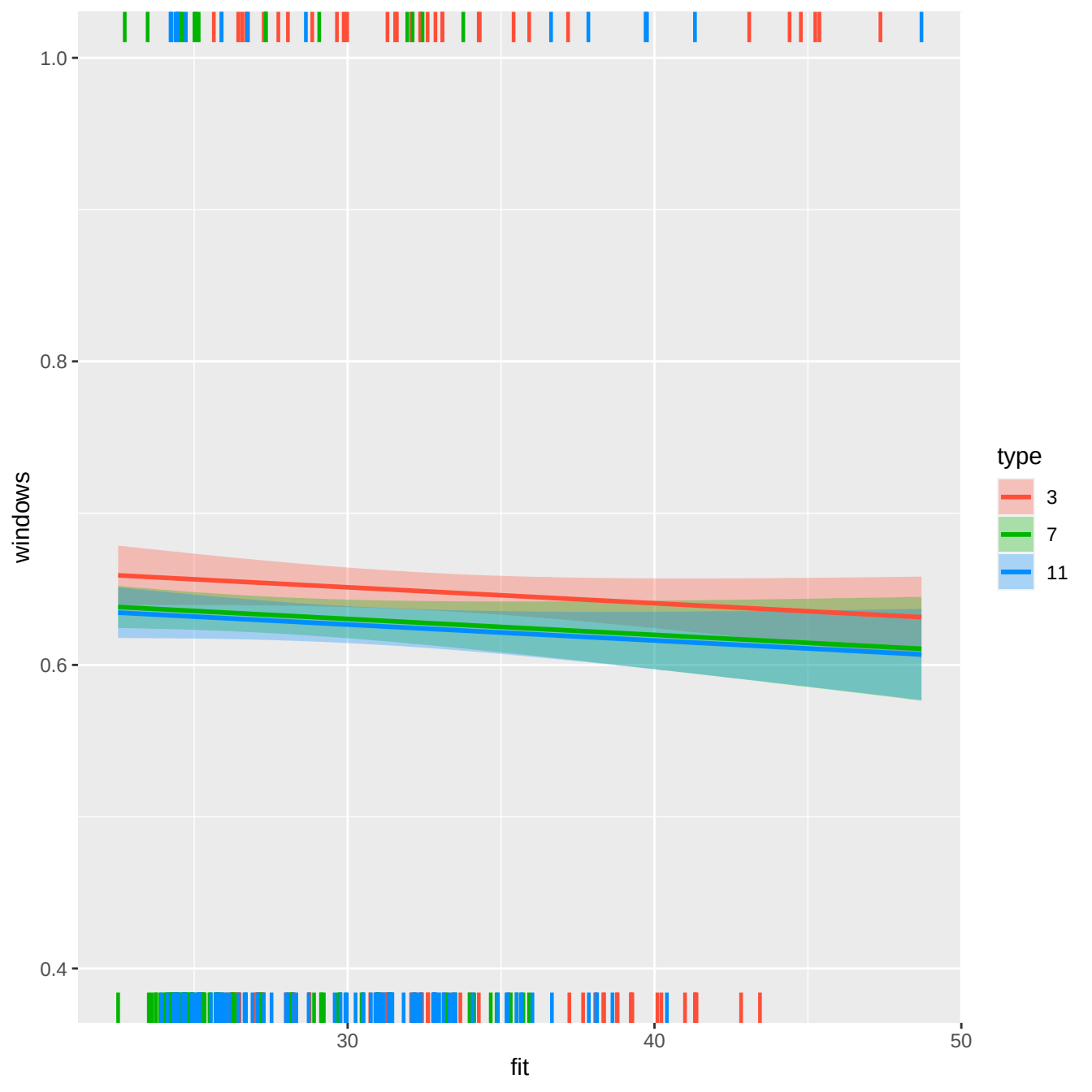

Sec_FGF4_secretion_rate.1 1G Pleiotropic upper
1 genome
2 genomes
2 genomes high mutation
Parameter range
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

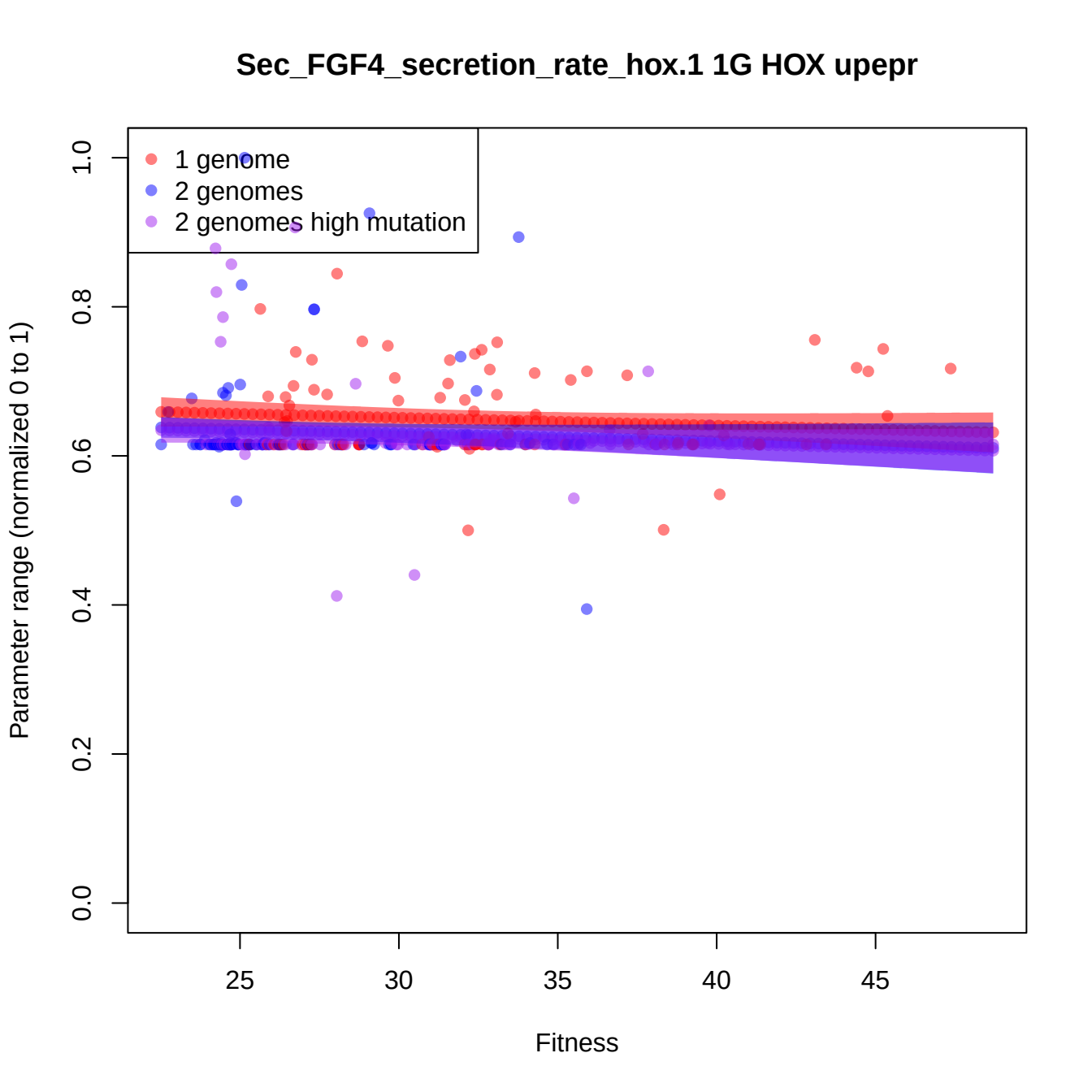

Sec_FGF4_secretion_rate.1
1G Pleiotropic upper
Counts
1.2
1.0
0.8
0.6
0.4
0.2
6.80062823131948e−10
7.05828976045017e−07
0.363266808047459
1 − 5
1 − 9
5 − 9
1 genome
2 genomes
2 genomes high mutation

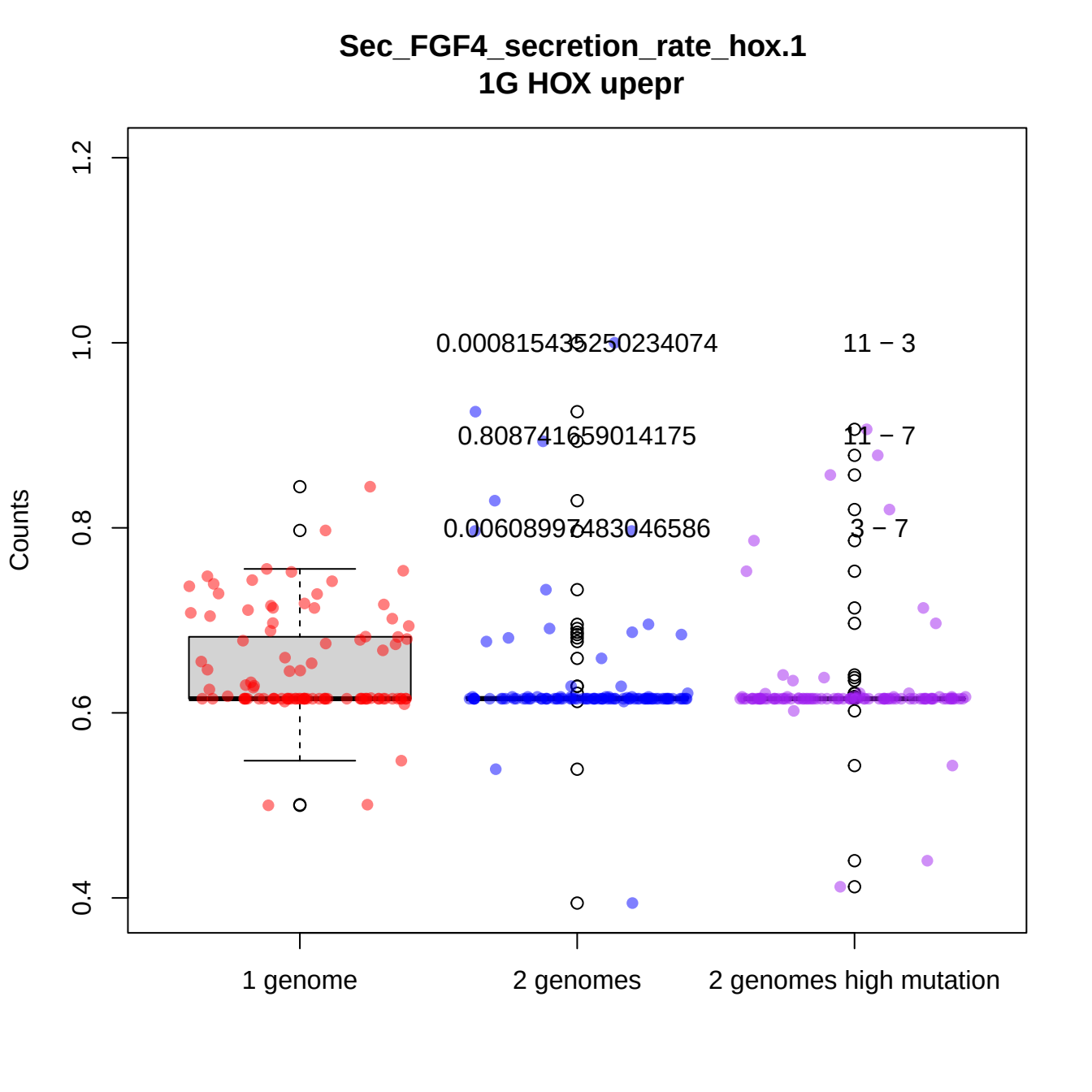

1.00
0.75
0.50
0.25
0.00
windows
30
40
50
fit
type
2
6
10

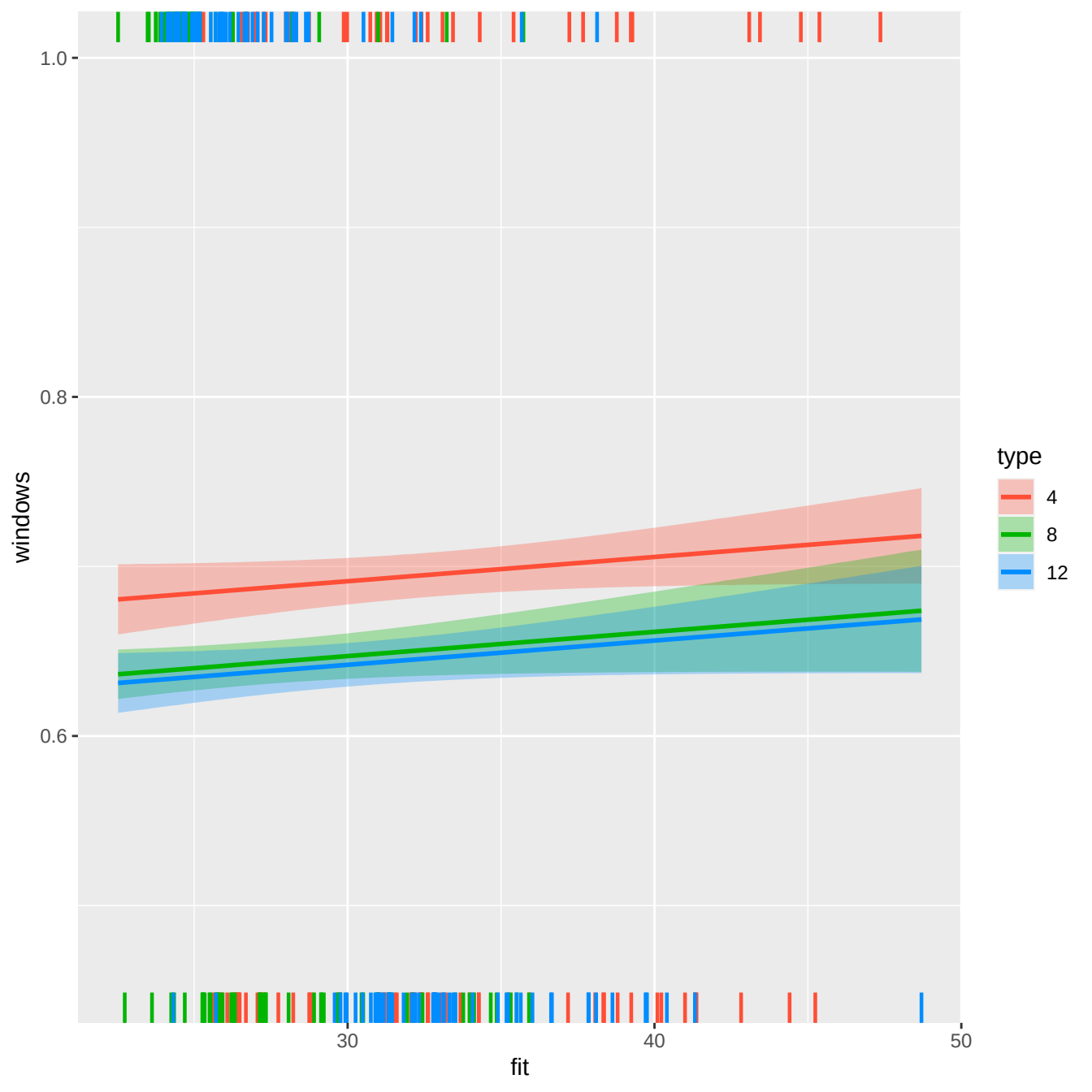

Sec_FGF4_secretion_rate.1 1G Pleiotropic lower
1 genome
2 genomes
2 genomes high mutation
Parameter range
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

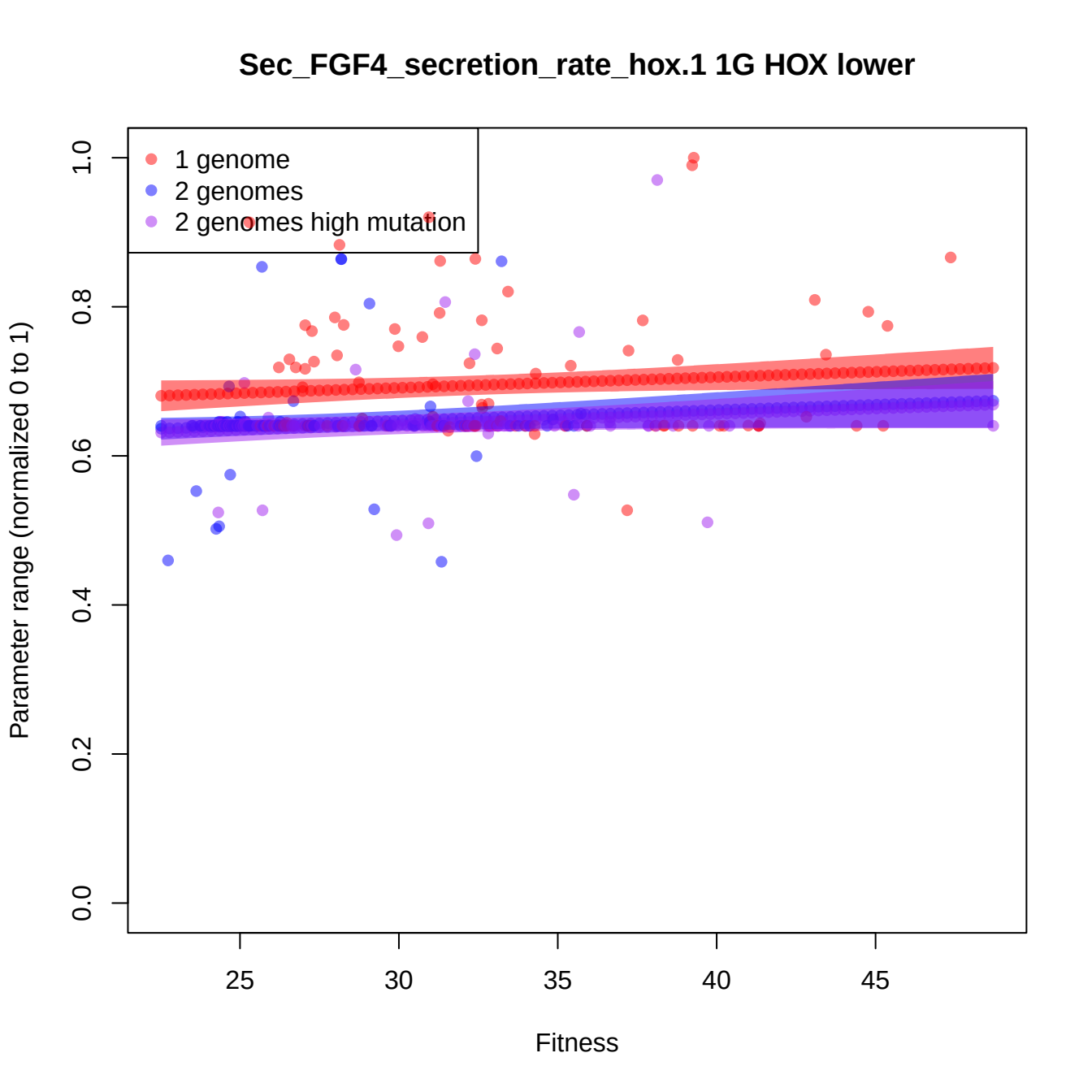

Sec_FGF4_secretion_rate.1
1G Pleiotropic lower
Counts
0.0
0.2
0.4
0.6
0.8
1.0
1.2
5.05686213329916e−13
0.824075664968528
3.43412219743612e−15
10 − 2
10 − 6
2 − 6
1 genome
2 genomes
2 genomes high mutation

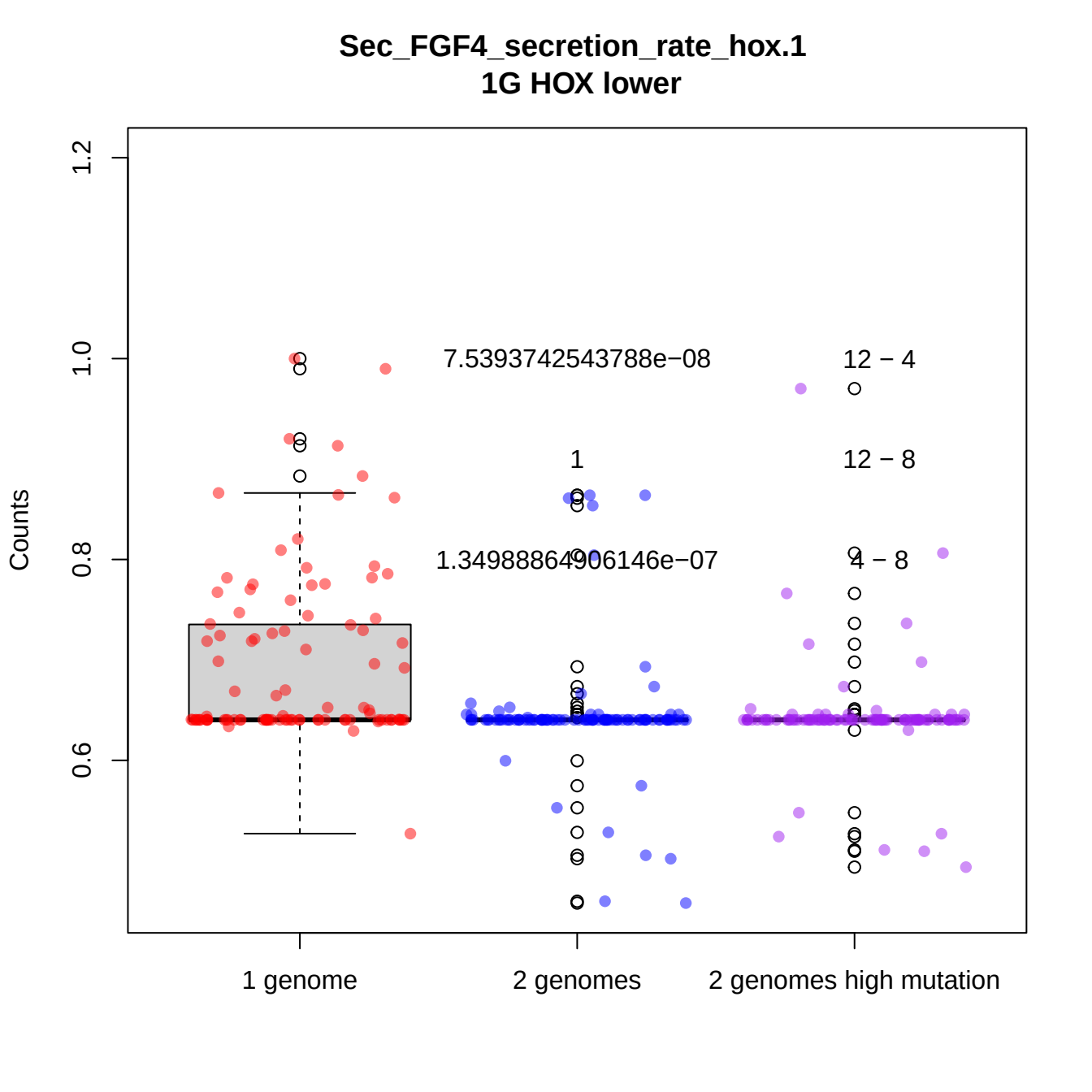

1.0
0.8
0.6
0.4
windows
30
40
50
fit
type
3
7
11

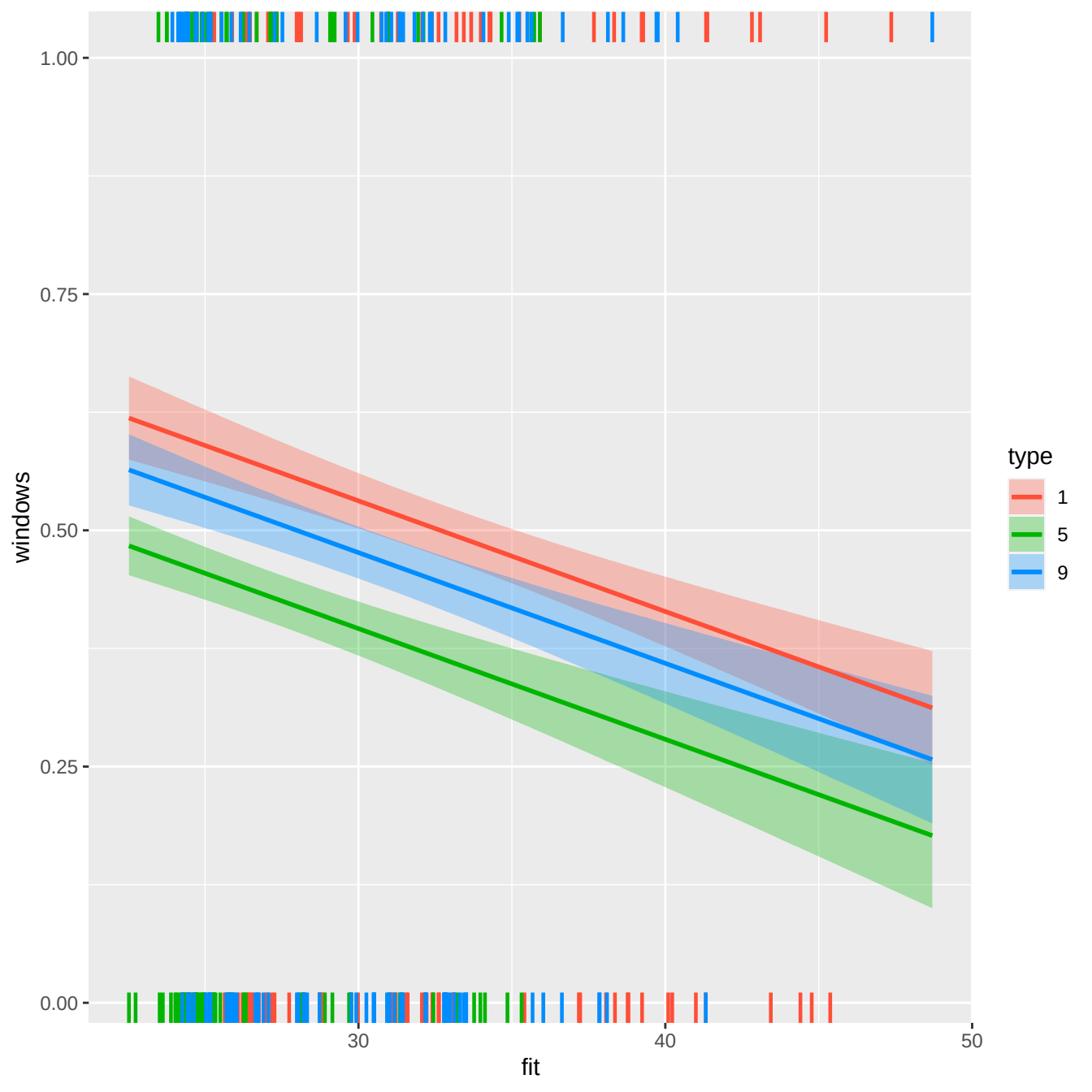

Sec_FGF4_secretion_rate_hox.1 1G HOX upepr
1 genome
2 genomes
2 genomes high mutation
Parameter range (normalized 0 to 1)
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

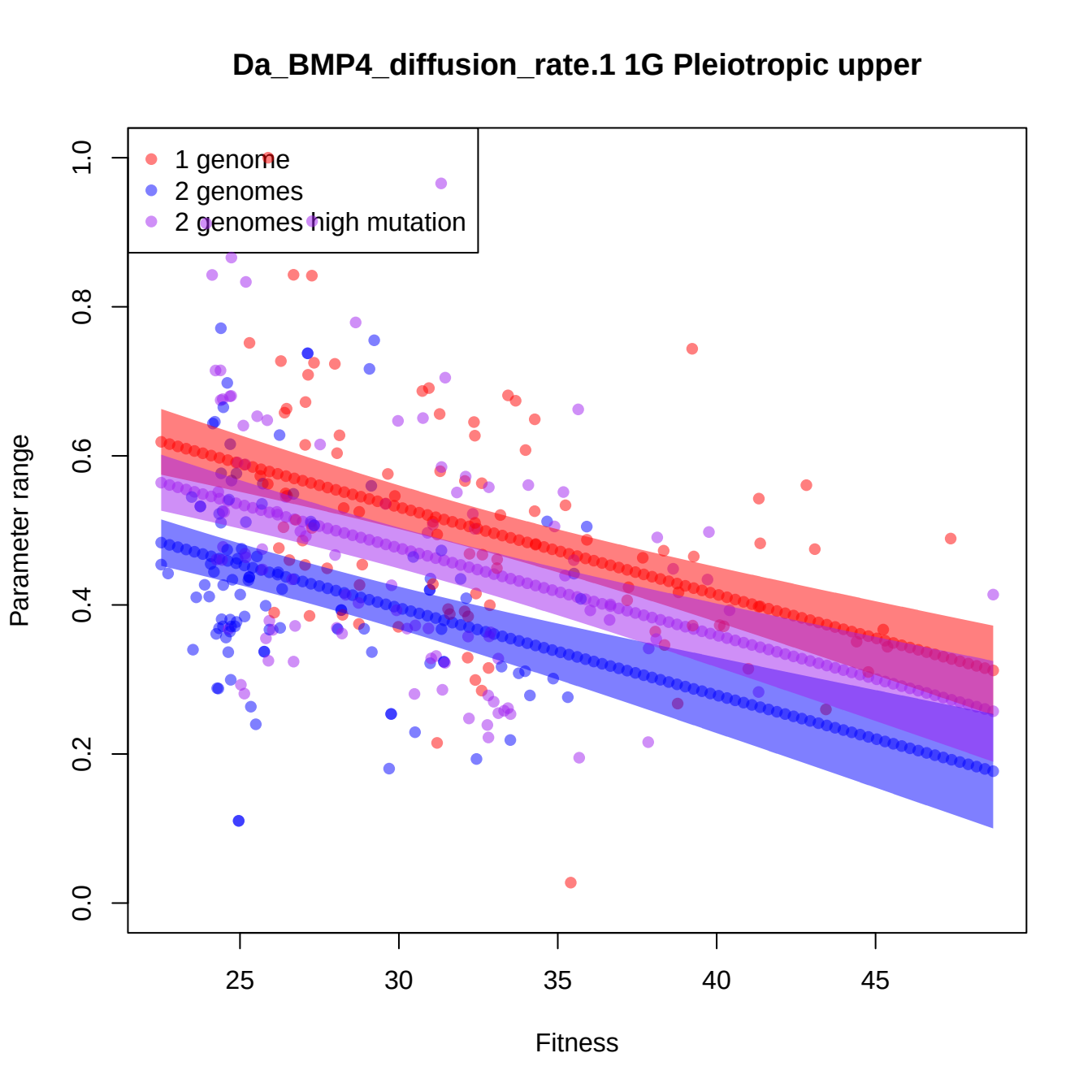

Sec_FGF4_secretion_rate_hox.1
1G HOX upepr
Counts
1.2
1.0
0.8
0.6
0.4
0.0008154335250234074
0.8087416590141 75
0.006089974830 46586
11 − 3
11 − 7
3 − 7
1 genome
2 genomes
2 genomes high mutation

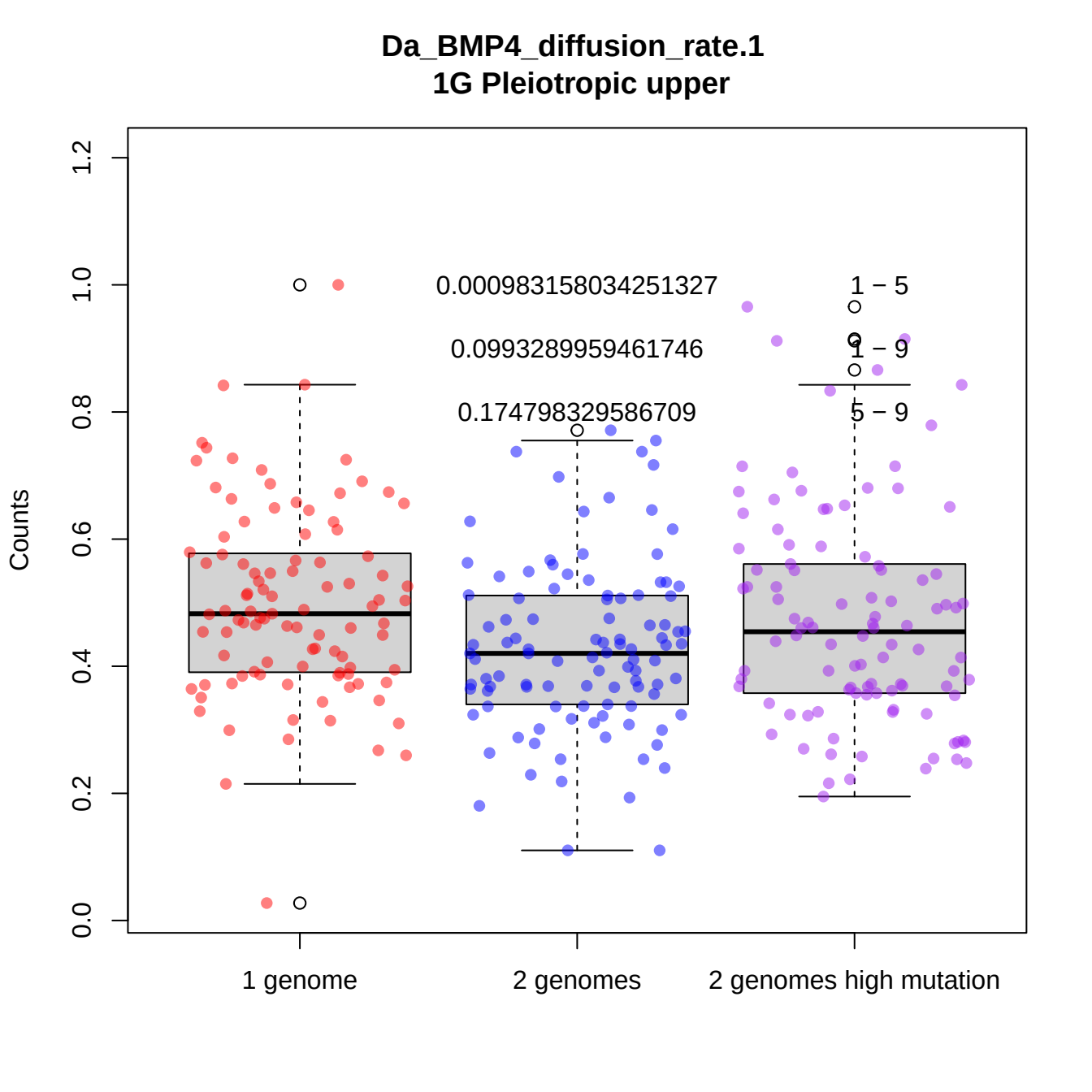

1.0
0.8
0.6
windows
30
40
50
fit
type
4
8
12

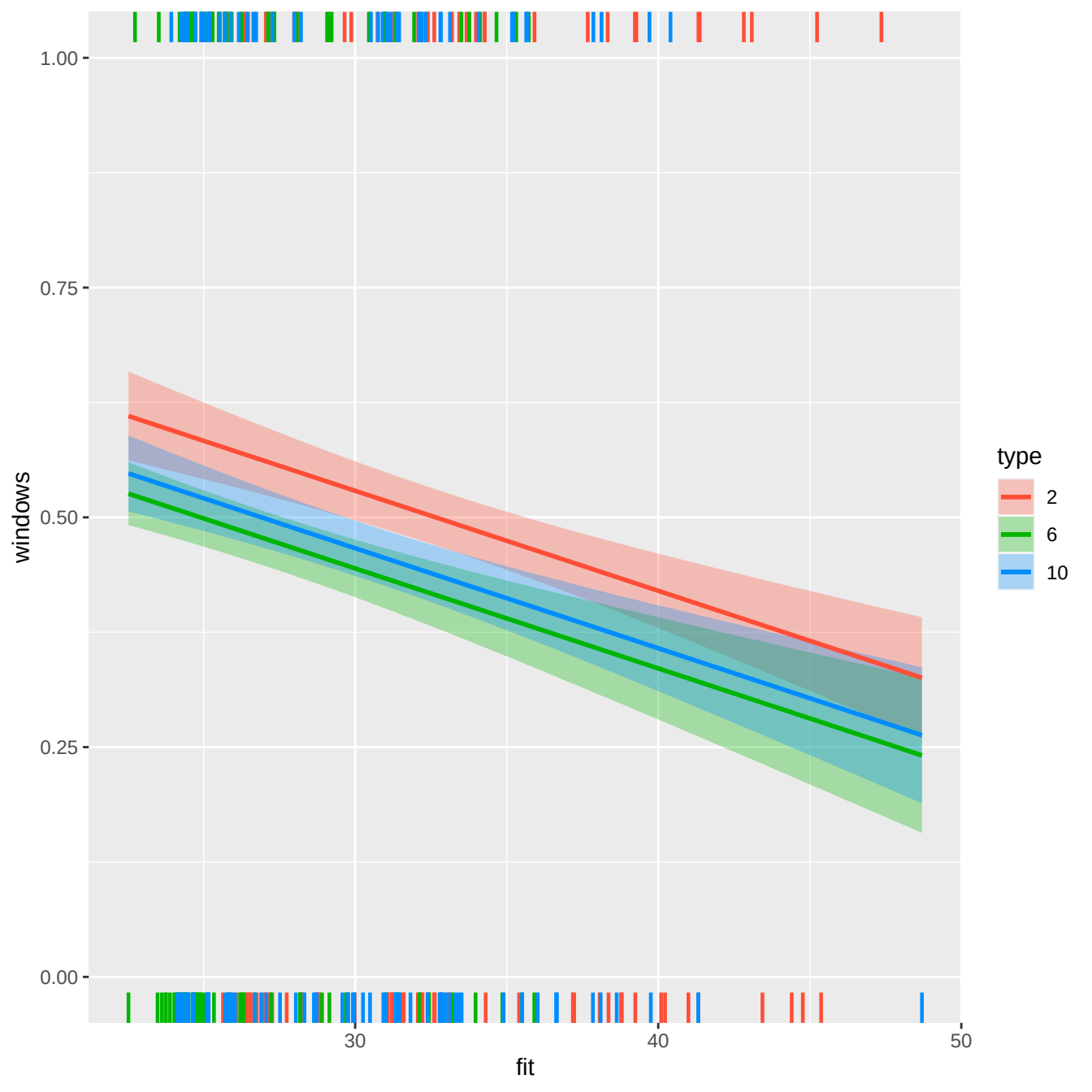

Sec_FGF4_secretion_rate_hox.1 1G HOX lower
1 genome
2 genomes
2 genomes high mutation
Parameter range (normalized 0 to 1)
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

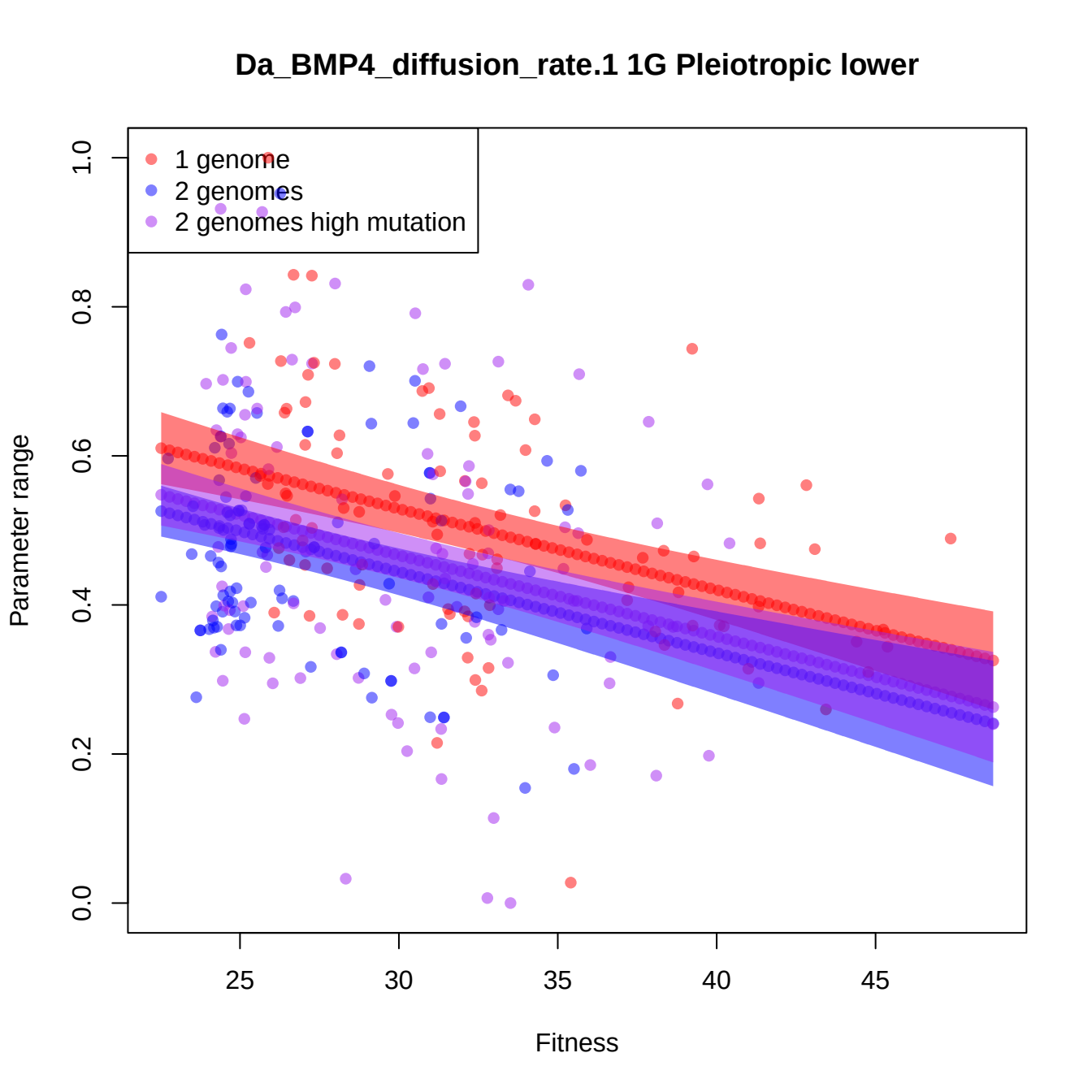

Sec_FGF4_secretion_rate_hox.1
1G HOX lower
Counts
1.2
1.0
0.8
0.6
7.5393742543788e−08
1
1.3498886906146e−07
12 − 4
12 − 8
4 − 8
1 genome
2 genomes
2 genomes high mutation

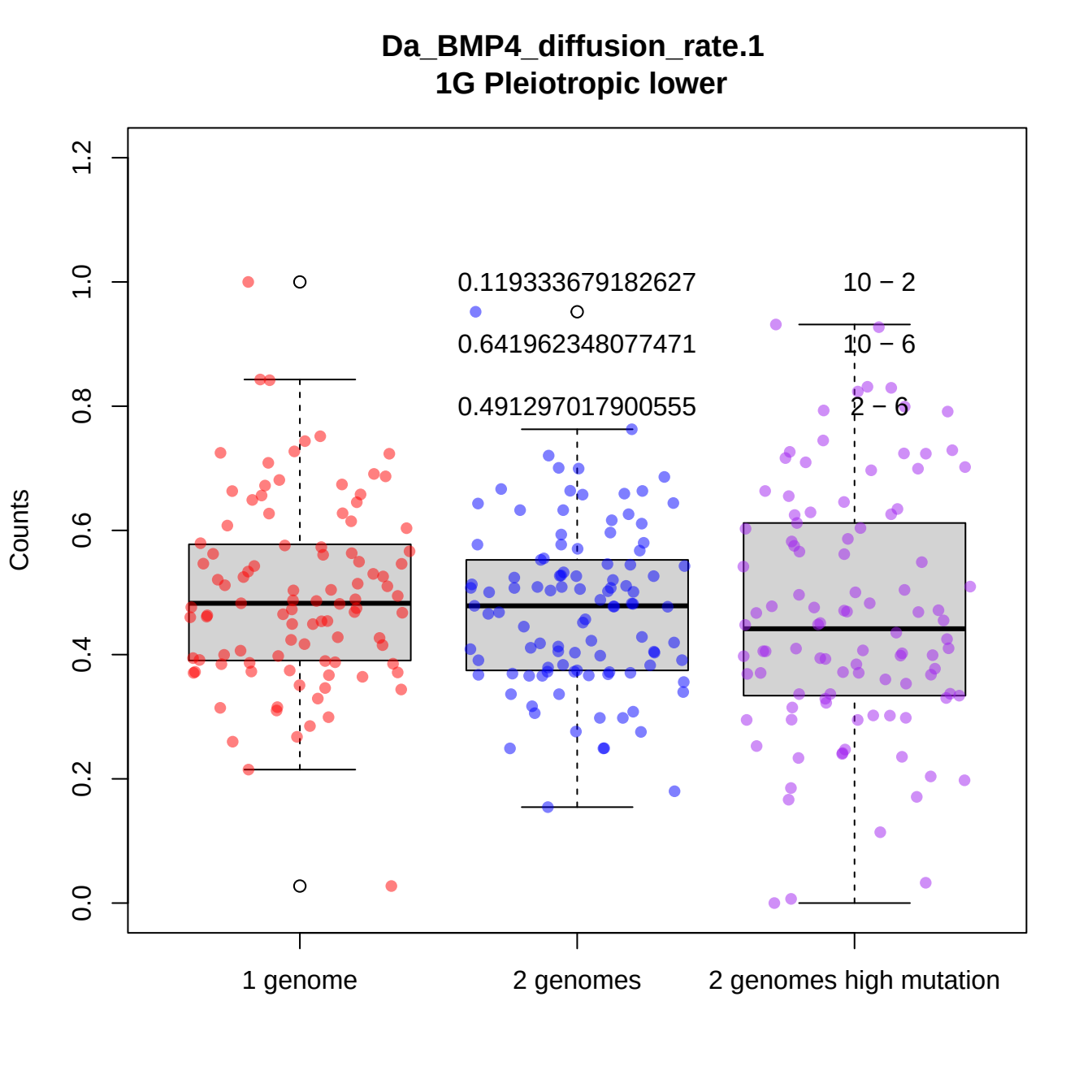

1.00
0.75
0.50
0.25
0.00
windows
30
40
50
fit
type
1
5
9

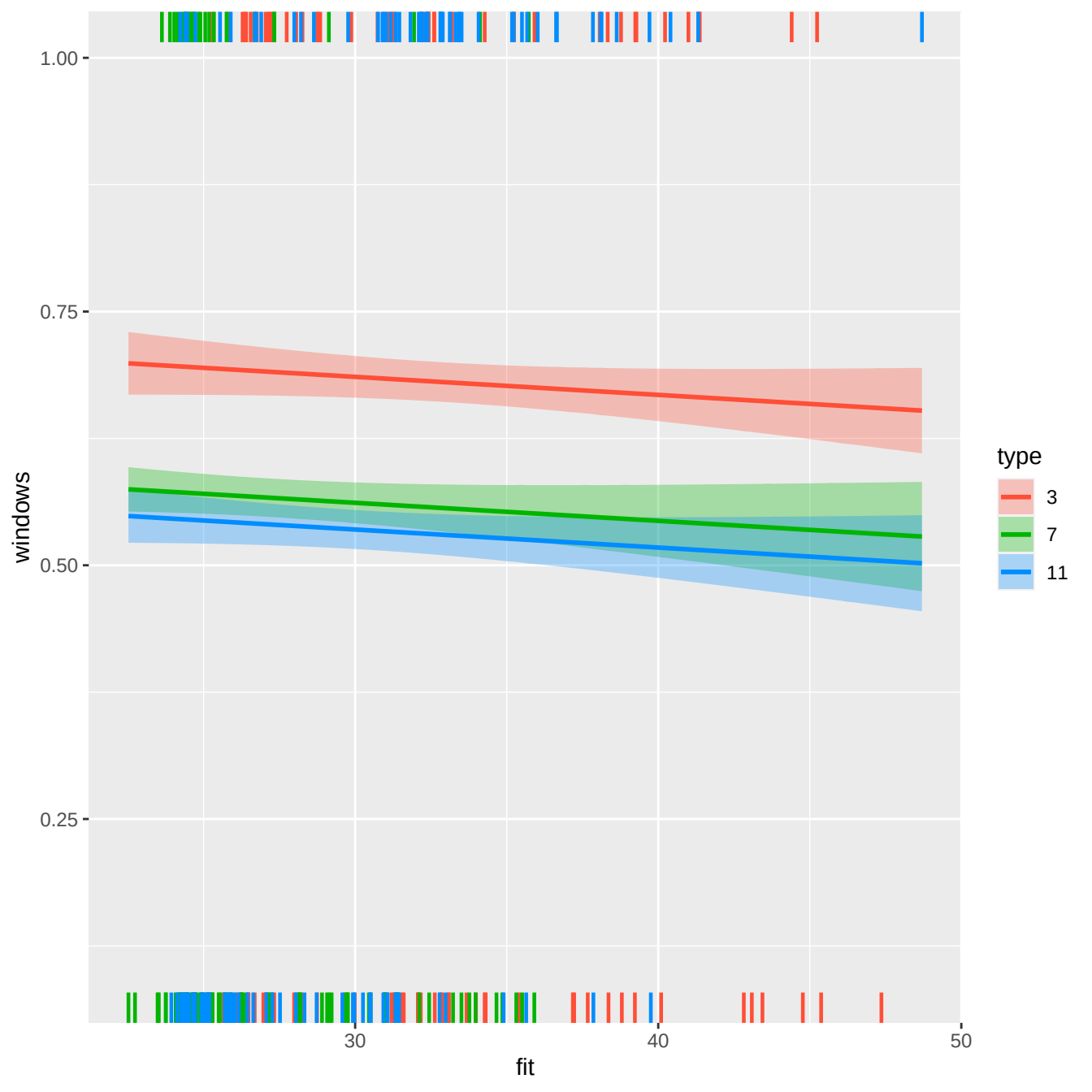

Da_BMP4_diffusion_rate.1 1G Pleiotropic upper
1 genome
2 genomes
2 genomes high mutation
Parameter range
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

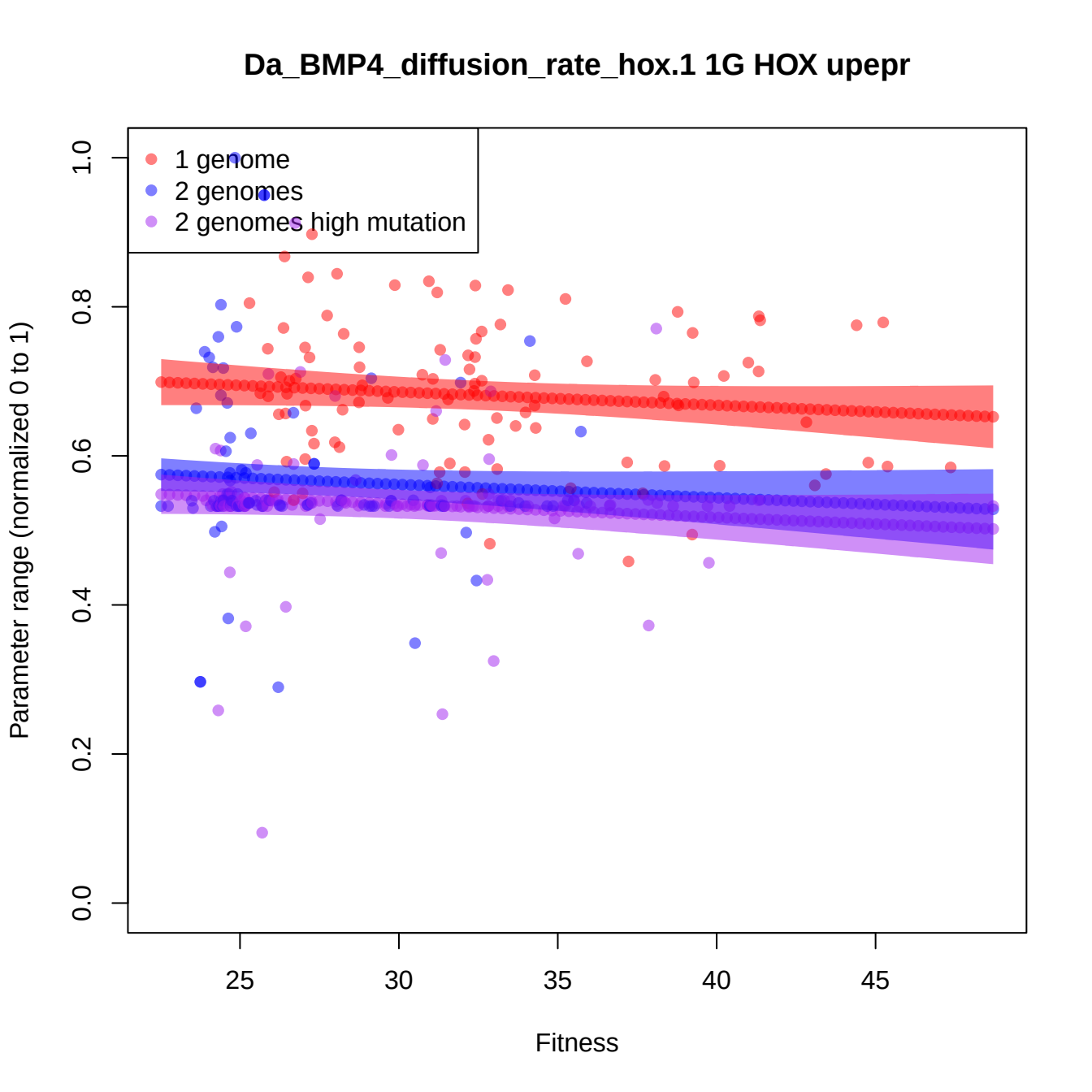

Da_BMP4_diffusion_rate.1
1G Pleiotropic upper
0.000983158034251327
0.0993289959461746
0.174798329586709
1 − 5
1 − 9
5 − 9
Counts
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1 genome
2 genomes
2 genomes high mutation

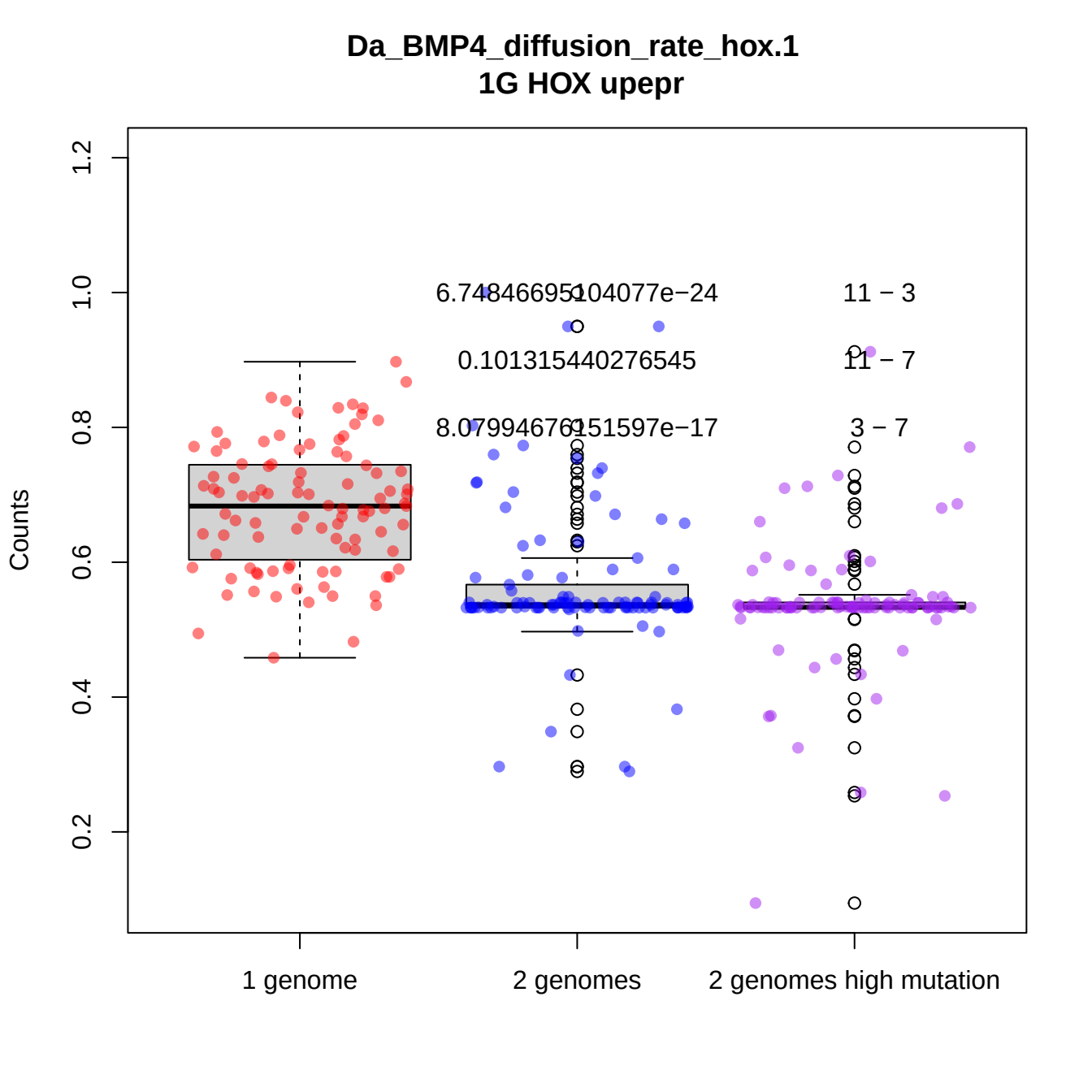

1.00
0.75
0.50
0.25
0.00
windows
30
40
50
fit
type
2
6
10

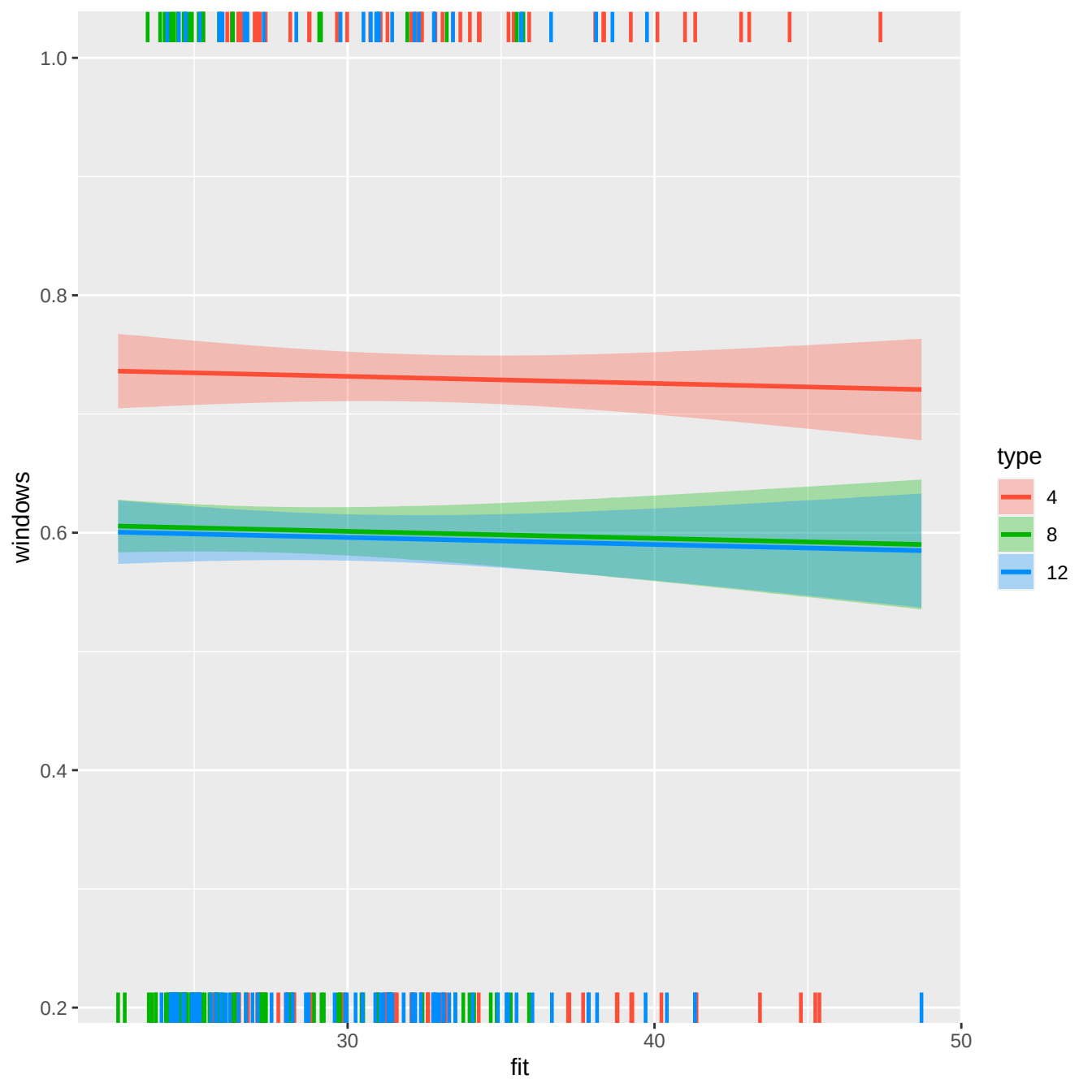

Da_BMP4_diffusion_rate.1 1G Pleiotropic lower
1 genome
2 genomes
2 genomes high mutation
Parameter range
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

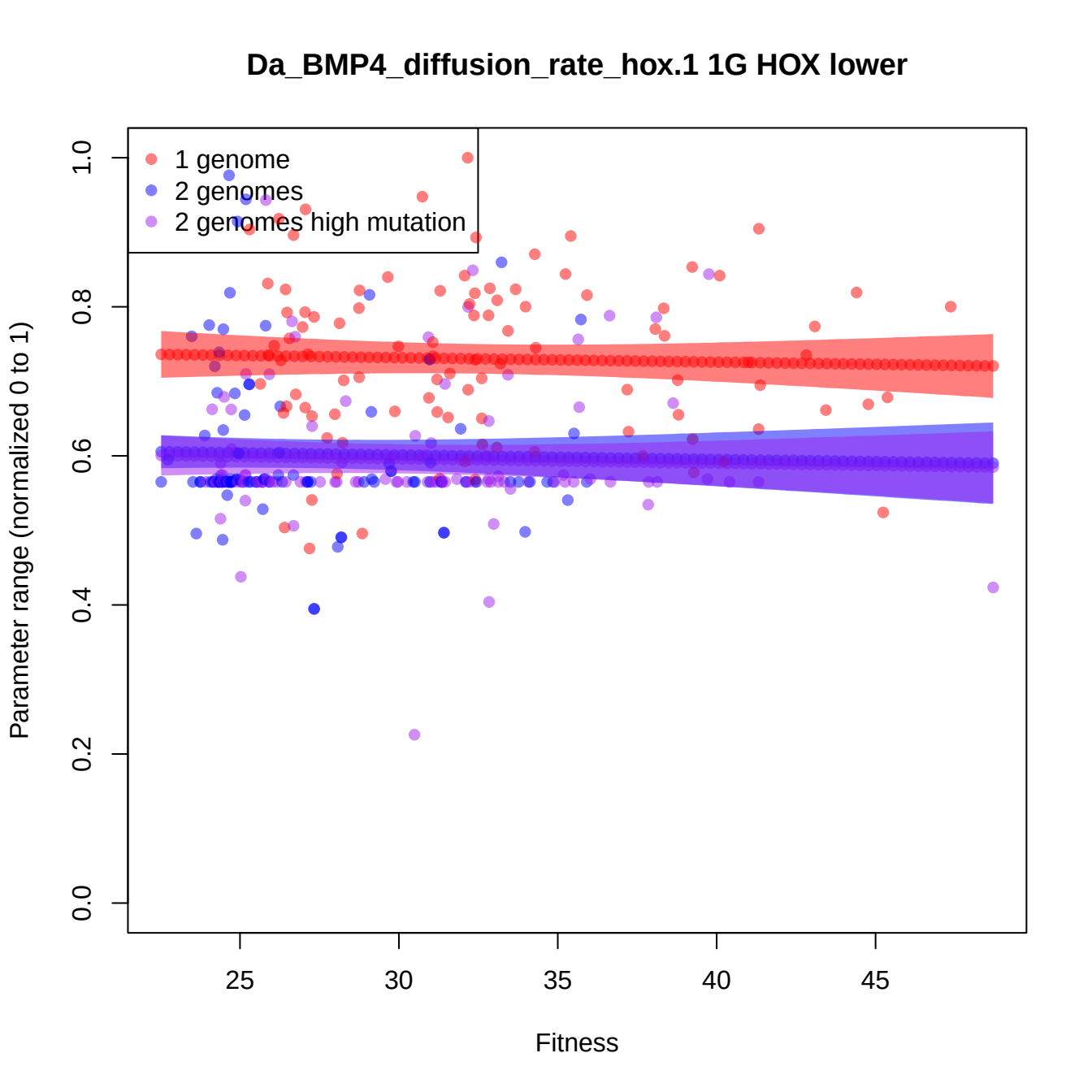

Da_BMP4_diffusion_rate.1
1G Pleiotropic lower
Counts
0.0
0.2
0.4
0.6
0.8
1.0
1.2
0.119333679182627
0.641962348077471
0.491297017900555
10 − 2
10 − 6
2 − 6
1 genome
2 genomes
2 genomes high mutation

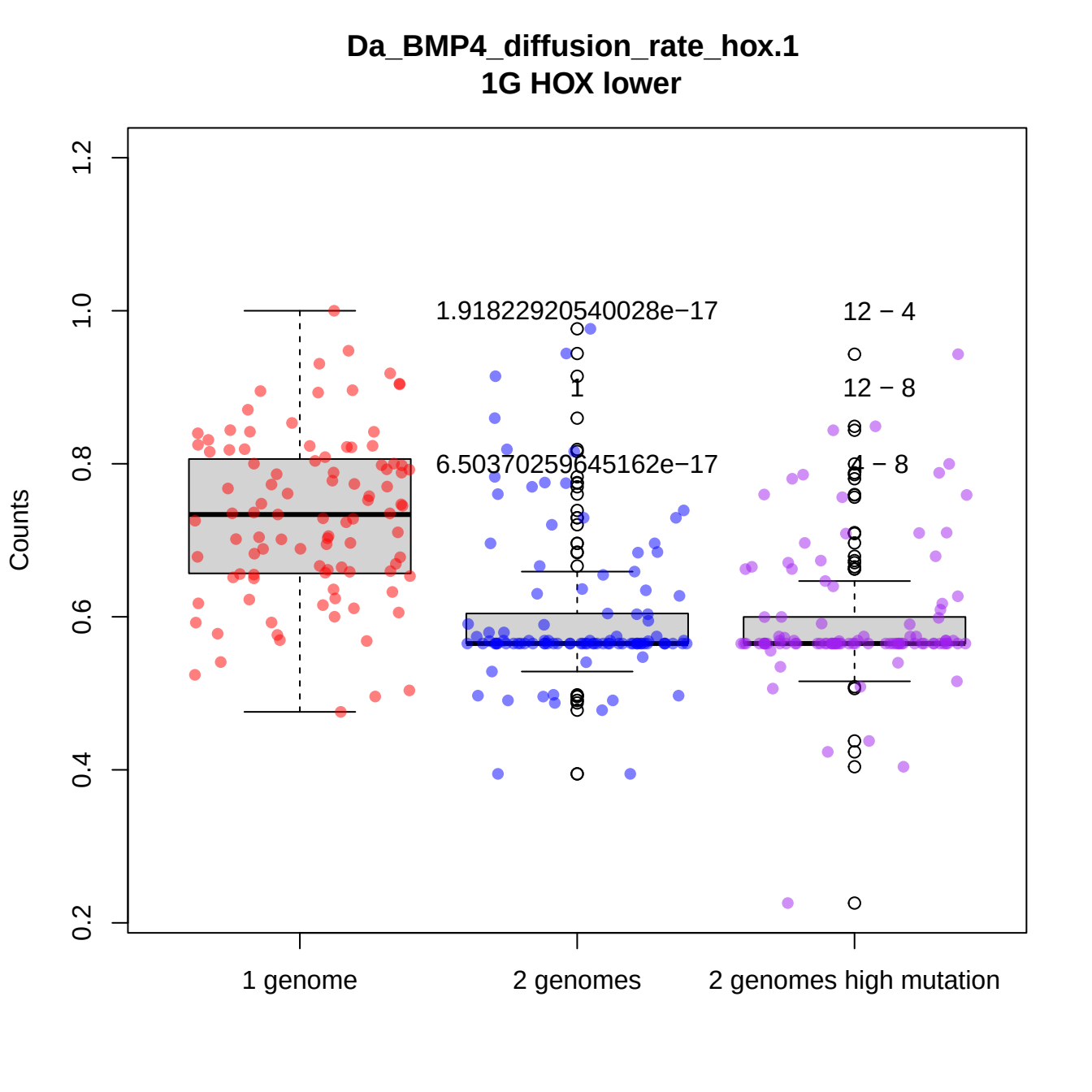

1.00
0.75
0.50
0.25
windows
30
40
50
fit
type
3
7
11

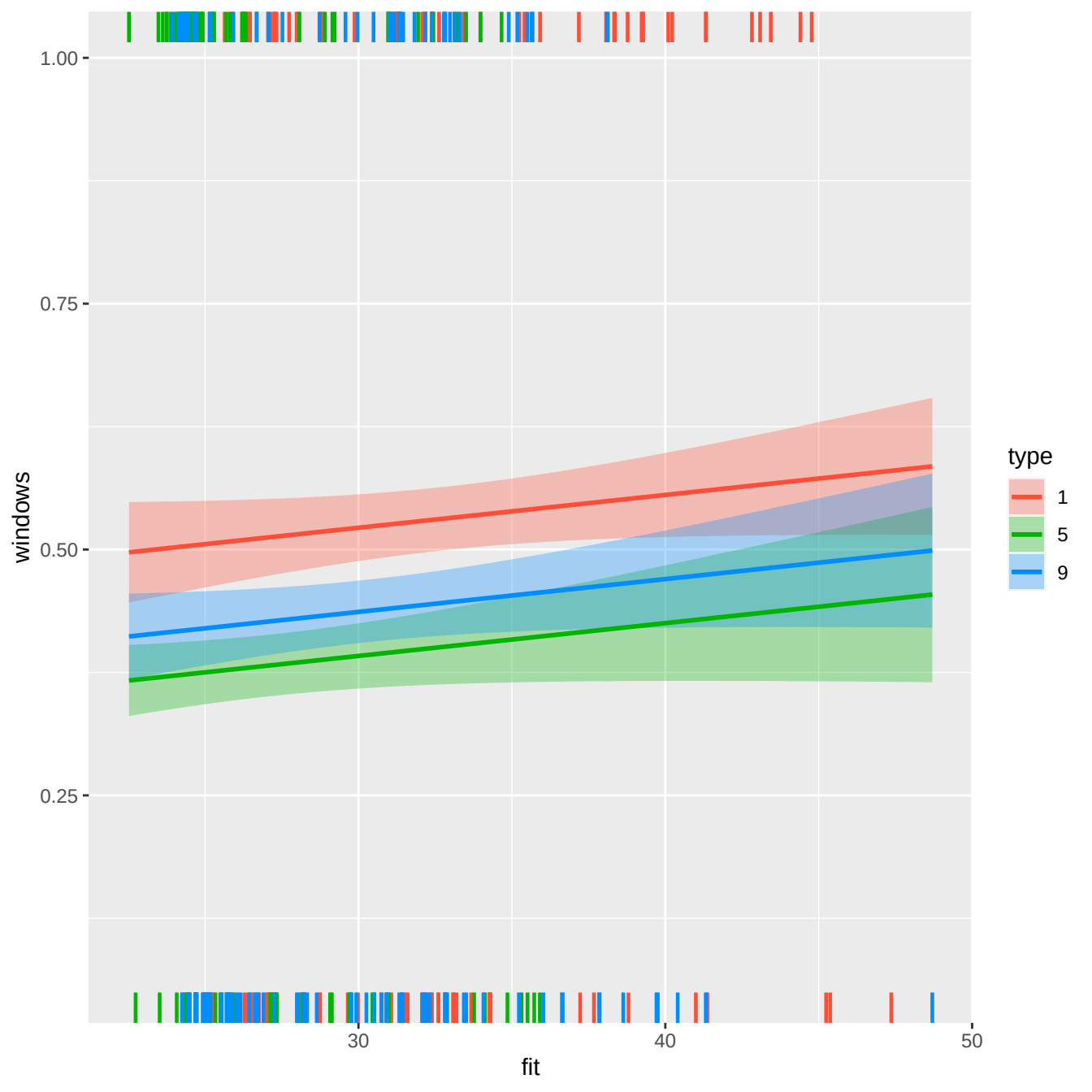

Da_BMP4_diffusion_rate_hox.1 1G HOX upepr
1 genome
2 genomes
2 genomes high mutation
Parameter range (normalized 0 to 1)
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

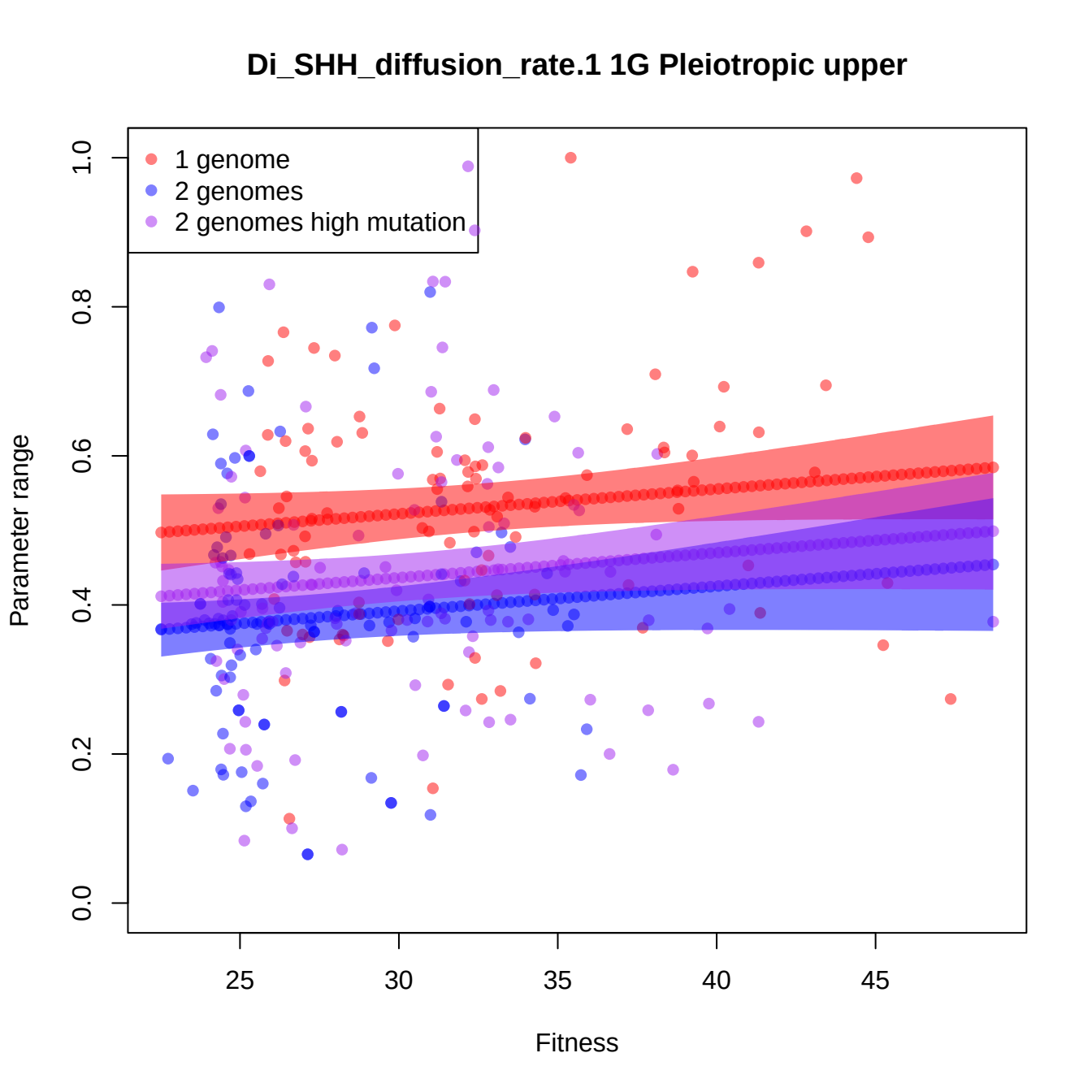

Da_BMP4_diffusion_rate_hox.1
1G HOX upepr
Counts
1.2
1.0
0.8
0.6
0.4
0.2
6.748466951104077e−24
0.101315440276545
8.079946761151597e−17
11 − 3
11 − 7
3 − 7
1 genome
2 genomes
2 genomes high mutation

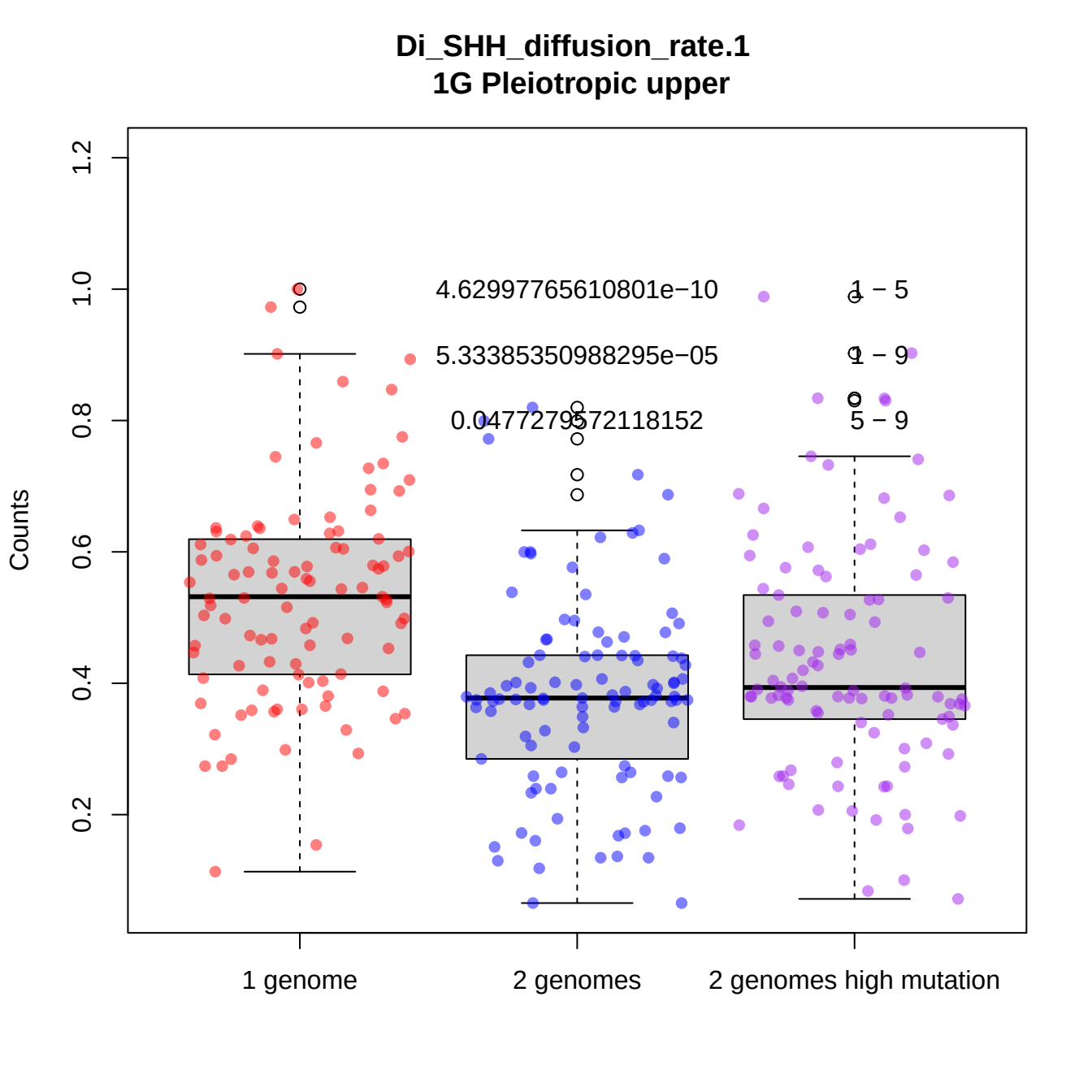

1.0
0.8
0.6
0.4
0.2
windows
30
40
50
fit
type
4
8
12

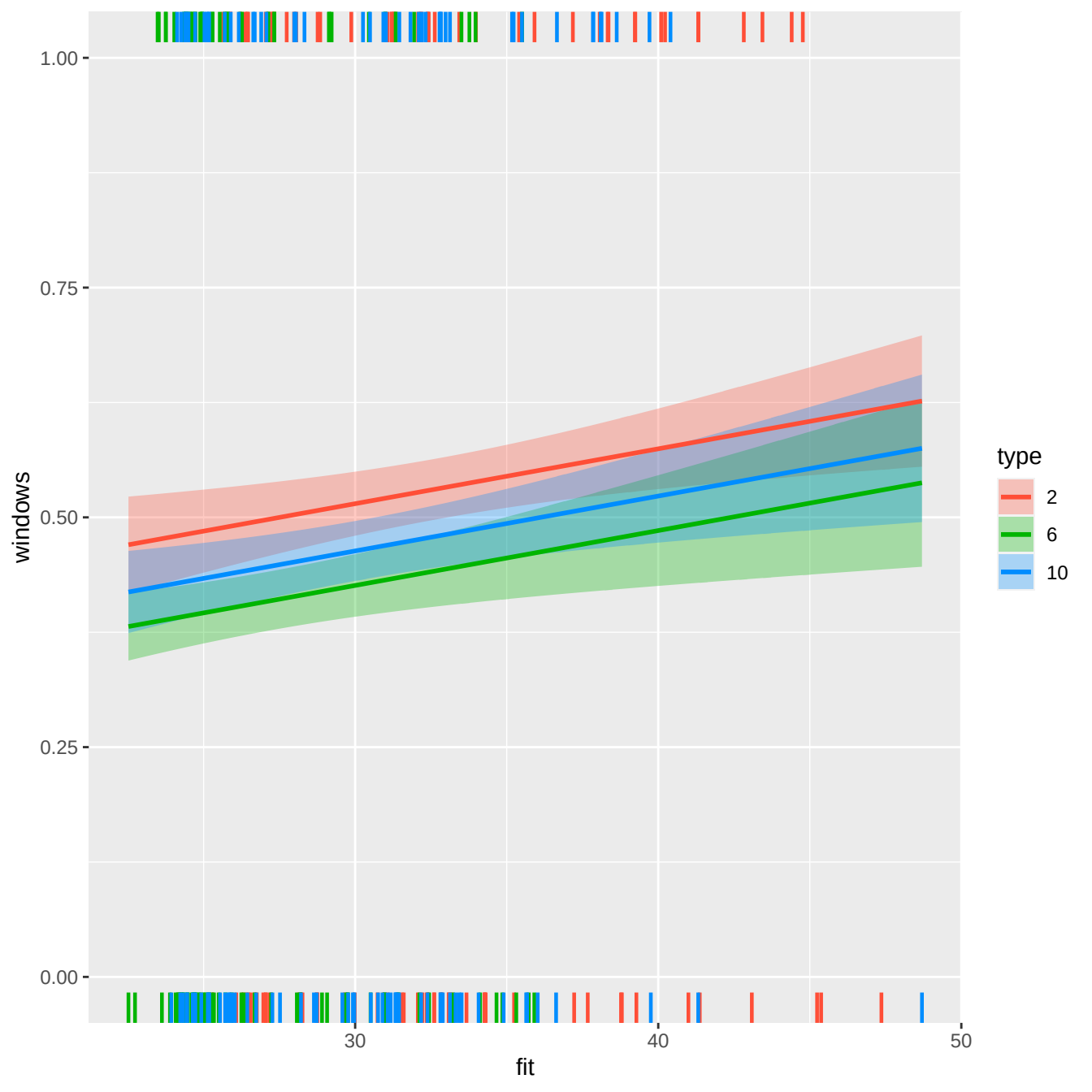

Da_BMP4_diffusion_rate_hox.1 1G HOX lower
1 genome
2 genomes
2 genomes high mutation
Parameter range (normalized 0 to 1)
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

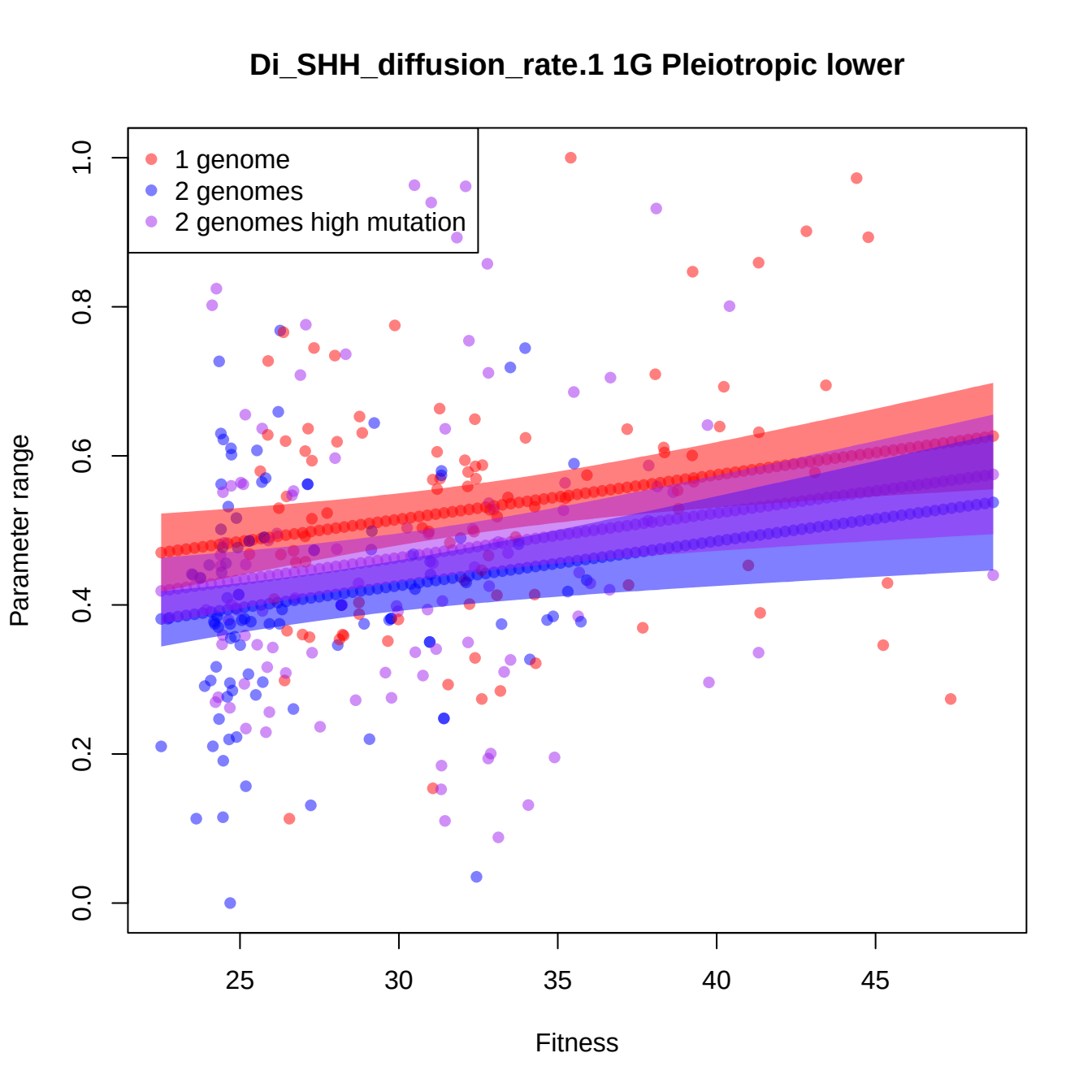

Da_BMP4_diffusion_rate_hox.1
1G HOX lower
Counts
1.2
1.0
0.8
0.6
0.4
0.2
1.91822920540028e−17
1
6.50370259645162e−17
12 − 4
12 − 8
4 − 8
1 genome
2 genomes
2 genomes high mutation

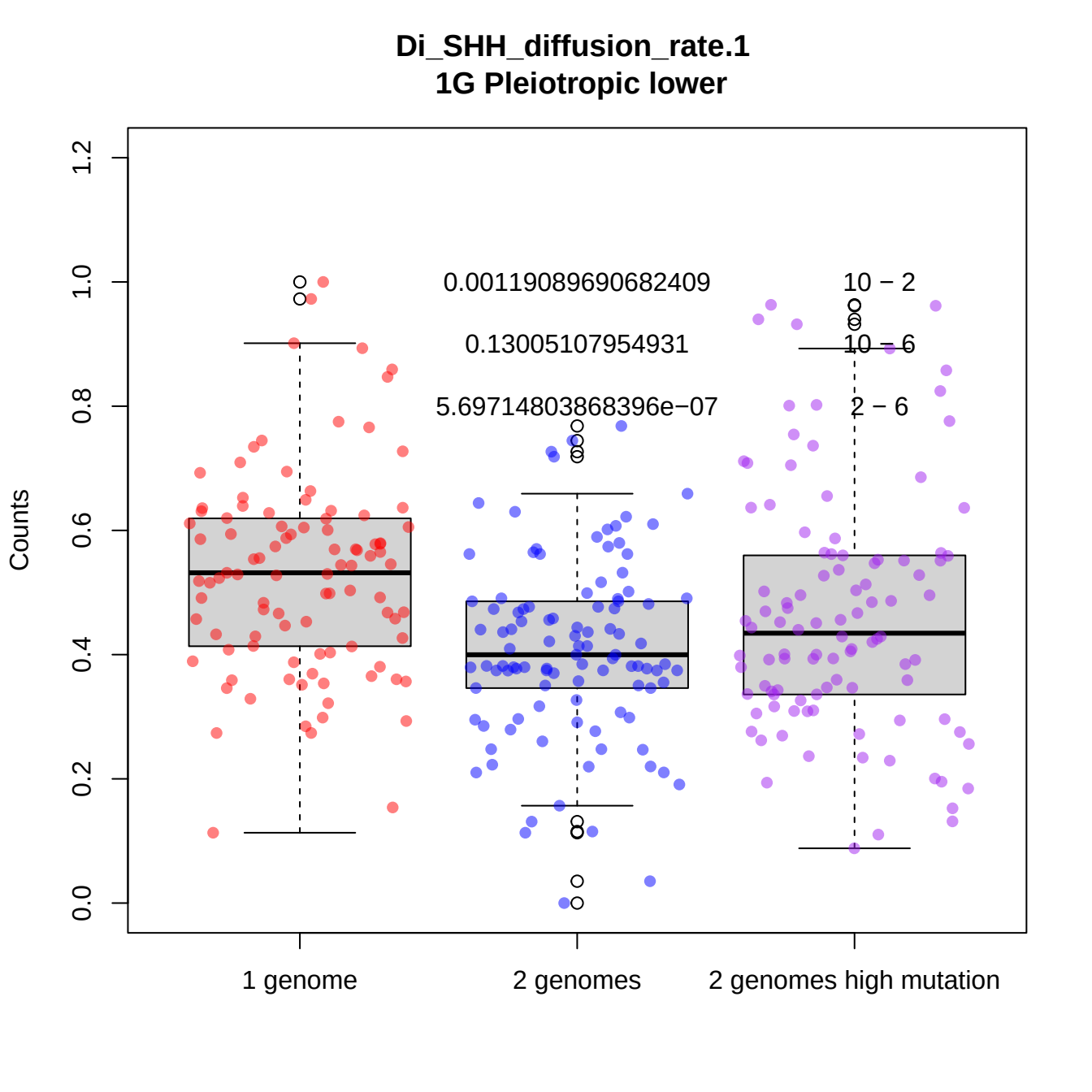

1.00
0.75
0.50
0.25
windows
30
40
50
fit
type
1
5
9

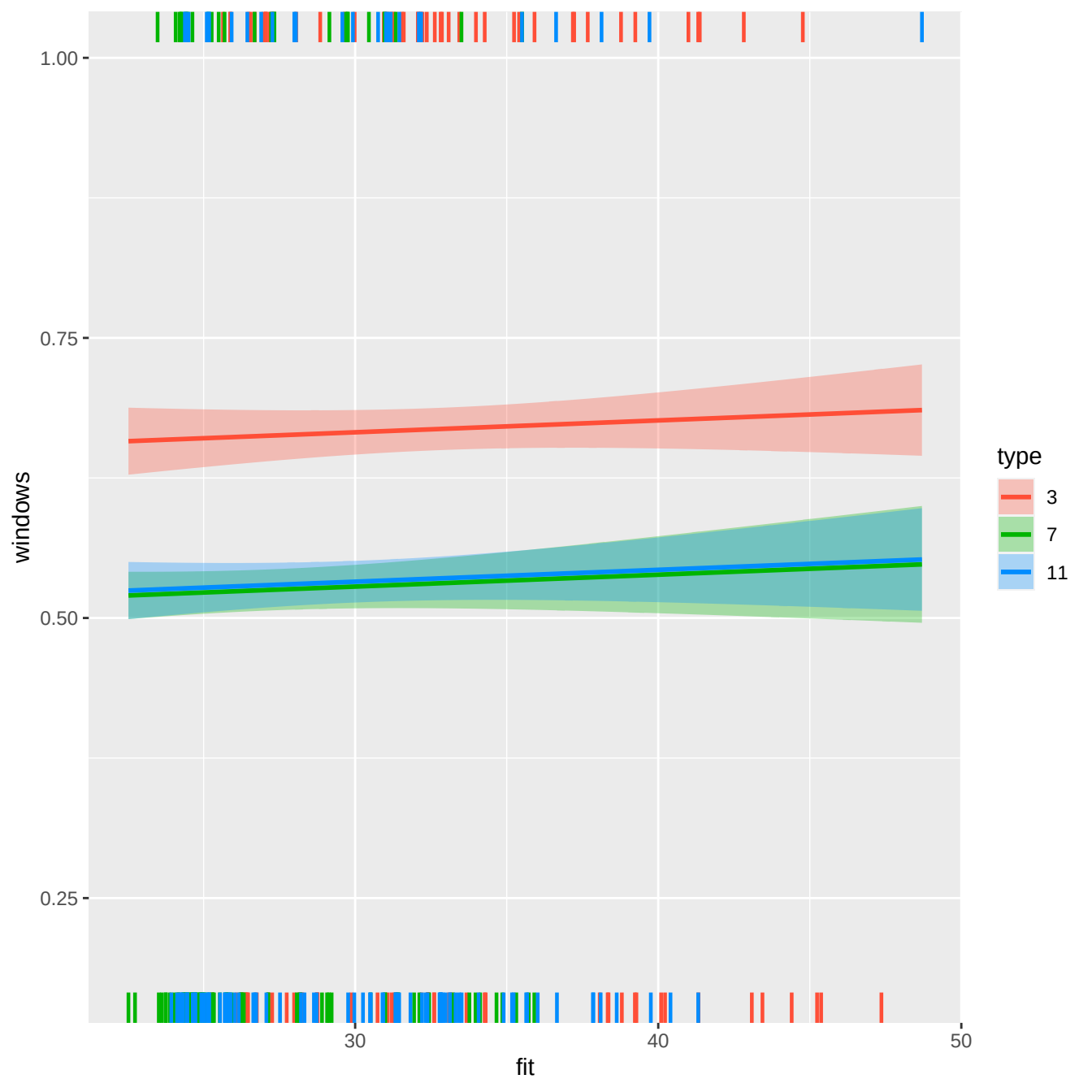

Di_SHH_diffusion_rate.1 1G Pleiotropic upper
1 genome
2 genomes
2 genomes high mutation
Parameter range
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

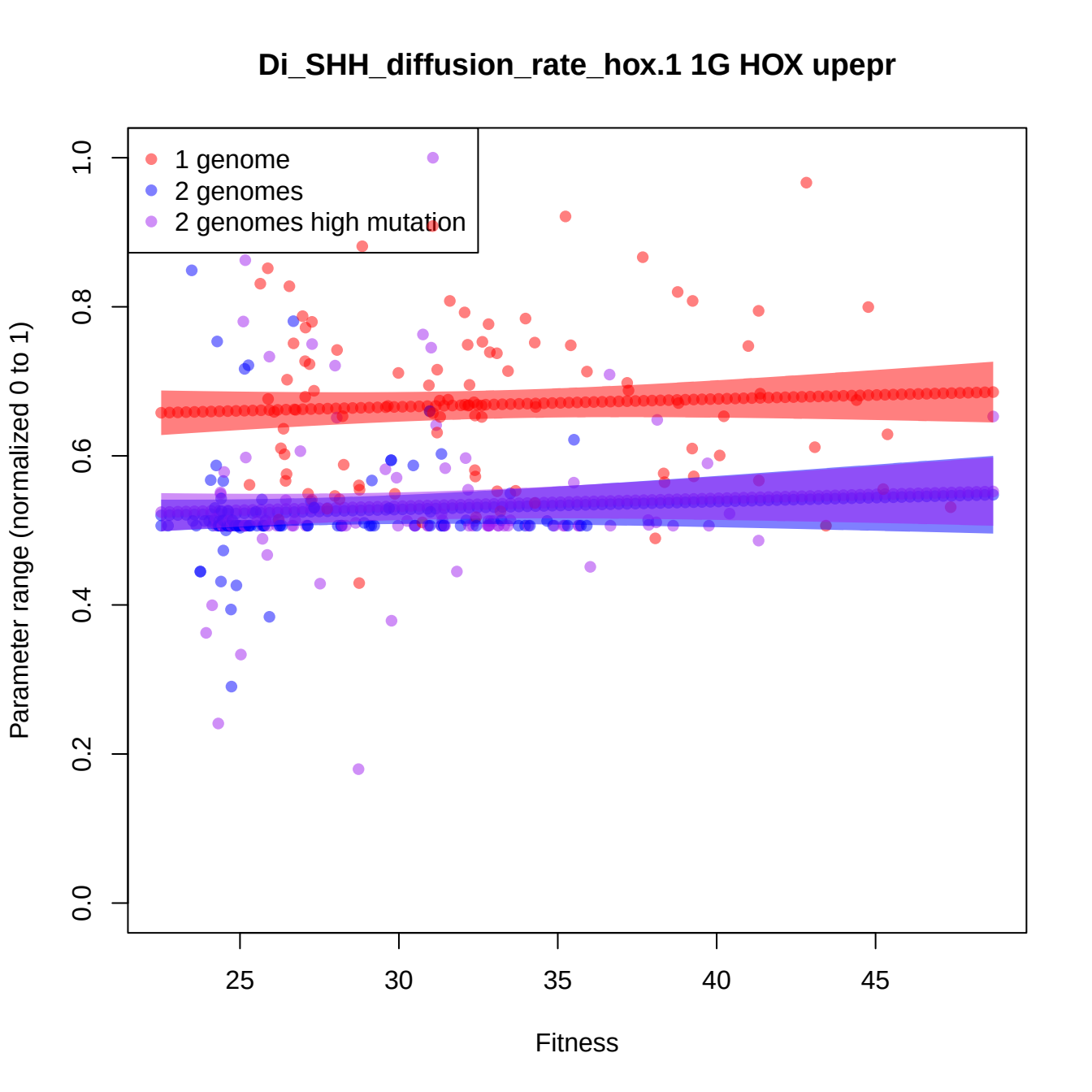

Di_SHH_diffusion_rate.1
1G Pleiotropic upper
Counts
1.2
1.0
0.8
0.6
0.4
0.2
4.62997765610801e−10
5.33385350988295e−05
0.0477279572118152
1 − 5
1 − 9
5 − 9
1 genome
2 genomes
2 genomes high mutation

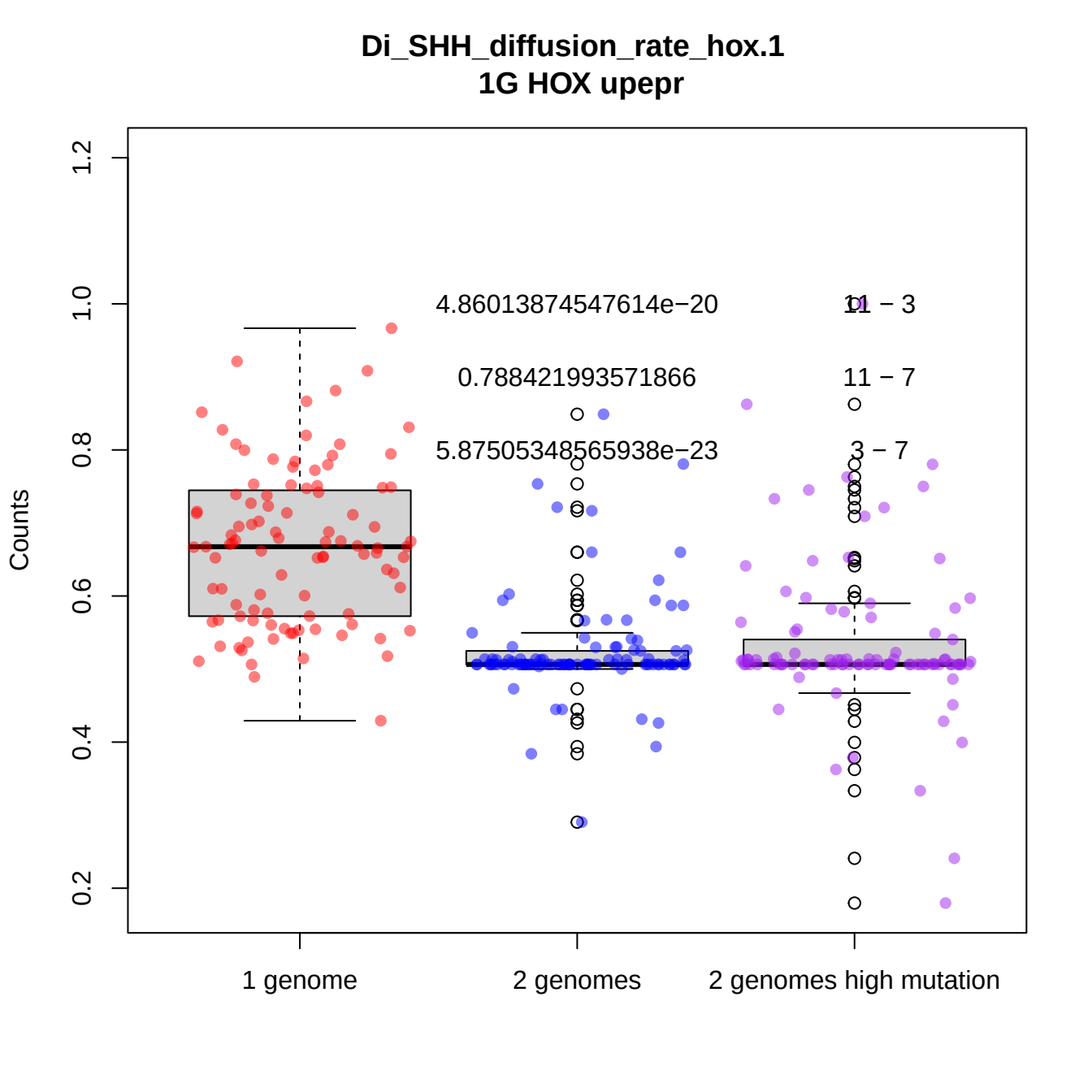

1.00
0.75
0.50
0.25
0.00
windows
30
40
50
fit
type
2
6
10

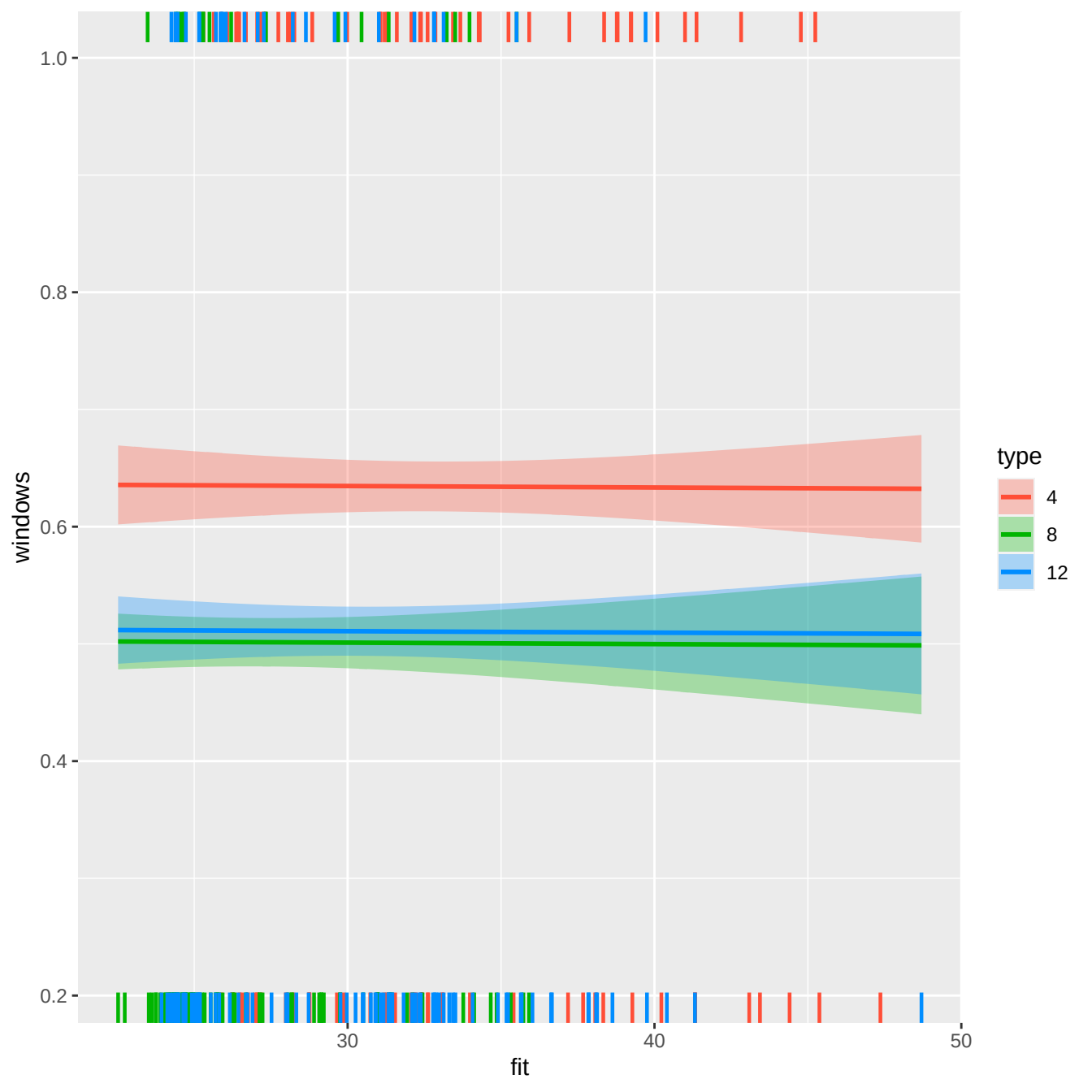

Di_SHH_diffusion_rate.1 1G Pleiotropic lower
1 genome
2 genomes
2 genomes high mutation
Parameter range
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

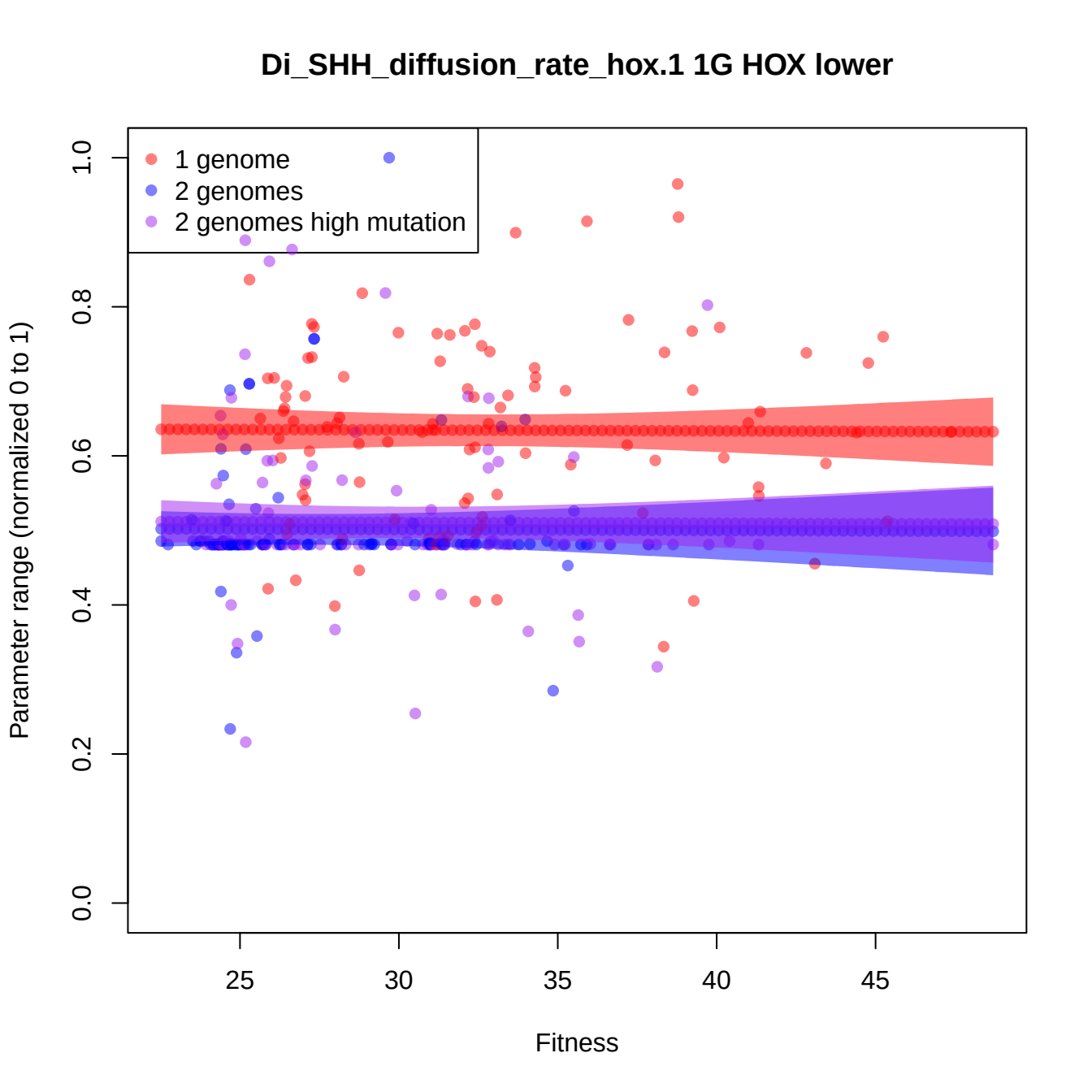

Di_SHH_diffusion_rate.1
1G Pleiotropic lower
Counts
0.0
0.2
0.4
0.6
0.8
1.0
1.2
0.00119089690682409
0.13005107954931
5.69714803868396e−07
10 − 2
10 − 6
2 − 6
1 genome
2 genomes
2 genomes high mutation

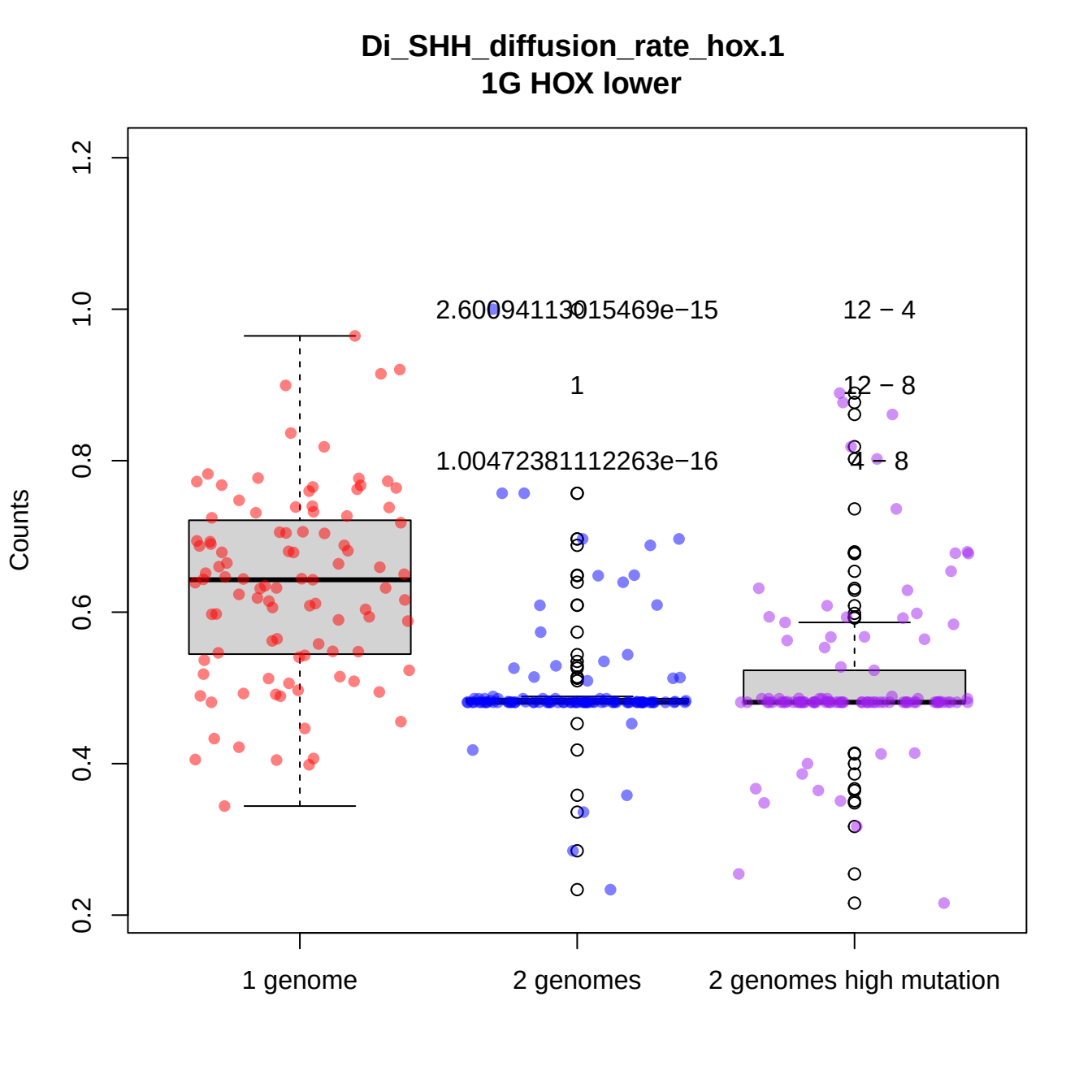

1.00
0.75
0.50
0.25
windows
30
40
50
fit
type
3
7
11

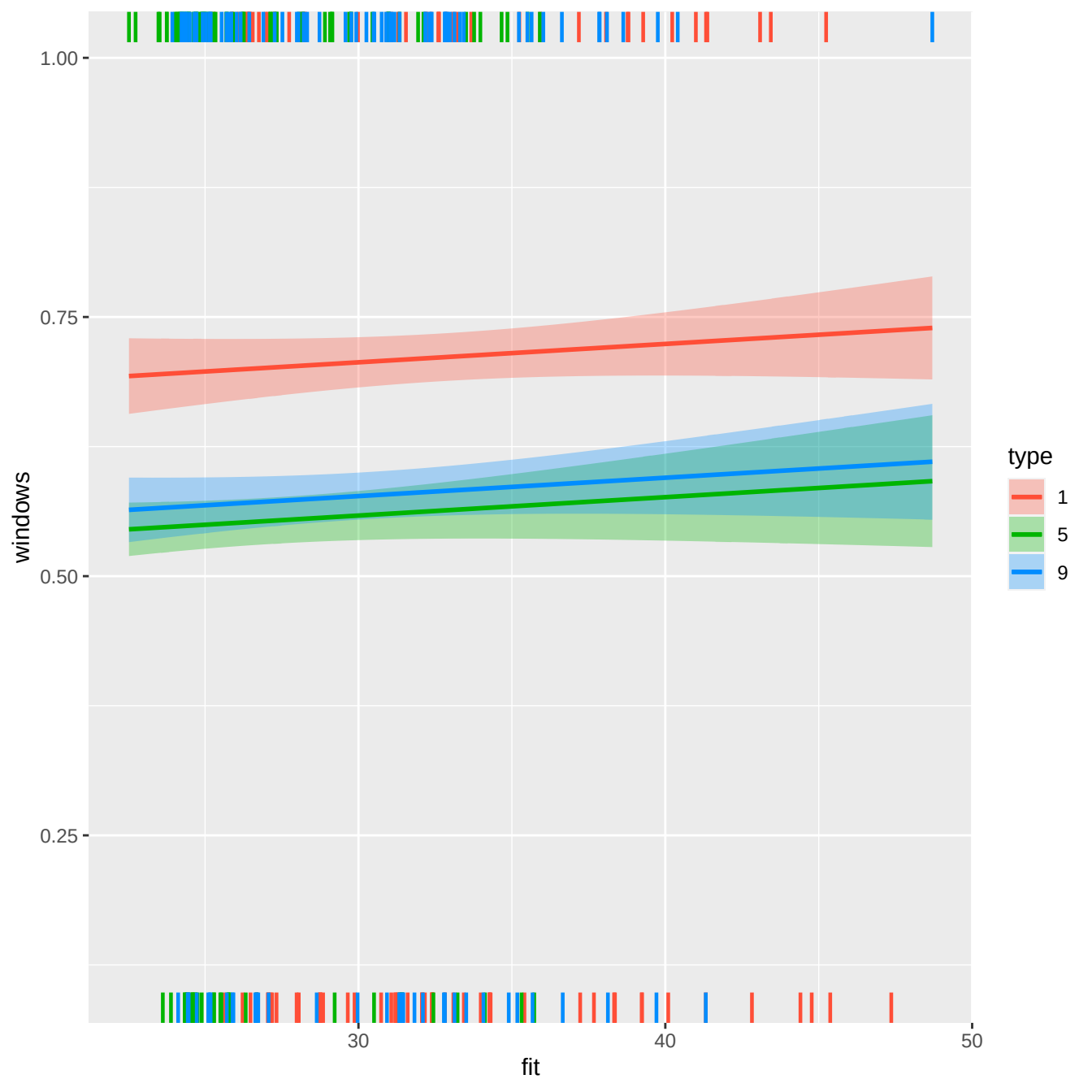

Di_SHH_diffusion_rate_hox.1 1G HOX upepr
1 genome
2 genomes
2 genomes high mutation
Parameter range (normalized 0 to 1)
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

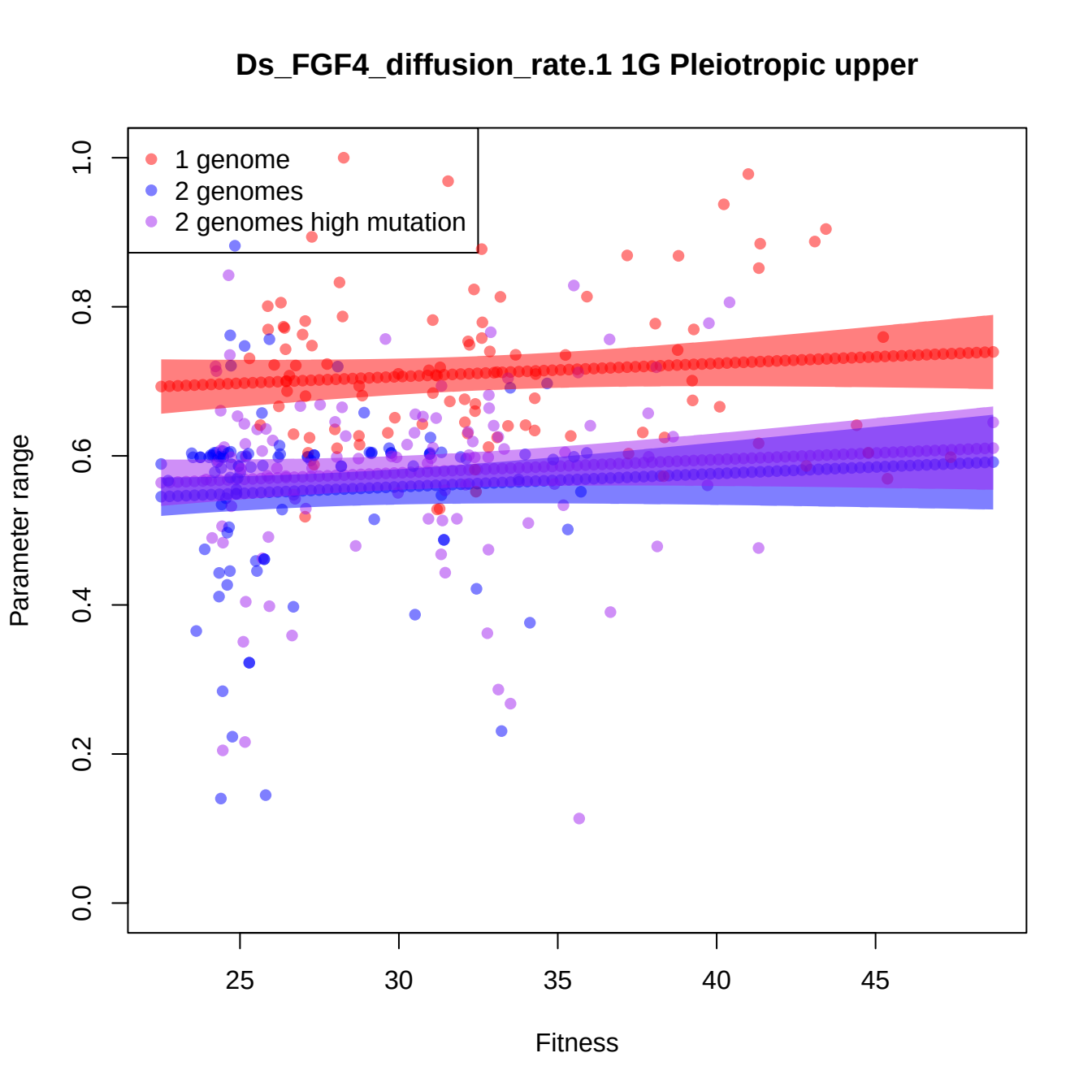

Di_SHH_diffusion_rate_hox.1
1G HOX upepr
Counts
1.2
1.0
0.8
0.6
0.4
0.2
4.86013874547614e−20
0.788421993571866
5.87505348565938e−23
11 − 3
11 − 7
3 − 7
1 genome
2 genomes
2 genomes high mutation

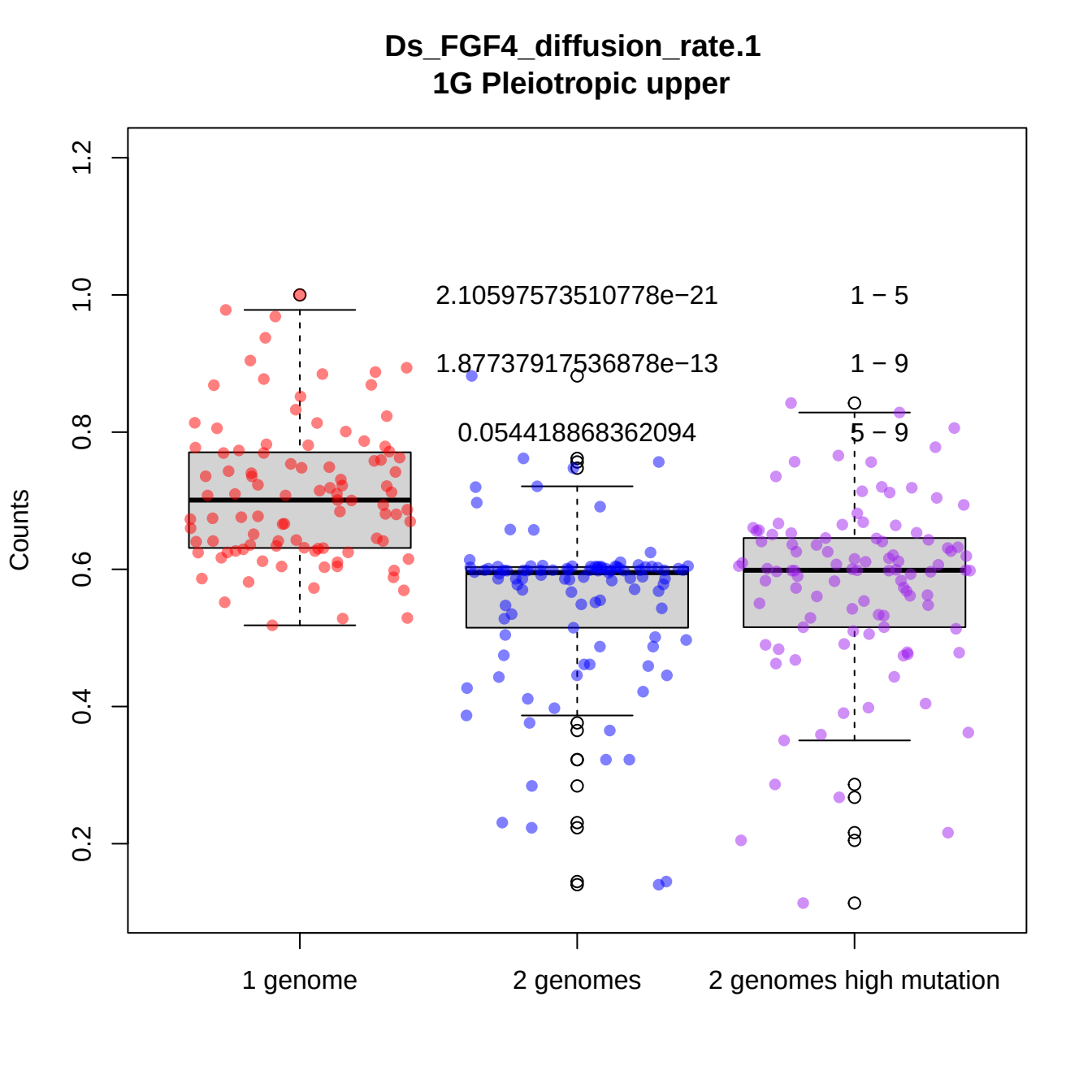

1.0
0.8
0.6
0.4
0.2
windows
30
40
50
fit
type
4
8
12

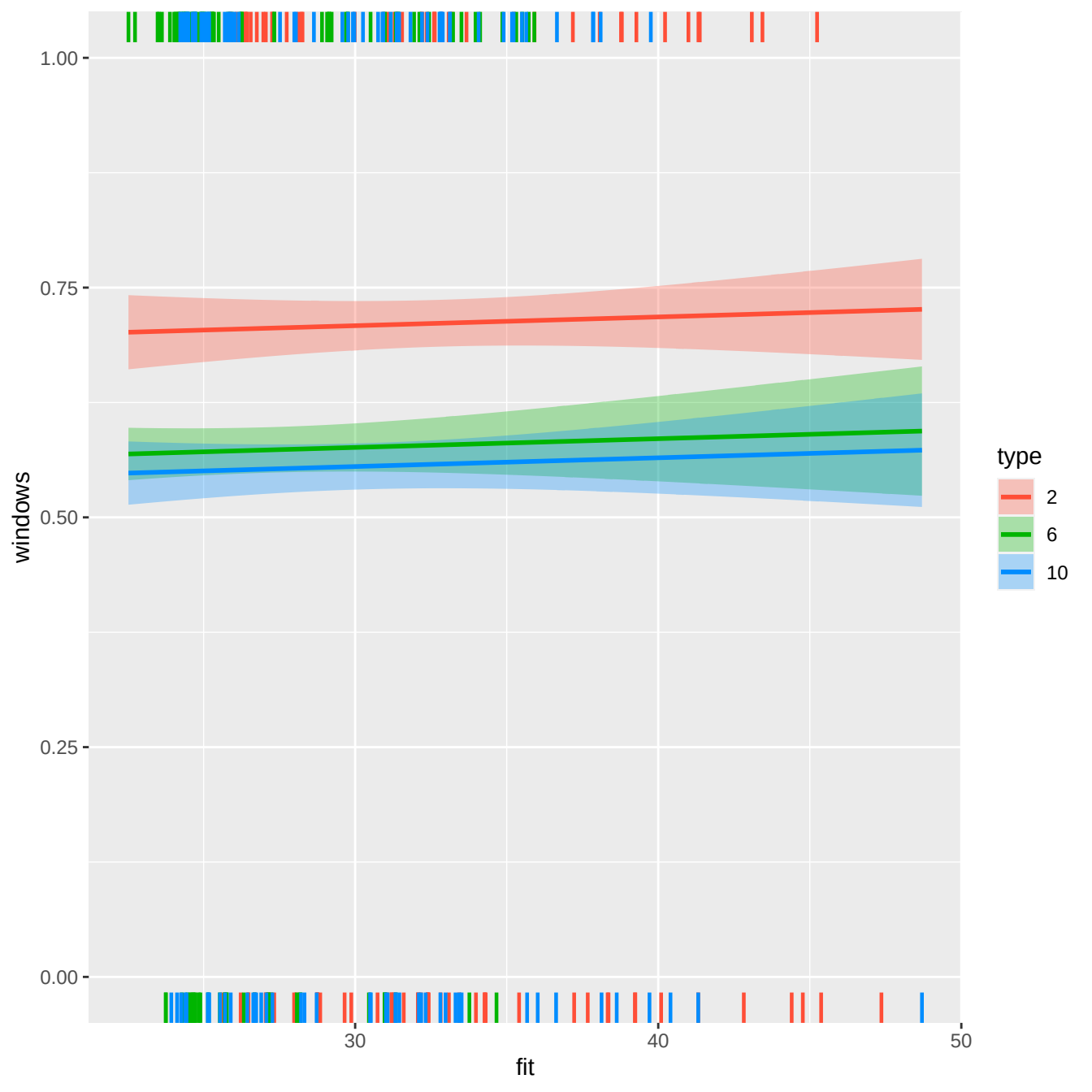

Di_SHH_diffusion_rate_hox.1 1G HOX lower
1 genome
2 genomes
2 genomes high mutation
Parameter range (normalized 0 to 1)
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

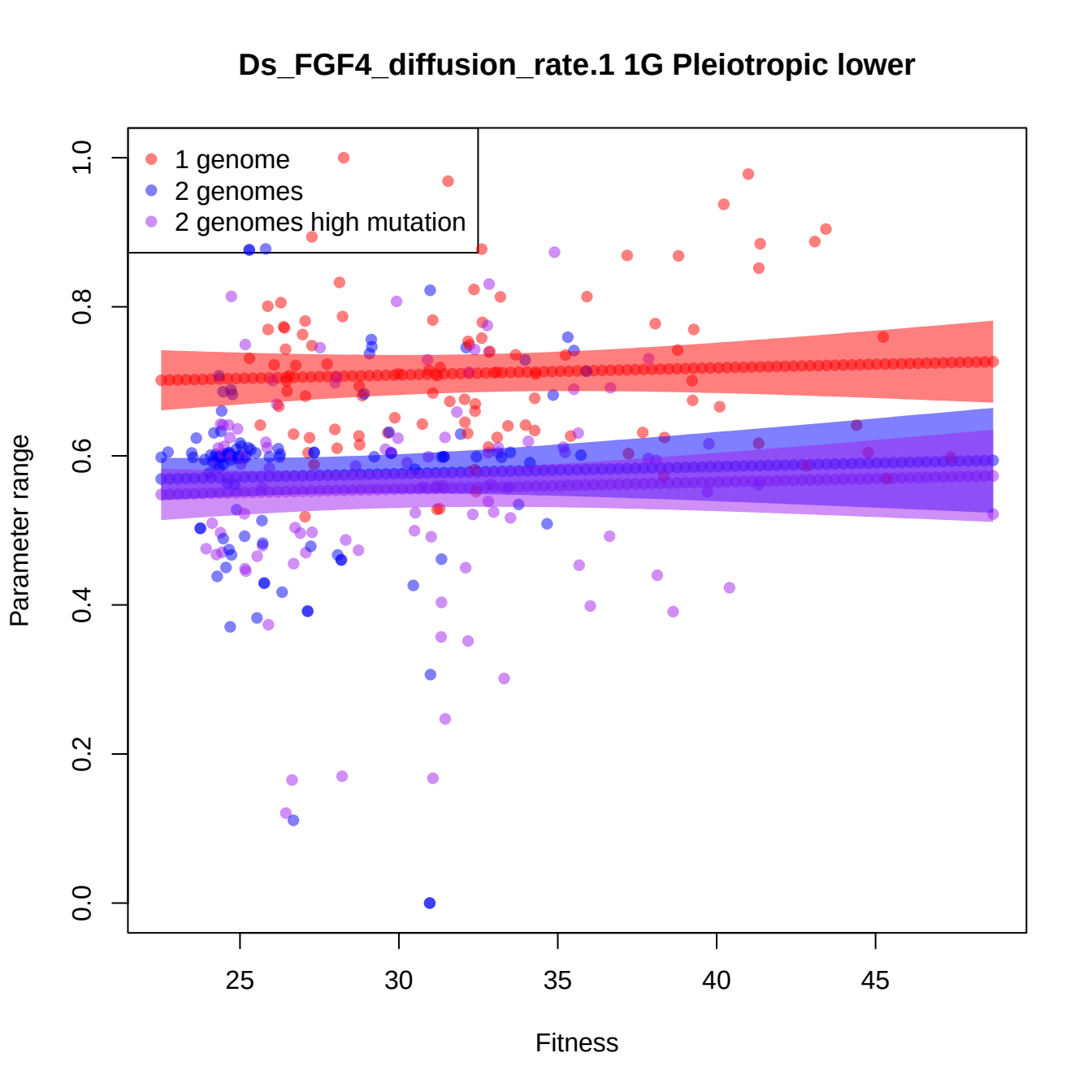

Di_SHH_diffusion_rate_hox.1
1G HOX lower
Counts
1.2
1.0
0.8
0.6
0.4
0.2
2.60094113015469e−15
1
1.00472381112263e−16
12 − 4
12 − 8
4 − 8
1 genome
2 genomes
2 genomes high mutation

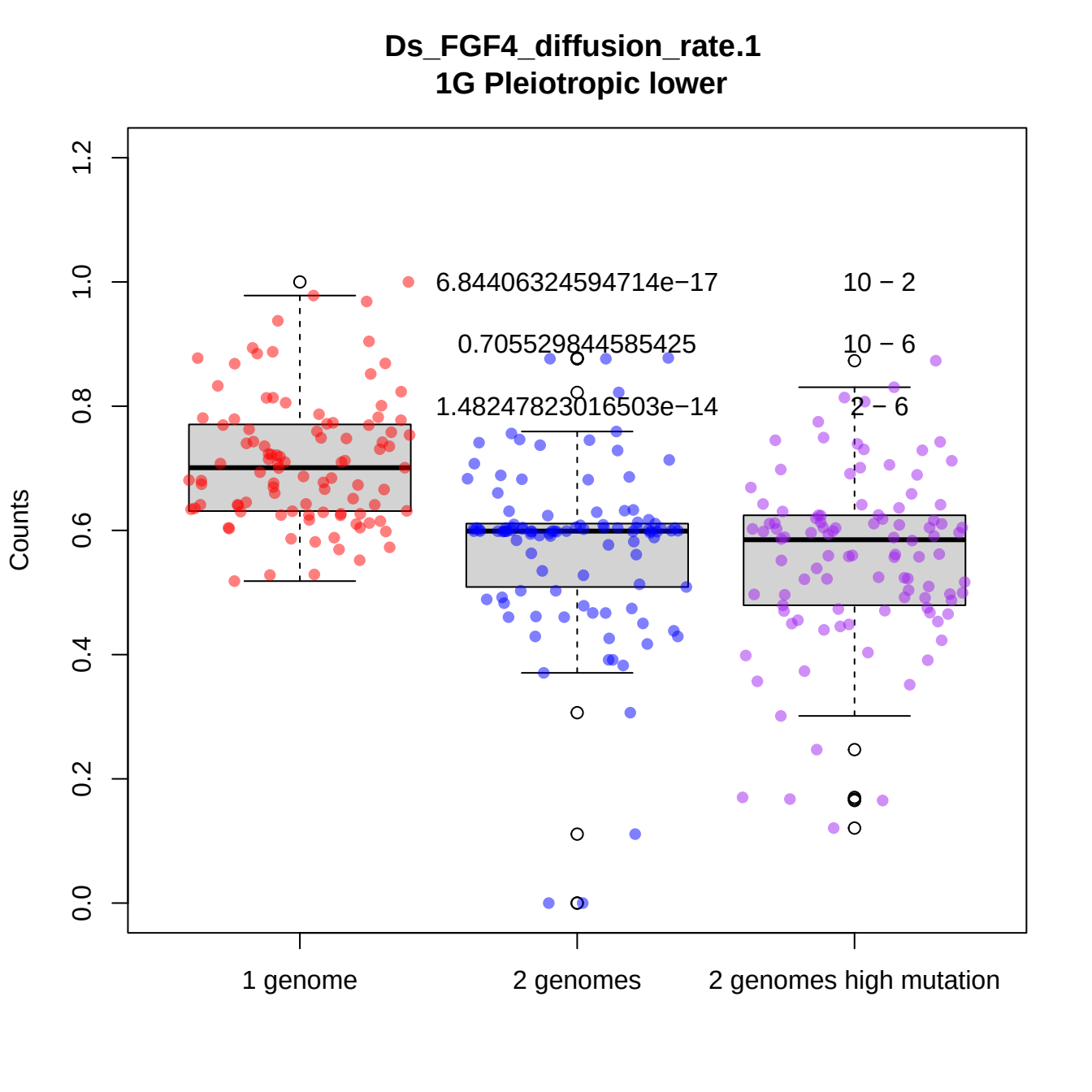

1.00
0.75
0.50
0.25
windows
30
40
50
fit
type
1
5
9

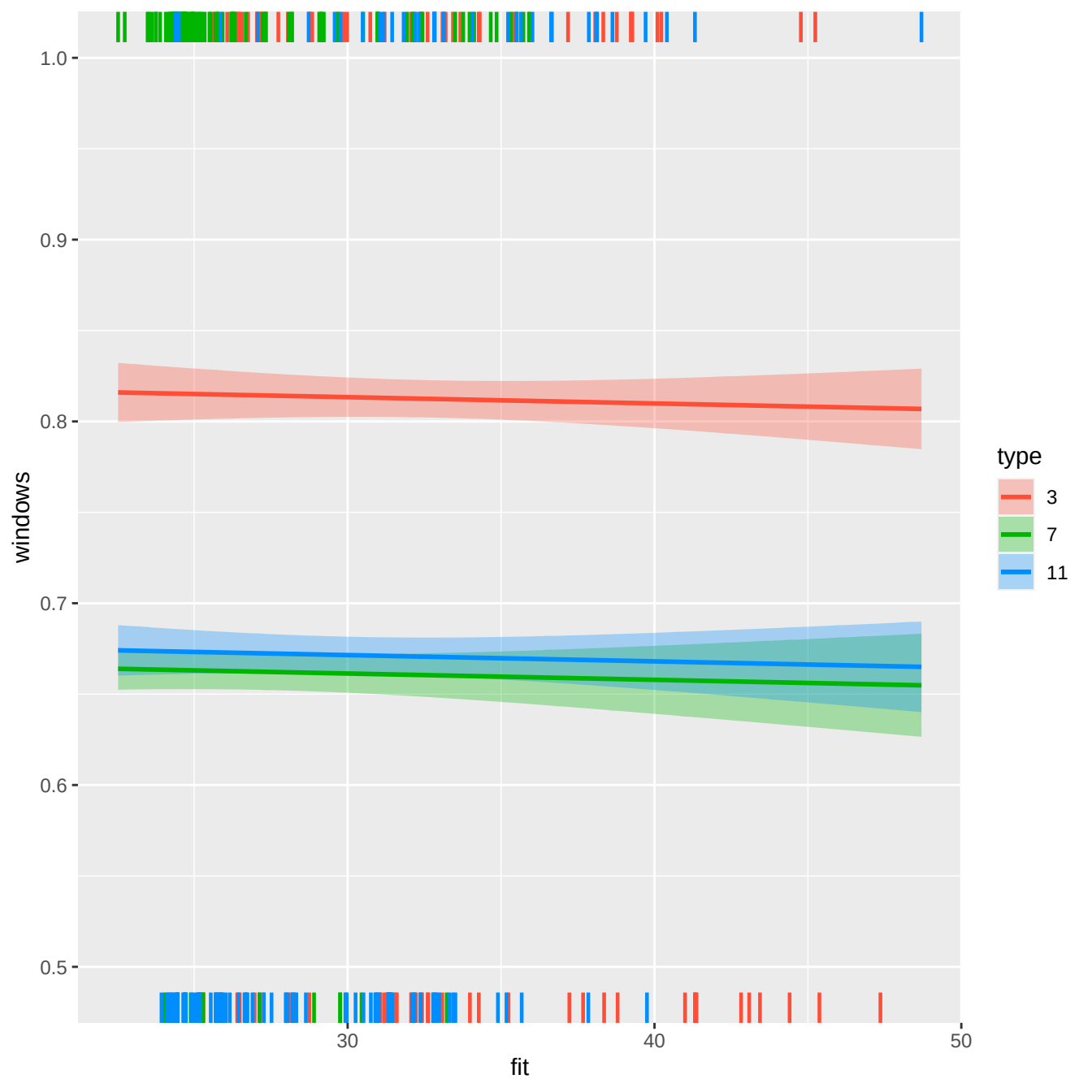

Ds_FGF4_diffusion_rate.1 1G Pleiotropic upper
1 genome
2 genomes
2 genomes high mutation
Parameter range
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

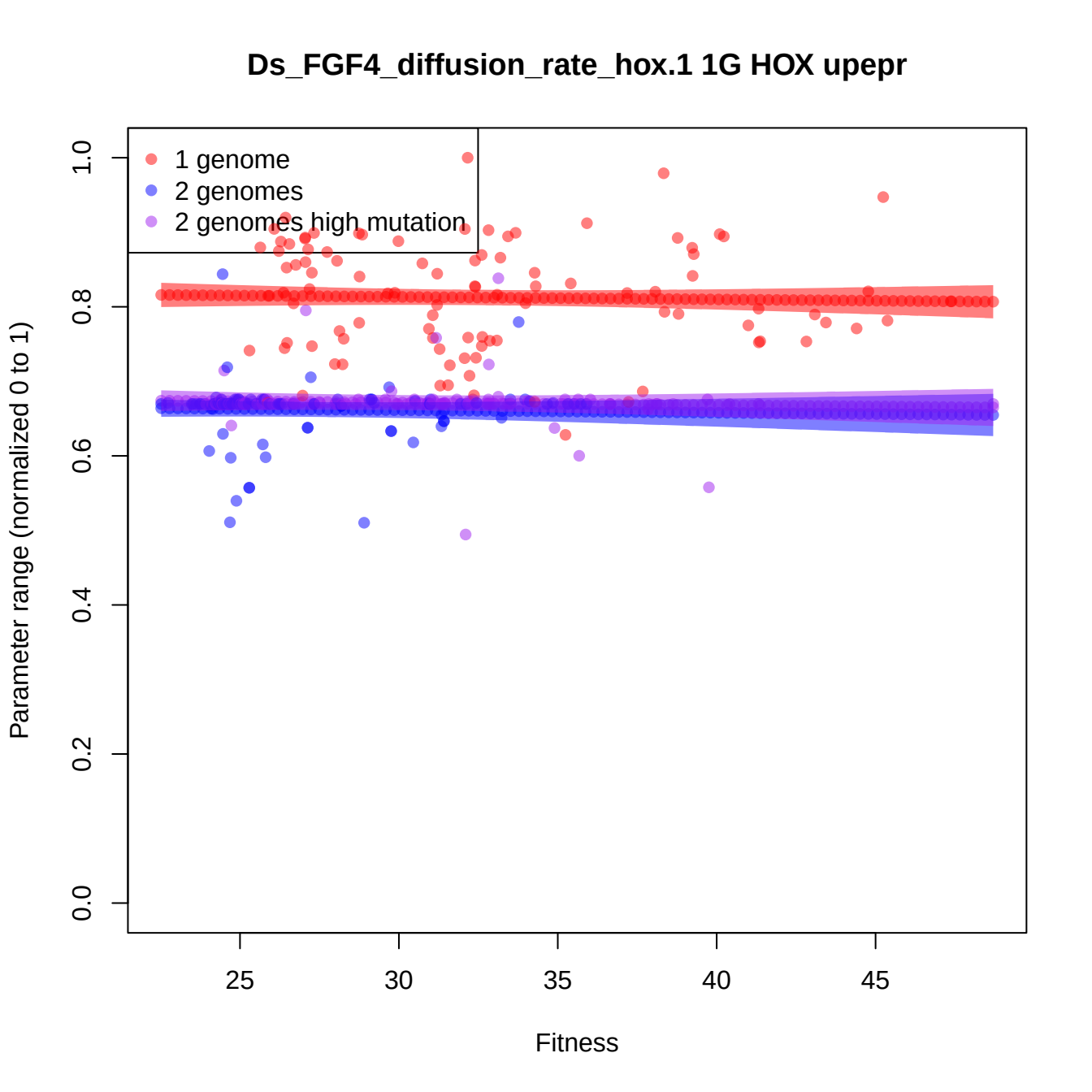

Ds_FGF4_diffusion_rate.1
1G Pleiotropic upper
Counts
1.2
1.0
0.8
0.6
0.4
0.2
2.10597573510778e−21
1.87737917536878e−13
0.054418868362094
1 − 5
1 − 9
5 − 9
1 genome
2 genomes
2 genomes high mutation

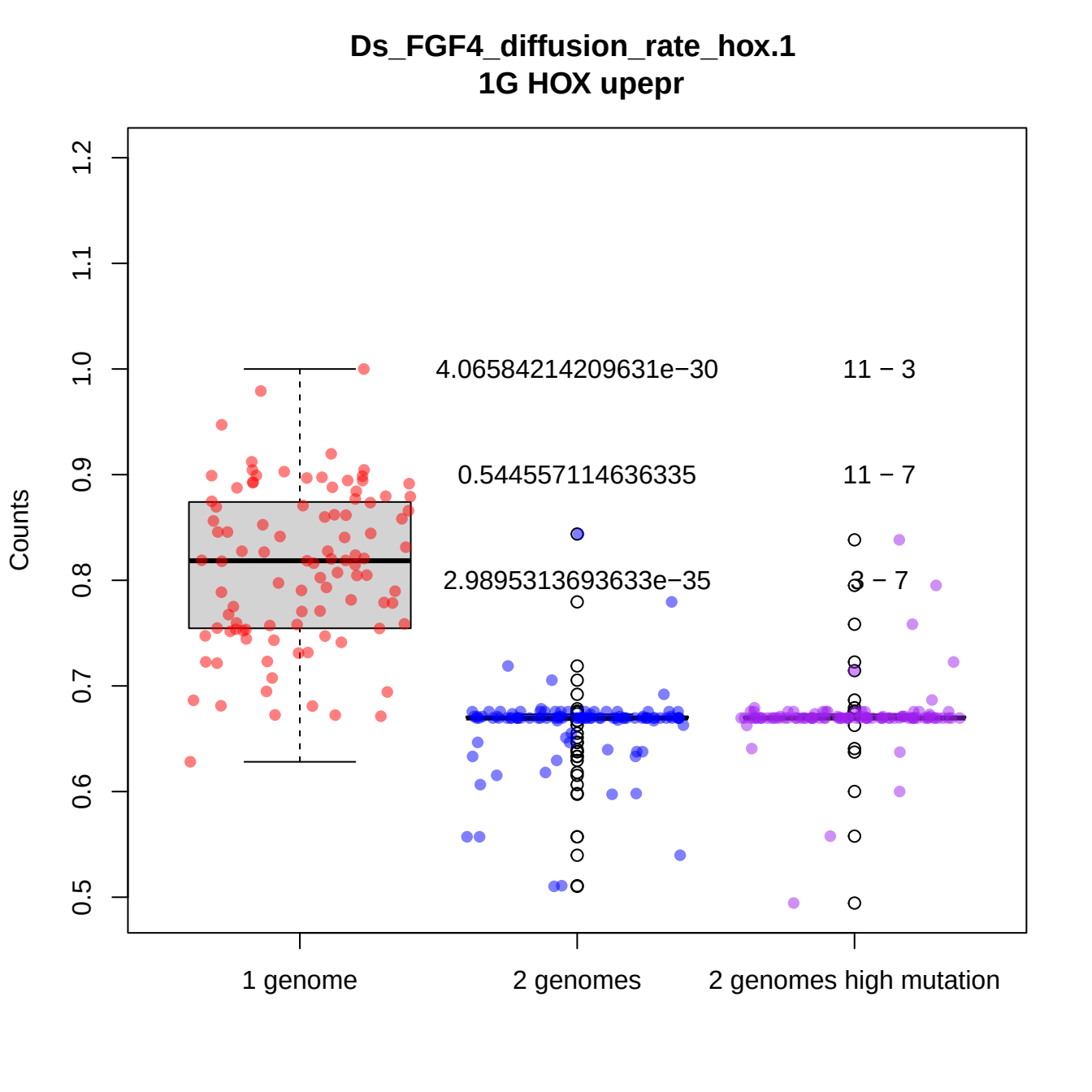

1.00
0.75
0.50
0.25
0.00
windows
30
40
50
fit
type
2
6
10

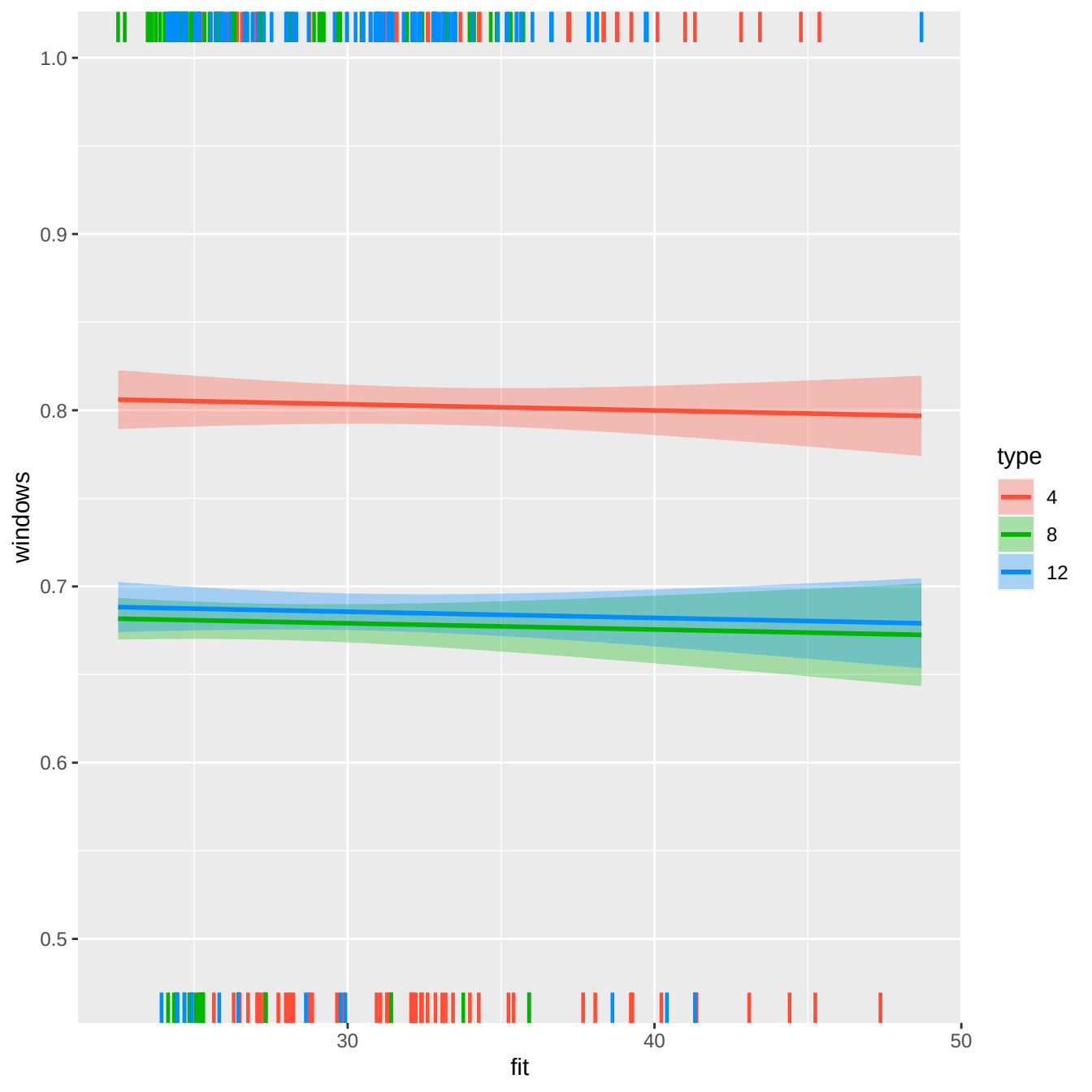

Ds_FGF4_diffusion_rate.1 1G Pleiotropic lower
1 genome
2 genomes
2 genomes high mutation
Parameter range
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

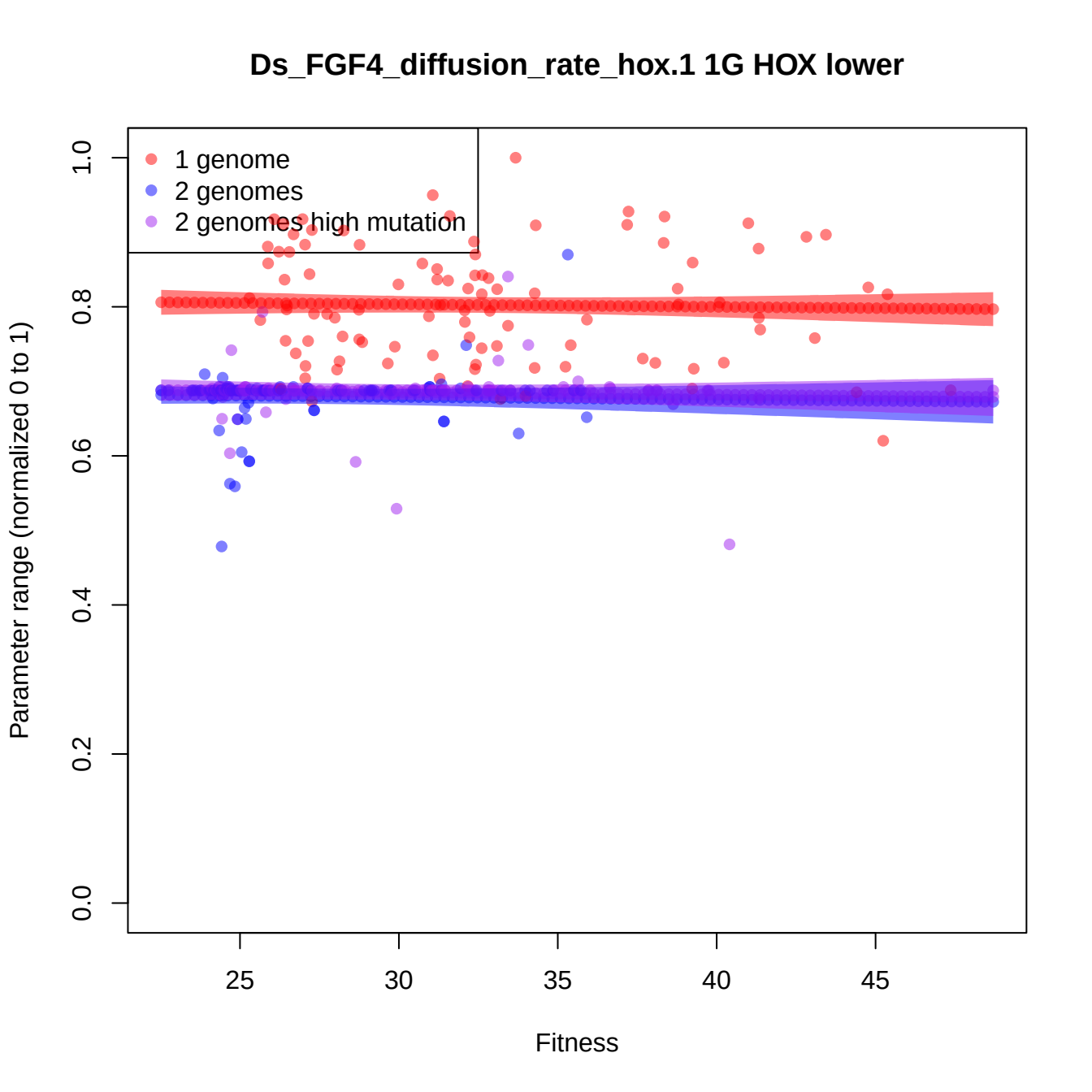

Ds_FGF4_diffusion_rate.1
1G Pleiotropic lower
Counts
0.0
0.2
0.4
0.6
0.8
1.0
1.2
6.84406324594714e−17
0.705529844585425
1.48247823016503e−14
10 − 2
10 − 6
2 − 6
1 genome
2 genomes
2 genomes high mutation

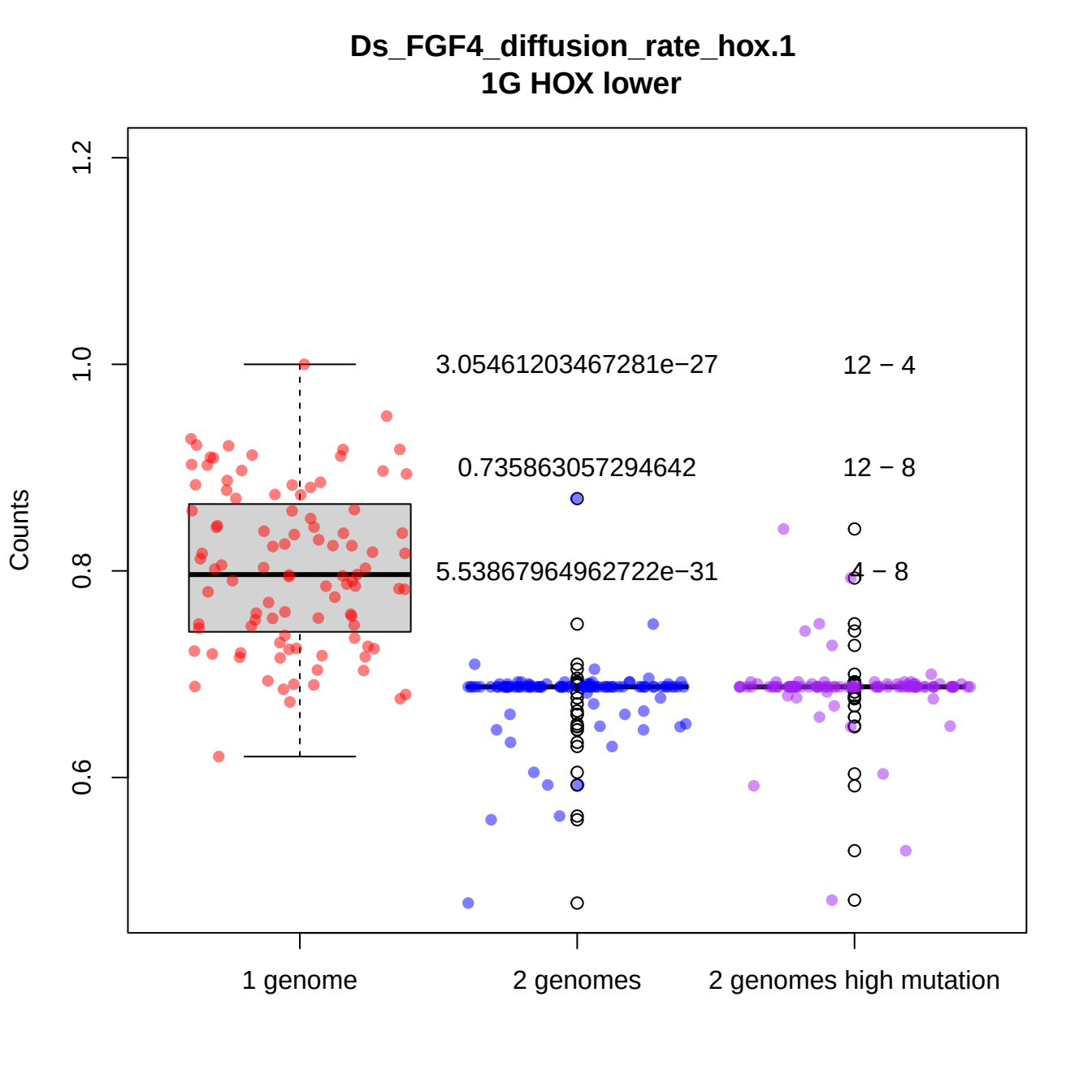

1.0
0.9
0.8
0.7
0.6
0.5
windows
30
40
50
fit
type
3
7
11

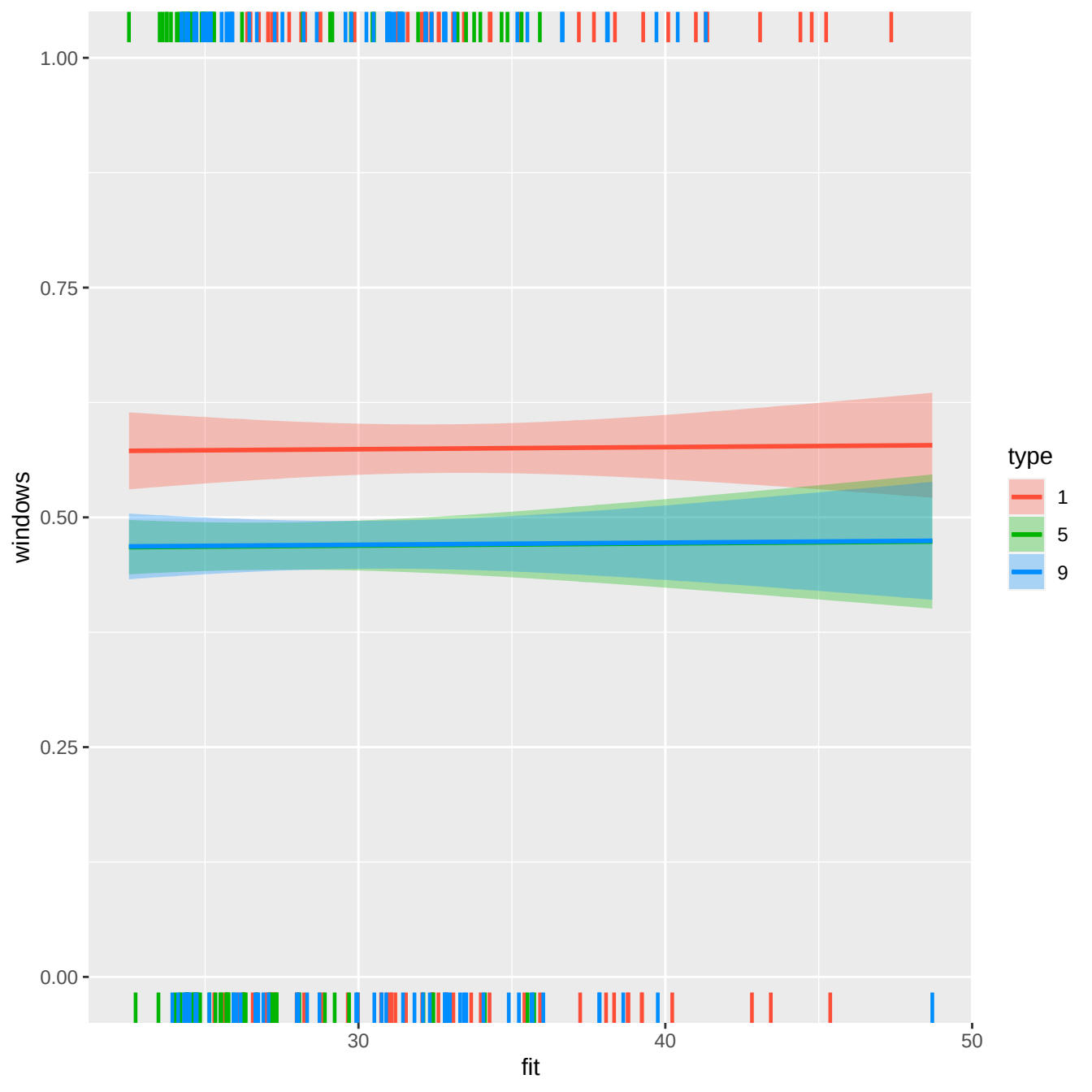

Ds_FGF4_diffusion_rate_hox.1 1G HOX upepr
1 genome
2 genomes
2 genomes high mutation
Parameter range (normalized 0 to 1)
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

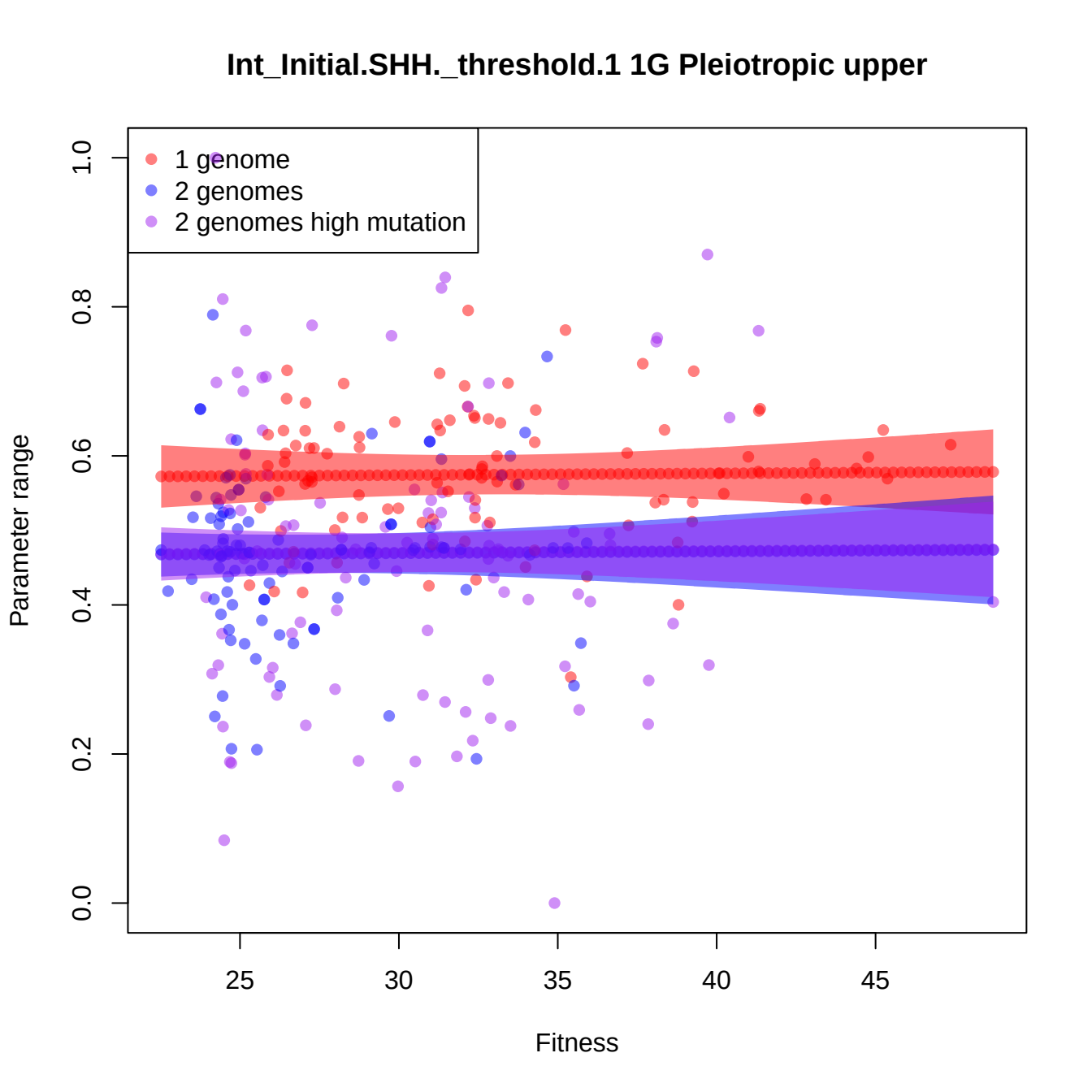

Ds_FGF4_diffusion_rate_hox.1
1G HOX upepr
Counts
1.2
1.1
1.0
0.9
0.8
0.7
0.6
0.5
4.06584214209631e−30
11 − 3
0.544557114636335
11 − 7
2.9895313693633e−35
3 − 7
1 genome
2 genomes
2 genomes high mutation

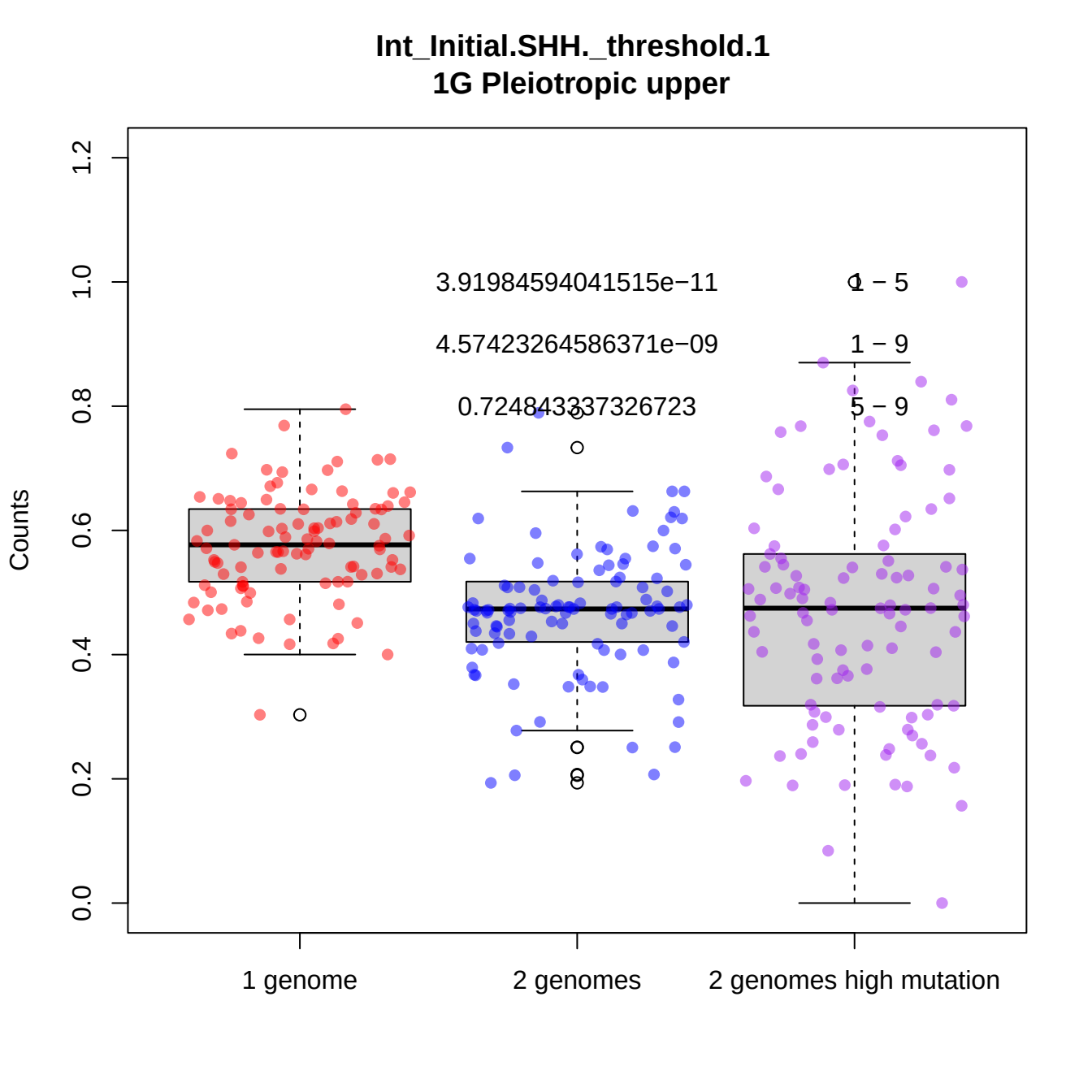

1.0
0.9
0.8
0.7
0.6
0.5
windows
30
40
50
fit
type
4
8
12

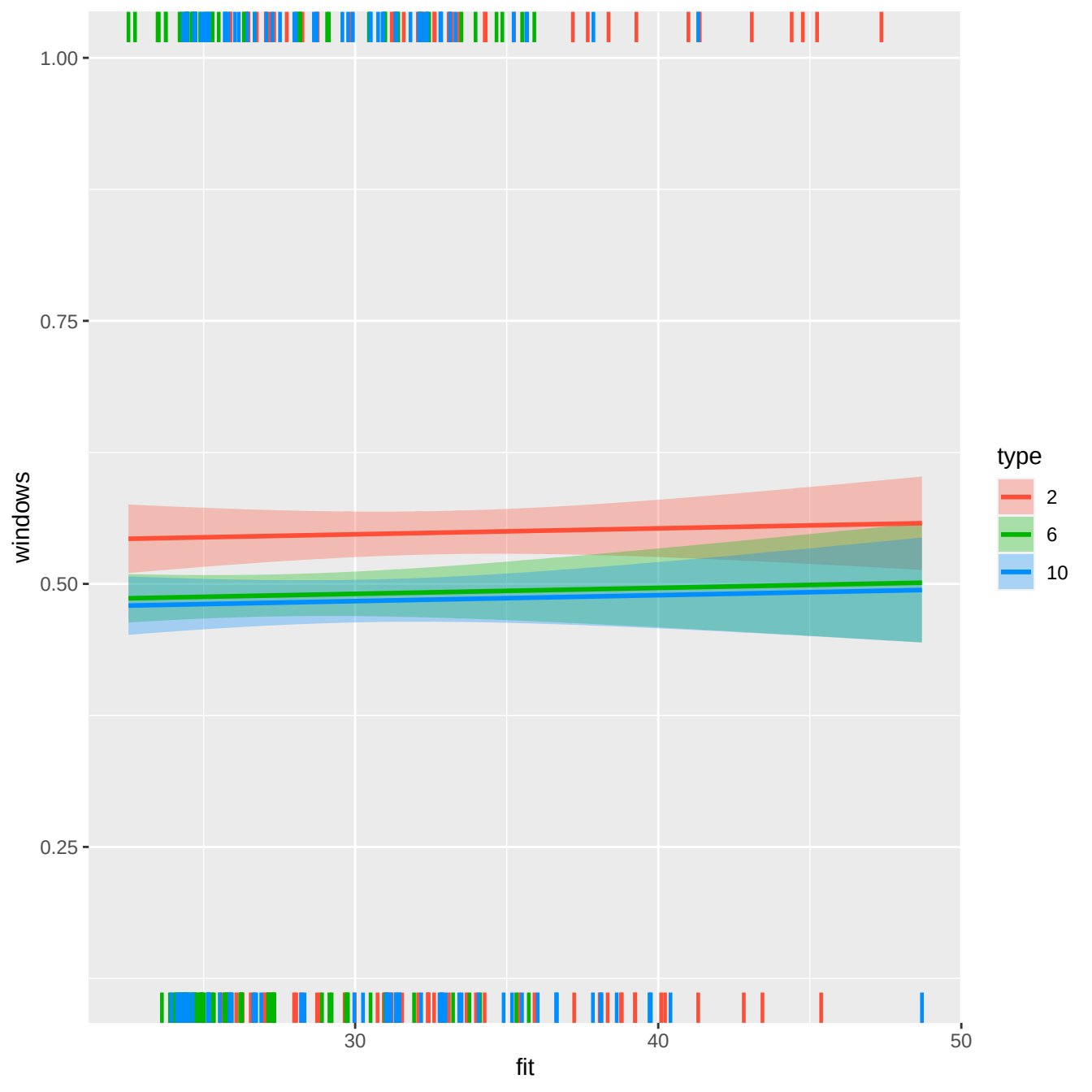

Ds_FGF4_diffusion_rate_hox.1 1G HOX lower
1 genome
2 genomes
2 genomes high mutation
Parameter range (normalized 0 to 1)
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

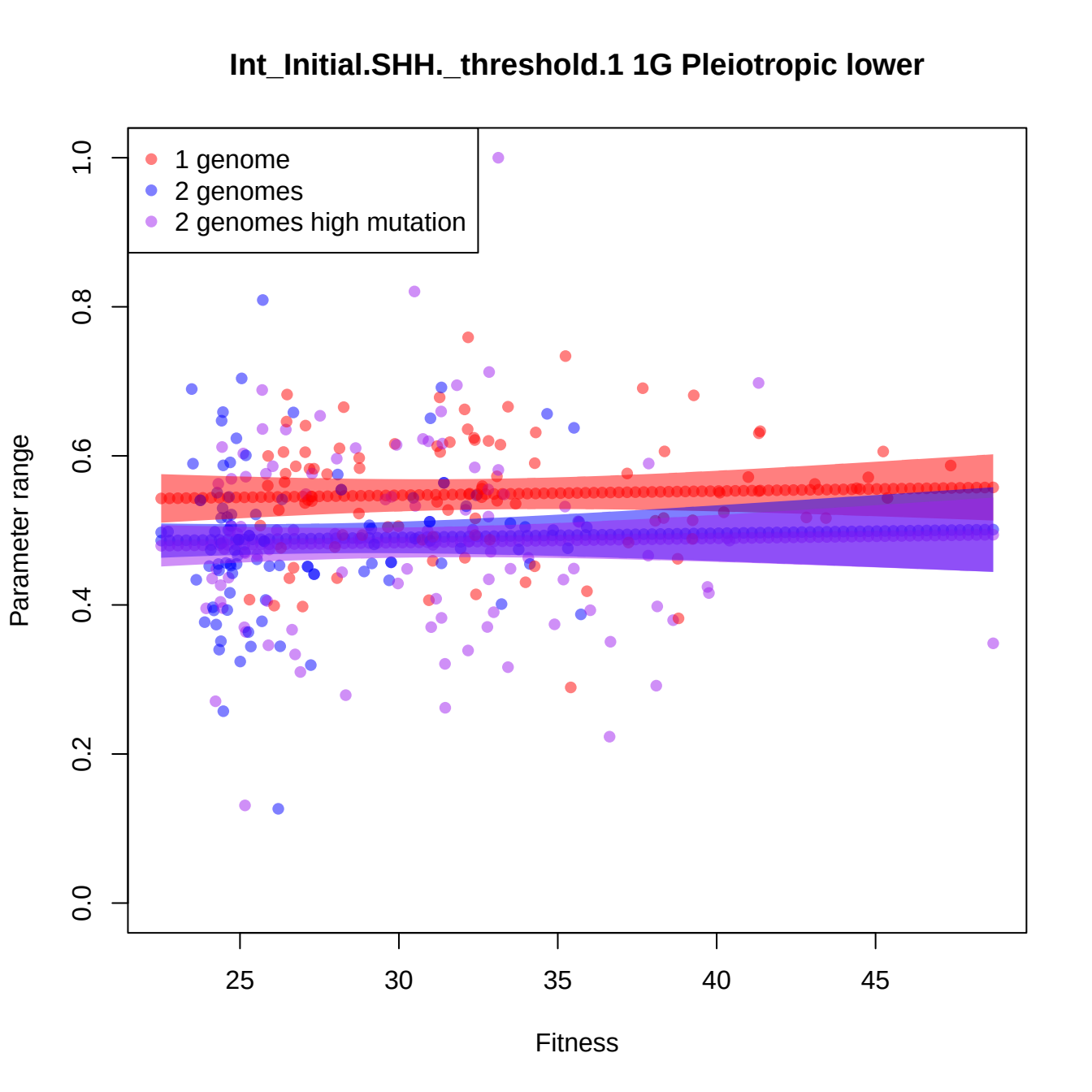

Ds_FGF4_diffusion_rate_hox.1
1G HOX lower
Counts
1.2
1.0
0.8
0.6
3.05461203467281e−27
0.735863057294642
5.53867964962722e−31
12 − 4
12 − 8
4 − 8
1 genome
2 genomes
2 genomes high mutation

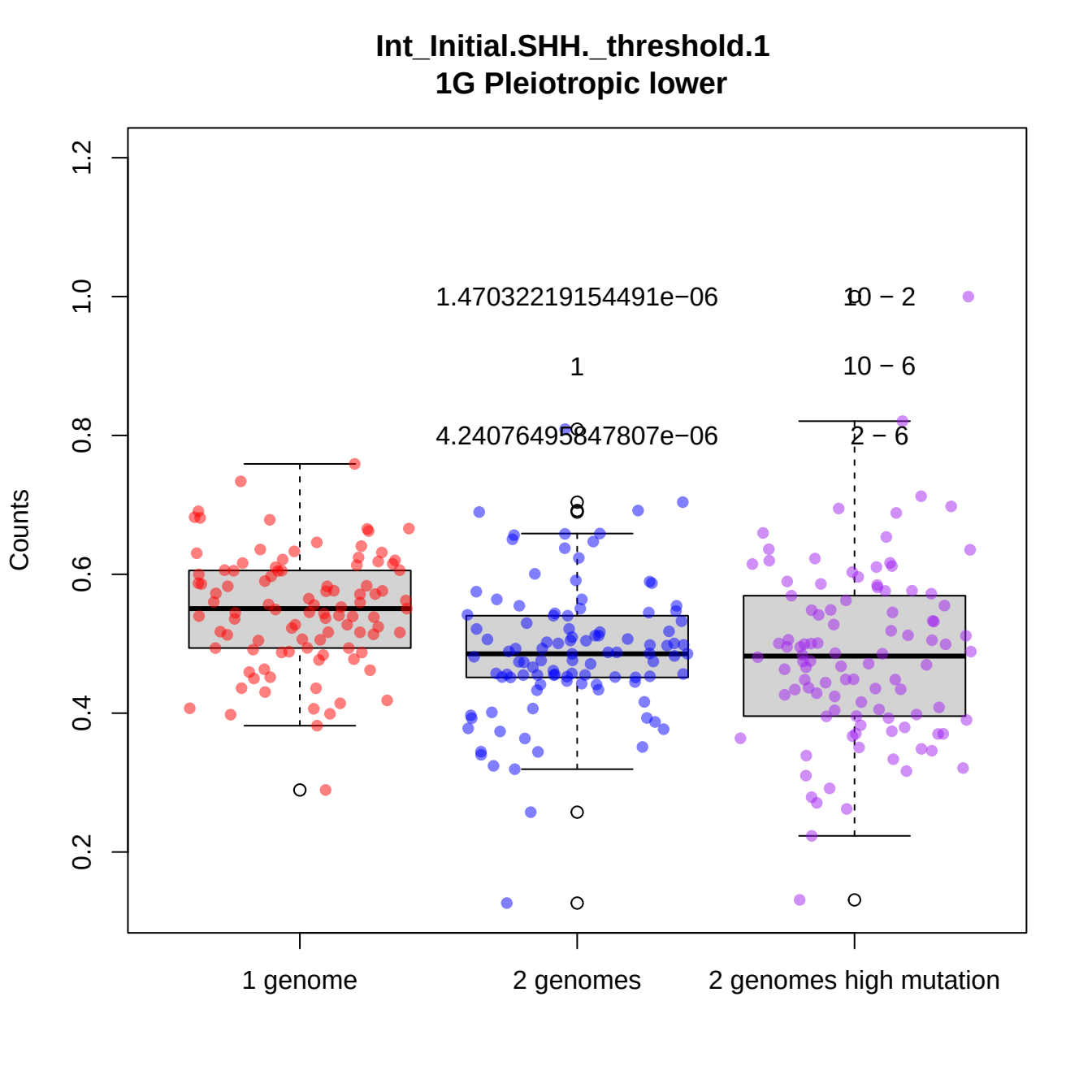

1.00
0.75
0.50
0.25
0.00
windows
30
40
50
fit
type
1
5
9

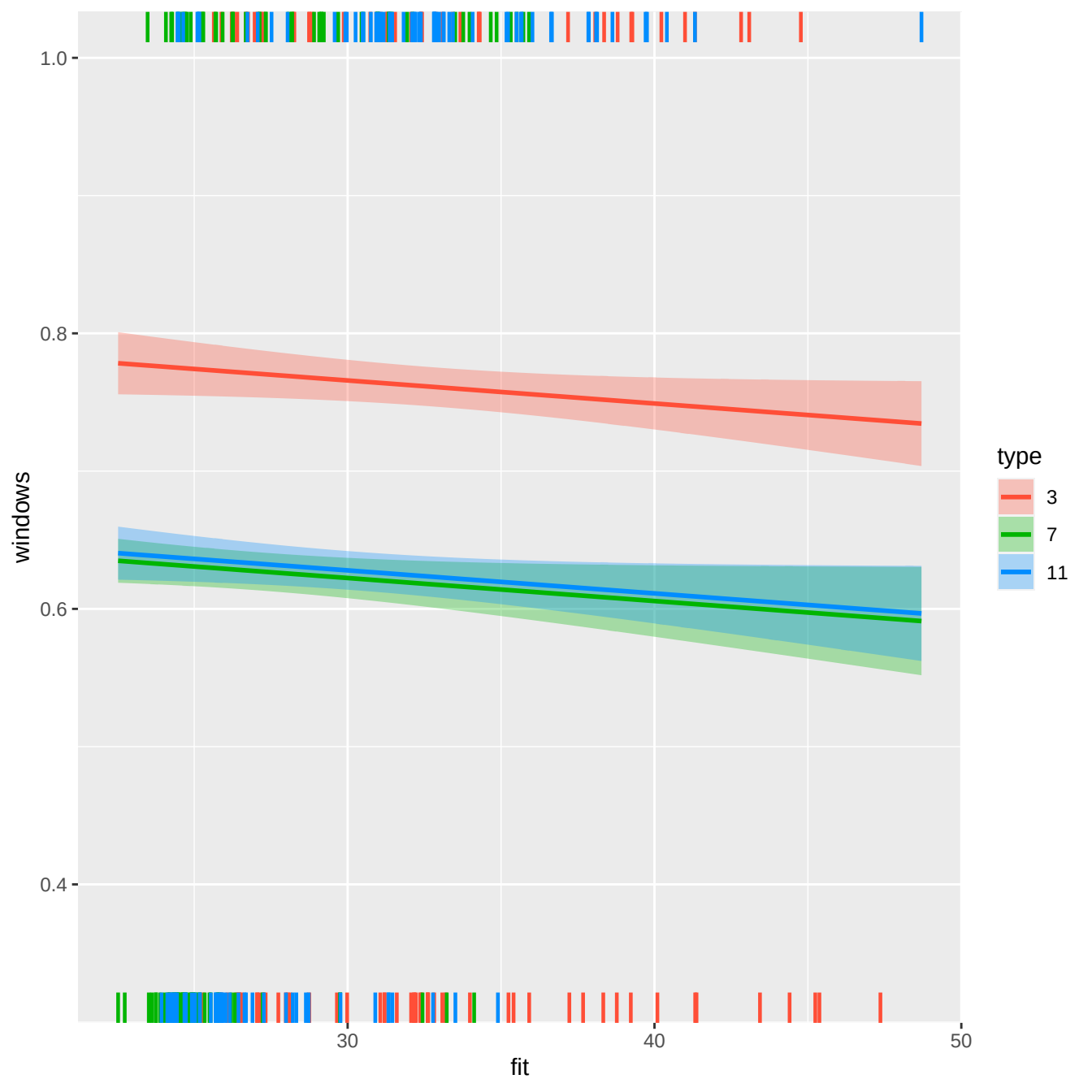

Int_Initial.SHH._threshold.1 1G Pleiotropic upper
1 genome
2 genomes
2 genomes high mutation
Parameter range
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

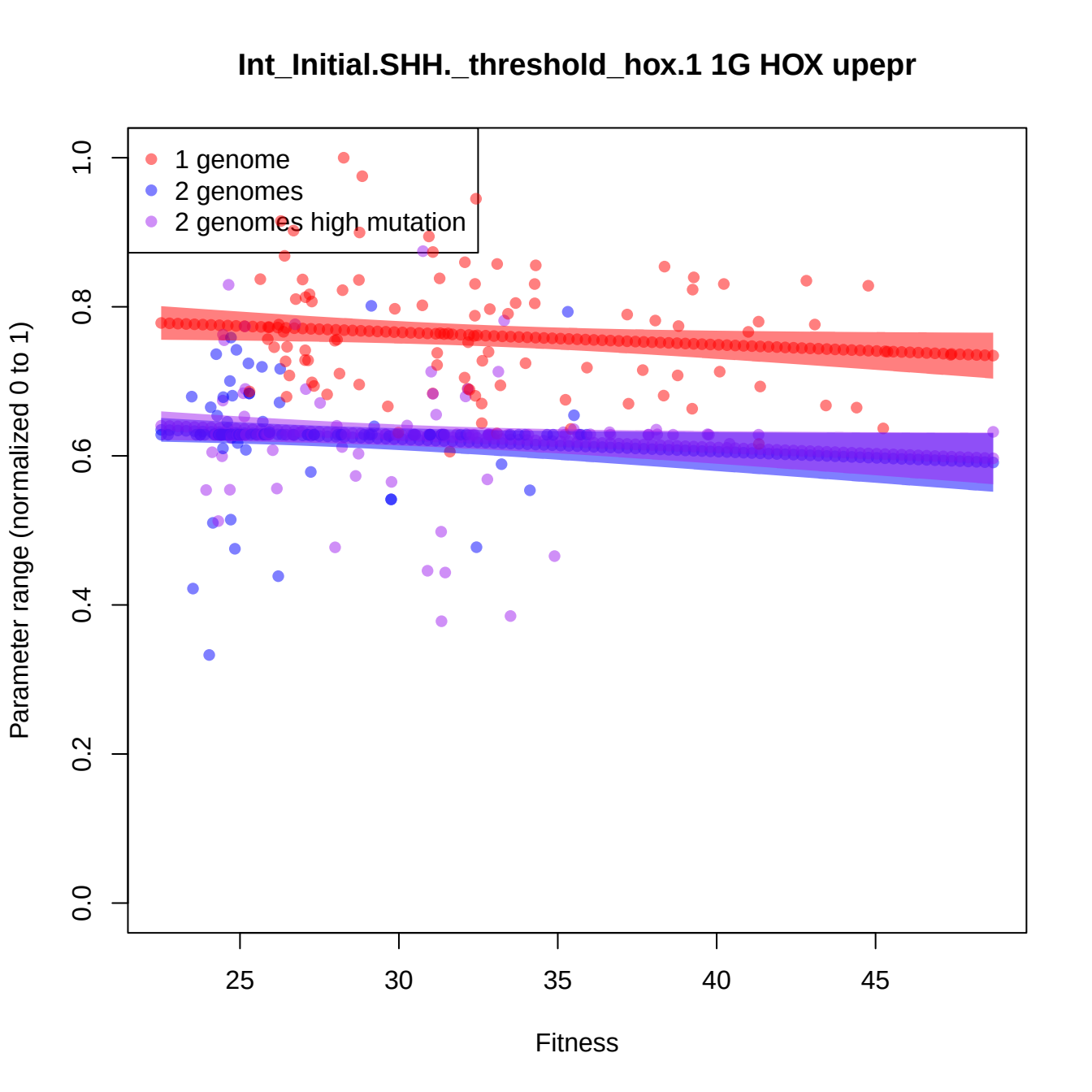

Int_Initial.SHH._threshold.1
1G Pleiotropic upper
Counts
0.0
0.2
0.4
0.6
0.8
1.0
1.2
3.91984594041515e−11
4.57423264586371e−09
0.724843337326723
1 − 5
1 − 9
5 − 9
1 genome
2 genomes
2 genomes high mutation

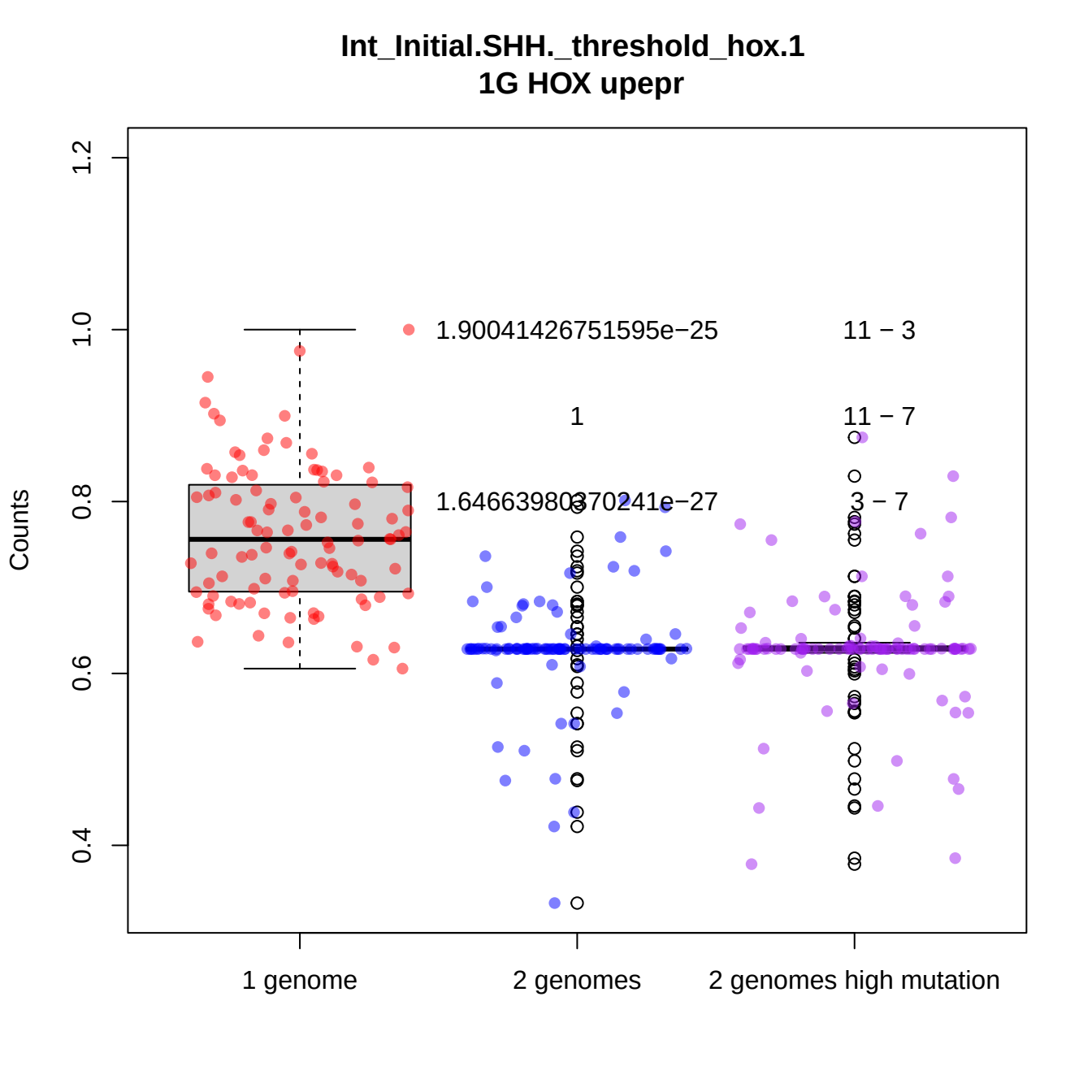

1.00
0.75
0.50
0.25
windows
30
40
50
fit
type
2
6
10

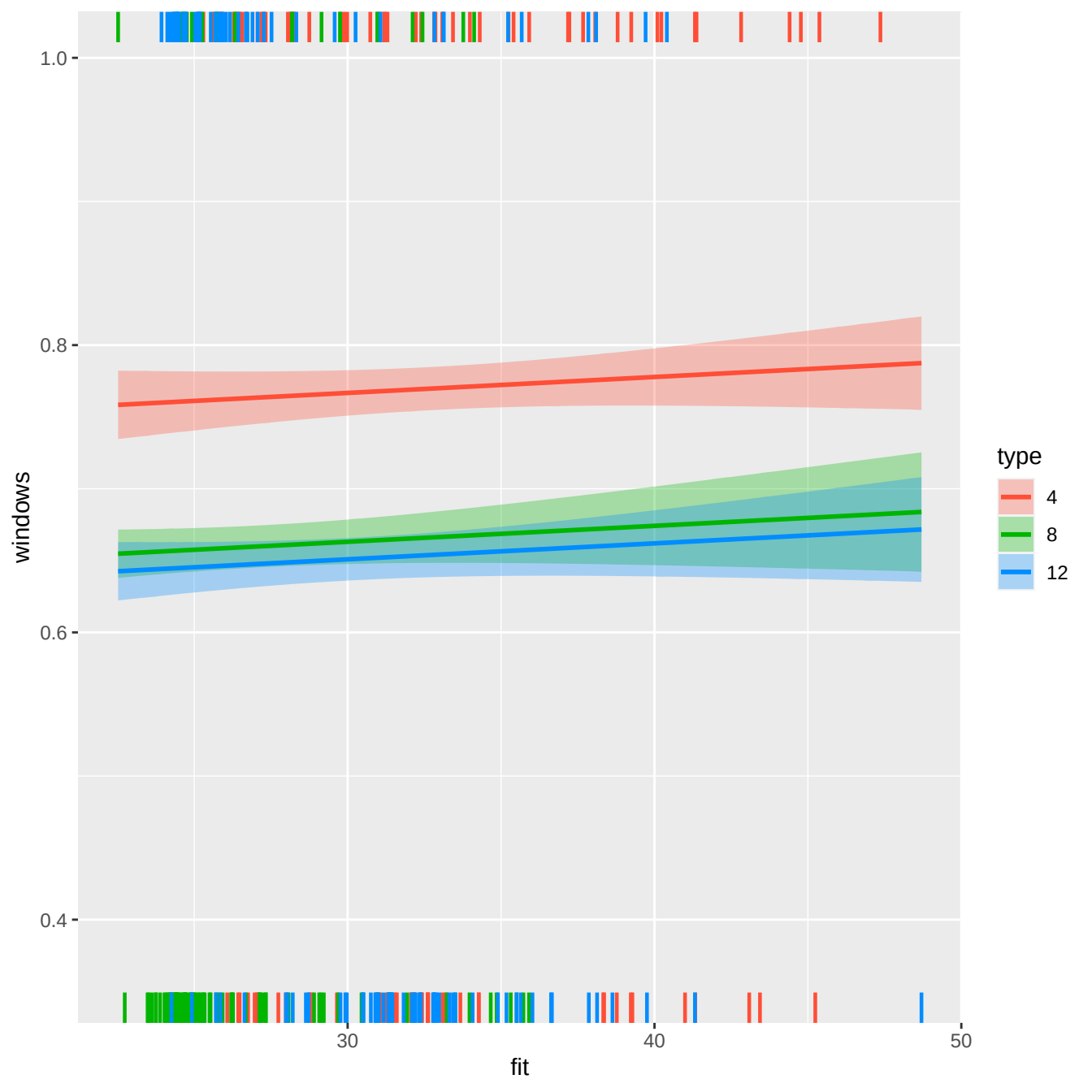

Int_Initial.SHH._threshold.1 1G Pleiotropic lower
1 genome
2 genomes
2 genomes high mutation
Parameter range
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

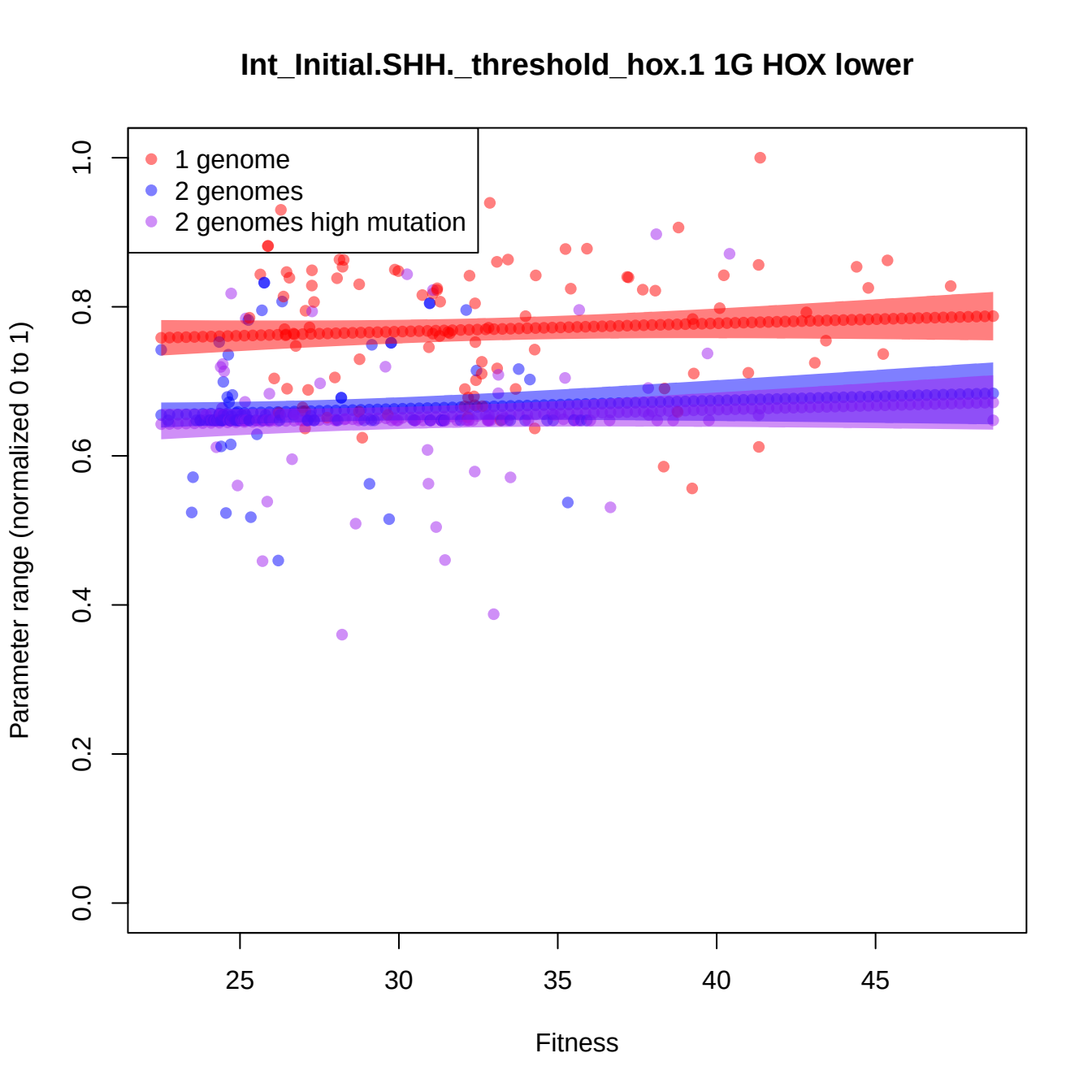

Int_Initial.SHH._threshold.1
1G Pleiotropic lower
Counts
1.2
1.0
0.8
0.6
0.4
0.2
1.47032219154491e−06
1
4.24076495847807e−06
10 − 2
10 − 6
2 − 6
1 genome
2 genomes
2 genomes high mutation

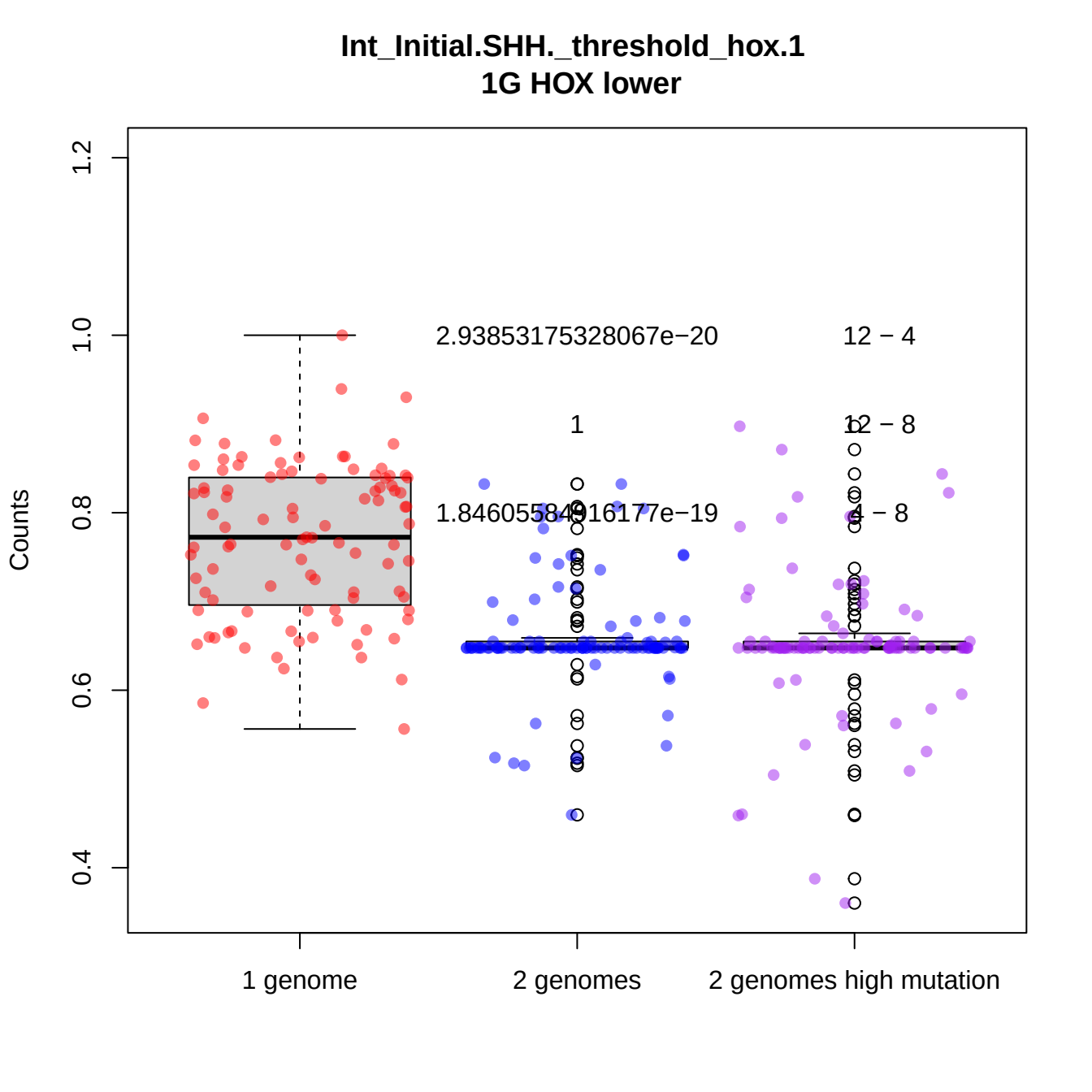

1.0
0.8
0.6
0.4
windows
30
40
50
fit
type
3
7
11

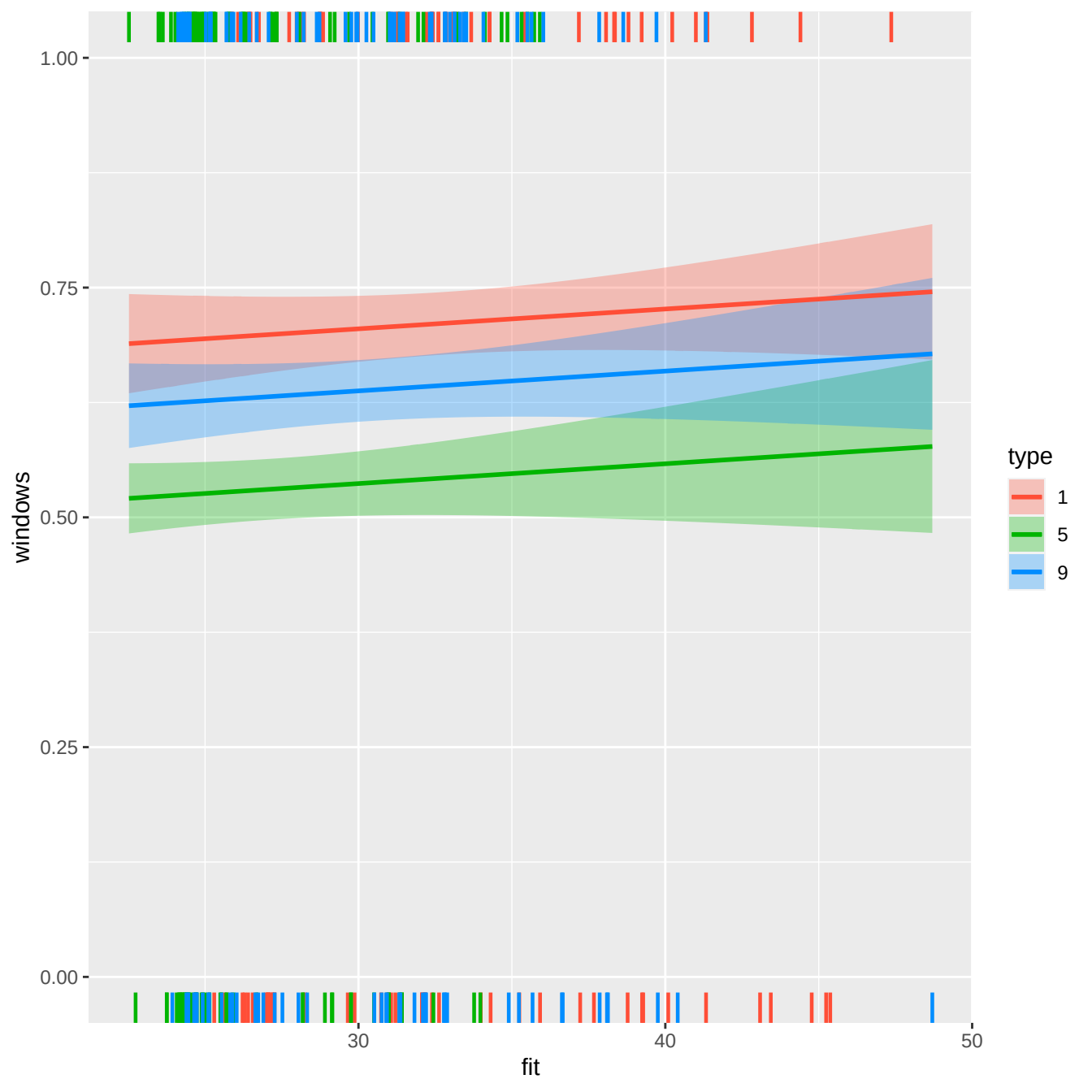

Int_Initial.SHH._threshold_hox.1 1G HOX upepr
1 genome
2 genomes
2 genomes high mutation
Parameter range (normalized 0 to 1)
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

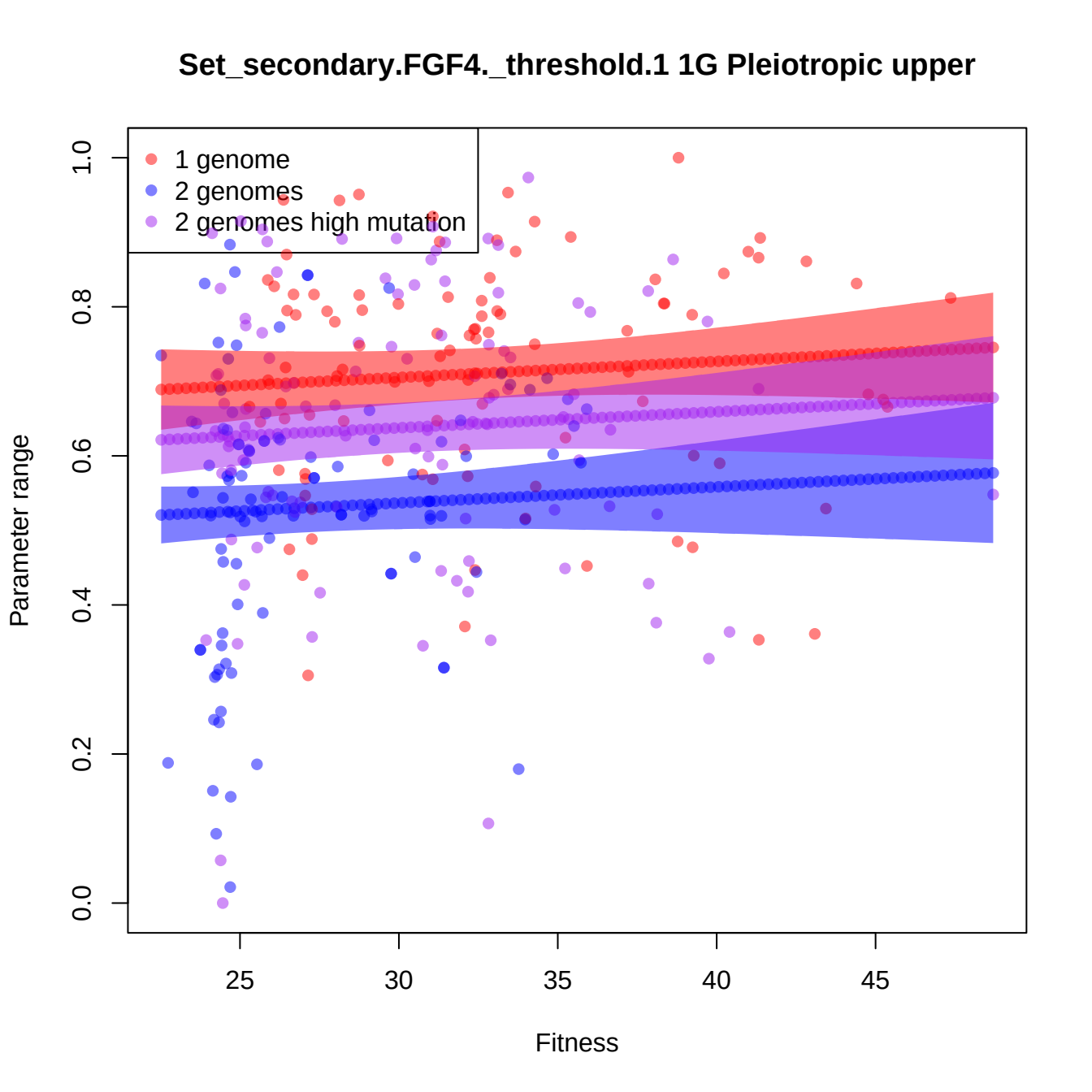

Int_Initial.SHH._threshold_hox.1
1G HOX upepr
Counts
1.2
1.0
0.8
0.6
0.4
1.90041426751595e−25
1
1.64663980370241e−27
11 − 3
11 − 7
3 − 7
1 genome
2 genomes
2 genomes high mutation

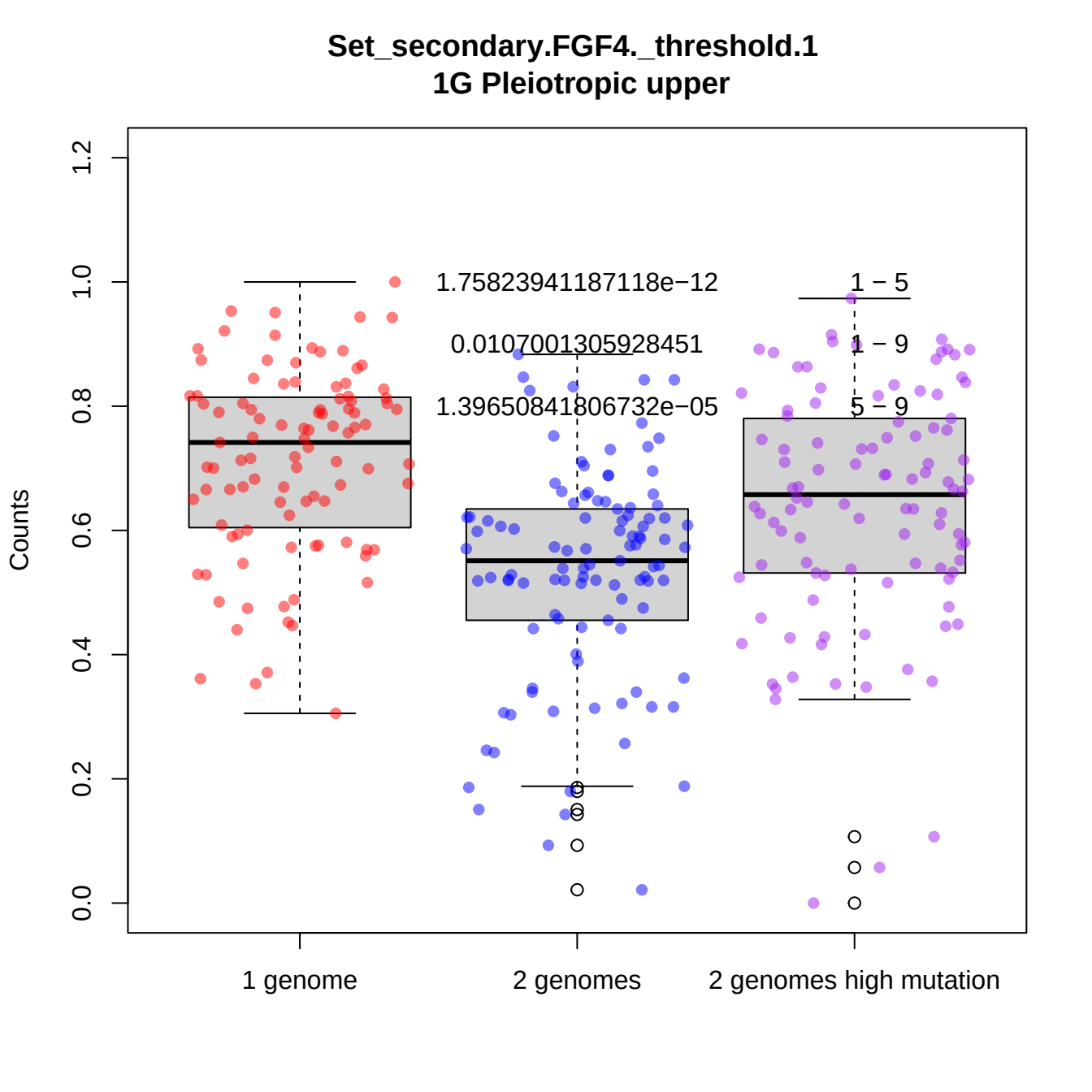

1.0
0.8
0.6
0.4
windows
30
40
50
fit
type
4
8
12

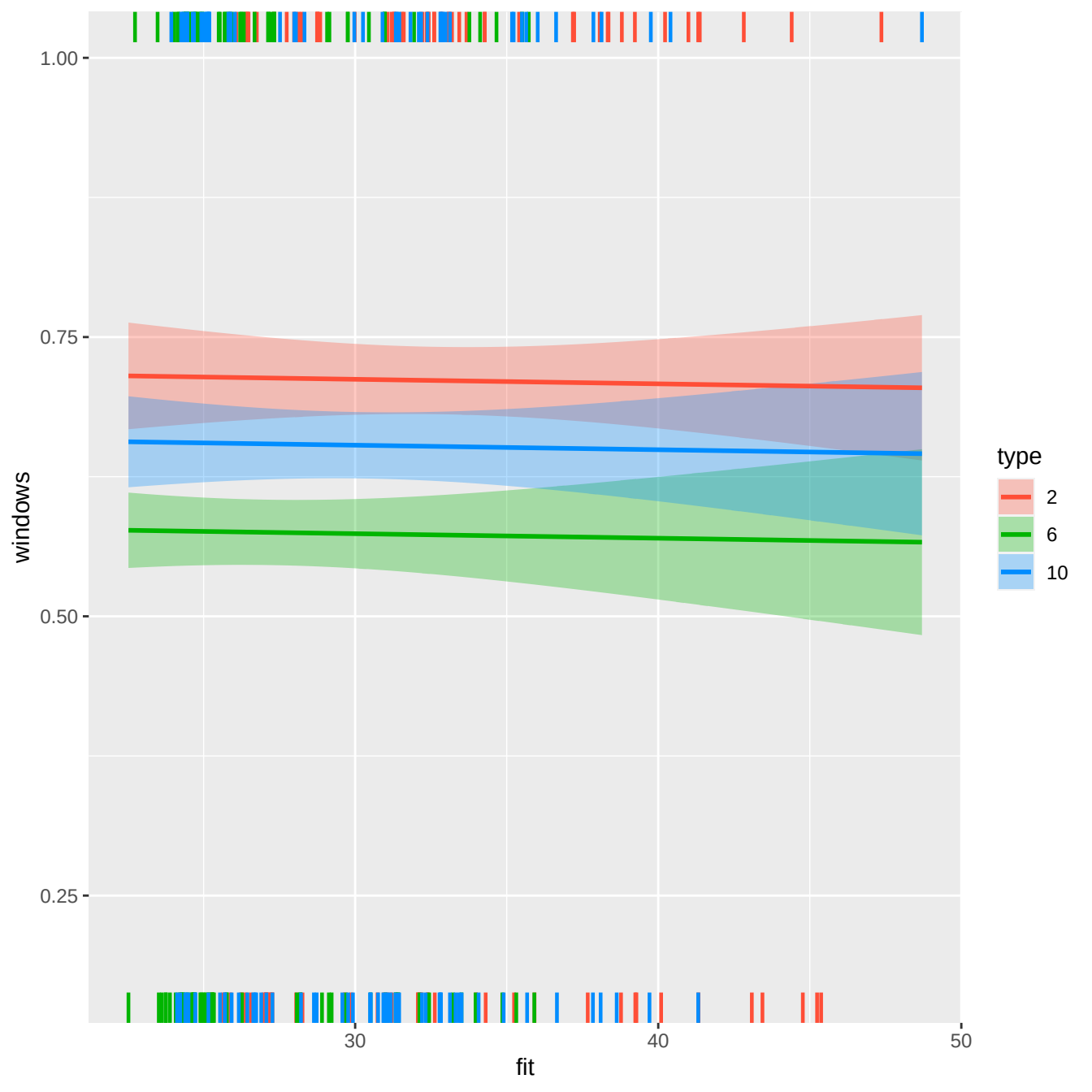

Int_Initial.SHH._threshold_hox.1 1G HOX lower
1 genome
2 genomes
2 genomes high mutation
Parameter range (normalized 0 to 1)
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

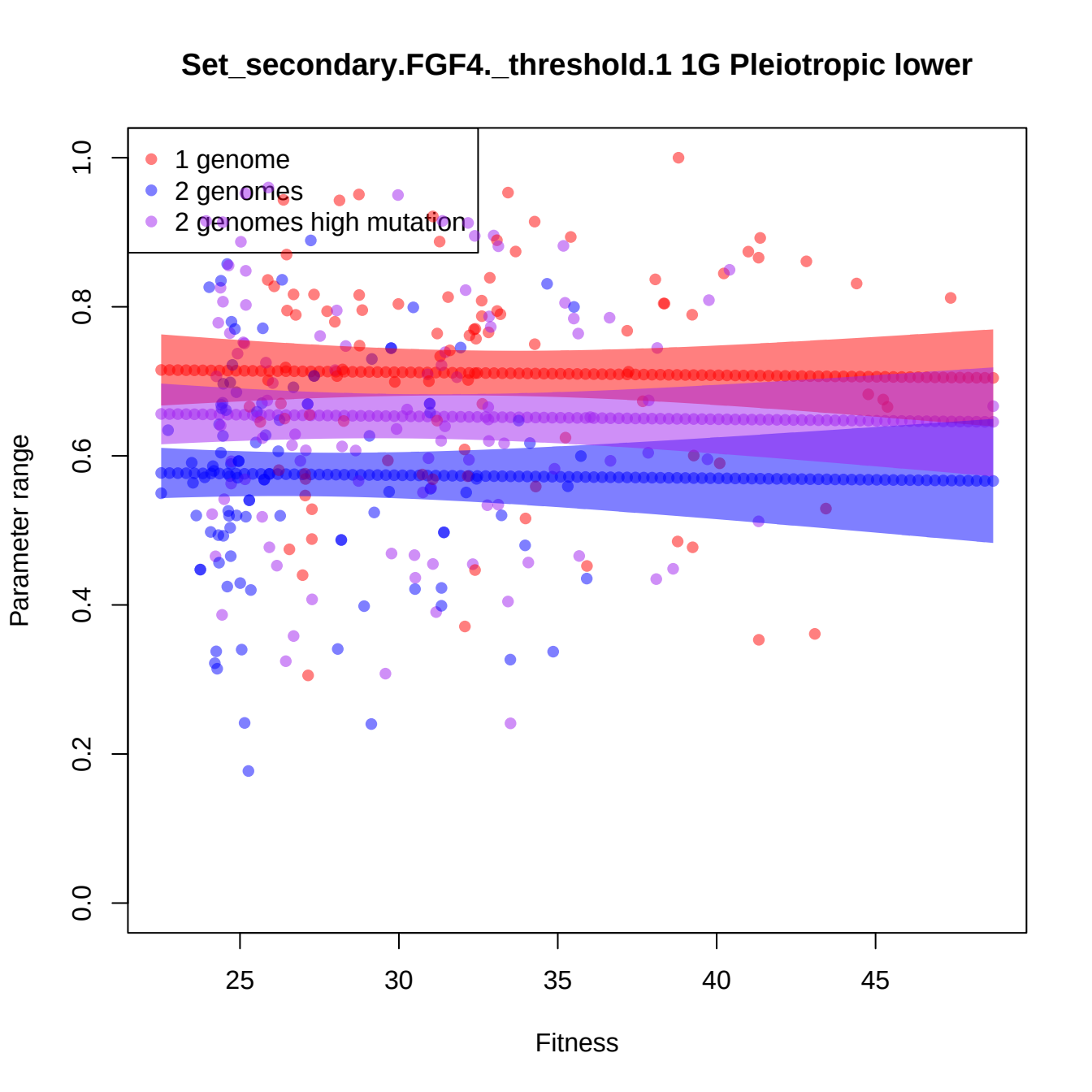

Int_Initial.SHH._threshold_hox.1
1G HOX lower
Counts
1.2
1.0
0.8
0.6
0.4
2.93853175328067e−20
1
1.84605584916177e−19
12 − 4
12 − 8
4 − 8
1 genome
2 genomes
2 genomes high mutation

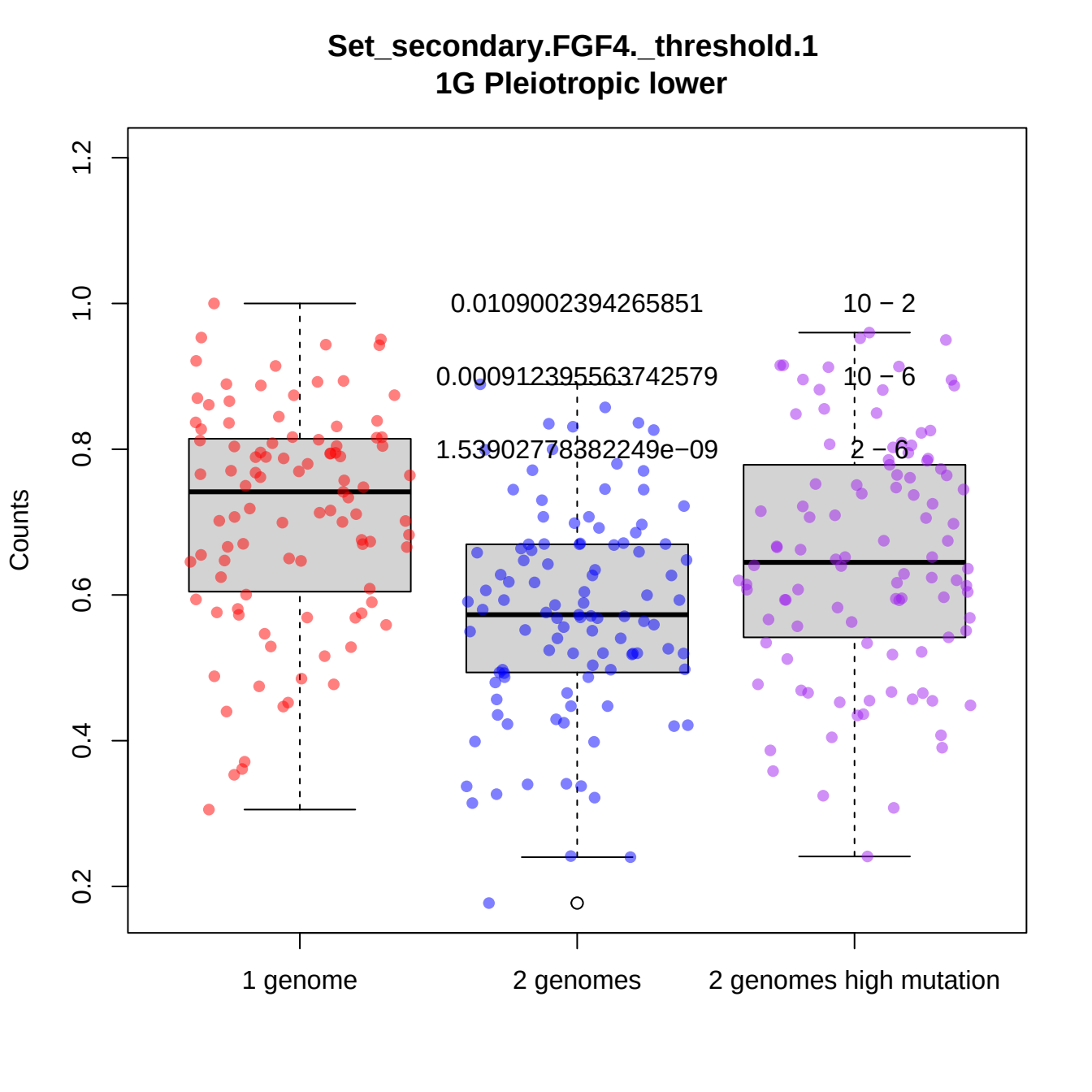

1.00
0.75
0.50
0.25
0.00
windows
30
40
50
fit
type
1
5
9

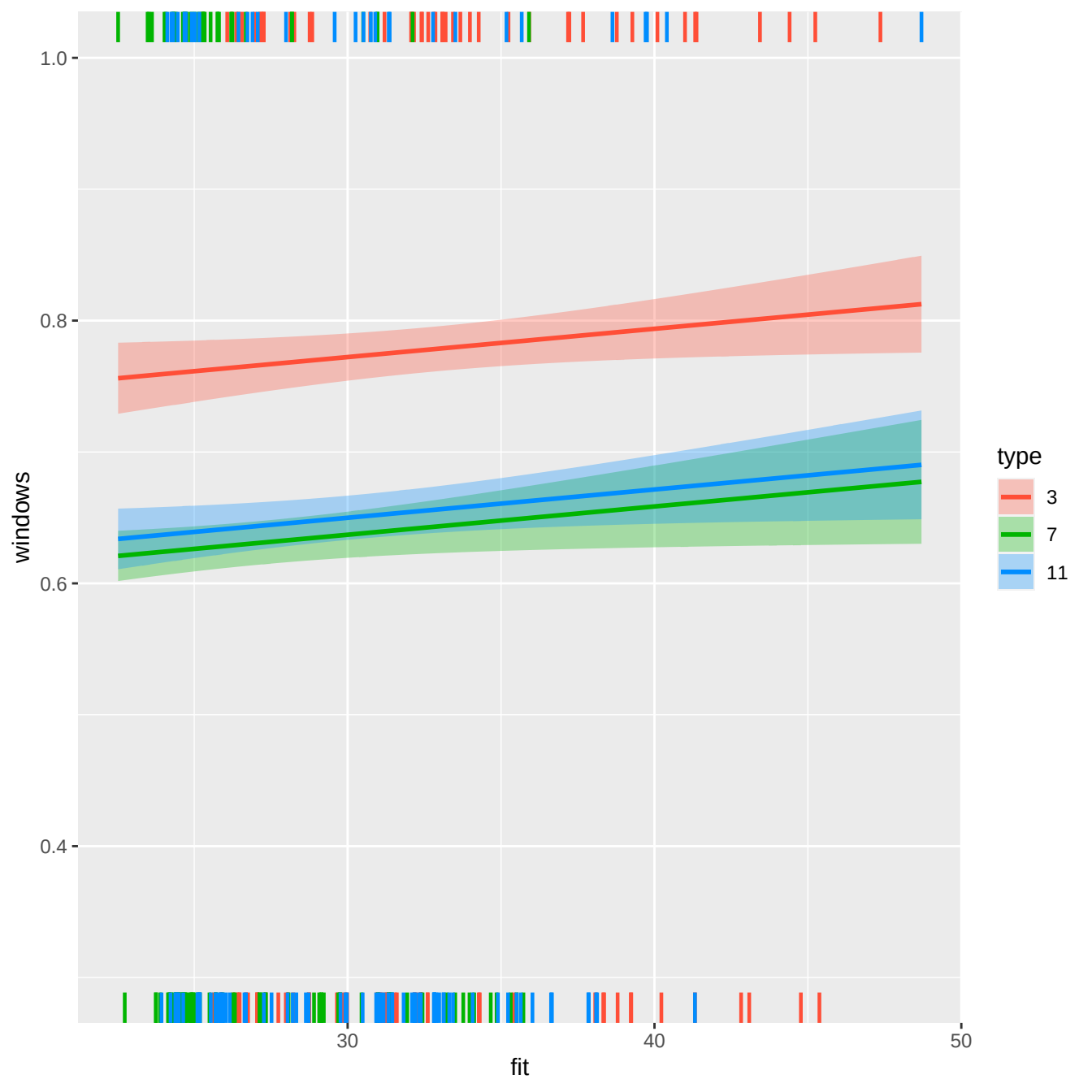

Set_secondary.FGF4._threshold.1 1G Pleiotropic upper
1 genome
2 genomes
2 genomes high mutation
Parameter range
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

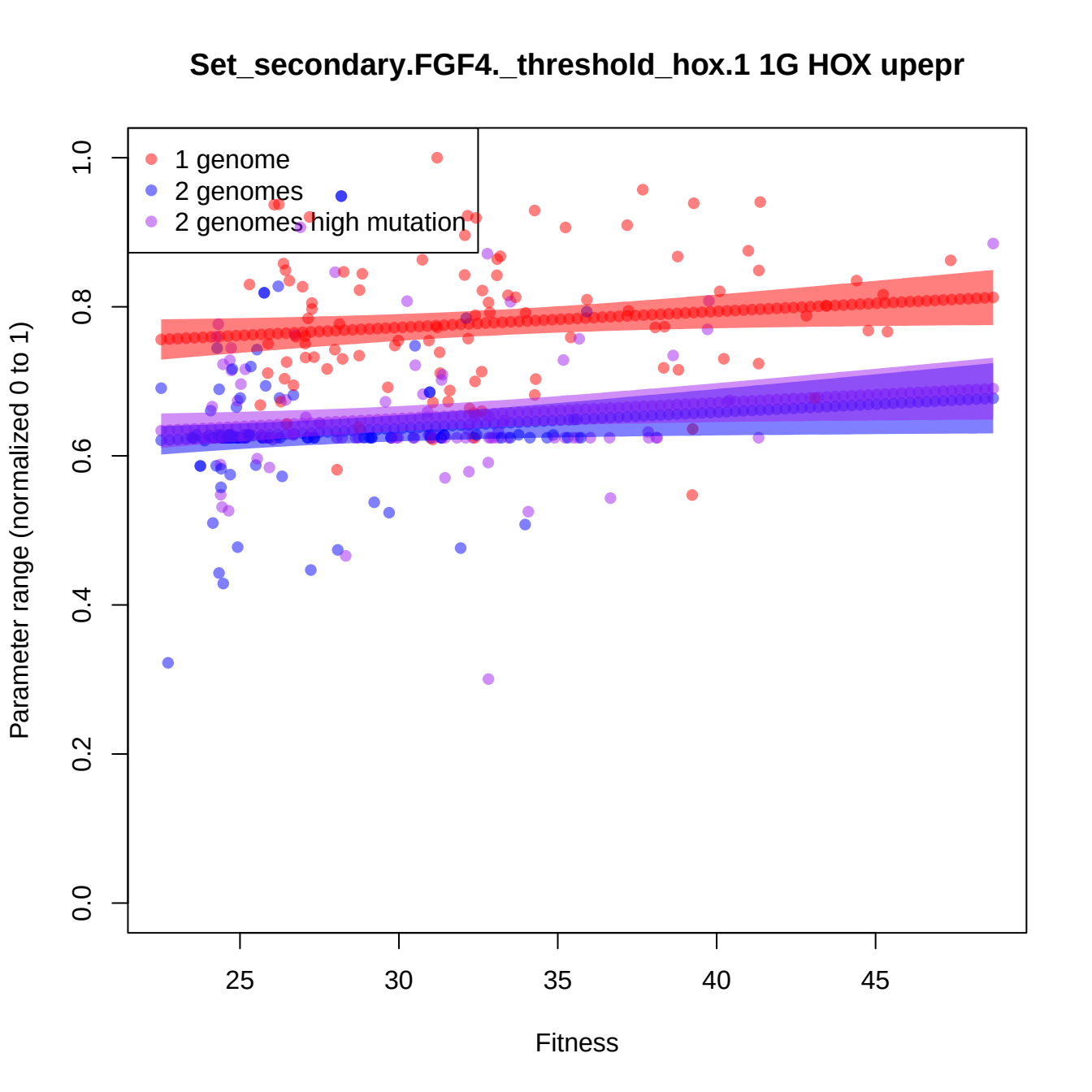

Set_secondary.FGF4._threshold.1
1G Pleiotropic upper
Counts
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.75823941187118e−12
0.0107001305928451
1.39650841806732e−05
1 − 5
1 − 9
5 − 9
1 genome
2 genomes
2 genomes high mutation

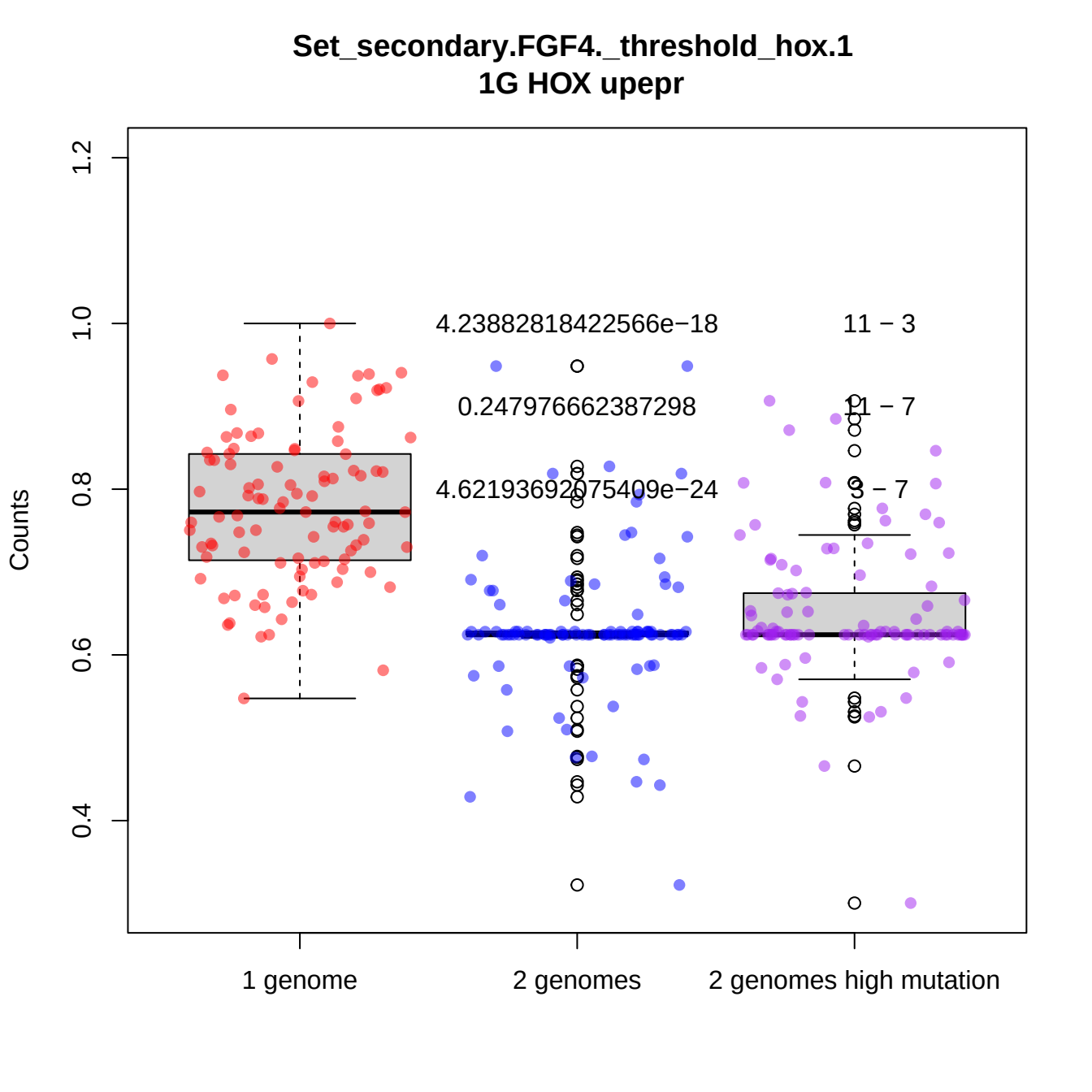

1.00
0.75
0.50
0.25
windows
30
40
50
fit
type
2
6
10

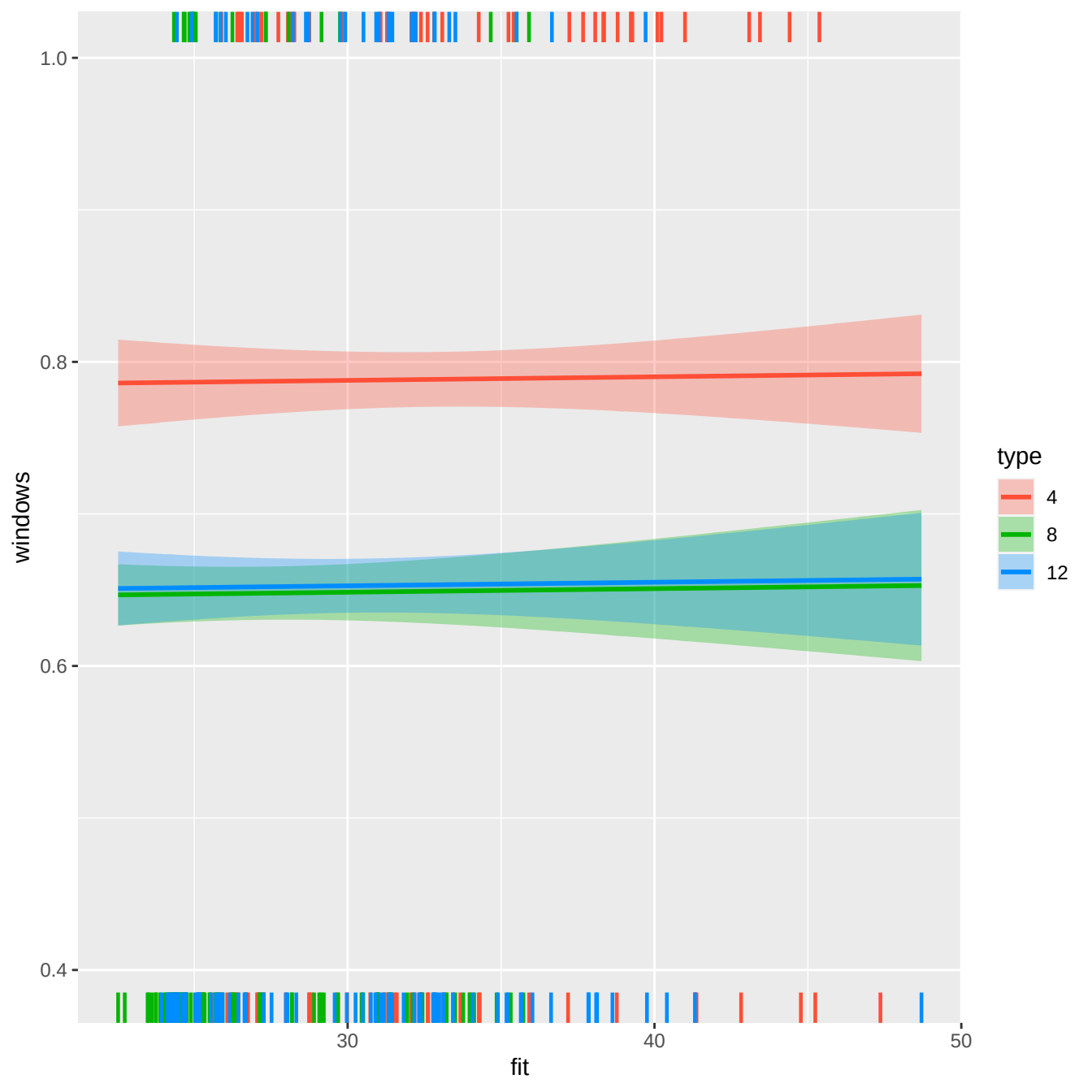

Set_secondary.FGF4._threshold.1 1G Pleiotropic lower
1 genome
2 genomes
2 genomes high mutation
Parameter range
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

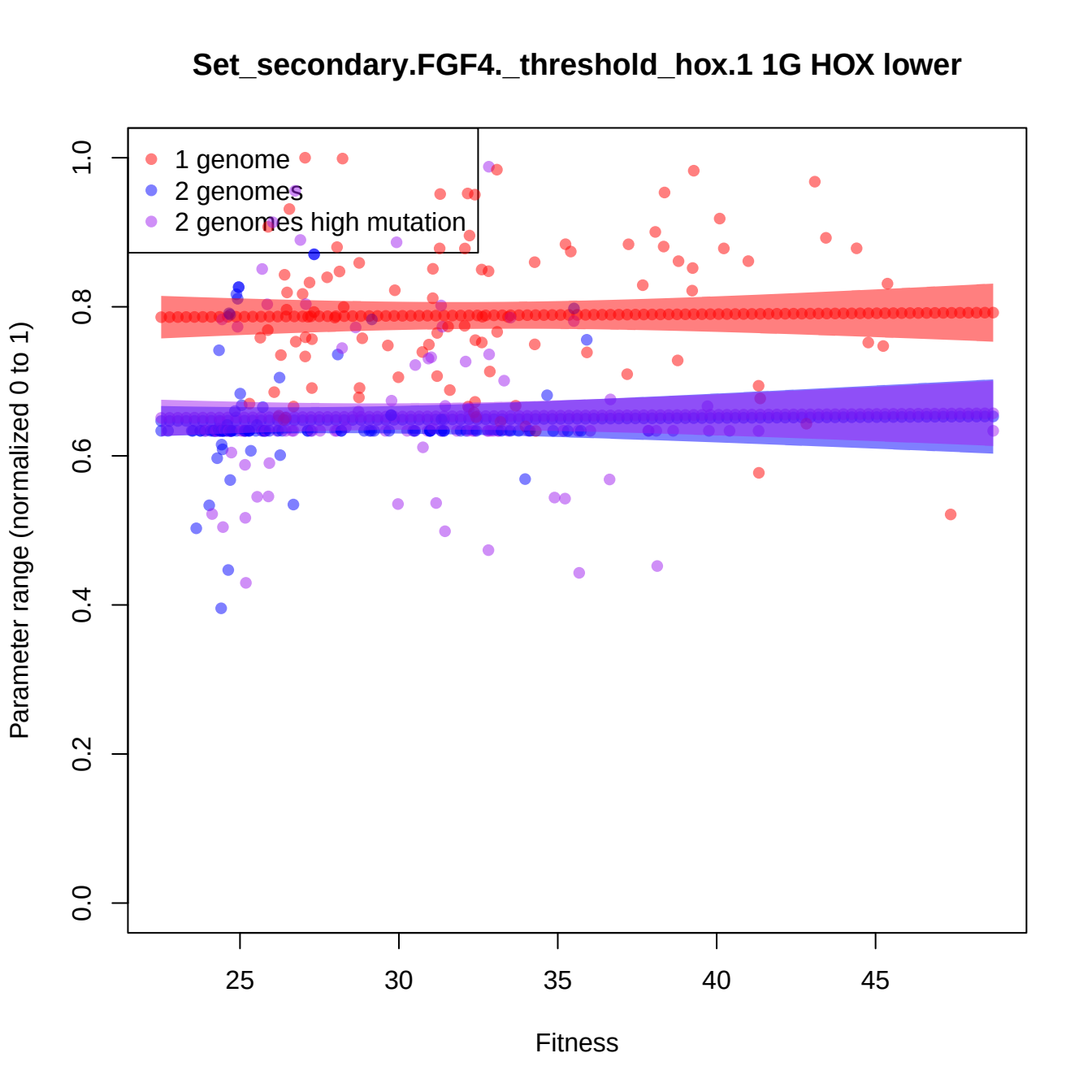

Set_secondary.FGF4._threshold.1
1G Pleiotropic lower
Counts
1.2
1.0
0.8
0.6
0.4
0.2
0.0109002394265851
0.000912395563742579
1.53902778382249e−09
10 − 2
10 − 6
2 − 6
1 genome
2 genomes
2 genomes high mutation

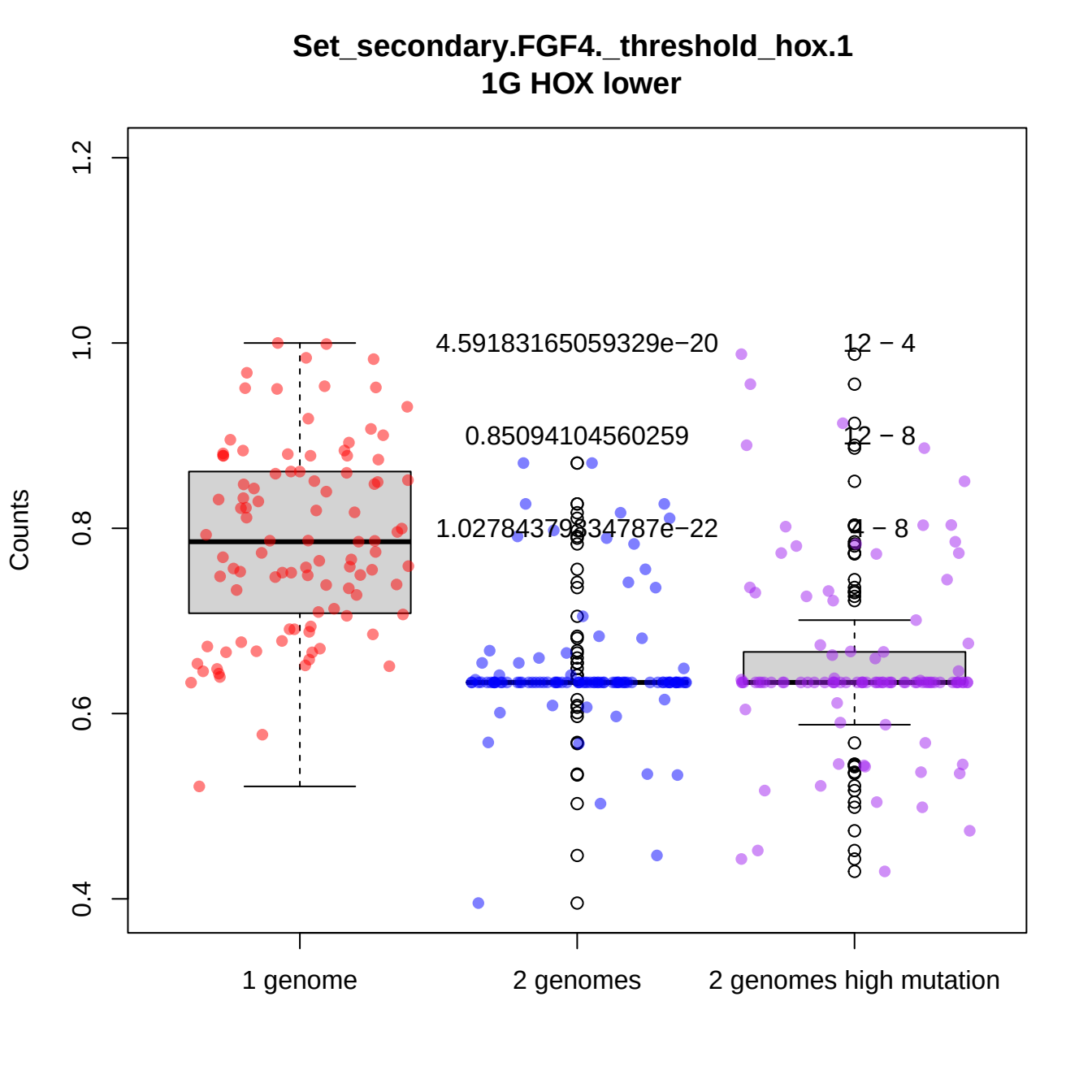

1.0
0.8
0.6
0.4
windows
30
40
50
fit
type
3
7
11

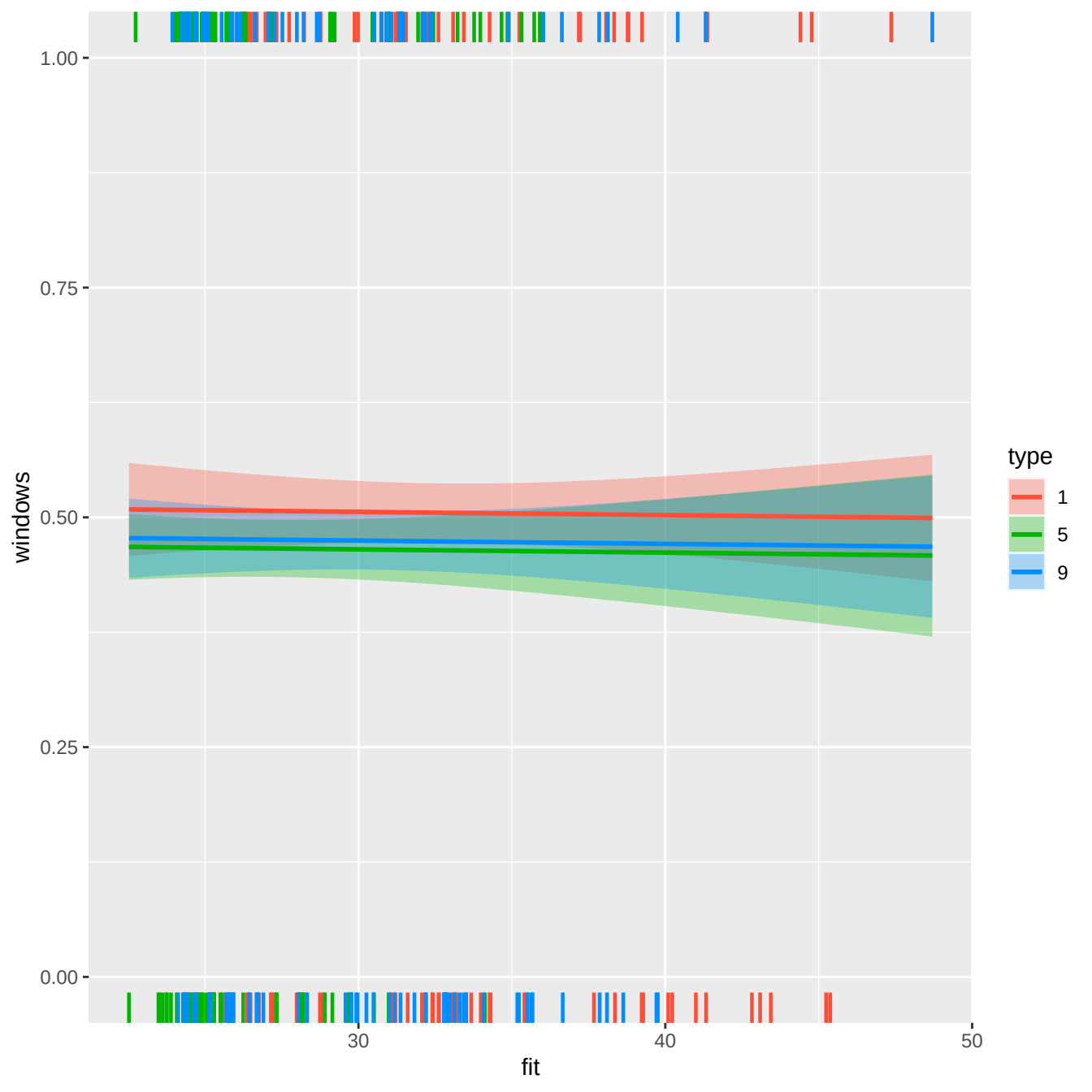

Set_secondary.FGF4._threshold_hox.1 1G HOX upepr
1 genome
2 genomes
2 genomes high mutation
Parameter range (normalized 0 to 1)
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

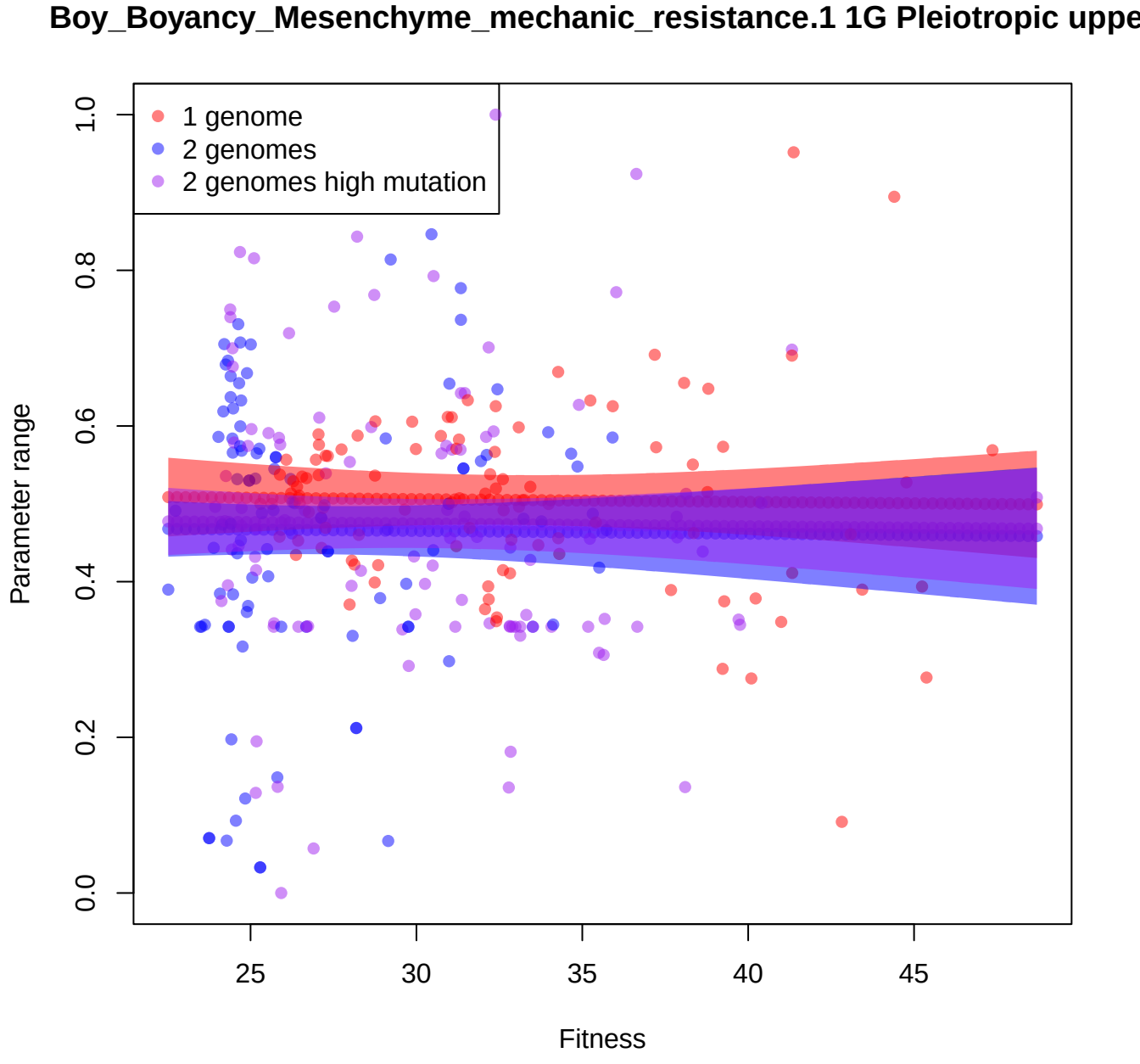

Set_secondary.FGF4._threshold_hox.1
1G HOX upepr
Counts
1.2
1.0
0.8
0.6
0.4
4.23882818422566e−18
0.247976662387298
4.62193692075409e−24
11 − 3
11 − 7
3 − 7
1 genome
2 genomes
2 genomes high mutation

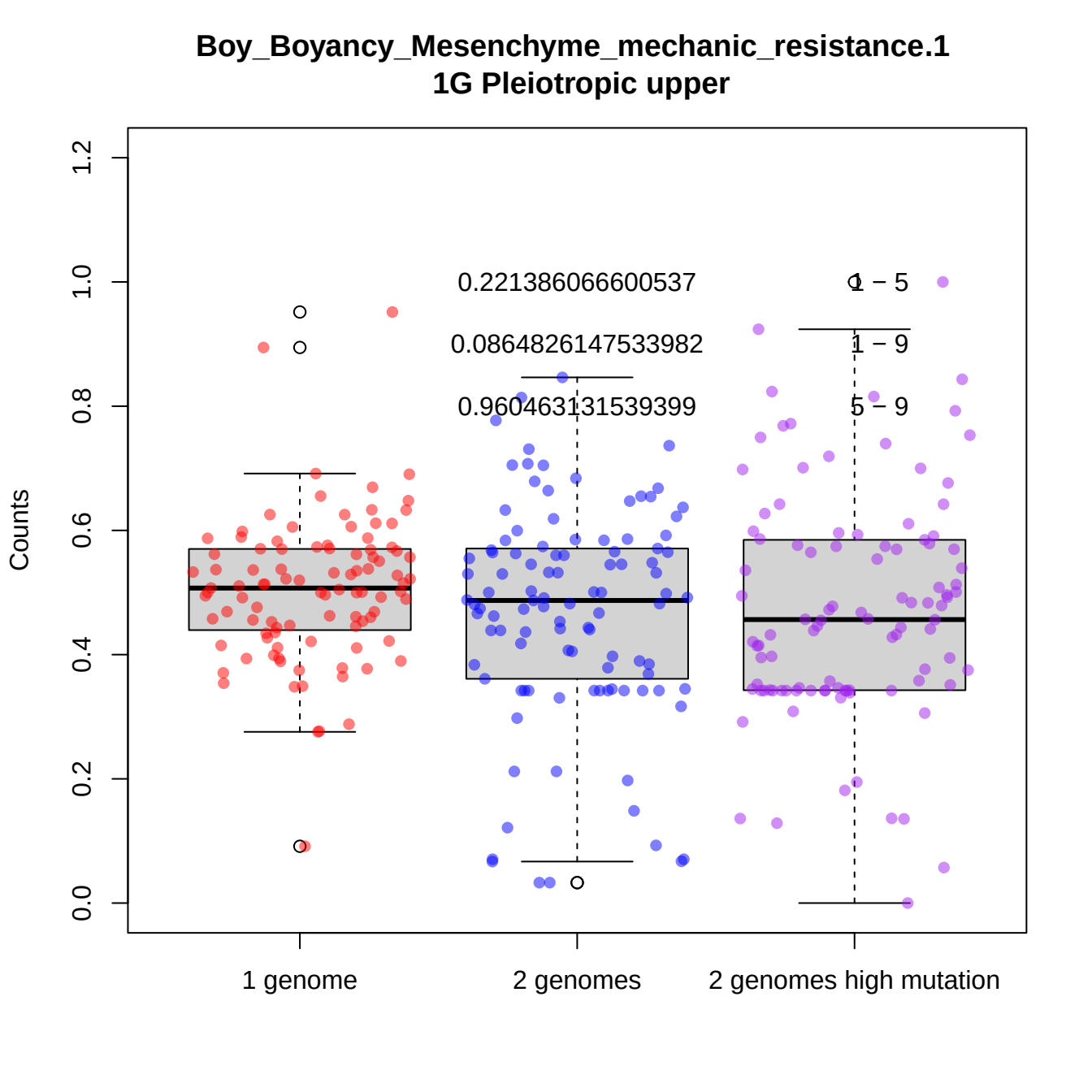

1.0
0.8
0.6
0.4
windows
30
40
50
fit
type
4
8
12

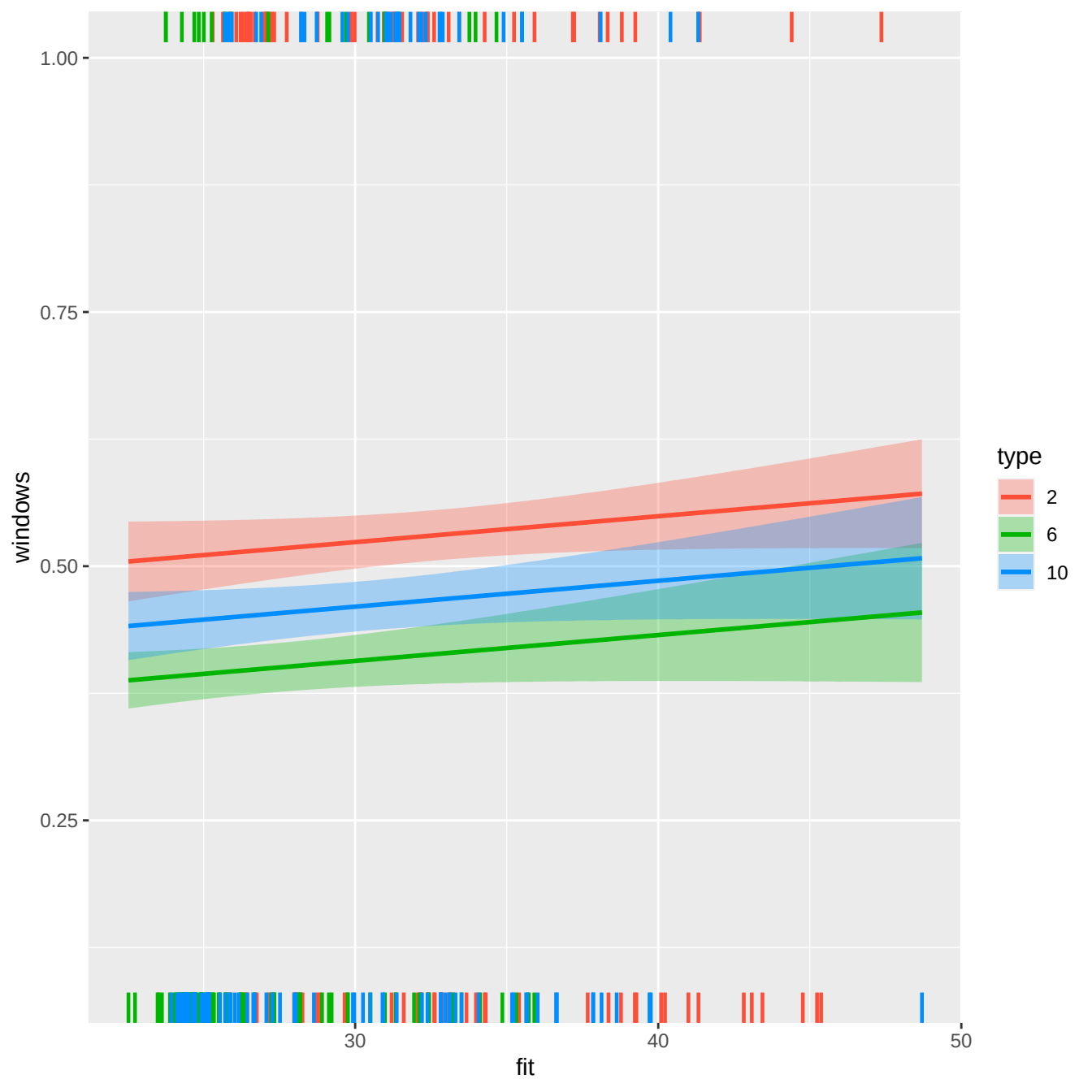

Set_secondary.FGF4._threshold_hox.1 1G HOX lower
1 genome
2 genomes
2 genomes high mutation
Parameter range (normalized 0 to 1)
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

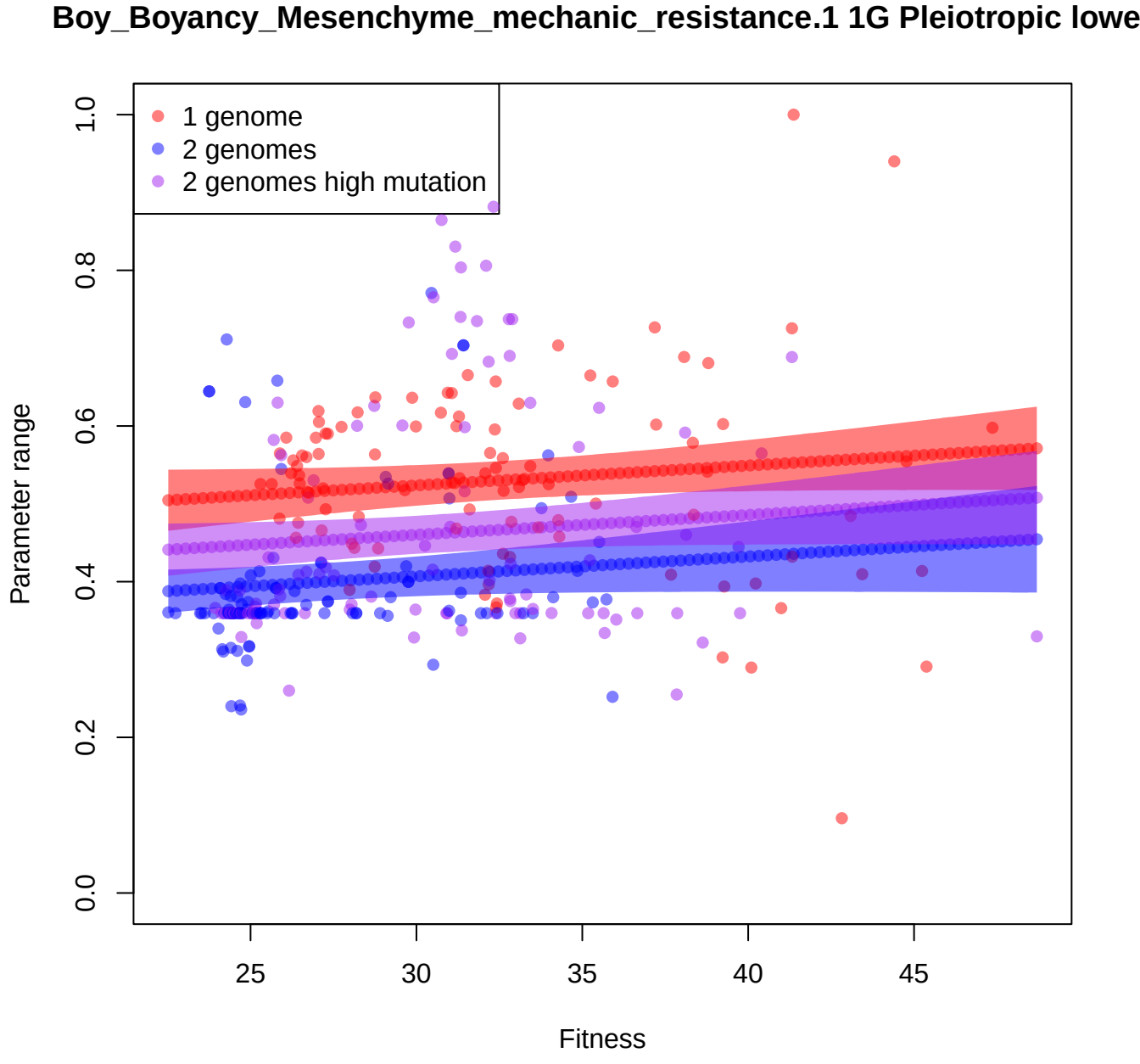

Set_secondary.FGF4._threshold_hox.1
1G HOX lower
Counts
1.2
1.0
0.8
0.6
0.4
4.59183165059329e−20
0.85094104560259
1.02784379834787e−22
12 − 4
12 − 8
4 − 8
1 genome
2 genomes
2 genomes high mutation

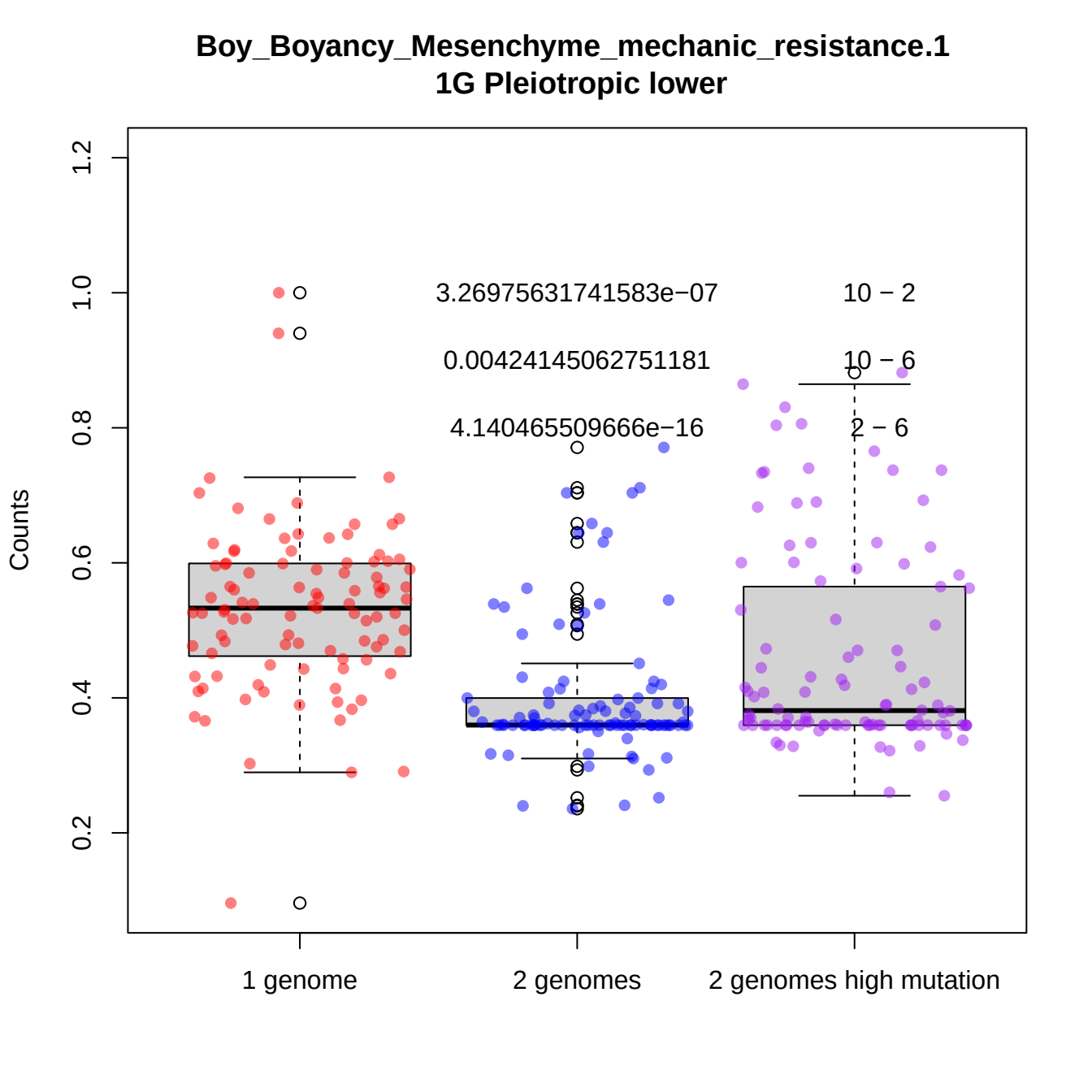

1.00
0.75
0.50
0.25
0.00
windows
30
40
50
fit
type
1
5
9

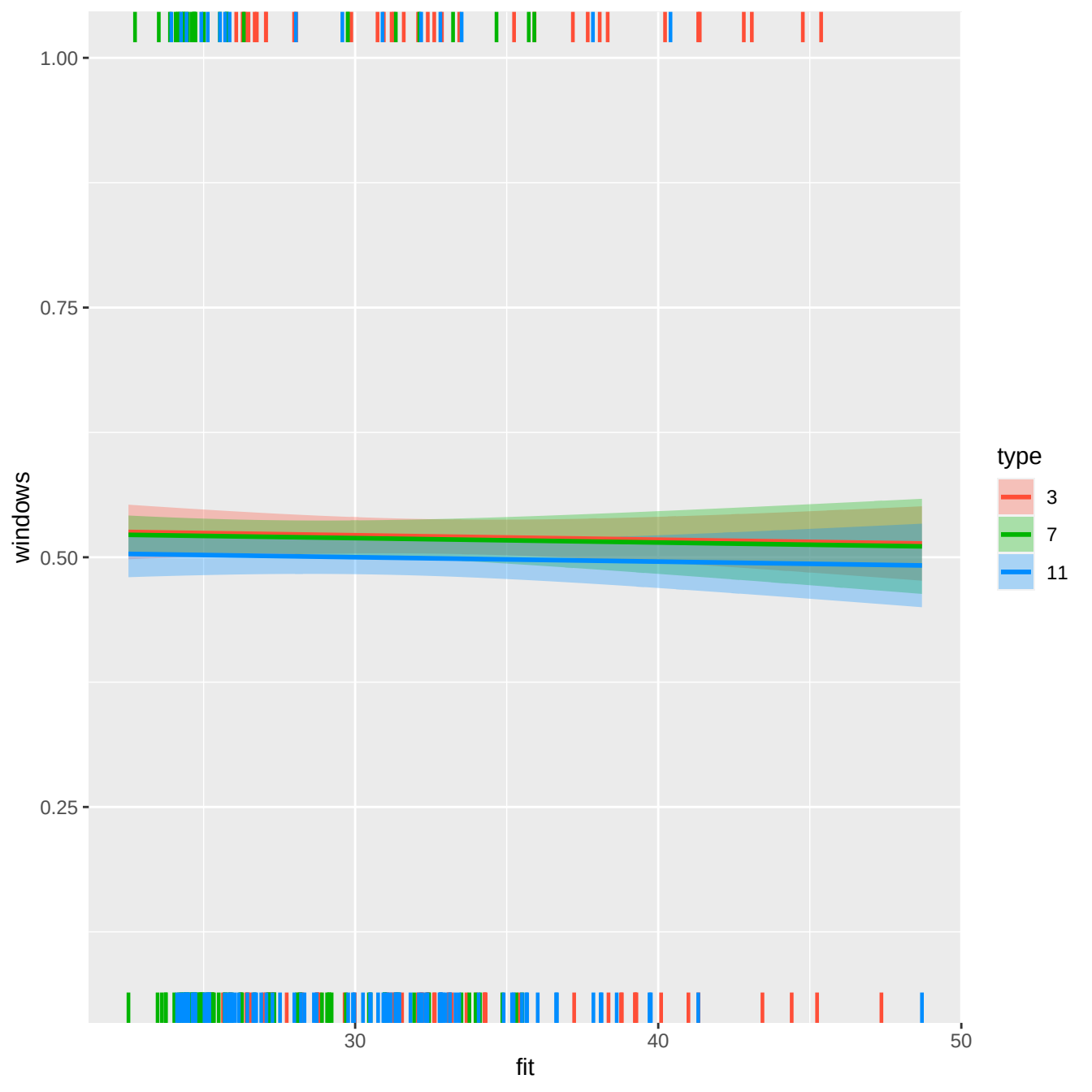

Boy_Boyancy_Mesenchyme_mechanic_resistance.1 1G Pleiotropic upper
1 genome
2 genomes
2 genomes high mutation
Parameter range
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

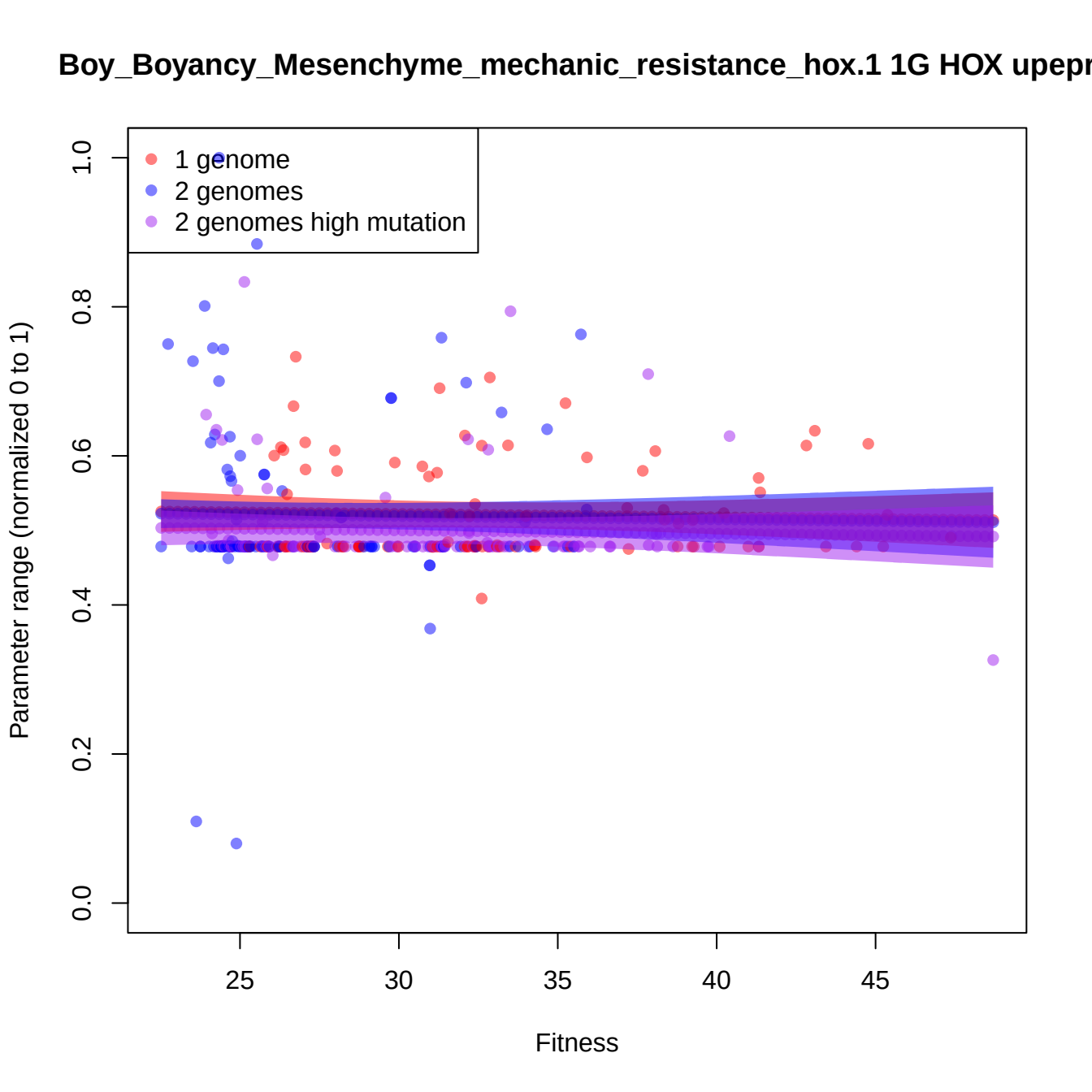

Boy_Boyancy_Mesenchyme_mechanic_resistance.1
1G Pleiotropic upper
Counts
0.0
0.2
0.4
0.6
0.8
1.0
1.2
0.221386066600537
0.0864826147533982
0.960463131539399
1 − 5
1 − 9
5 − 9
1 genome
2 genomes
2 genomes high mutation

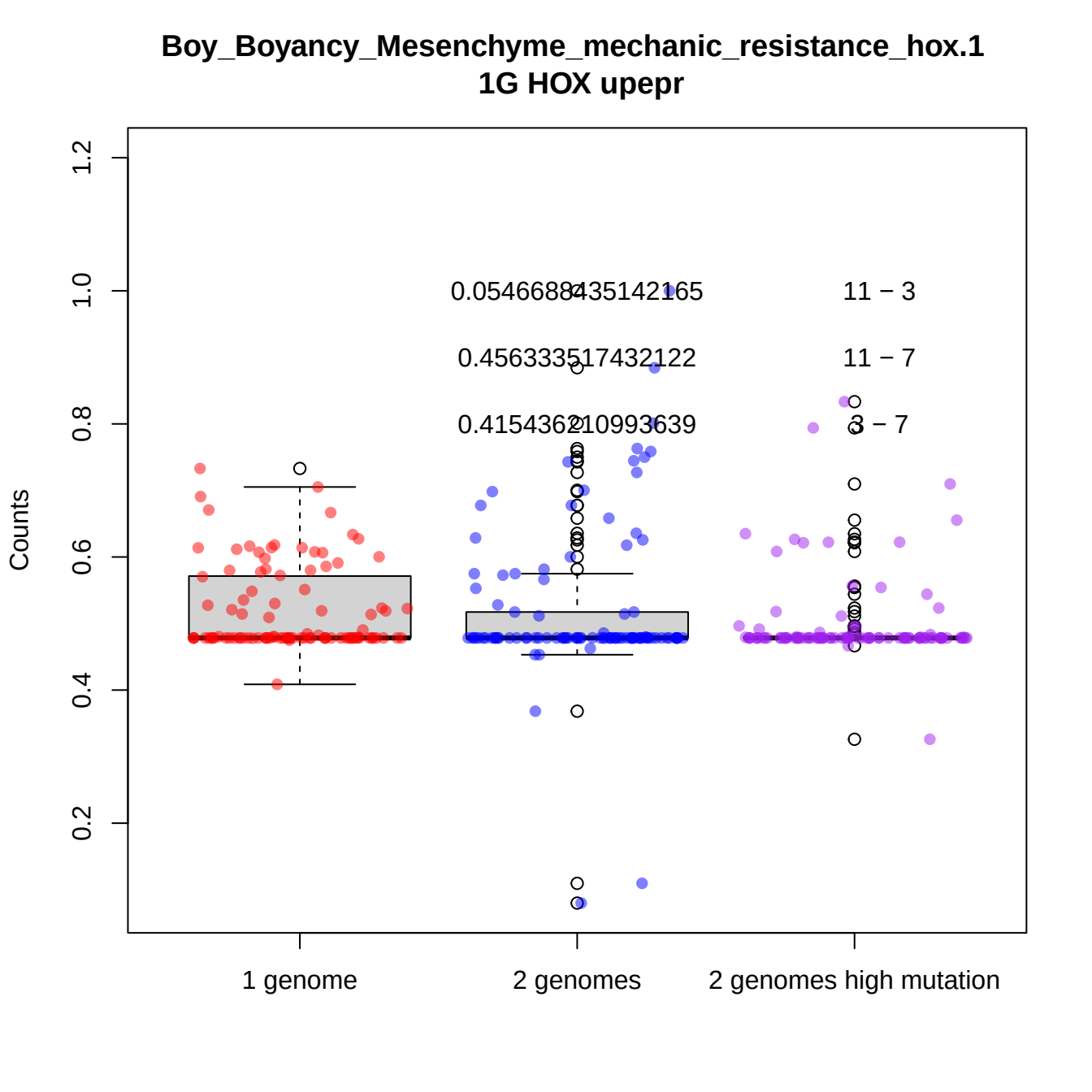

1.00
0.75
0.50
0.25
windows
30
40
50
fit
type
2
6
10

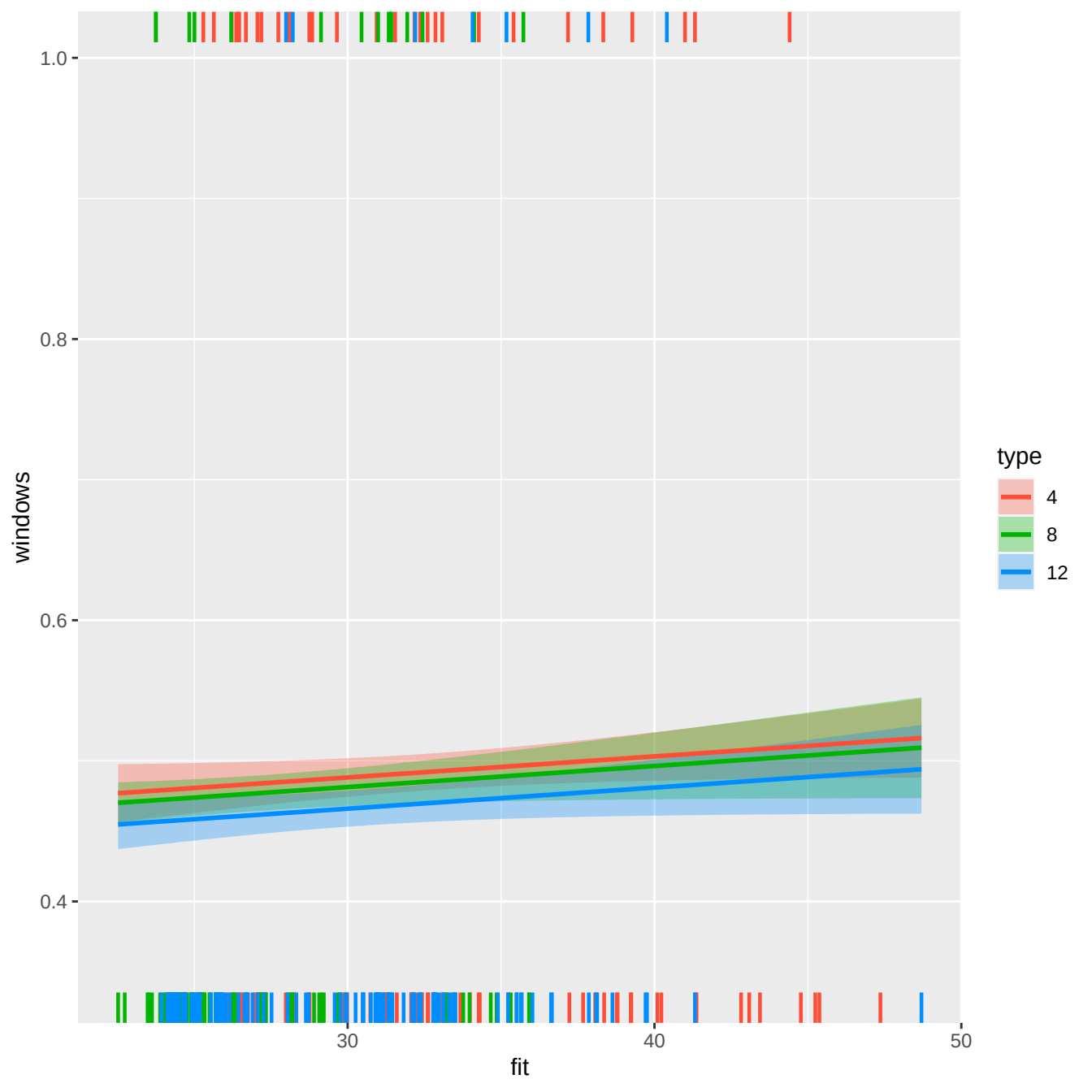

Boy_Boyancy_Mesenchyme_mechanic_resistance.1 1G Pleiotropic lowe
1 genome
2 genomes
2 genomes high mutation
Parameter range
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

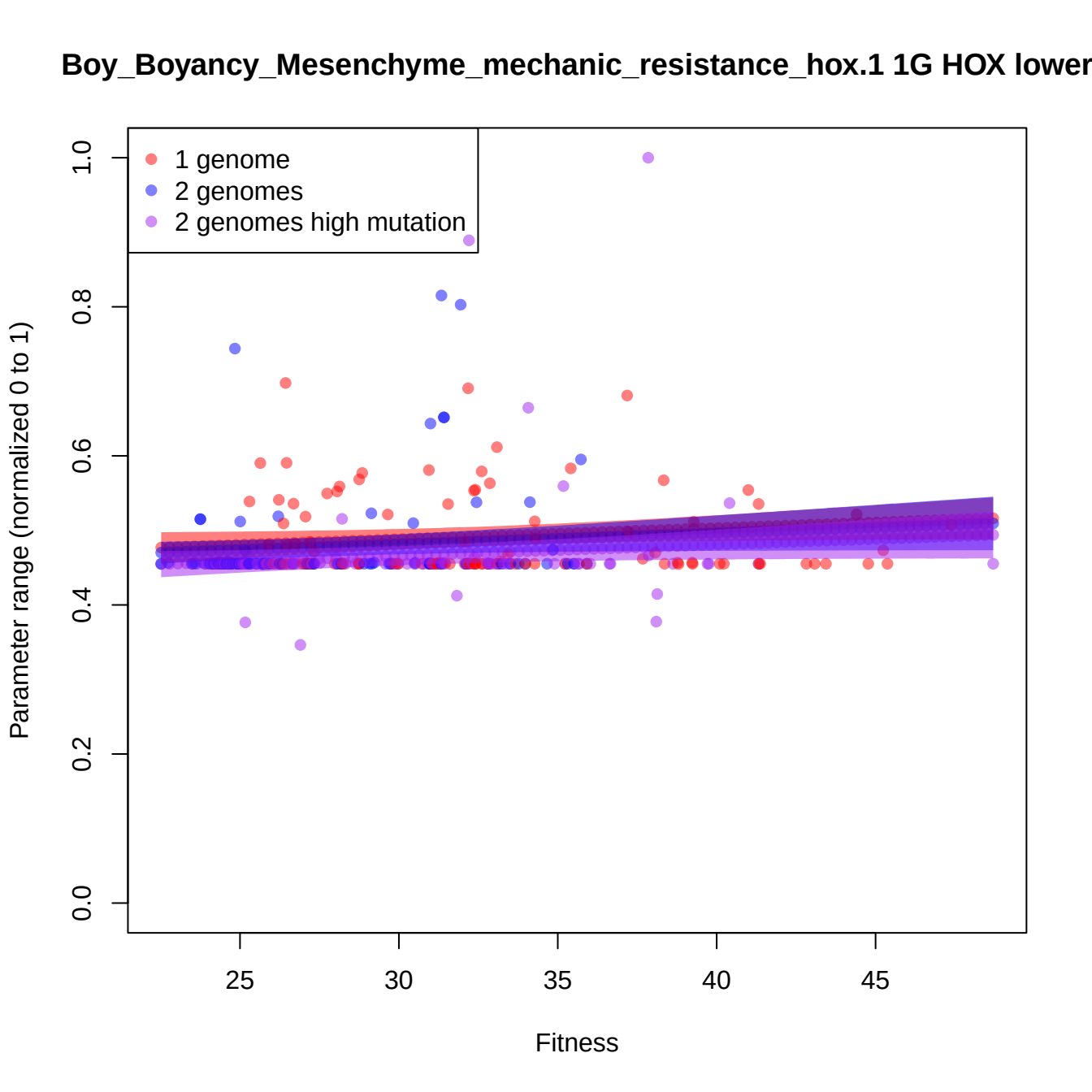

Boy_Boyancy_Mesenchyme_mechanic_resistance.1
1G Pleiotropic lower
Counts
1.2
1.0
0.8
0.6
0.4
0.2
3.26975631741583e−07
0.00424145062751181
4.140465509666e−16
10 − 2
10 − 6
2 − 6
1 genome
2 genomes
2 genomes high mutation

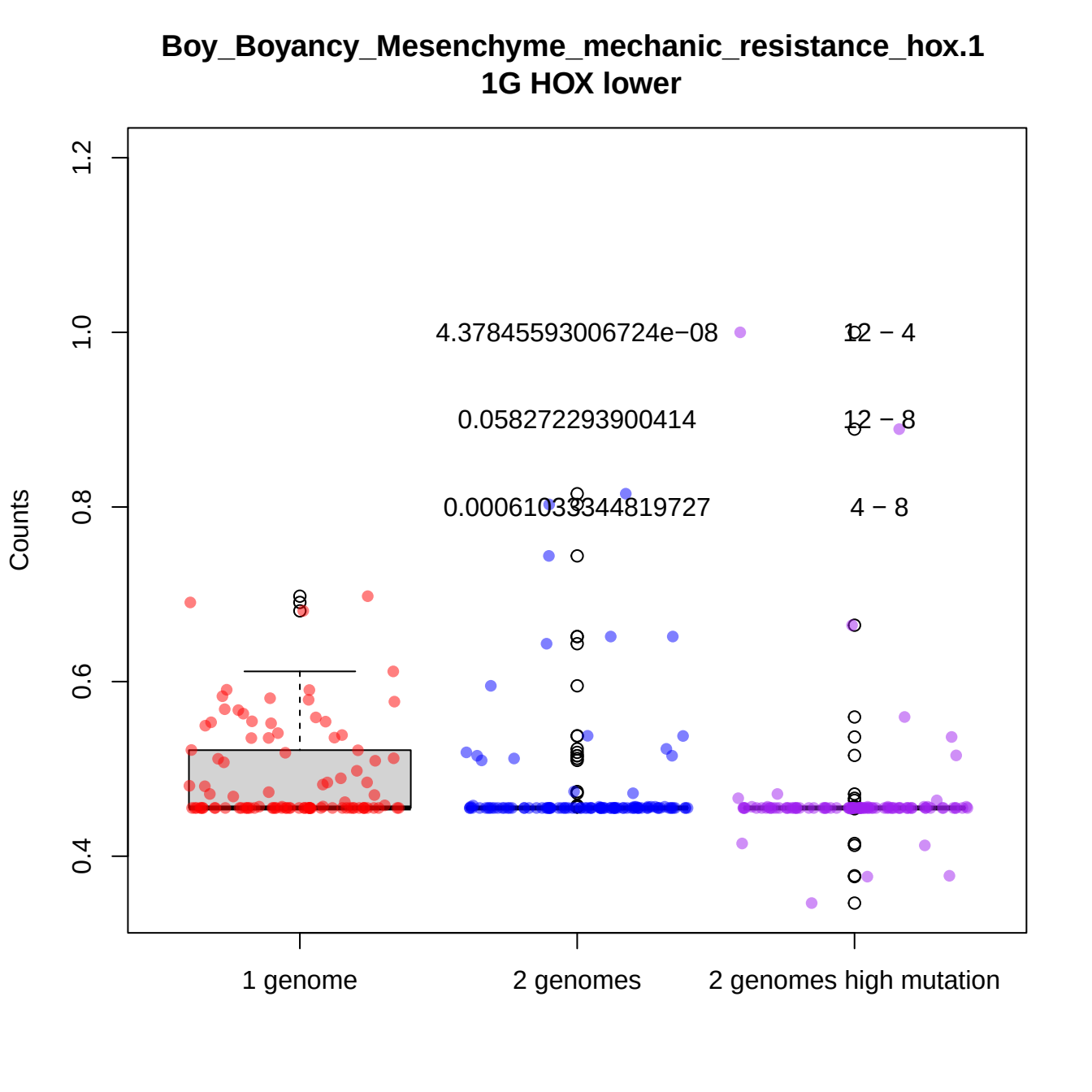

1.00
0.75
0.50
0.25
windows
30
40
50
fit
type
3
7
11

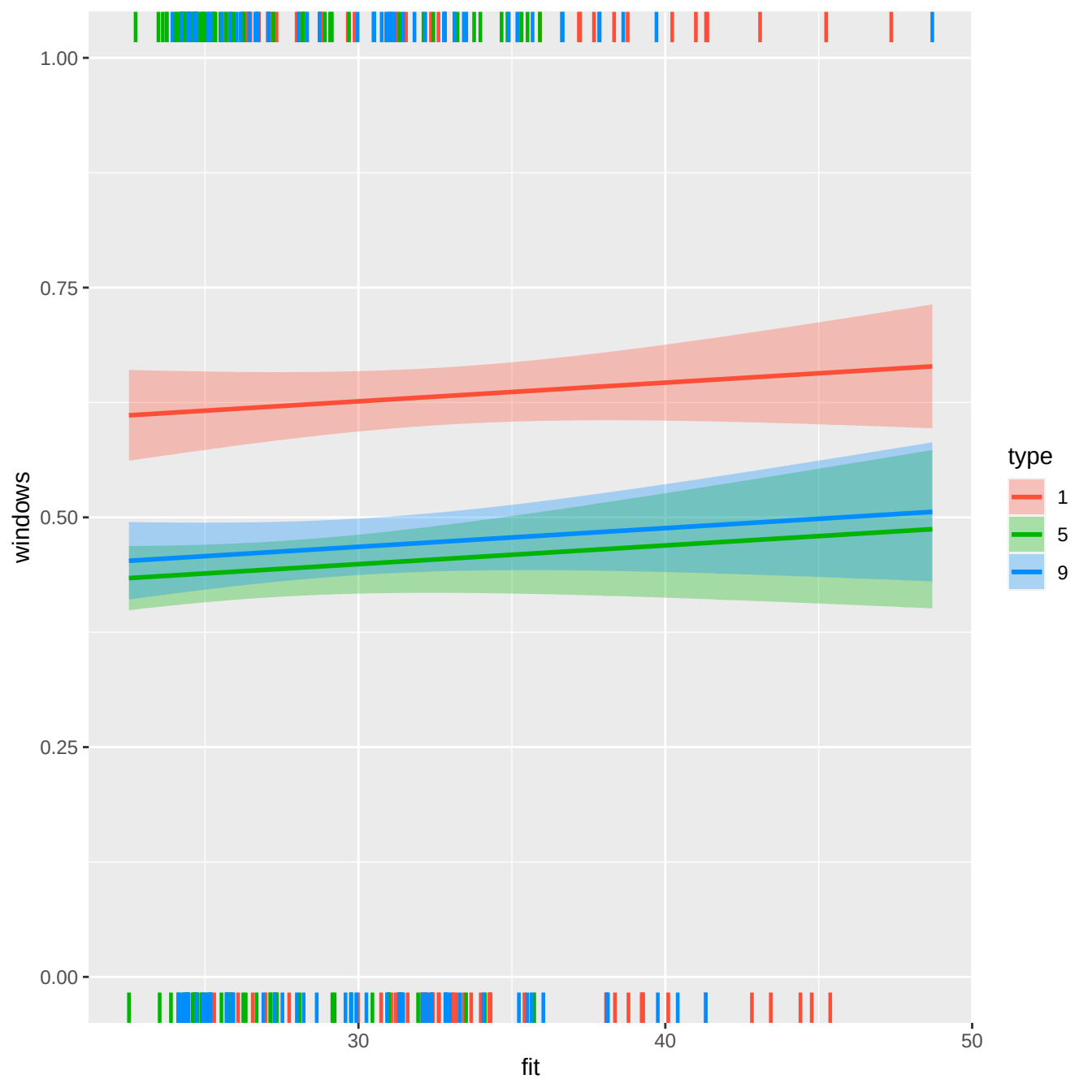

Boy_Boyancy_Mesenchyme_mechanic_resistance_hox.1 1G HOX upepr
1 genome
2 genomes
2 genomes high mutation
Parameter range (normalized 0 to 1)
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

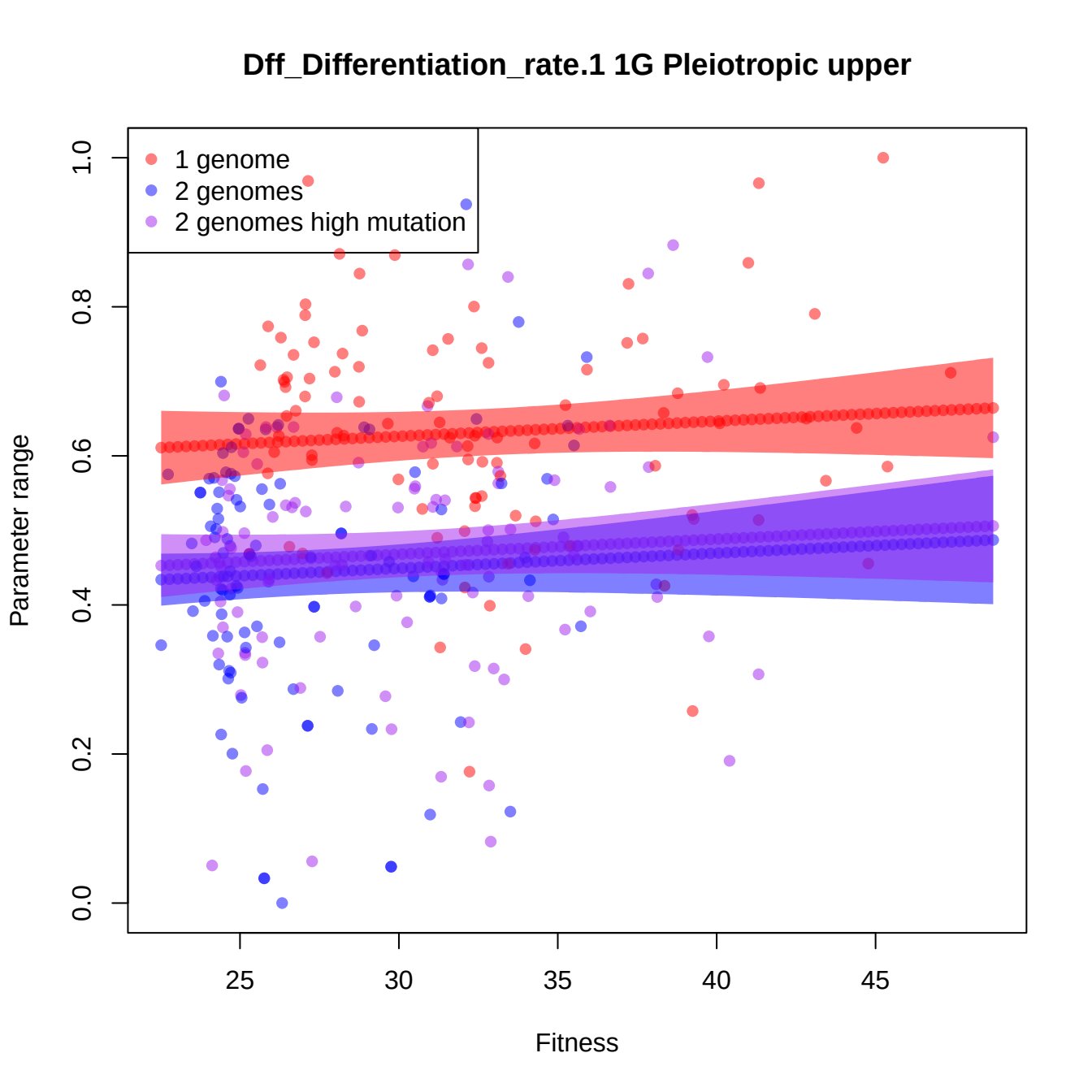

Boy_Boyancy_Mesenchyme_mechanic_resistance_hox.1
1G HOX upepr
Counts
1.2
1.0
0.8
0.6
0.4
0.2
0.05466884351421 65
0.45633351743212 2
0.41543621099363 9
11 − 3
11 − 7
3 − 7
1 genome
2 genomes
2 genomes high mutation

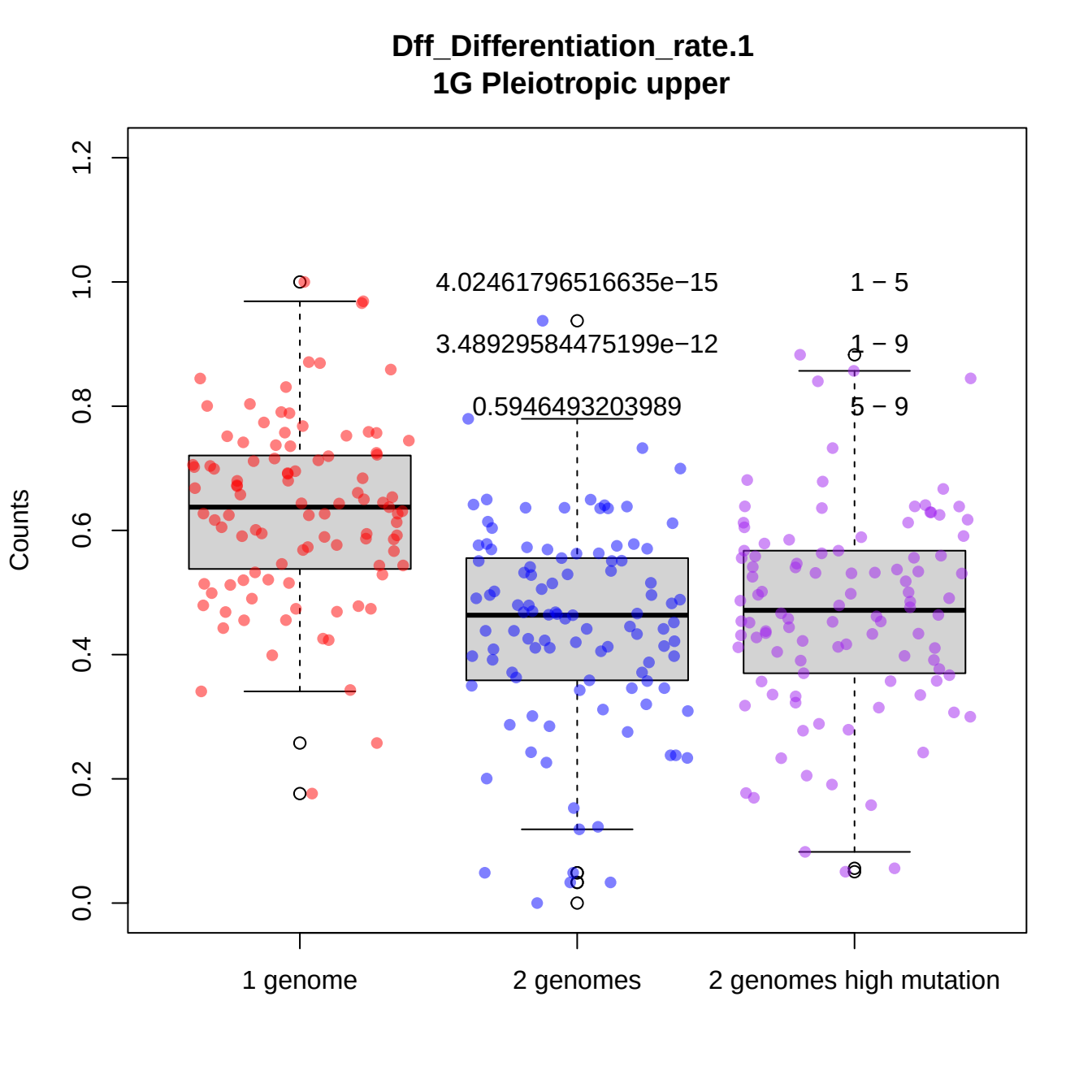

1.0
0.8
0.6
0.4
windows
30
40
50
fit
type
4
8
12

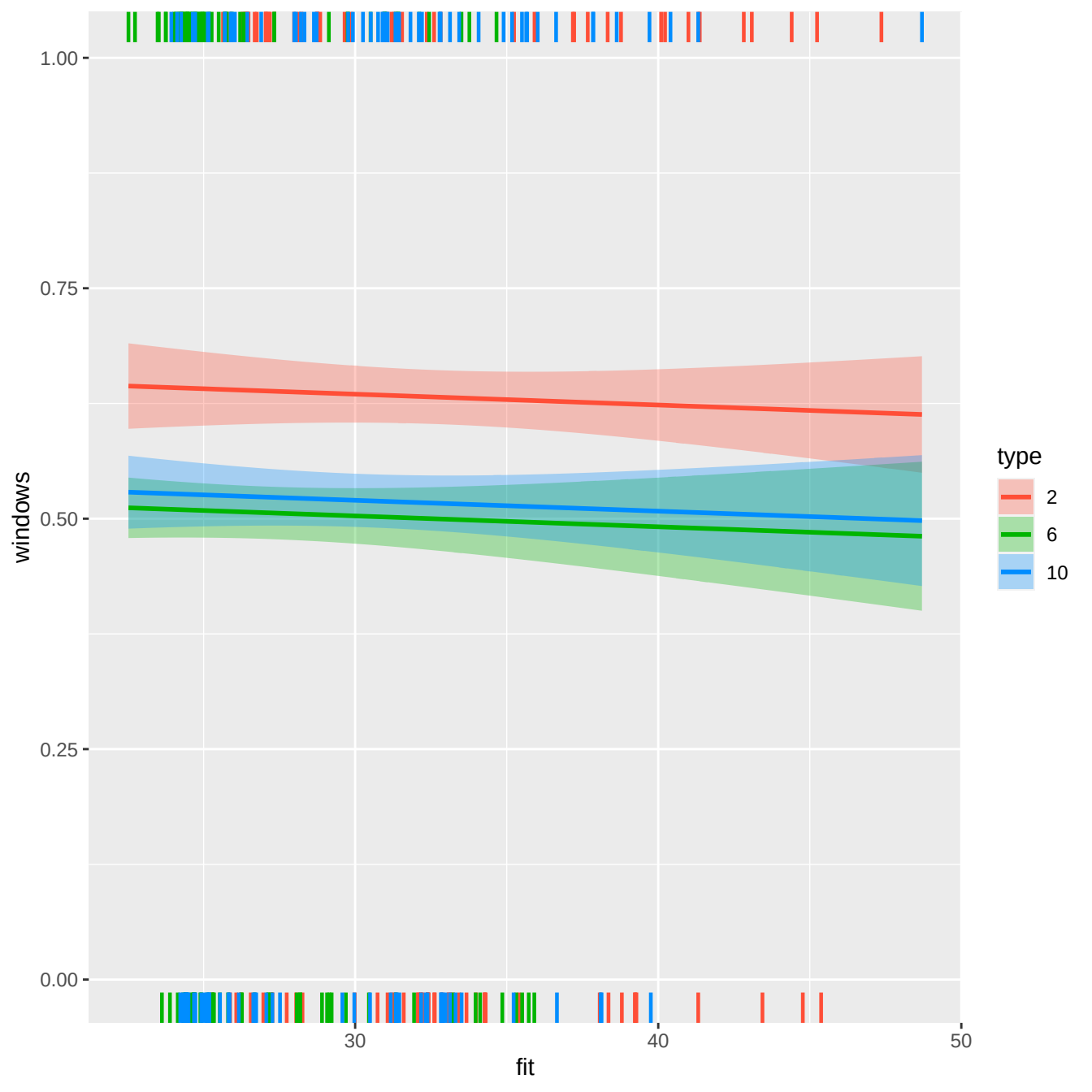

Boy_Boyancy_Mesenchyme_mechanic_resistance_hox.1 1G HOX lower
1 genome
2 genomes
2 genomes high mutation
Parameter range (normalized 0 to 1)
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

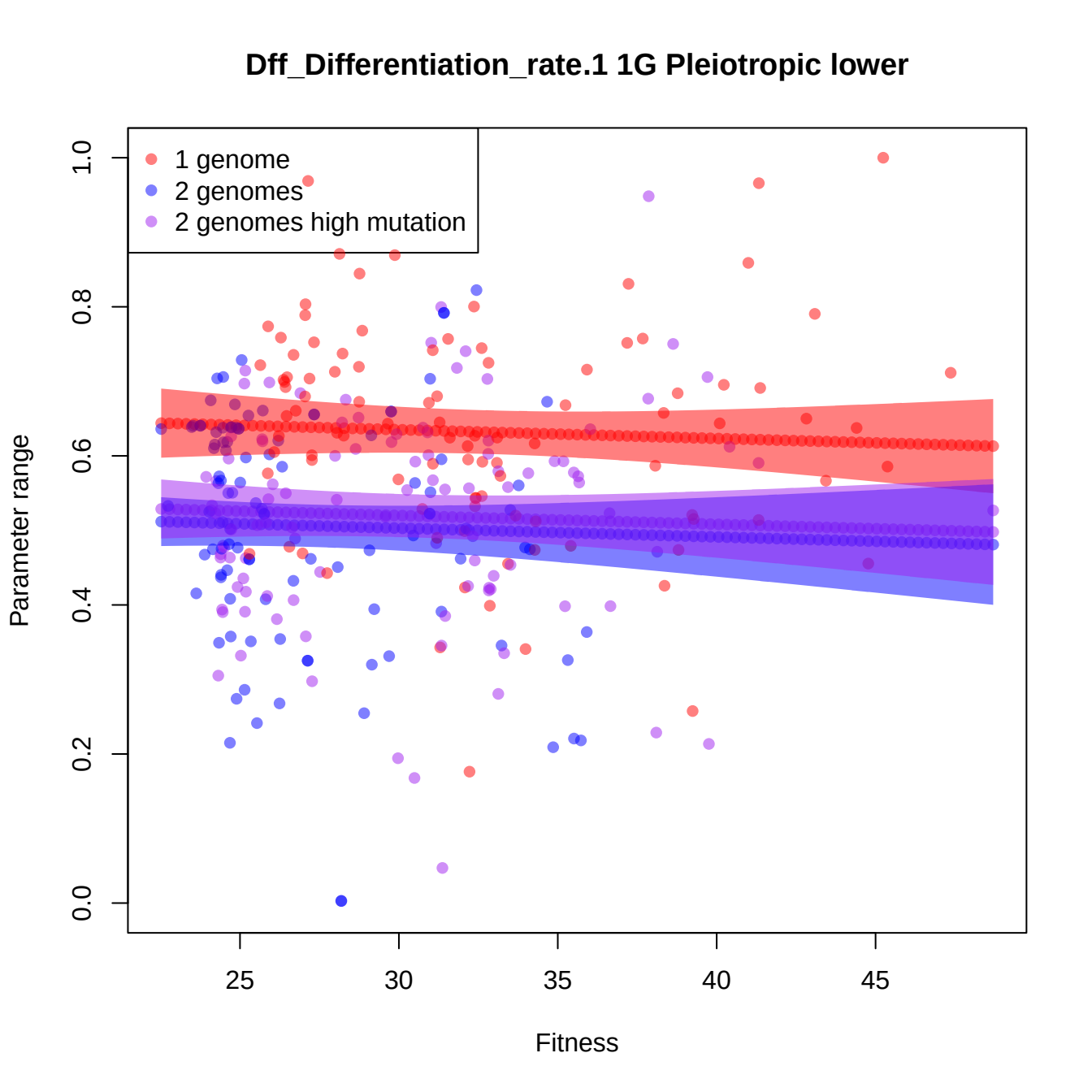

Boy_Boyancy_Mesenchyme_mechanic_resistance_hox.1
1G HOX lower
Counts
1.2
1.0
0.8
0.6
0.4
4.37845593006724e−08
0.058272293900414
0.000610333344819727
12 − 4
12 − 8
4 − 8
1 genome
2 genomes
2 genomes high mutation

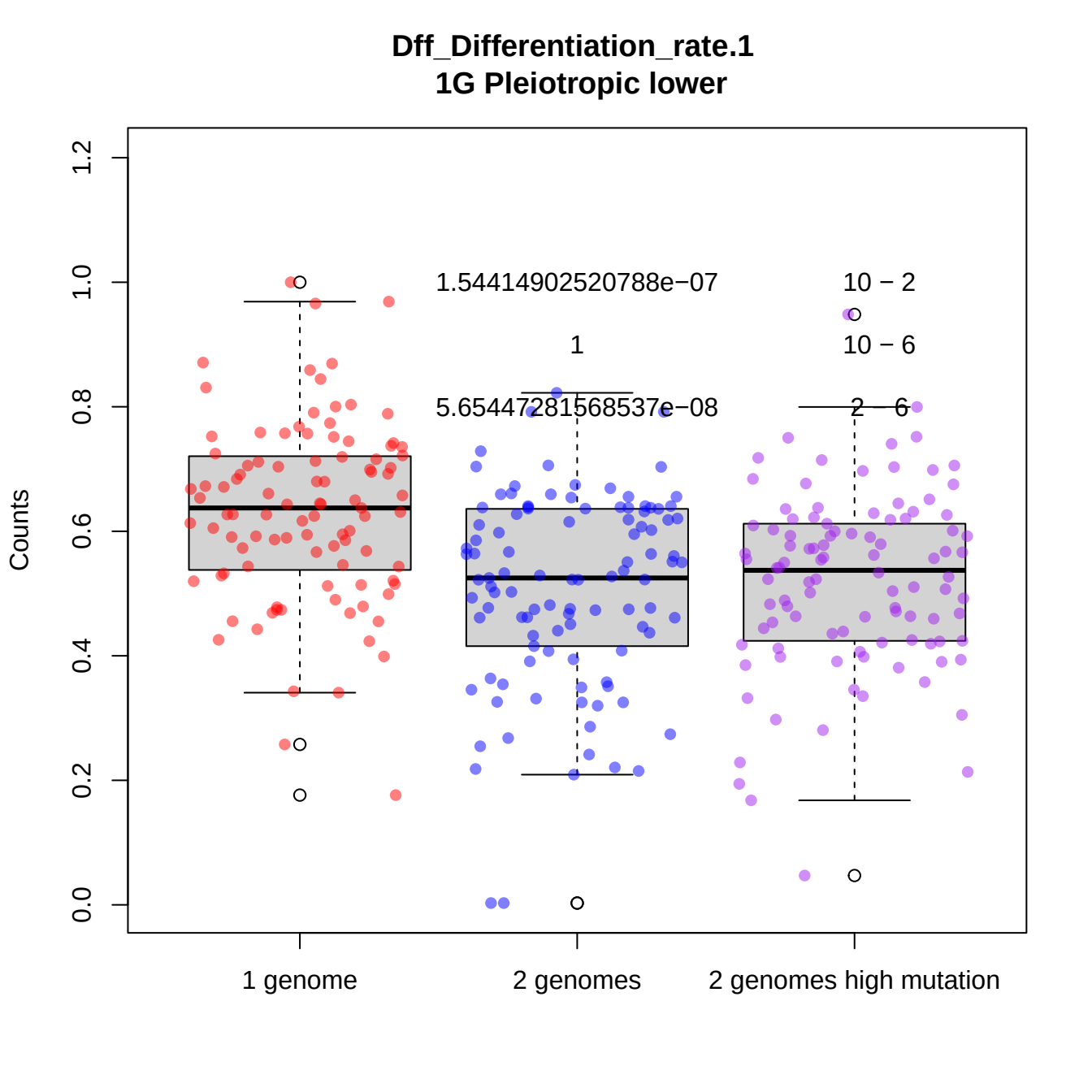

1.00
0.75
0.50
0.25
0.00
windows
30
40
50
fit
type
1
5
9

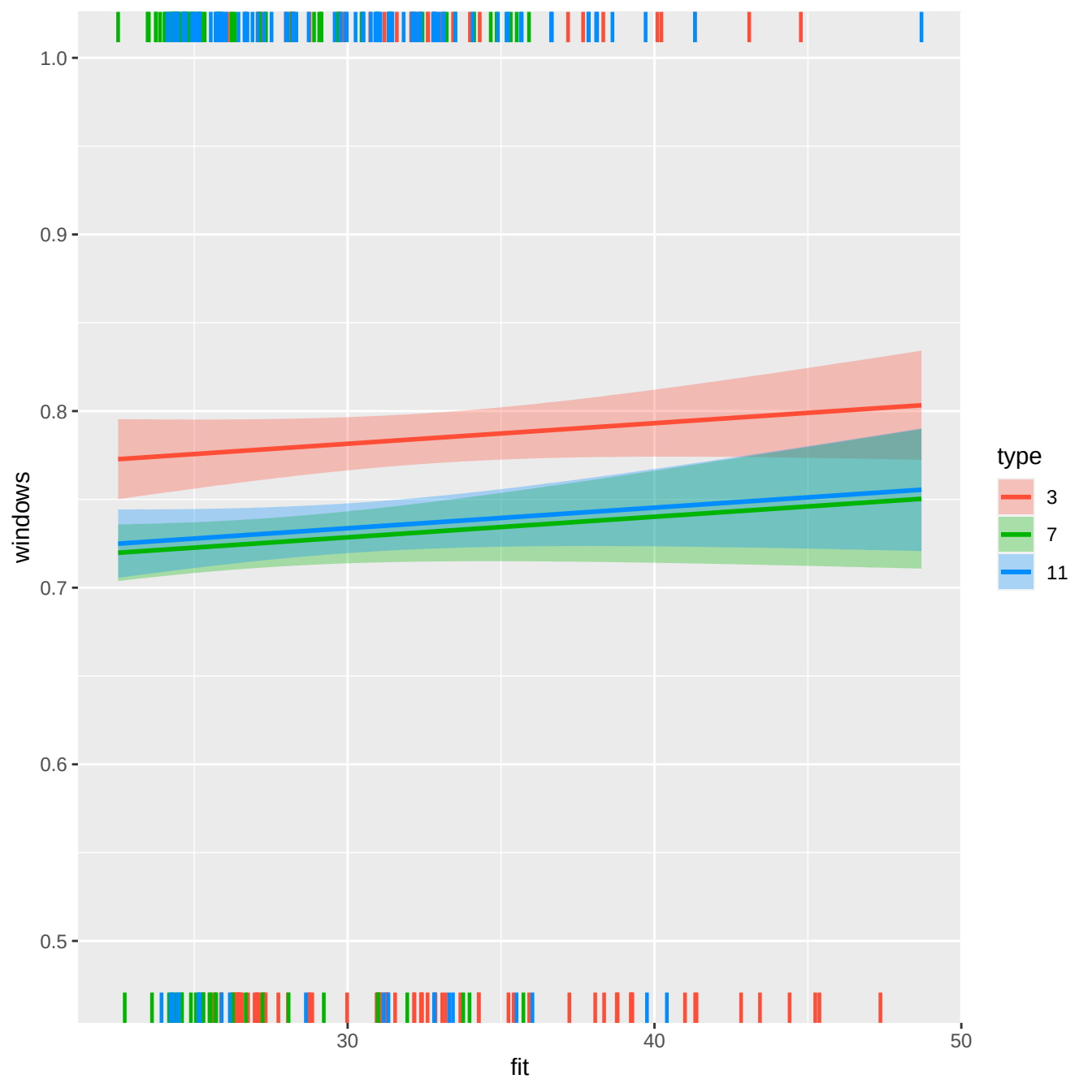

Dff_Differentiation_rate.1 1G Pleiotropic upper
1 genome
2 genomes
2 genomes high mutation
Parameter range
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

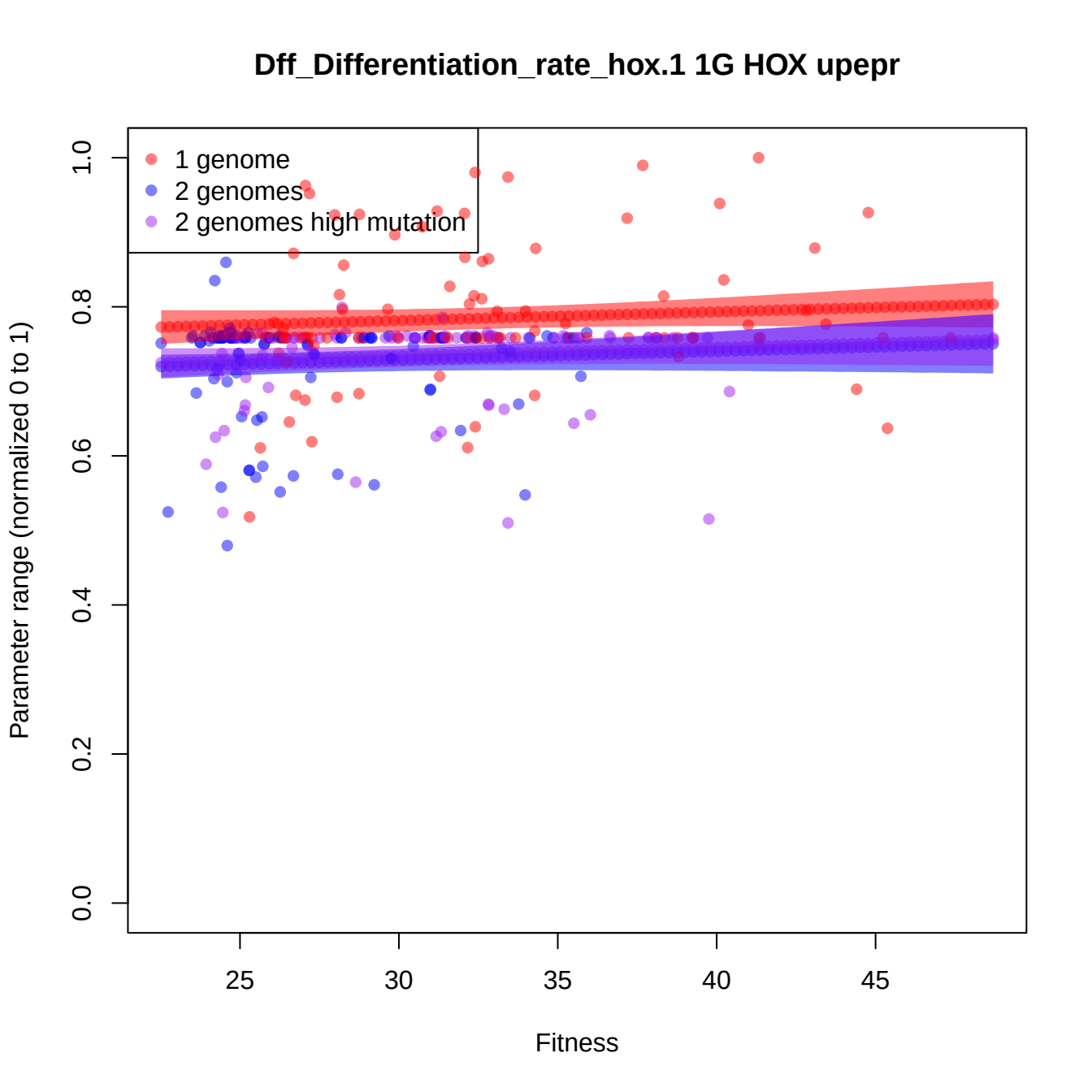

Dff_Differentiation_rate.1
1G Pleiotropic upper
Counts
0.0
0.2
0.4
0.6
0.8
1.0
1.2
4.02461796516635e−15
3.48929584475199e−12
0.5946493203989
1 − 5
1 − 9
5 − 9
1 genome
2 genomes
2 genomes high mutation

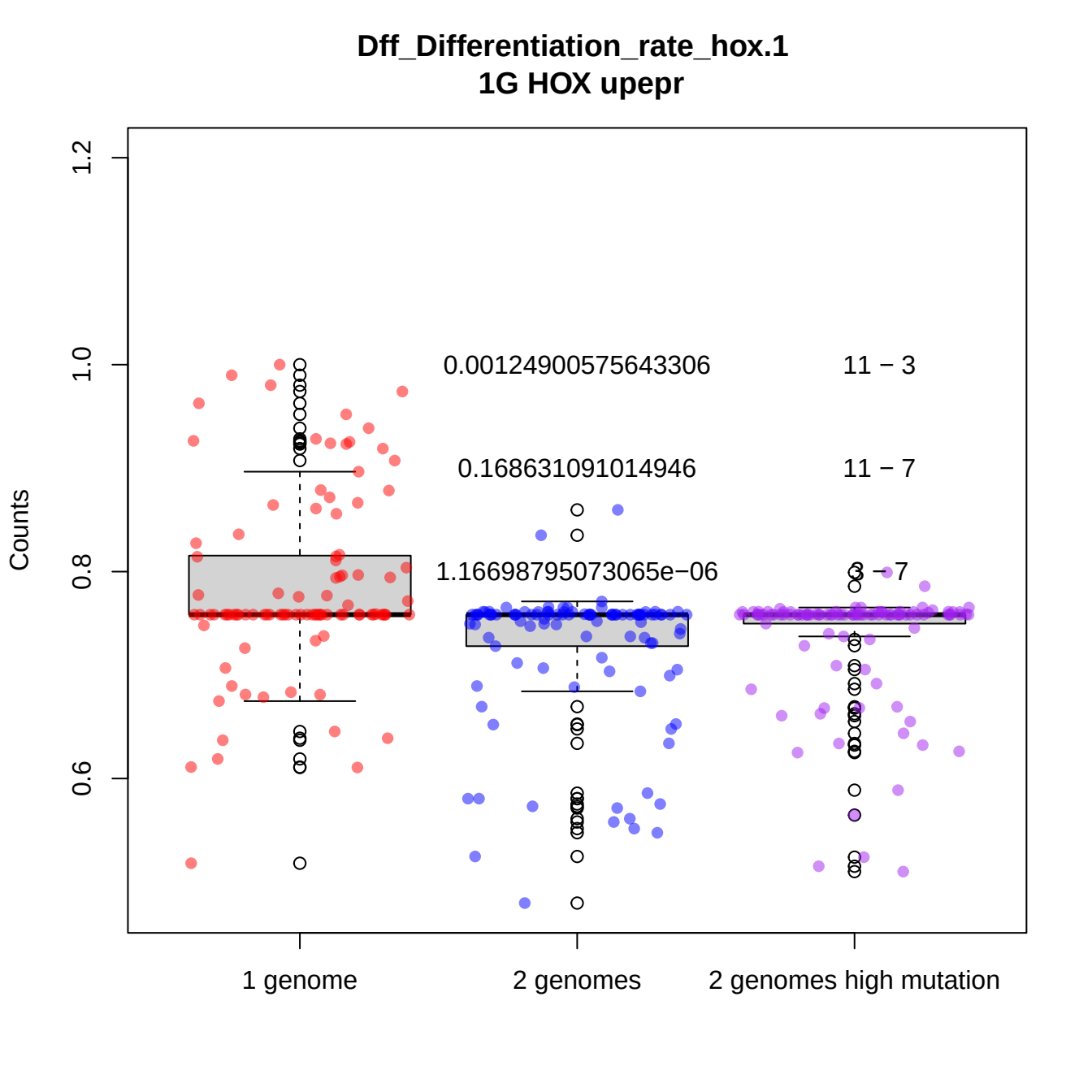

1.00
0.75
0.50
0.25
0.00
windows
30
40
50
fit
type
2
6
10

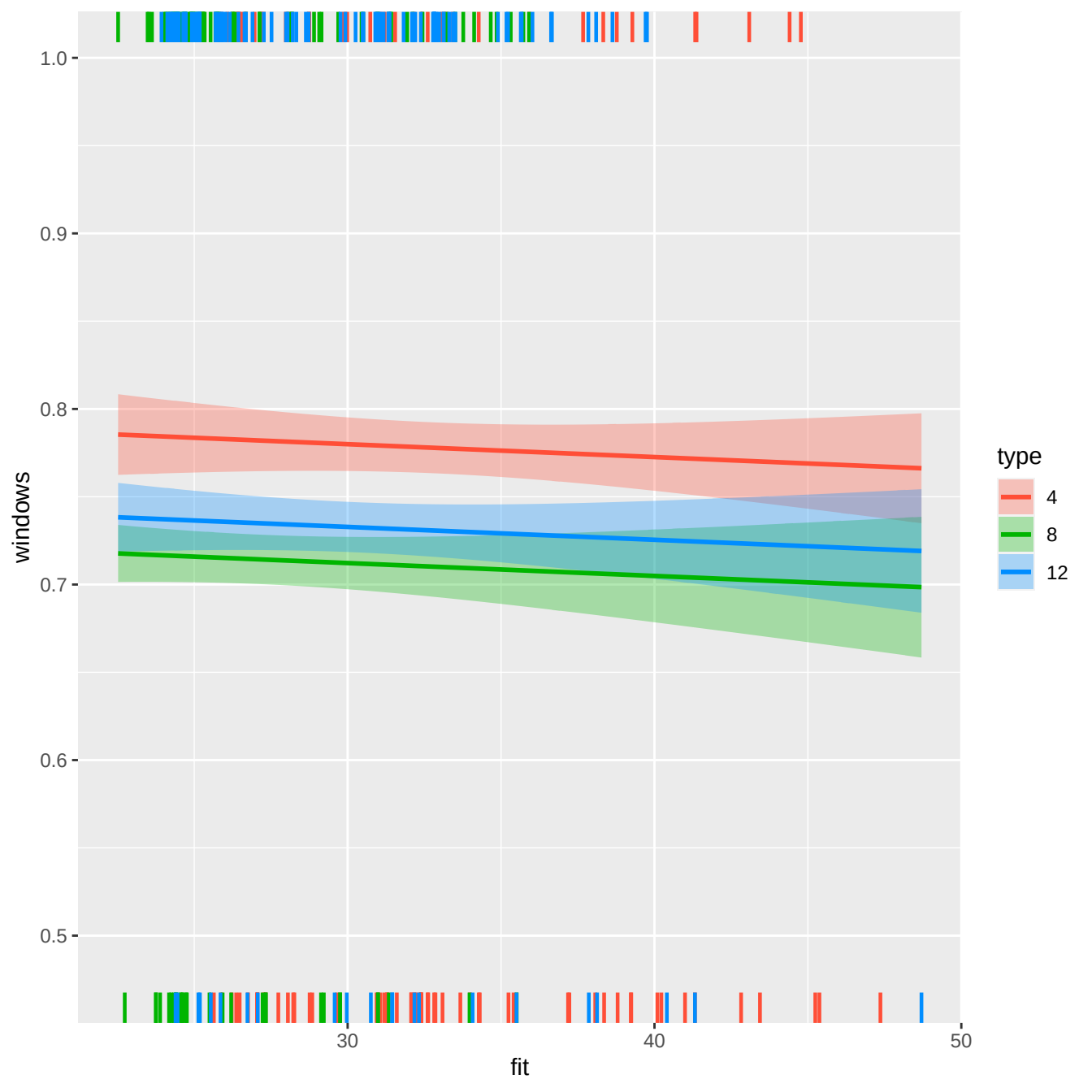

Dff_Differentiation_rate.1 1G Pleiotropic lower
1 genome
2 genomes
2 genomes high mutation
Parameter range
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

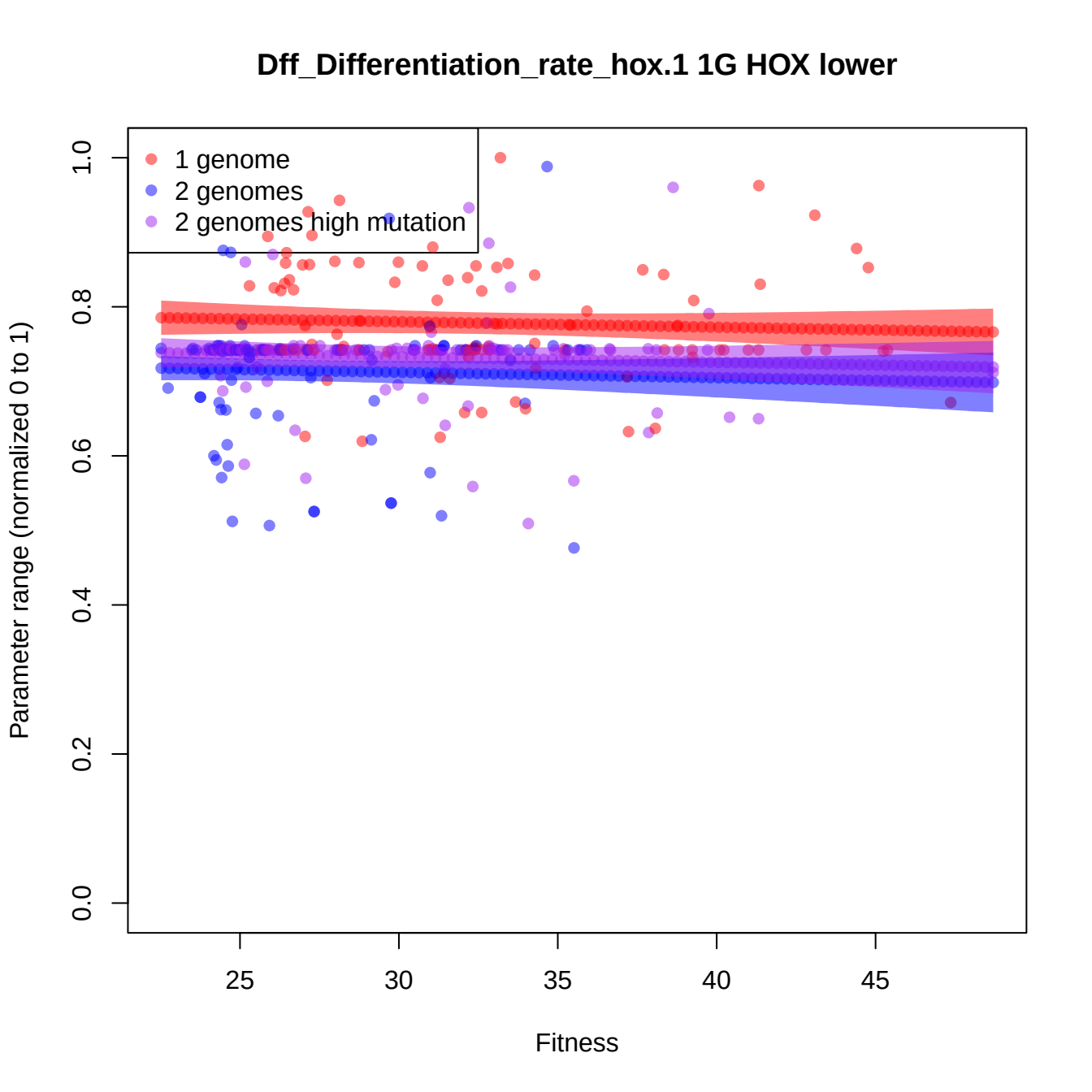

Dff_Differentiation_rate.1
1G Pleiotropic lower
Counts
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.54414902520788e−07
1
5.65447281568537e−08
10 − 2
10 − 6
2 − 6
1 genome
2 genomes
2 genomes high mutation

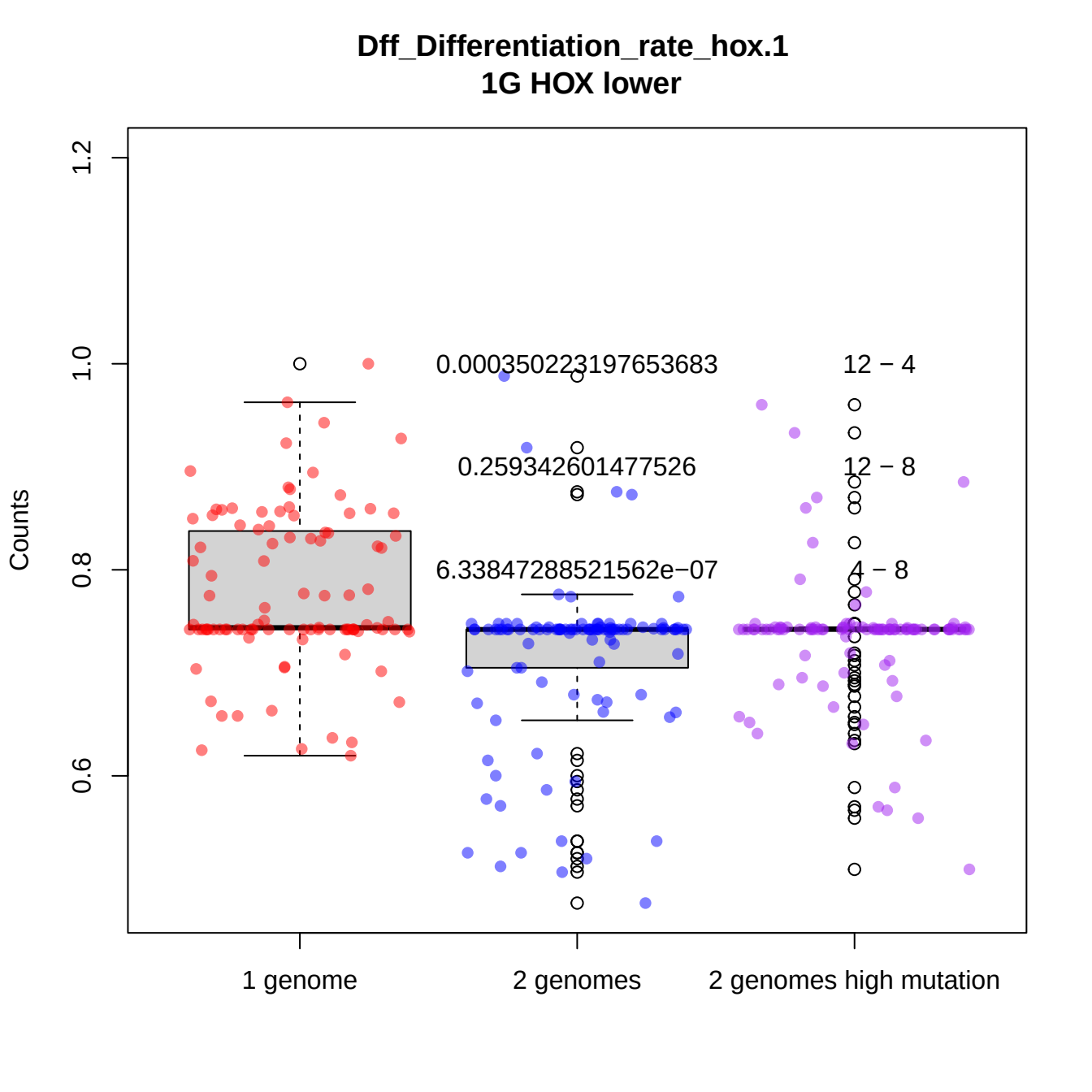

1.0
0.9
0.8
0.7
0.6
0.5
windows
30
40
50
fit
type
3
7
11

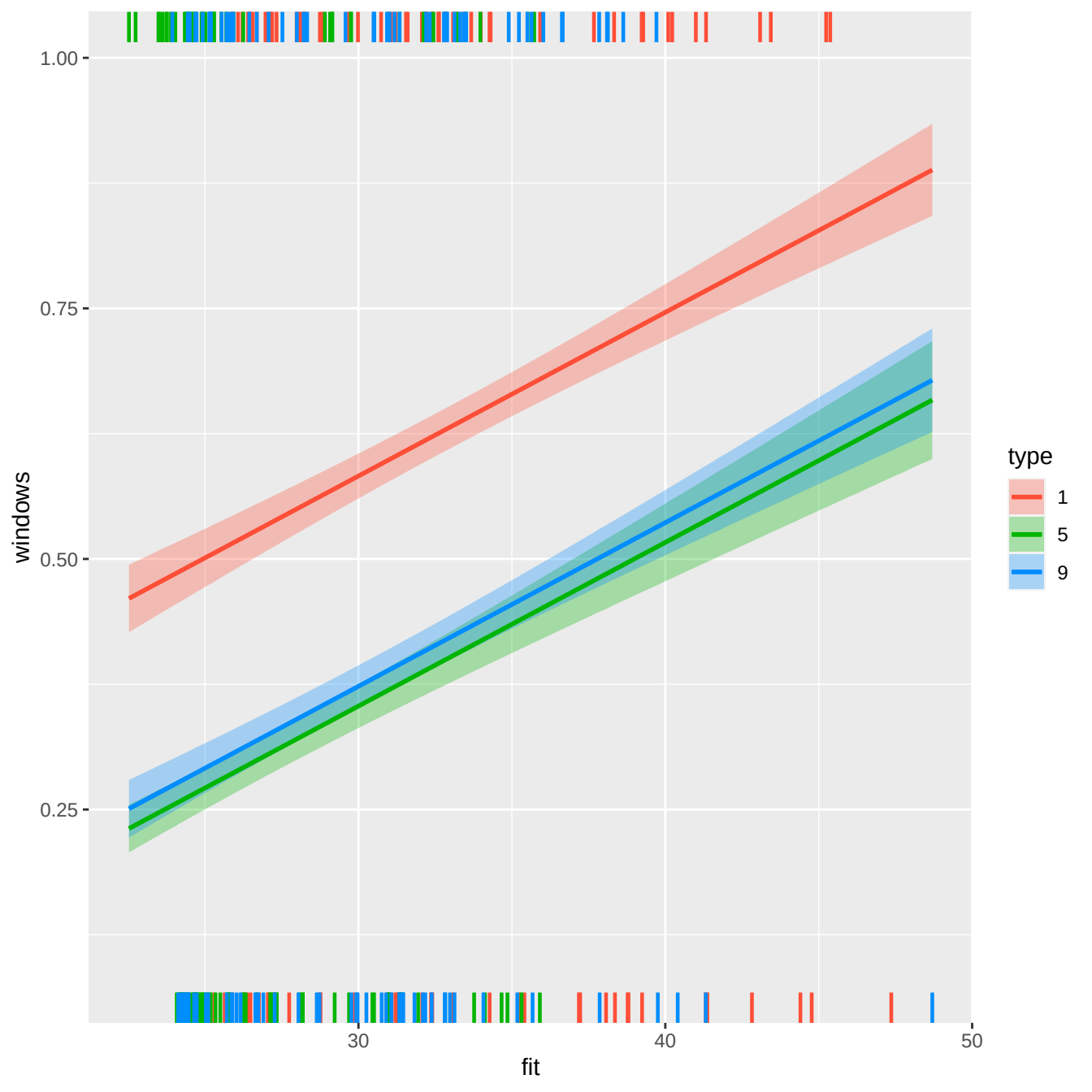

Dff_Differentiation_rate_hox.1 1G HOX upepr
1 genome
2 genomes
2 genomes high mutation
Parameter range (normalized 0 to 1)
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

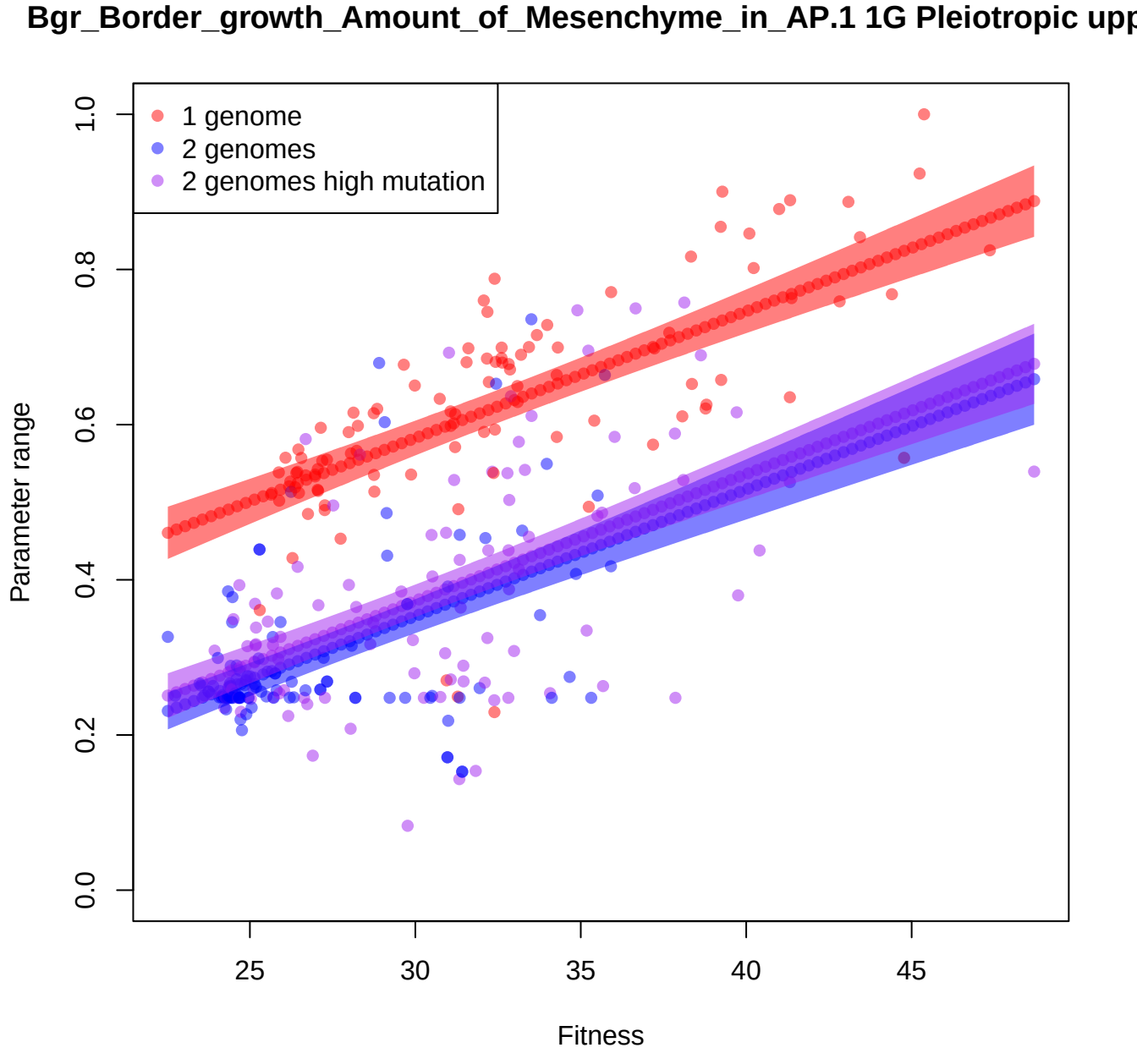

Dff_Differentiation_rate_hox.1
1G HOX upepr
Counts
1.2
1.0
0.8
0.6
0.00124900575643306
0.168631091014946
1.16698795073065e−06
11 − 3
11 − 7
3 − 7
1 genome
2 genomes
2 genomes high mutation

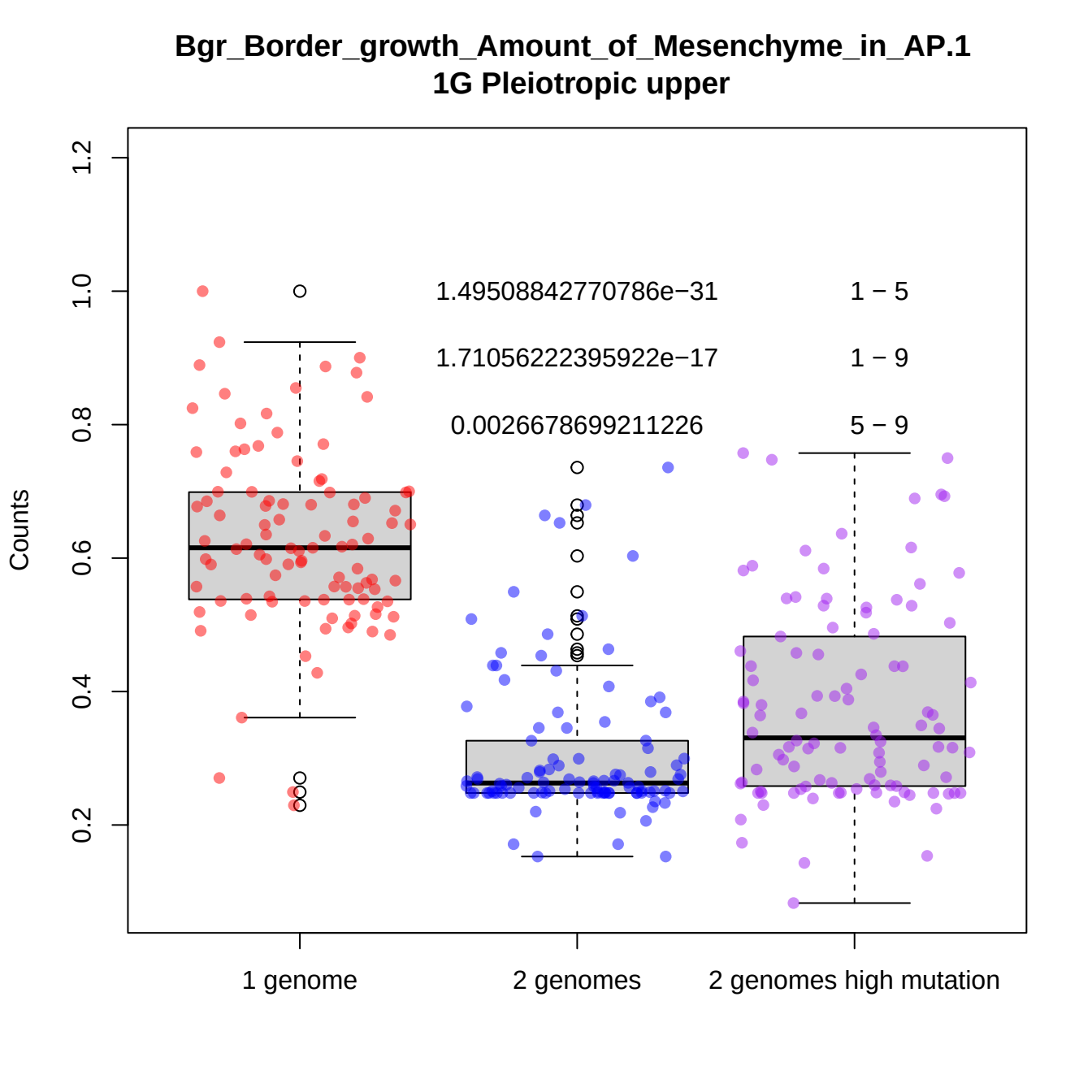

1.0
0.9
0.8
0.7
0.6
0.5
windows
30
40
50
fit
type
4
8
12

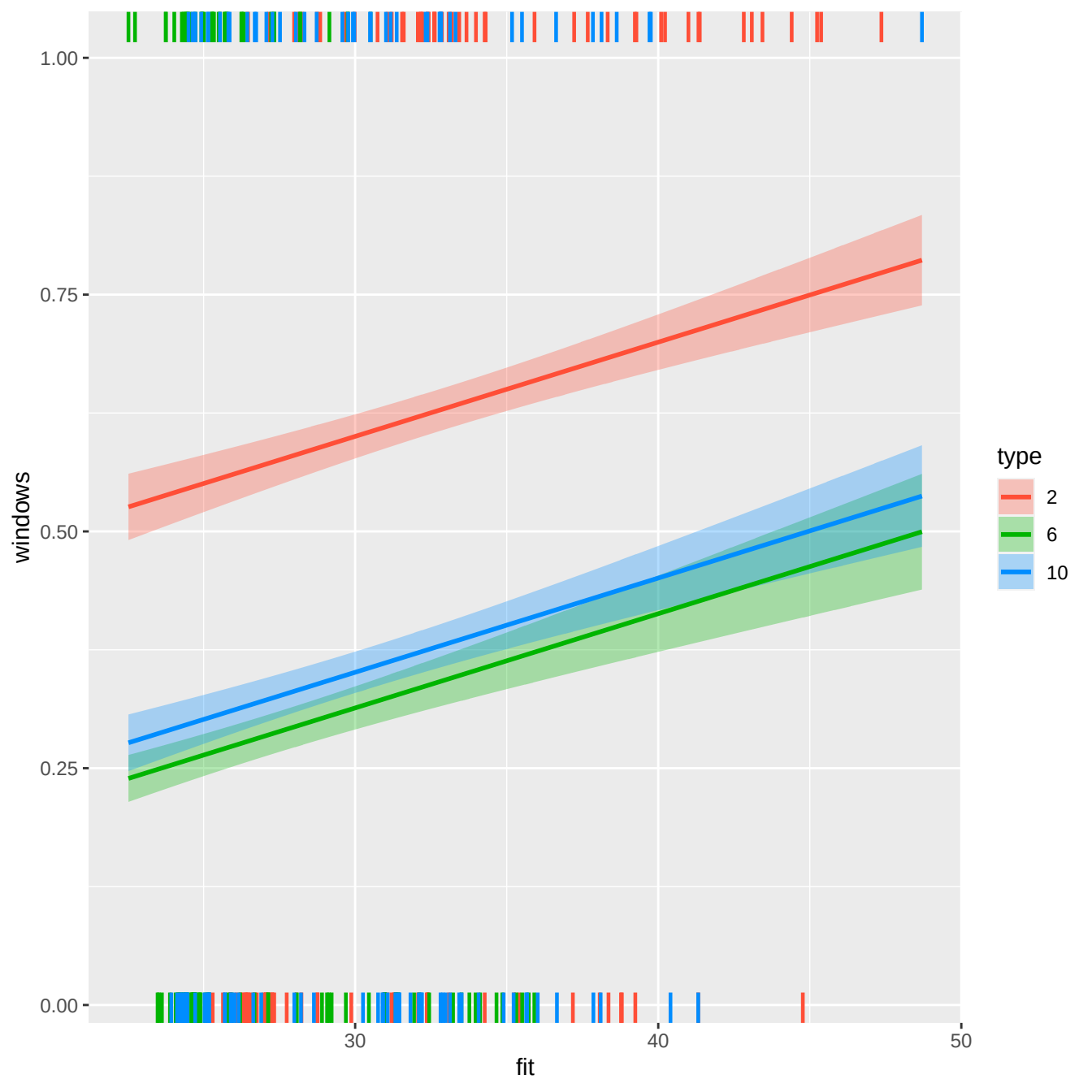

Dff_Differentiation_rate_hox.1 1G HOX lower
1 genome
2 genomes
2 genomes high mutation
Parameter range (normalized 0 to 1)
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

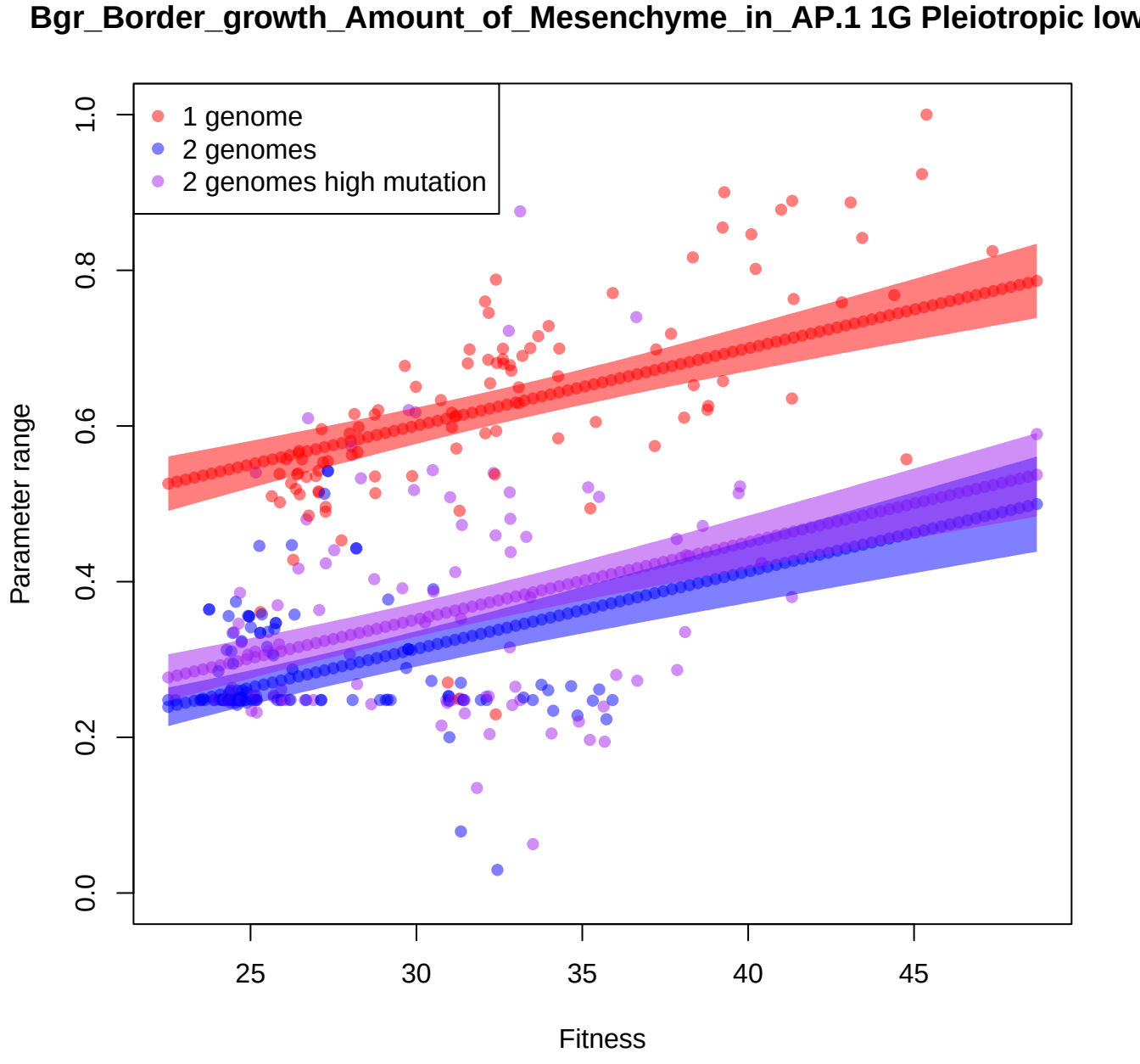

Dff_Differentiation_rate_hox.1
1G HOX lower
Counts
1.2
1.0
0.8
0.6
0.0003502231976536 83
0.259342601477526
6.33847288521562e−07
12 − 4
12 − 8
4 − 8
1 genome
2 genomes
2 genomes high mutation

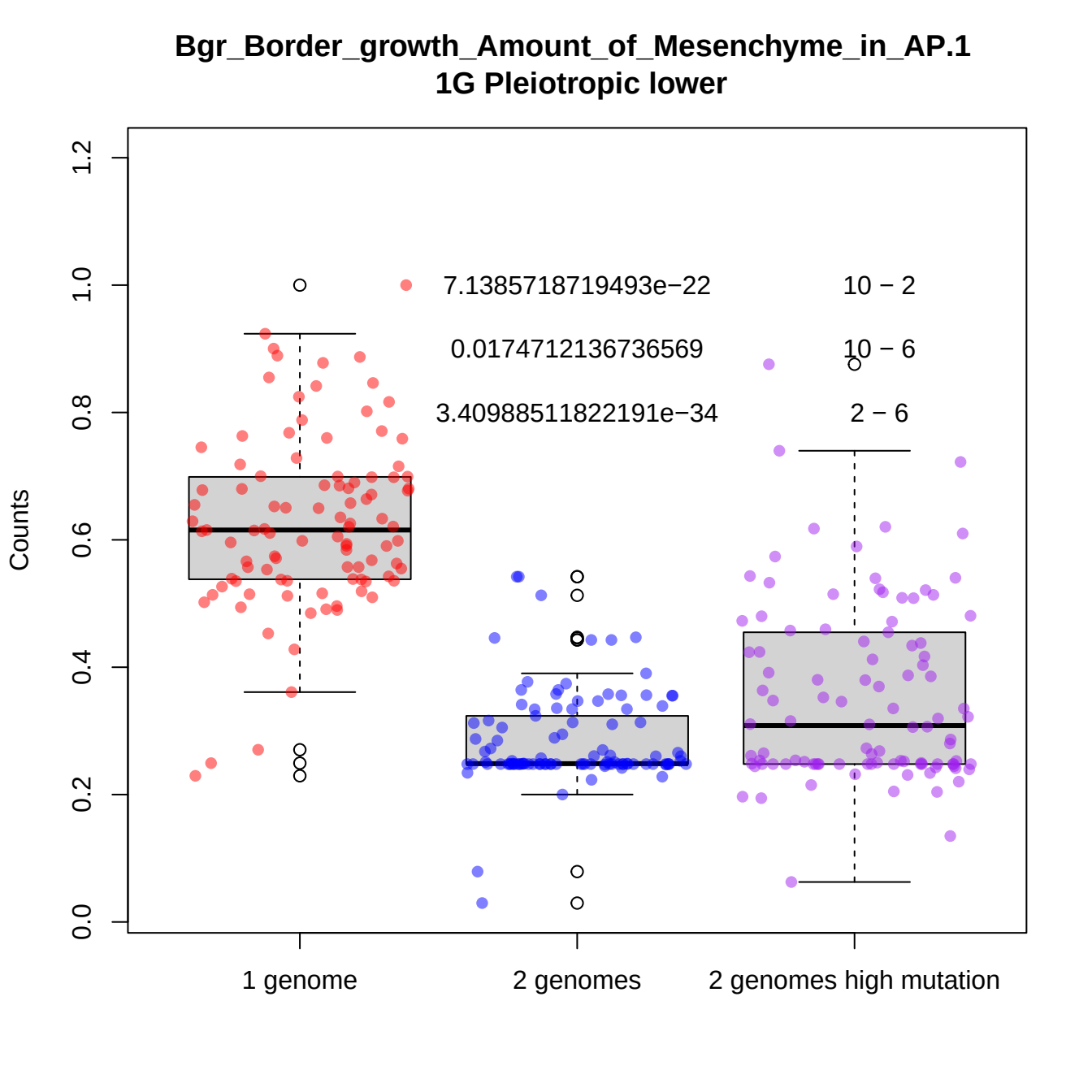

1.00
0.75
0.50
0.25
windows
30
40
50
fit
type
1
5
9

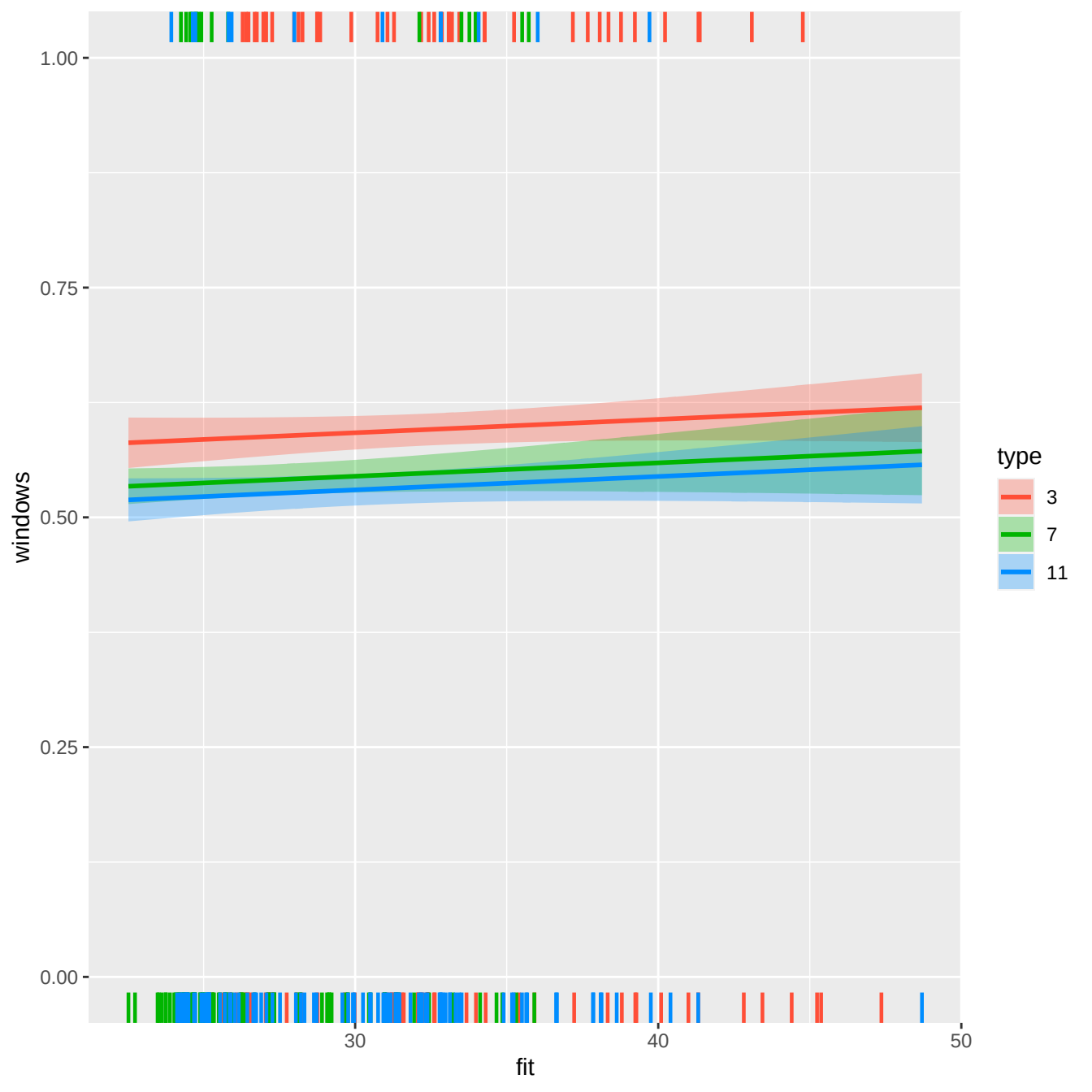

Bgr_Border_growth_Amount_of_Mesenchyme_in_AP.1 1G Pleiotropic upp
1 genome
2 genomes
2 genomes high mutation
Parameter range
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

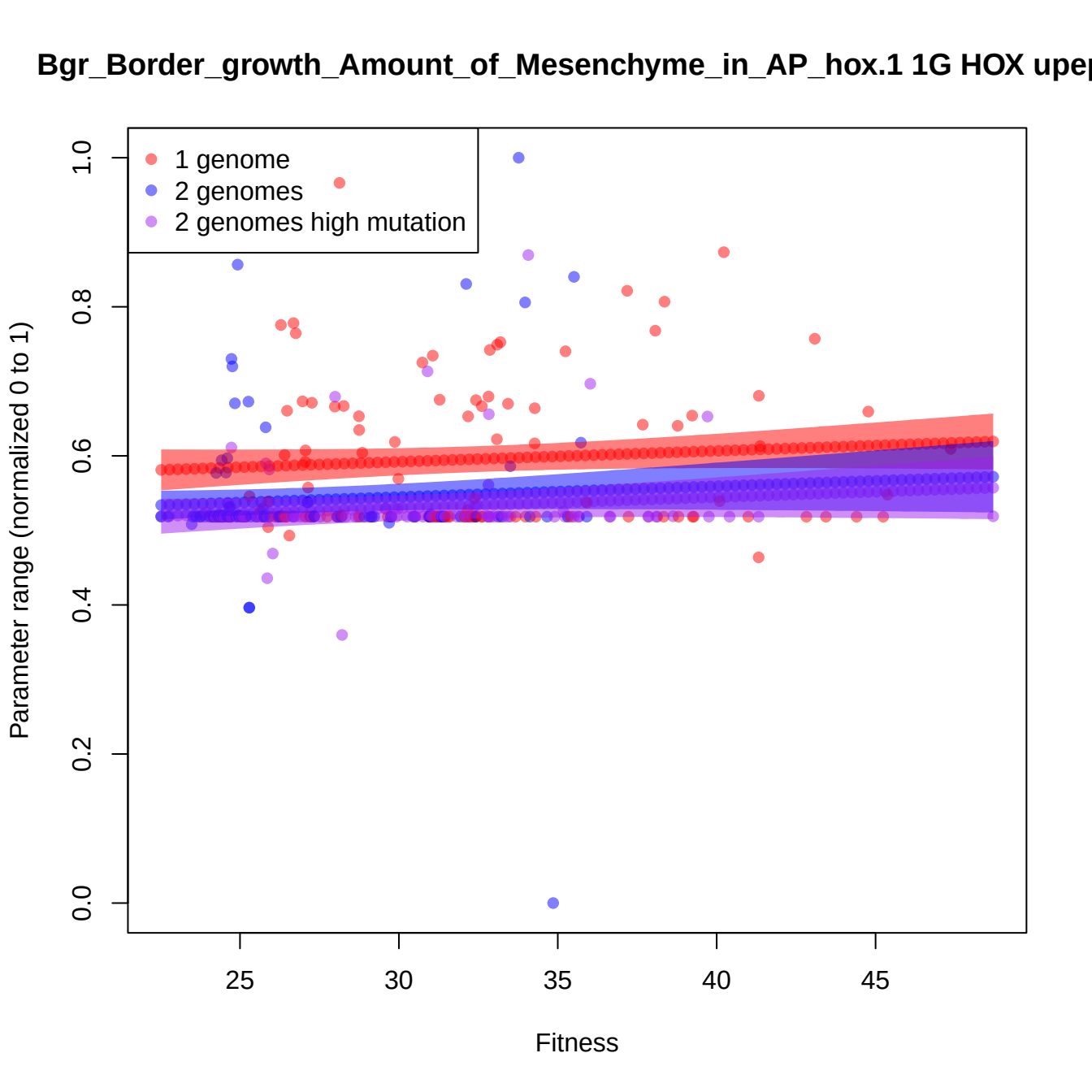

Bgr_Border_growth_Amount_of_Mesenchyme_in_AP.1
1G Pleiotropic upper
Counts
0.2
0.4
0.6
0.8
1.0
1.2
1.49508842770786e−31
1 − 5
1.71056222395922e−17
1 − 9
0.0026678699211226
5 − 9
1 genome
2 genomes
2 genomes high mutation

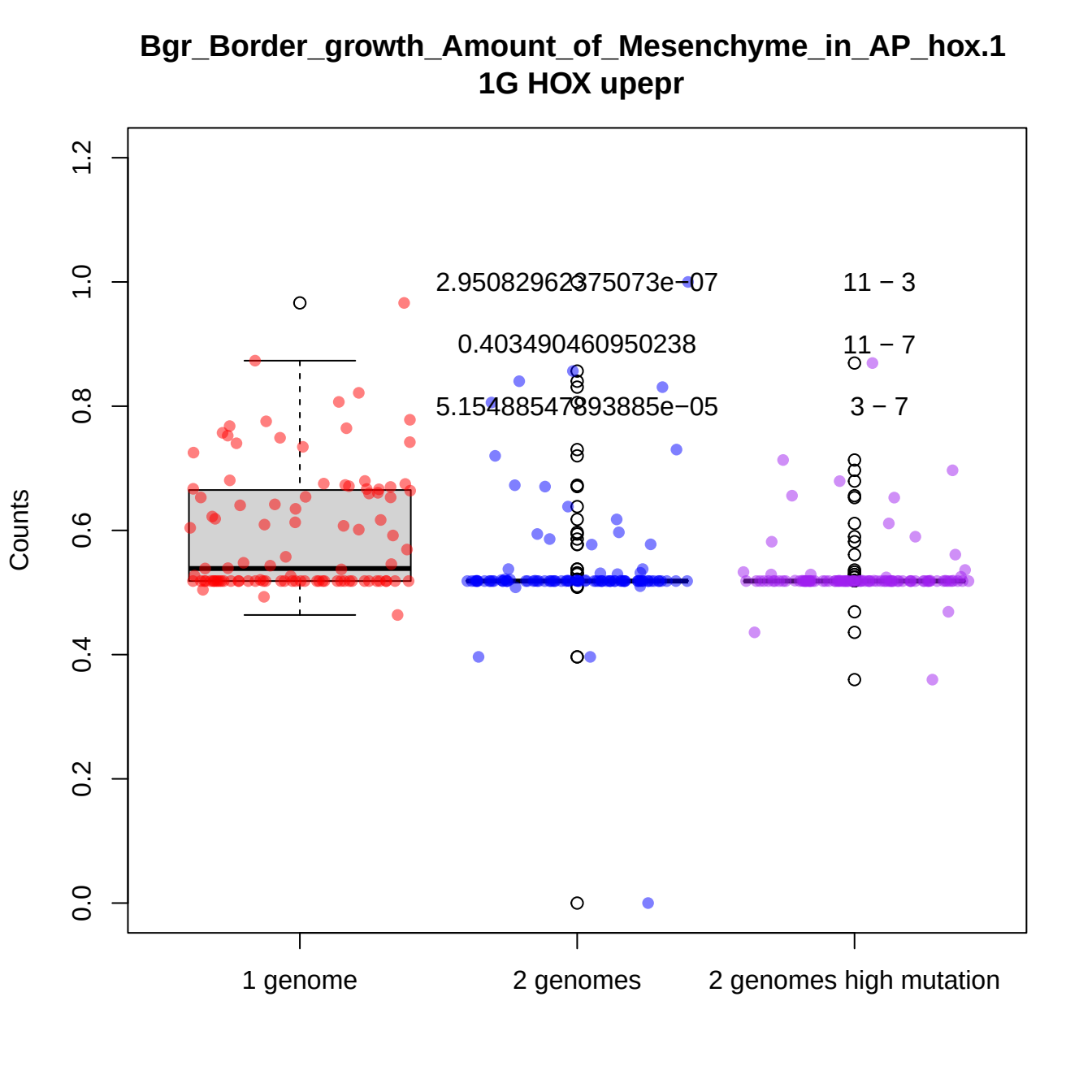

1.00
0.75
0.50
0.25
0.00
windows
30
40
50
fit
type
2
6
10

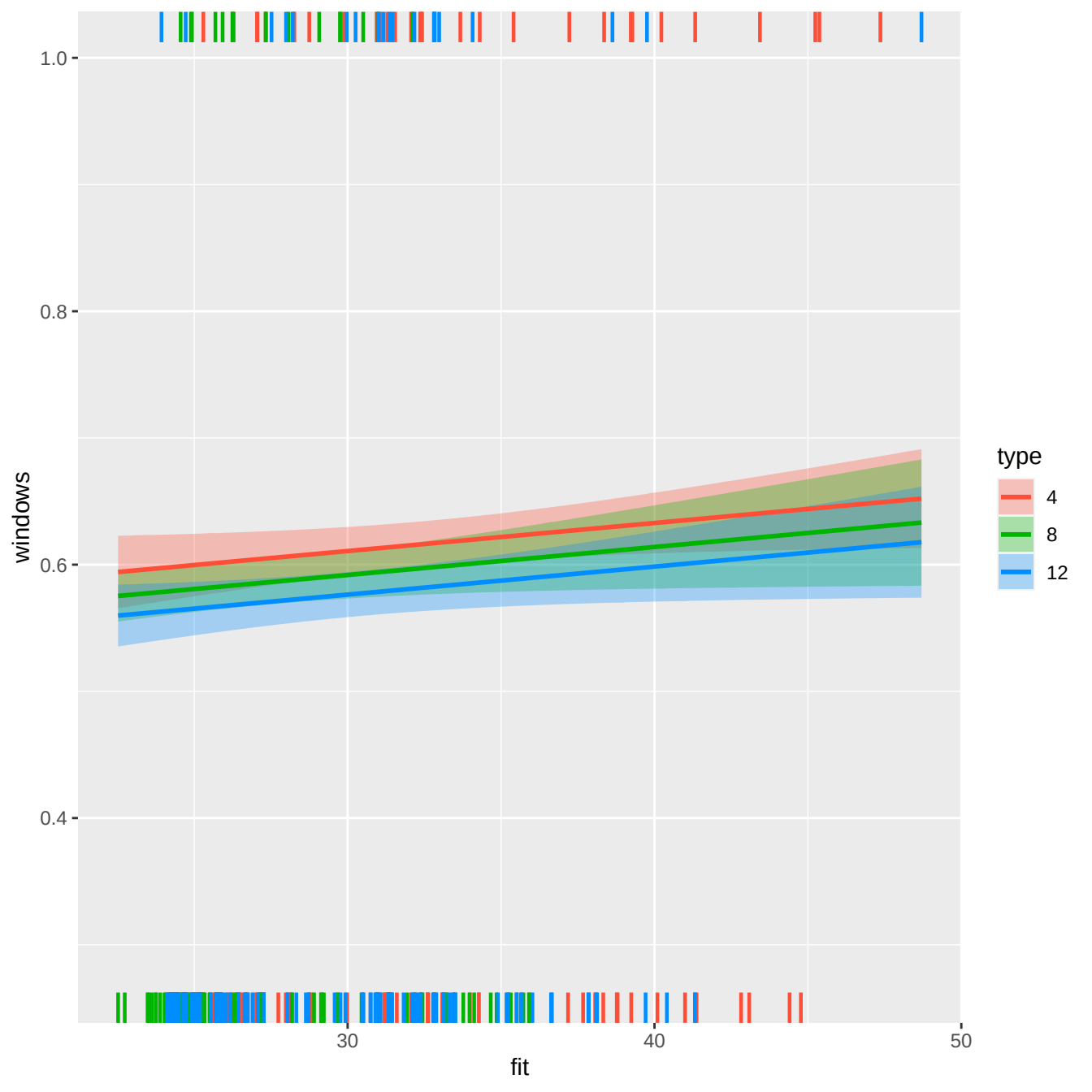

Bgr_Border_growth_Amount_of_Mesenchyme_in_AP.1 1G Pleiotropic low
1 genome
2 genomes
2 genomes high mutation
Parameter range
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

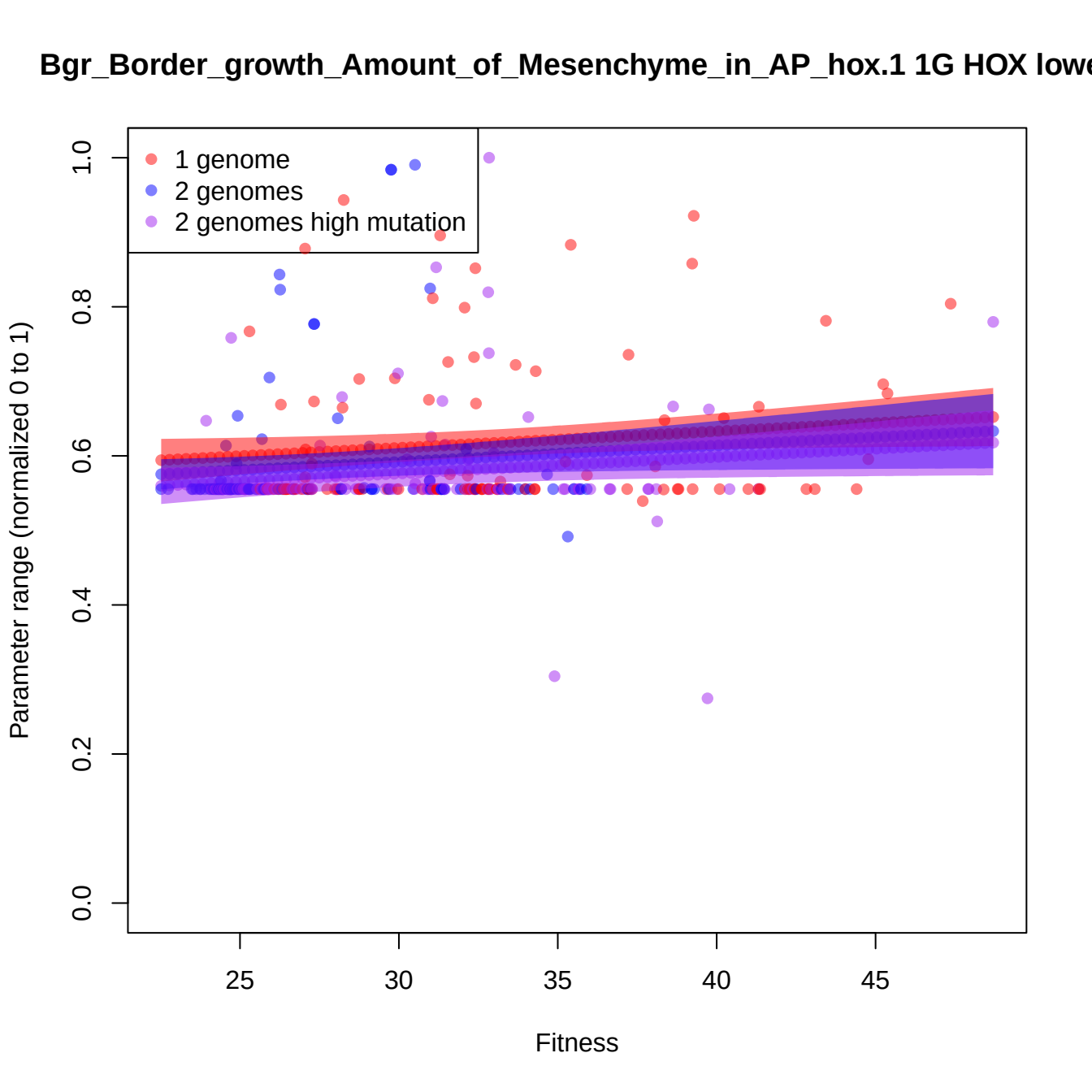

Bgr_Border_growth_Amount_of_Mesenchyme_in_AP.1
1G Pleiotropic lower
Counts
0.0
0.2
0.4
0.6
0.8
1.0
1.2
7.1385718719493e−22
0.0174712136736569
3.40988511822191e−34
10 − 2
10 − 6
2 − 6
1 genome
2 genomes
2 genomes high mutation

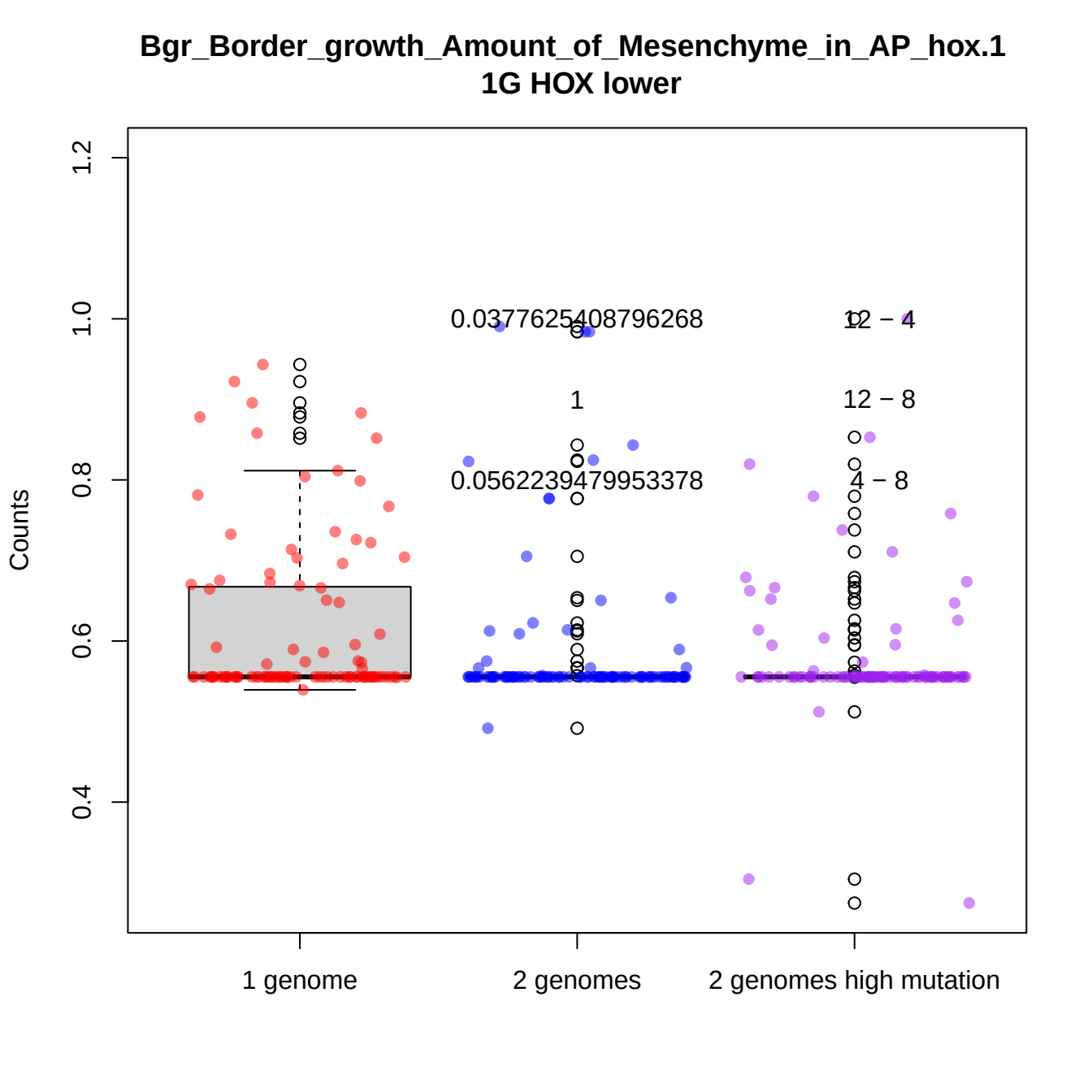

1.00
0.75
0.50
0.25
0.00
windows
30
40
50
fit
type
3
7
11

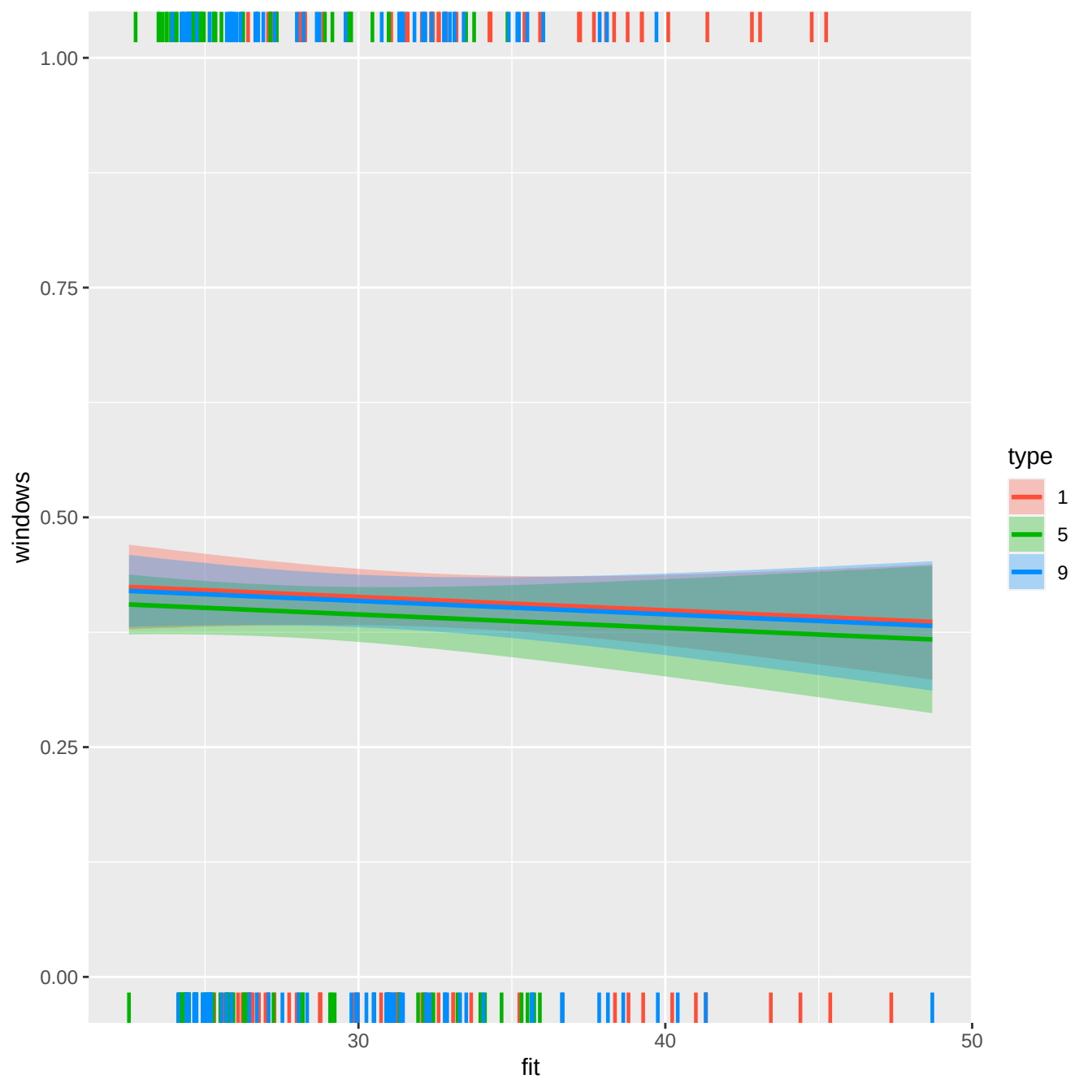

Bgr_Border_growth_Amount_of_Mesenchyme_in_AP_hox.1 1G HOX upep
1 genome
2 genomes
2 genomes high mutation
Parameter range (normalized 0 to 1)
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

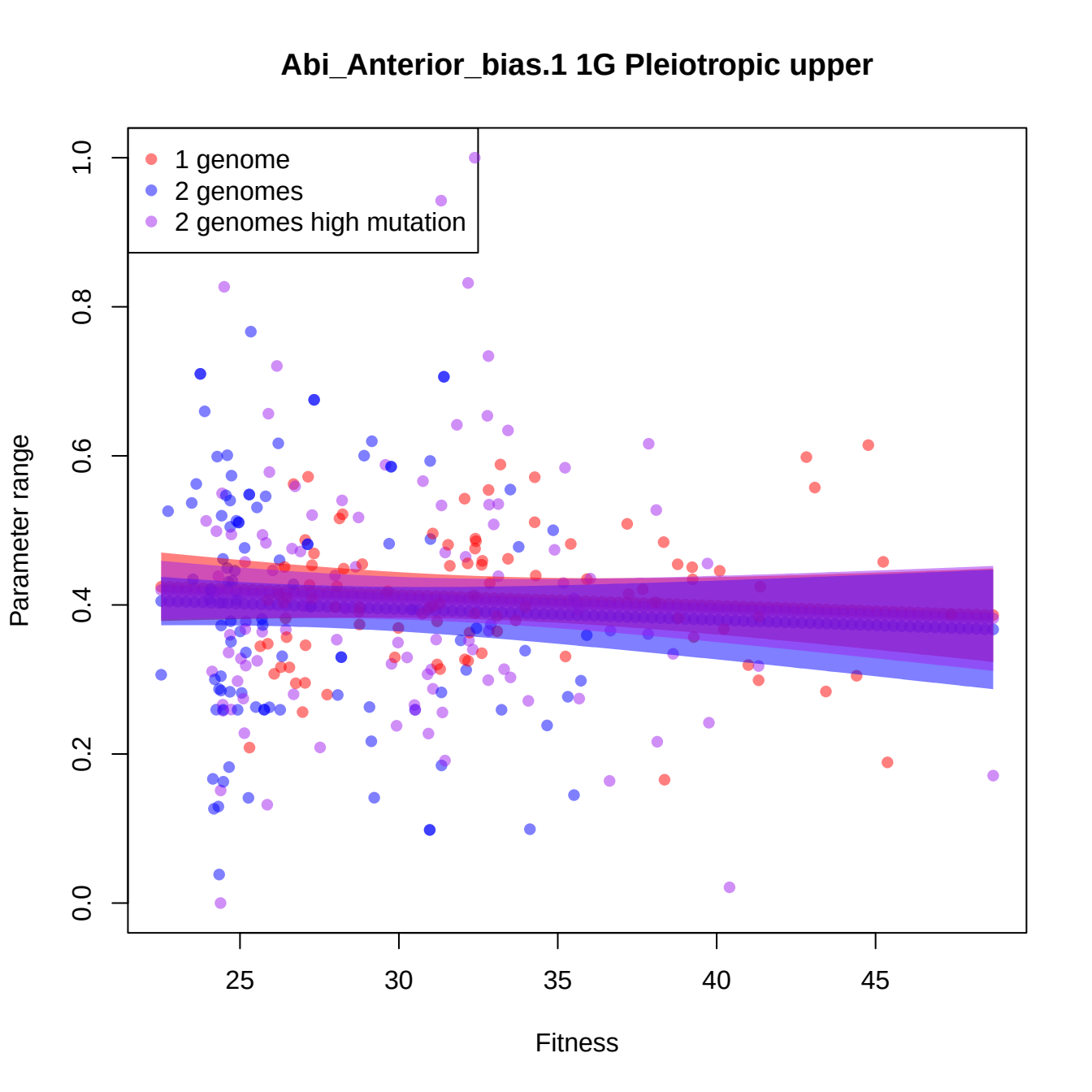

Bgr_Border_growth_Amount_of_Mesenchyme_in_AP_hox.1
1G HOX upepr
Counts
0.0
0.2
0.4
0.6
0.8
1.0
1.2
2.9508296237507 3e−07
0.403490460950238
5.154885478938 85e−05
11 − 3
11 − 7
3 − 7
1 genome
2 genomes
2 genomes high mutation

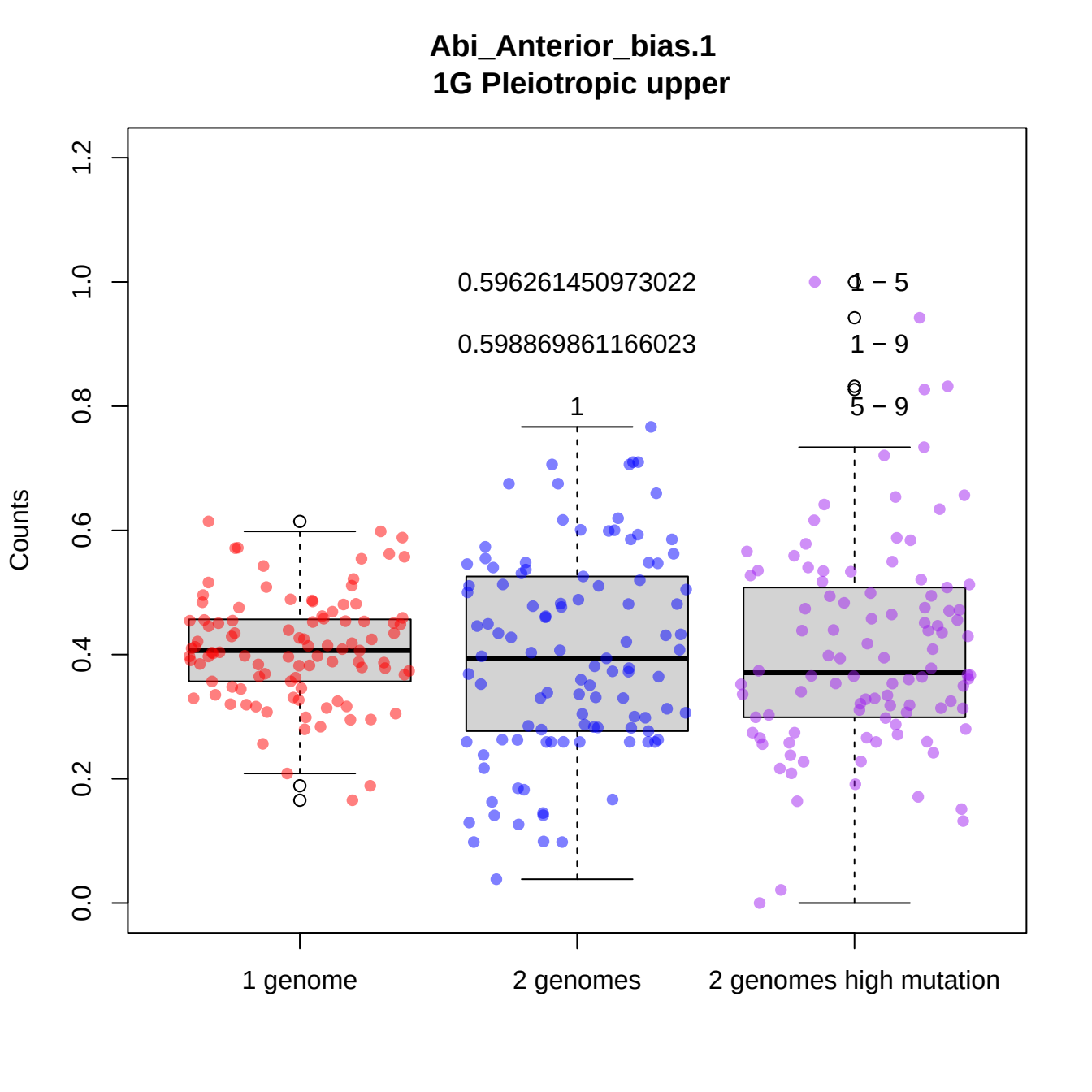

1.0
0.8
0.6
0.4
windows
30
40
50
fit
type
4
8
12

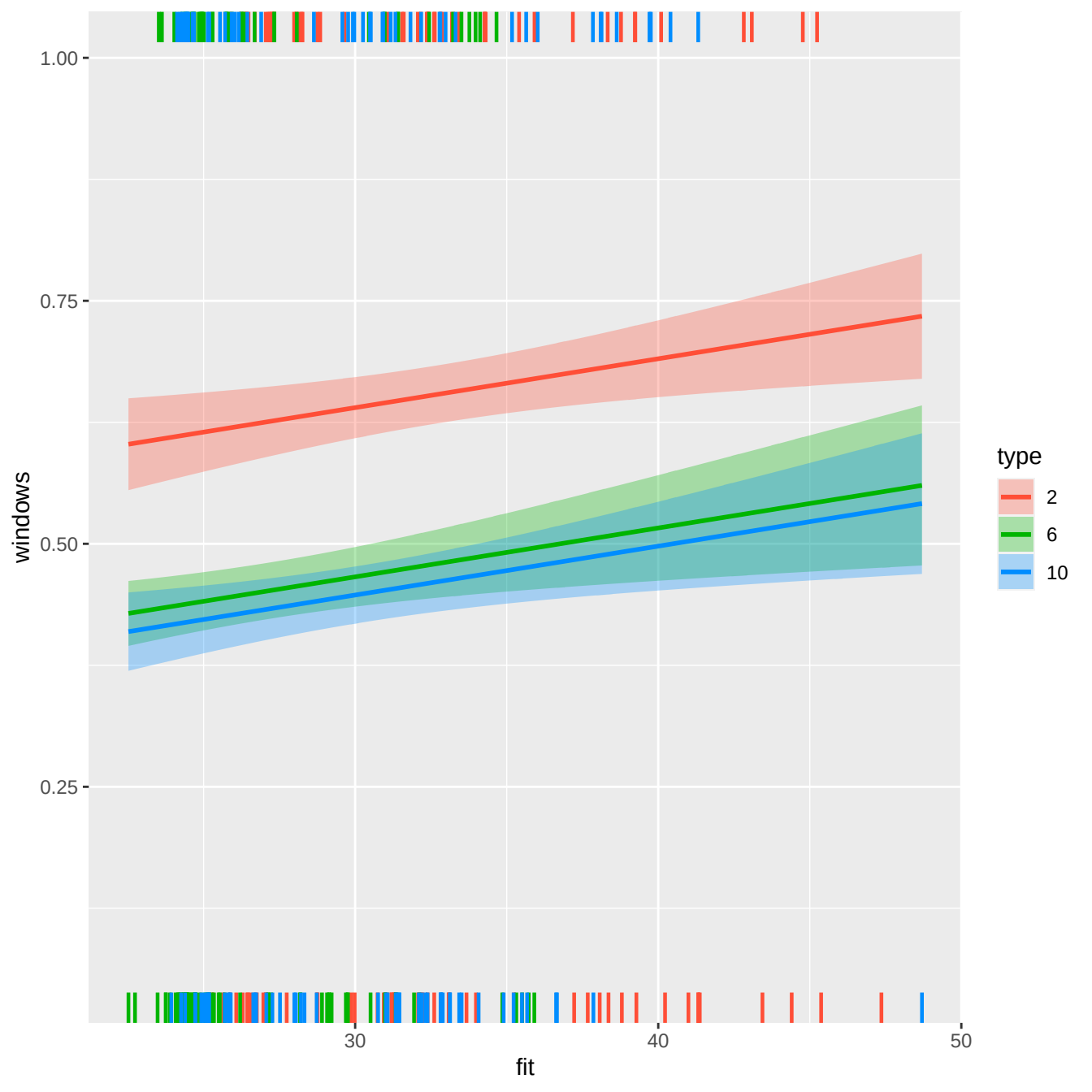

Bgr_Border_growth_Amount_of_Mesenchyme_in_AP_hox.1 1G HOX lowe
1 genome
2 genomes
2 genomes high mutation
Parameter range (normalized 0 to 1)
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

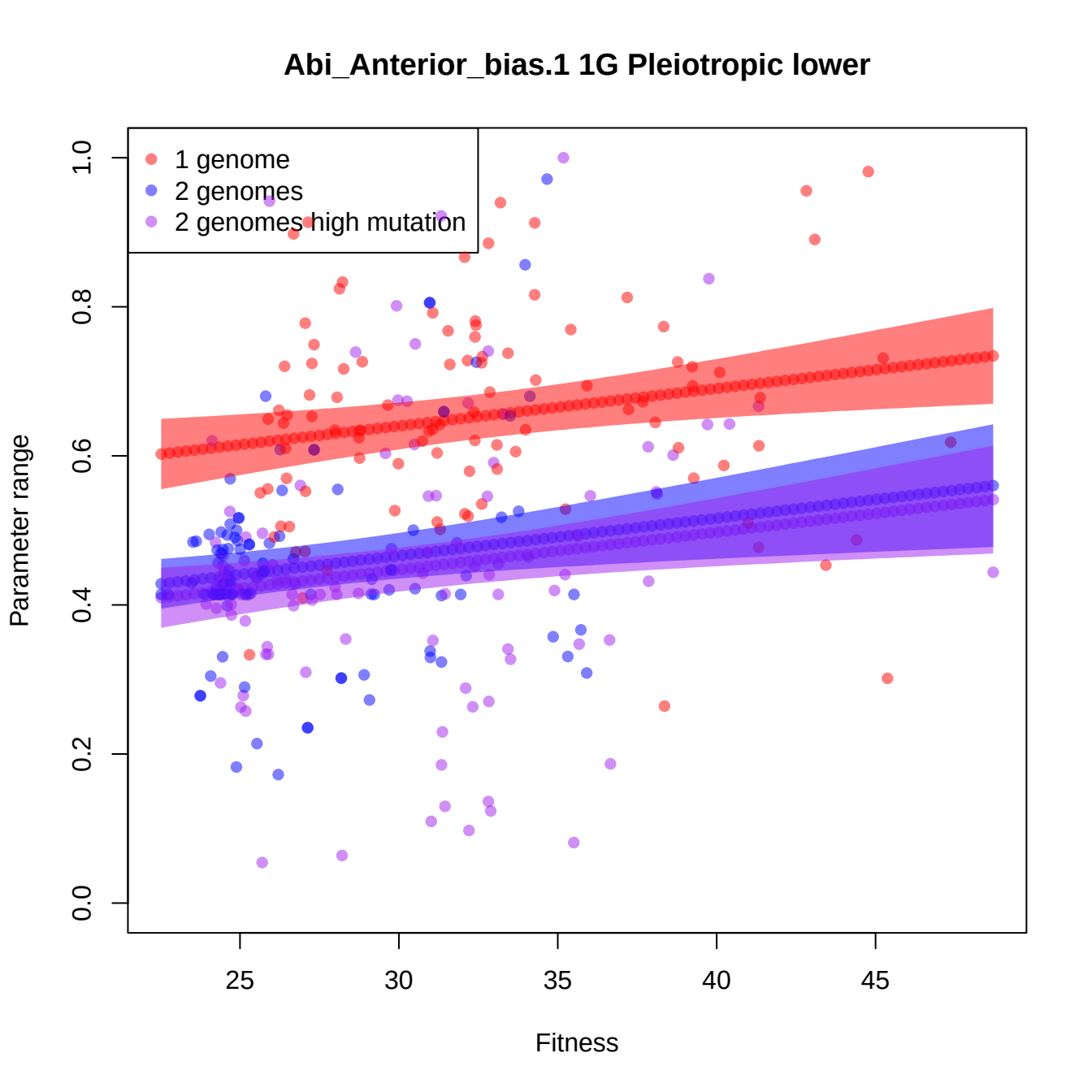

Bgr_Border_growth_Amount_of_Mesenchyme_in_AP_hox.1
1G HOX lower
Counts
1.2
1.0
0.8
0.6
0.4
0.0377625408796268
1
0.0562239479953378
12 − 4
12 − 8
4 − 8
1 genome
2 genomes
2 genomes high mutation

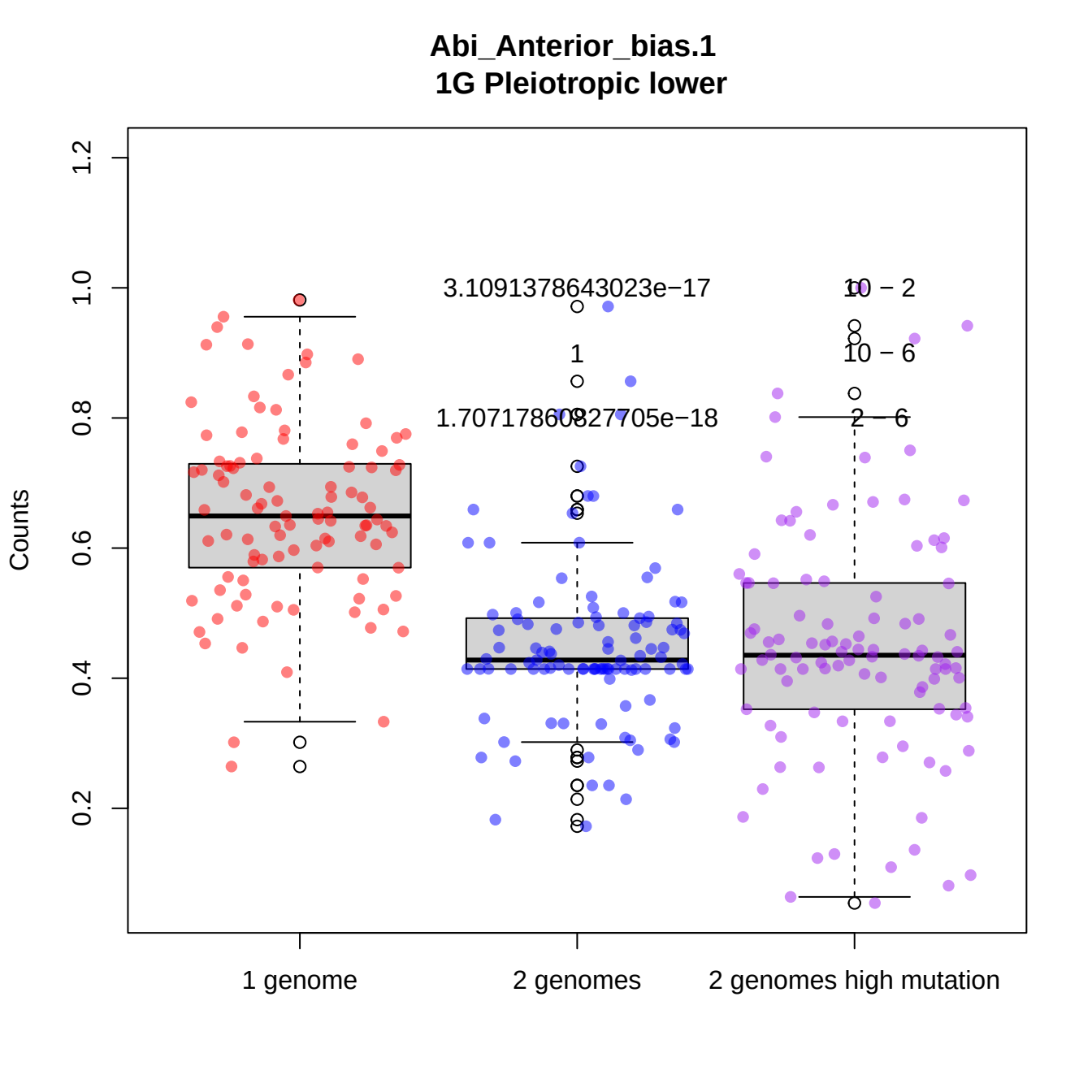

1.00
0.75
0.50
0.25
0.00
windows
30
40
50
fit
type
1
5
9

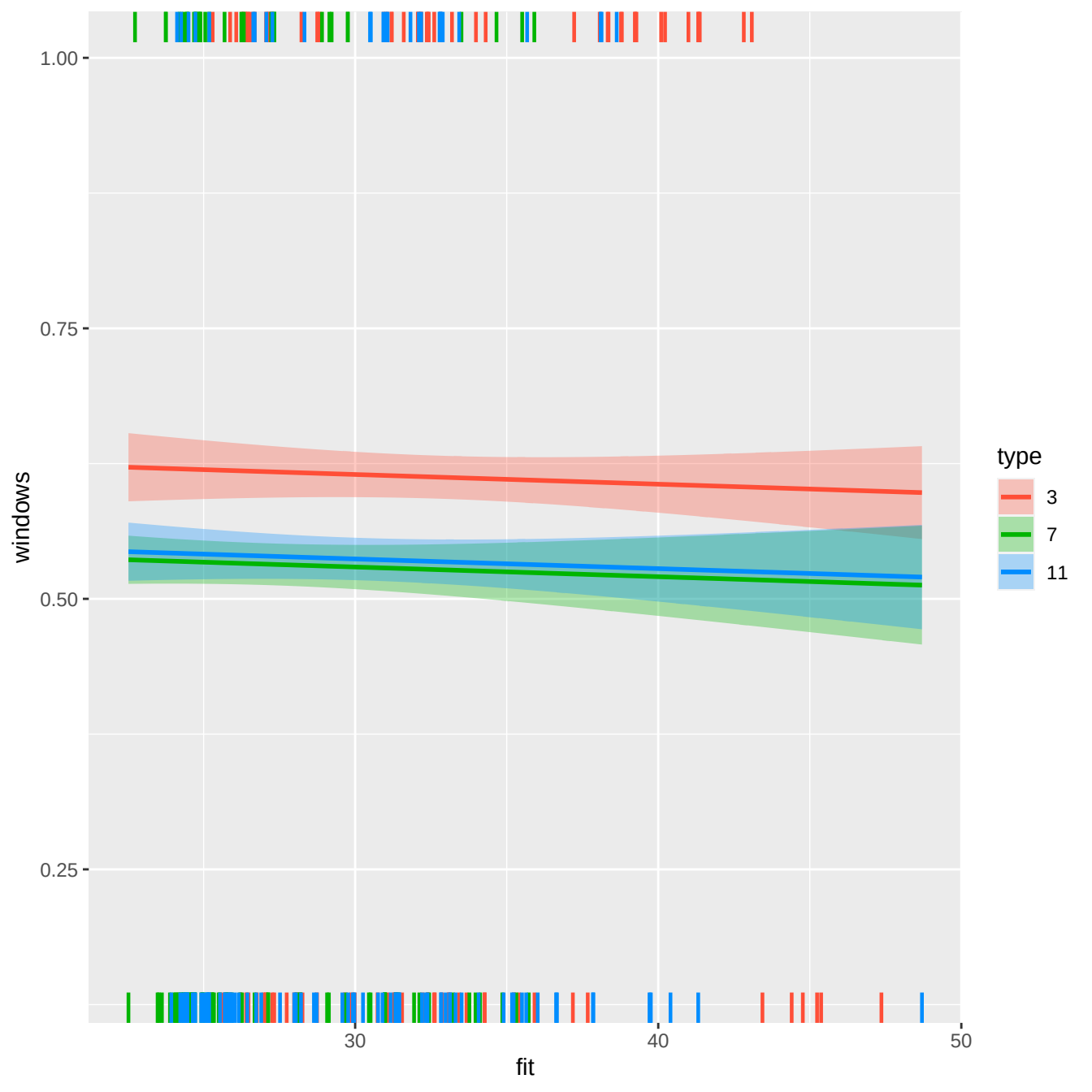

Abi_Anterior_bias.1 1G Pleiotropic upper
1 genome
2 genomes
2 genomes high mutation
Parameter range
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

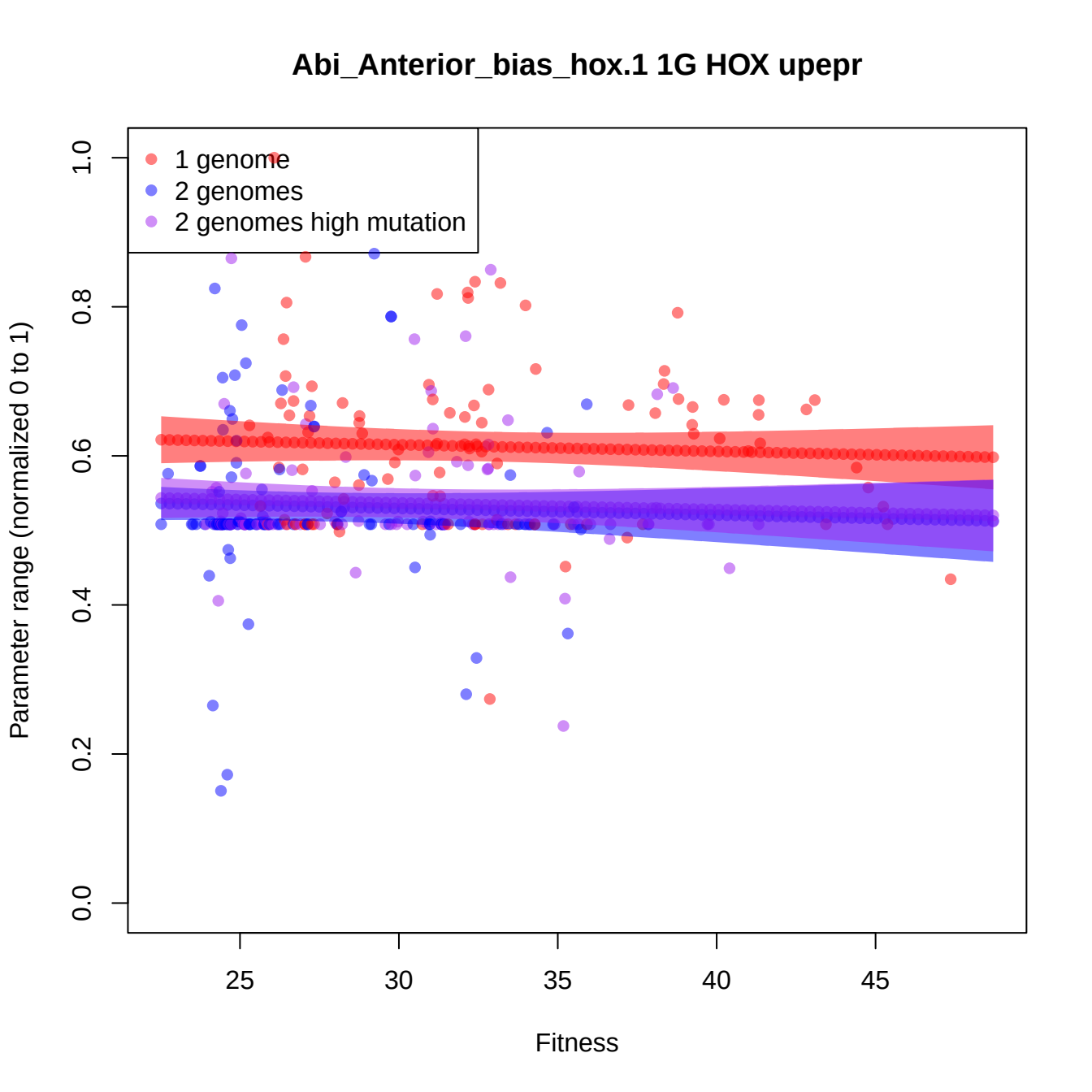

Abi_Anterior_bias.1
1G Pleiotropic upper
Counts
0.0
0.2
0.4
0.6
0.8
1.0
1.2
0.596261450973022
0.598869861166023
1
1 − 5
1 − 9
5 − 9
1 genome
2 genomes
2 genomes high mutation

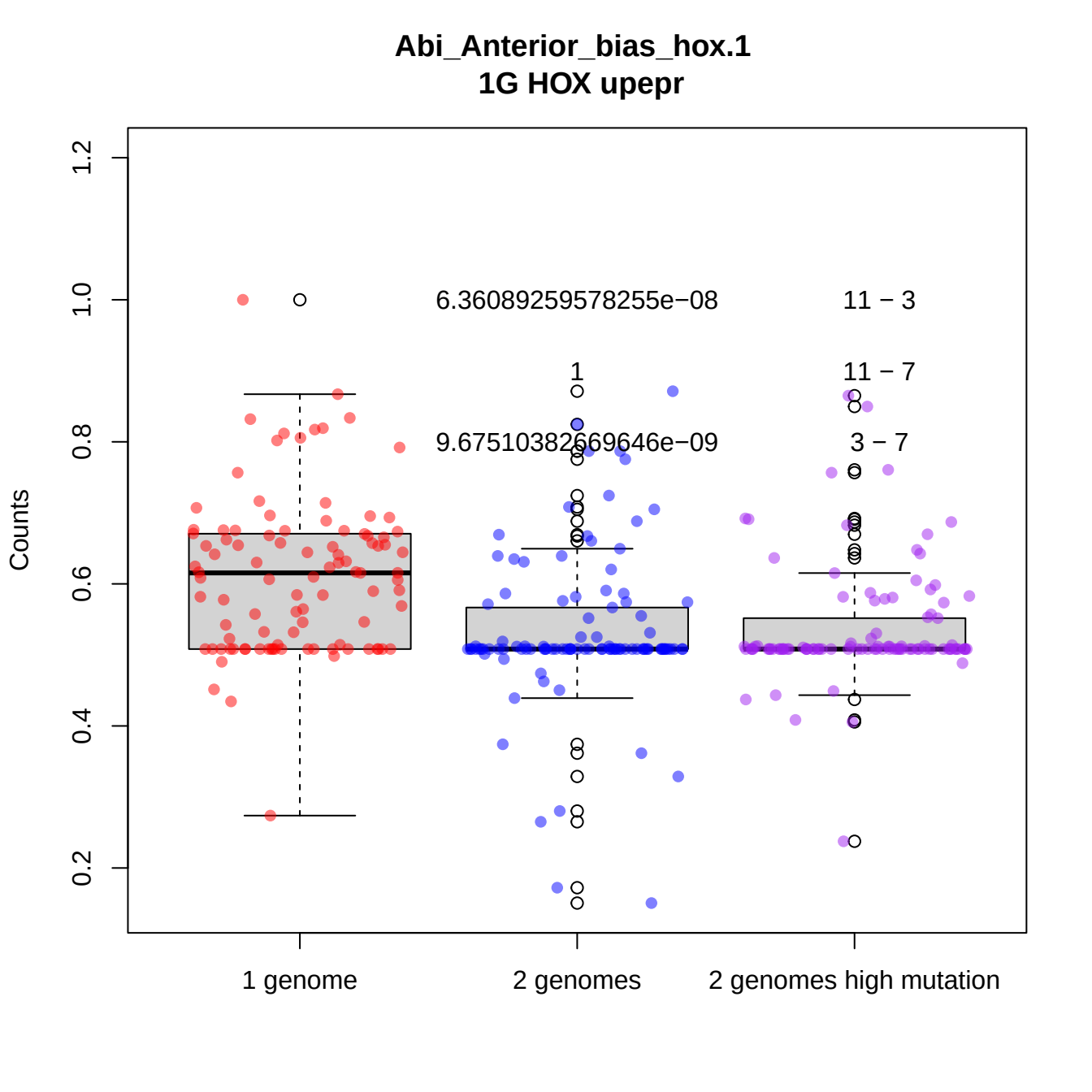

1.00
0.75
0.50
0.25
windows
30
40
50
fit
type
2
6
10

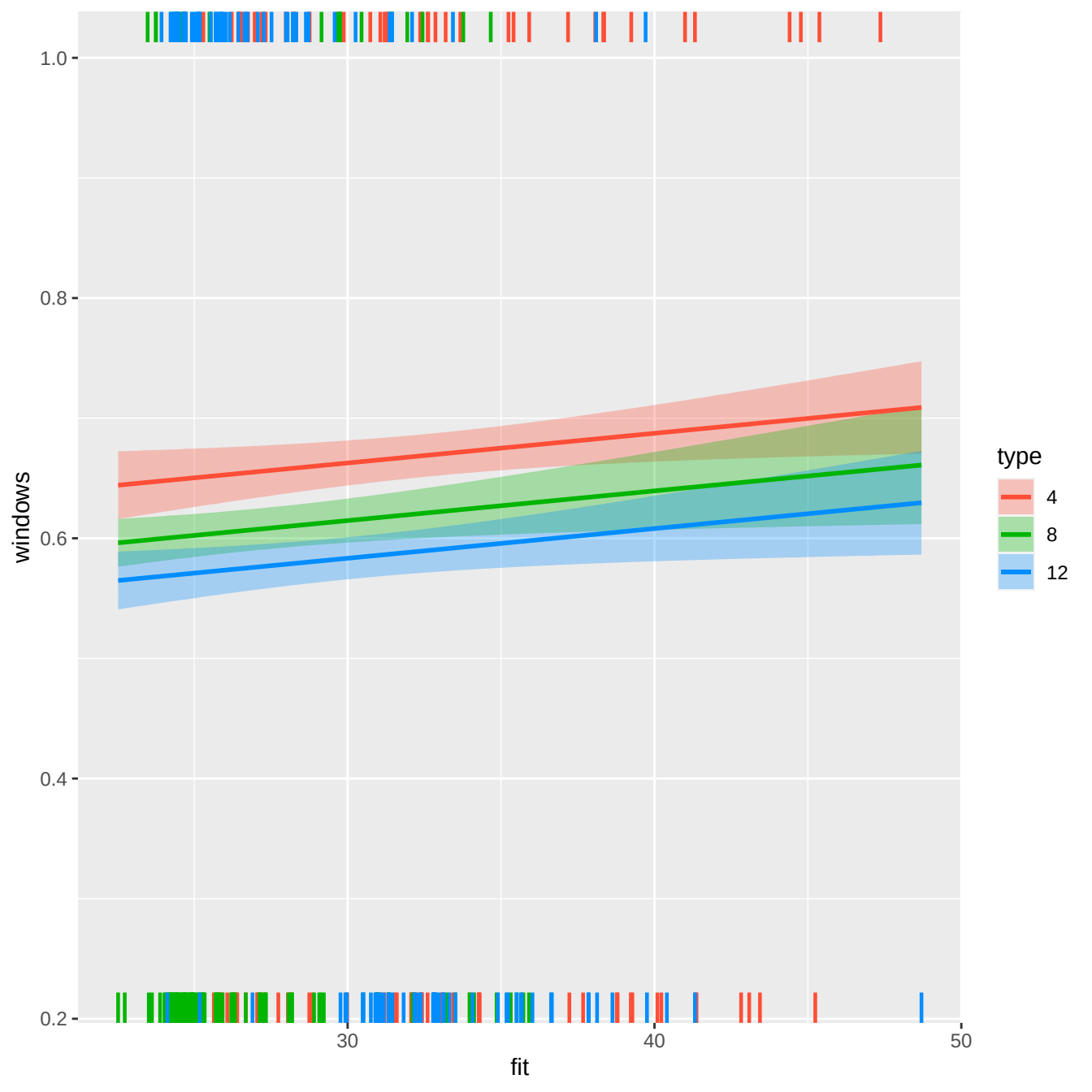

Abi_Anterior_bias.1 1G Pleiotropic lower
1 genome
2 genomes
2 genomes high mutation
Parameter range
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

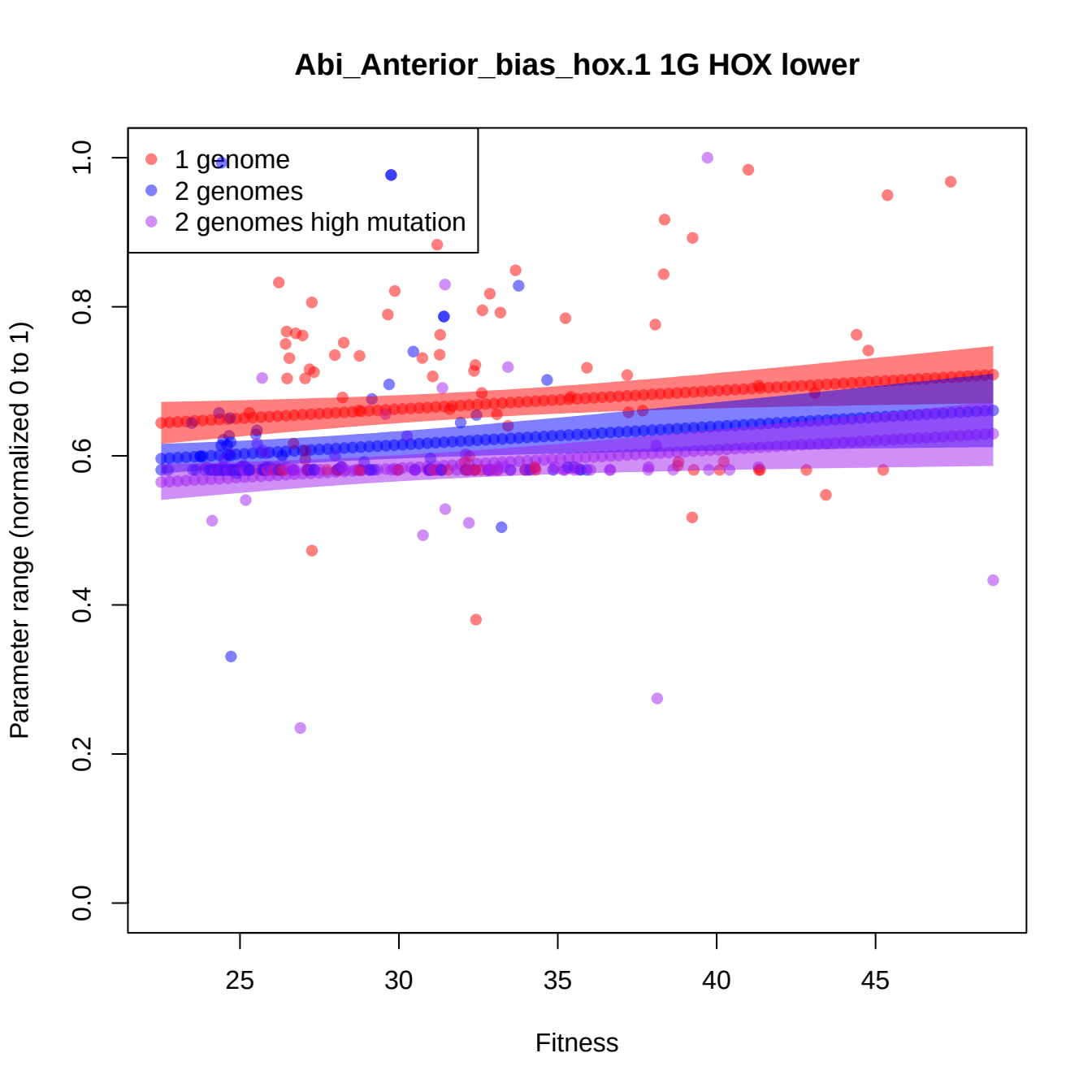

Abi_Anterior_bias.1
1G Pleiotropic lower
Counts
1.2
1.0
0.8
0.6
0.4
0.2
3.1091378643023e−17
1
1.7071786827705e−18
10 − 2
10 − 6
2 − 6
1 genome
2 genomes
2 genomes high mutation

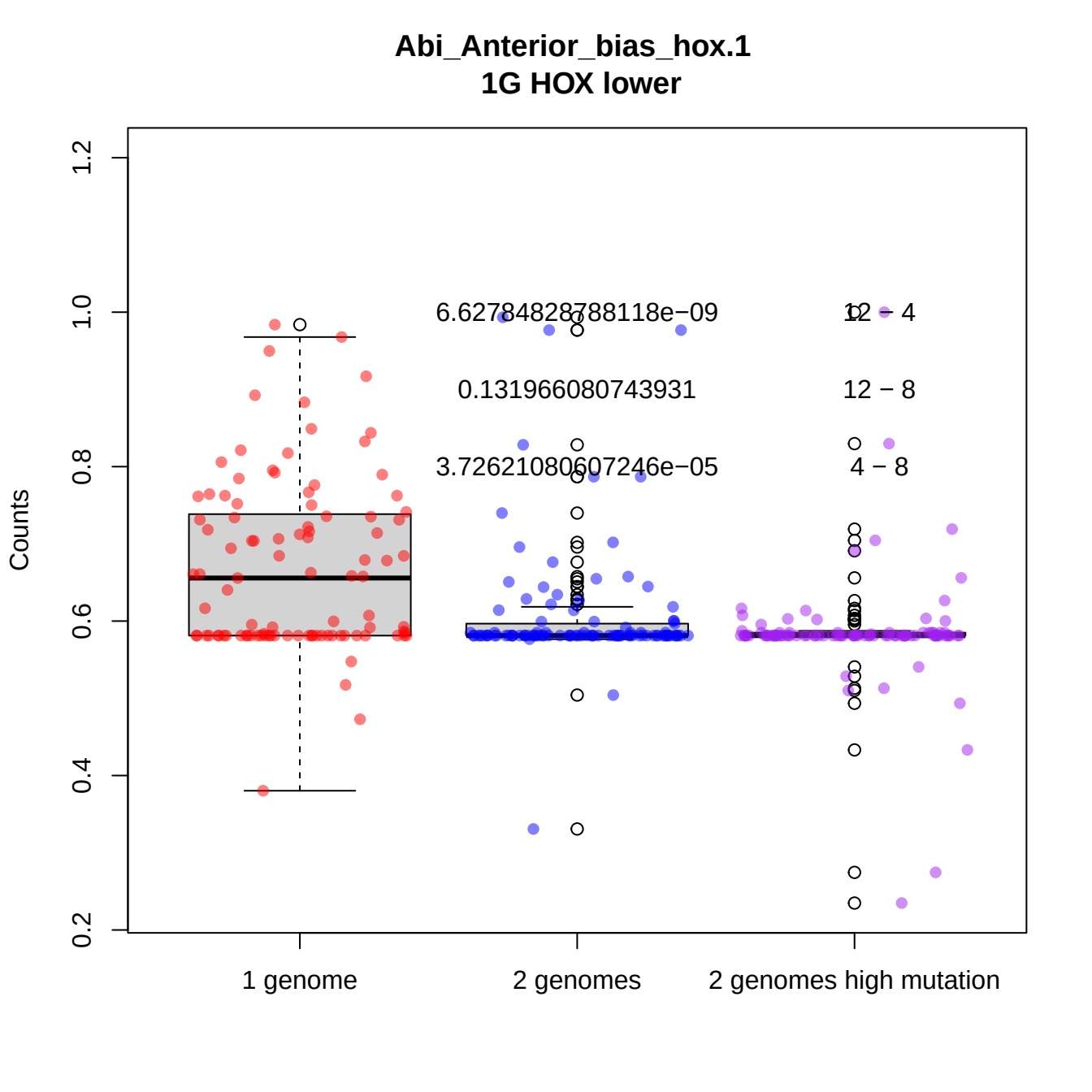

1.00
0.75
0.50
0.25
windows
30
40
50
fit
type
3
7
11

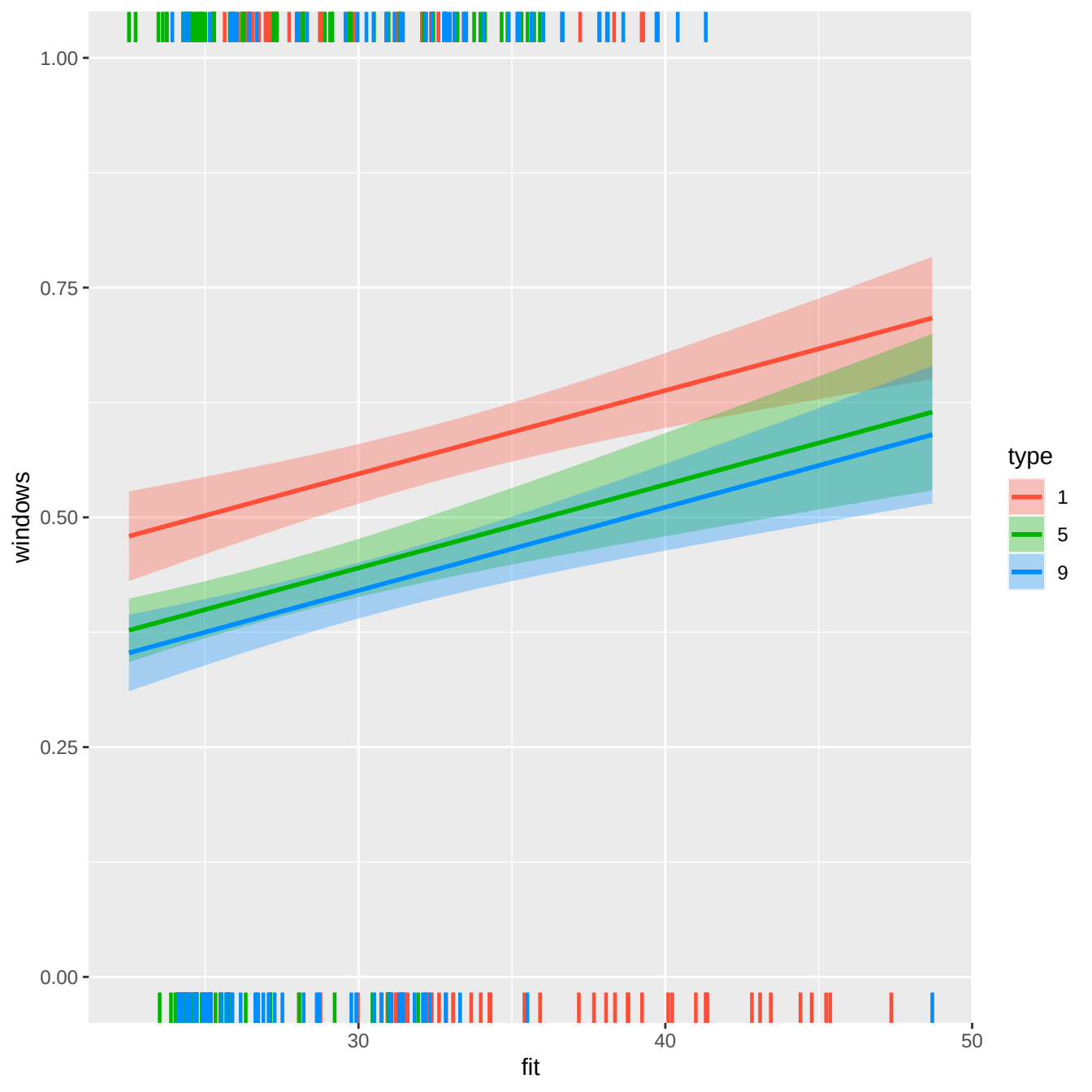

Abi_Anterior_bias_hox.1 1G HOX upepr
1 genome
2 genomes
2 genomes high mutation
Parameter range (normalized 0 to 1)
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

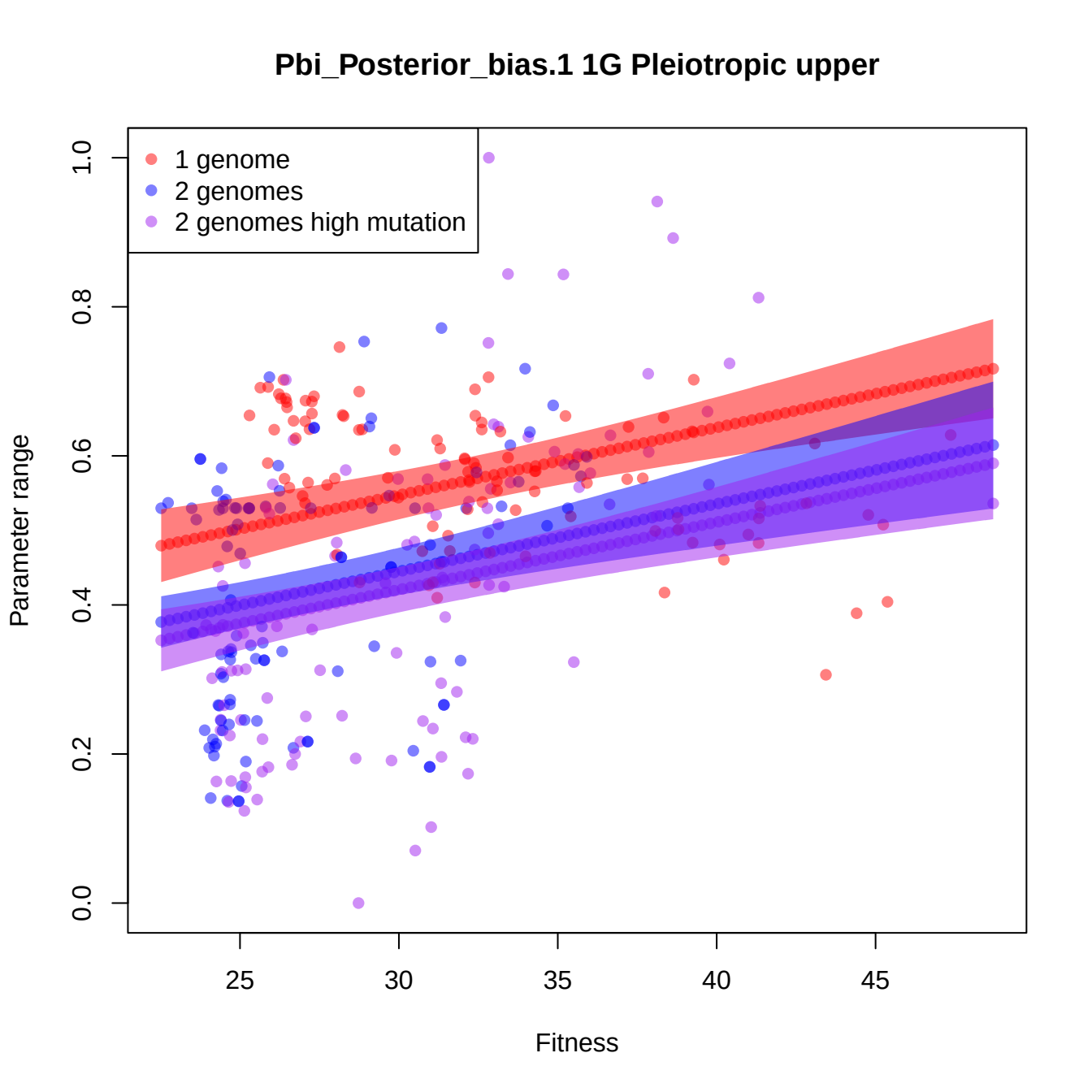

Abi_Anterior_bias_hox.1
1G HOX upepr
Counts
1.2
1.0
0.8
0.6
0.4
0.2
6.36089259578255e−08
1
9.67510382669646e−09
11 − 3
11 − 7
3 − 7
1 genome
2 genomes
2 genomes high mutation

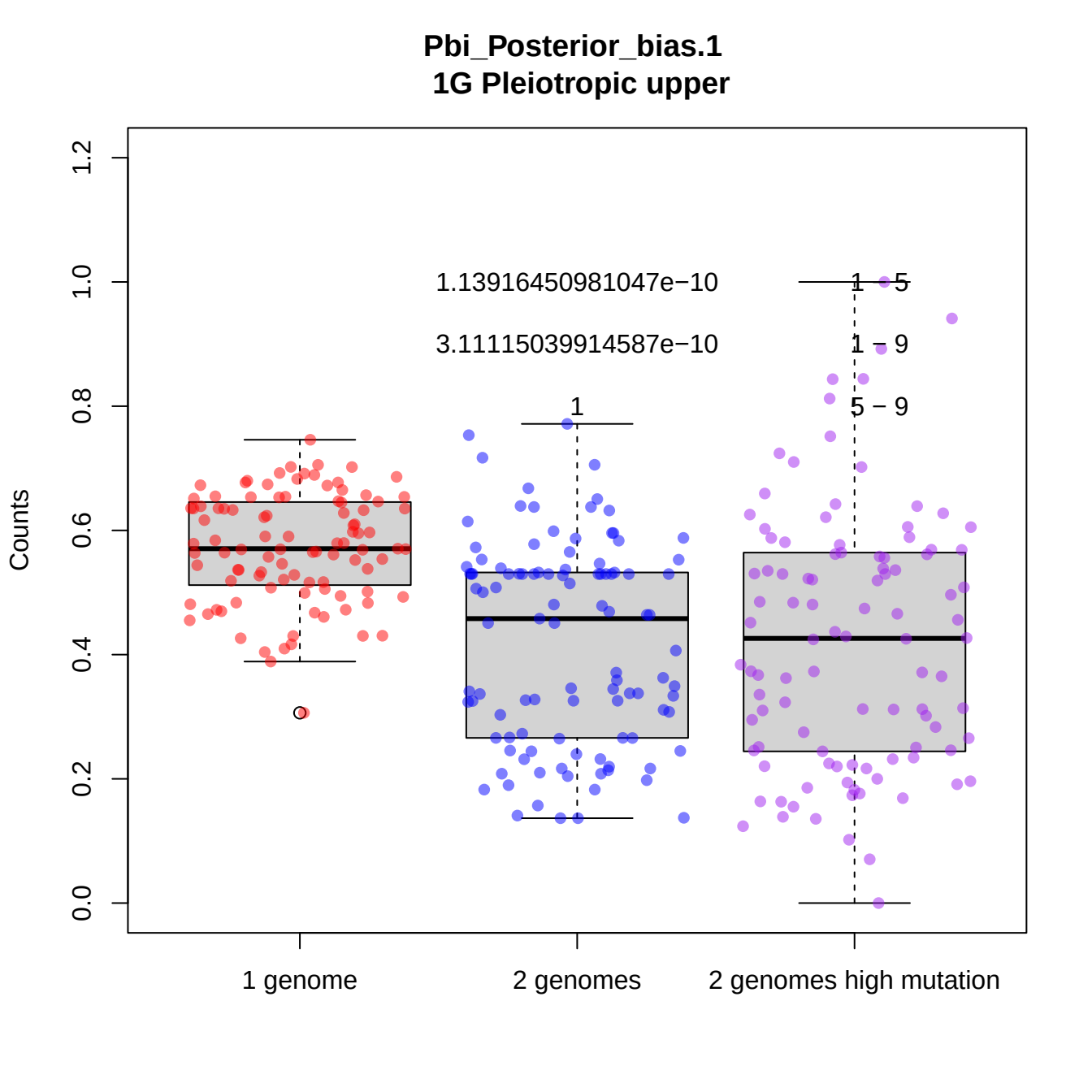

1.0
0.8
0.6
0.4
0.2
windows
30
40
50
fit
type
4
8
12

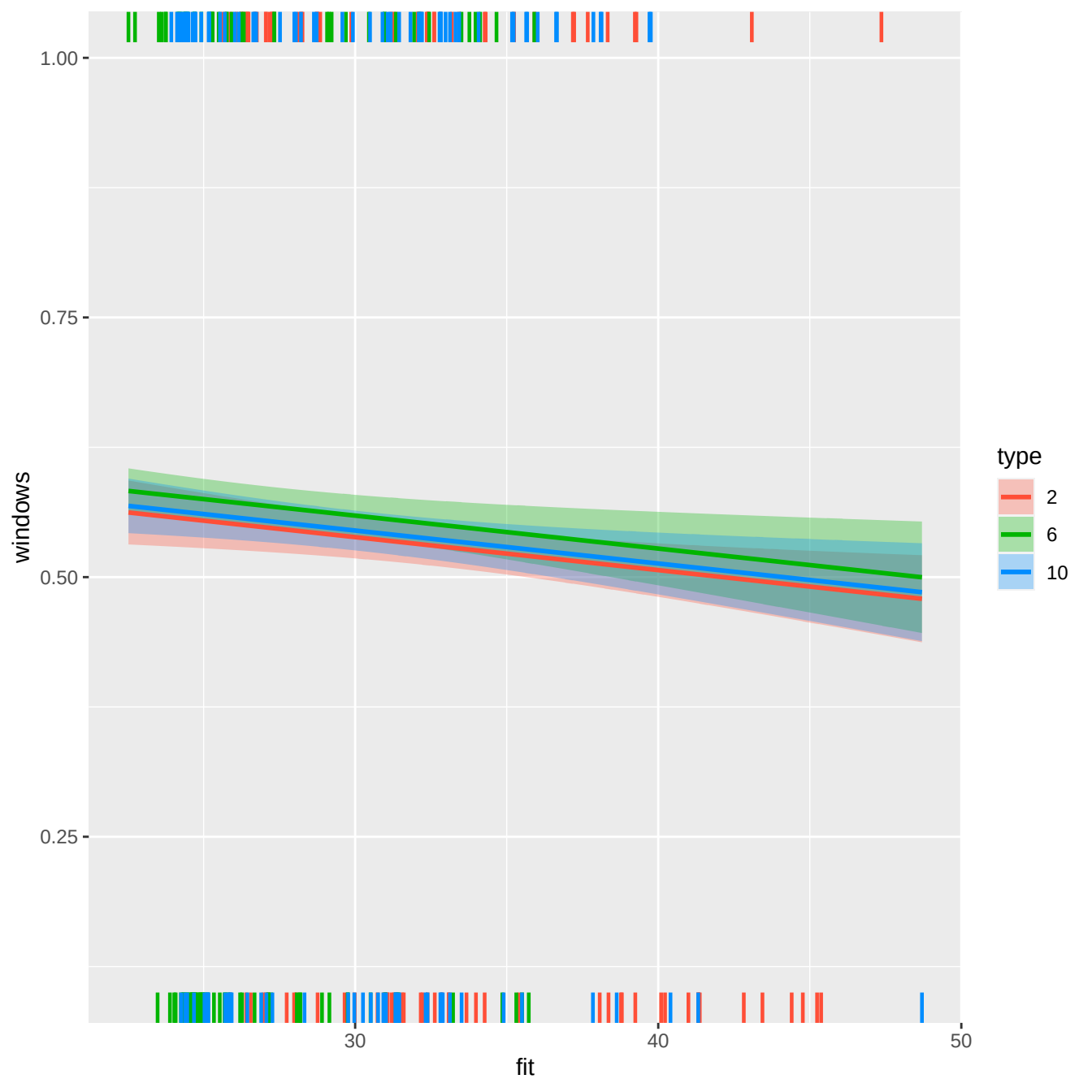

Abi_Anterior_bias_hox.1 1G HOX lower
1 genome
2 genomes
2 genomes high mutation
Parameter range (normalized 0 to 1)
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

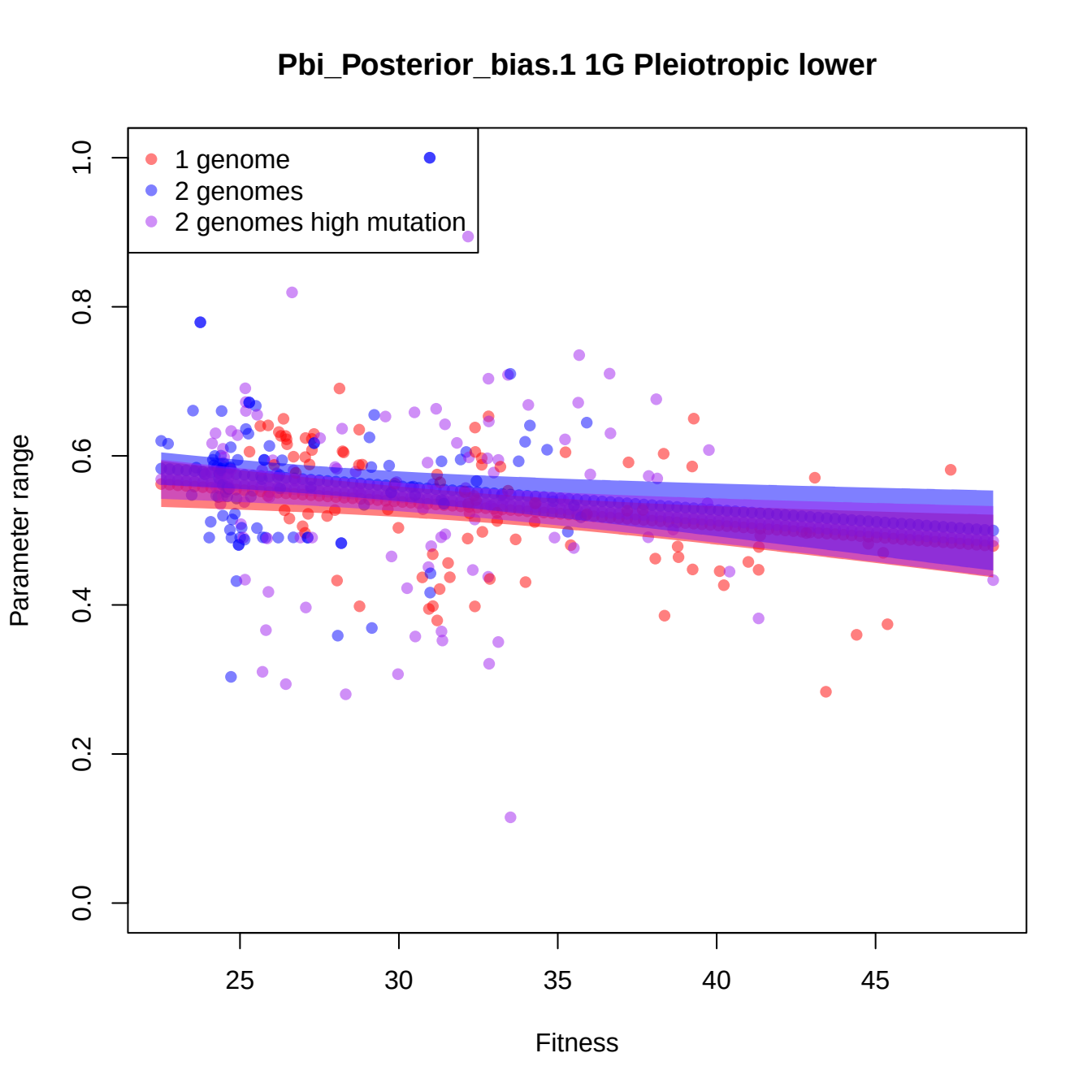

Abi_Anterior_bias_hox.1
1G HOX lower
Counts
1.2
1.0
0.8
0.6
0.4
0.2
6.627848287881l8e−09
0.131966080743931
3.726210806072 46e−05
12 − 4
12 − 8
4 − 8
1 genome
2 genomes
2 genomes high mutation

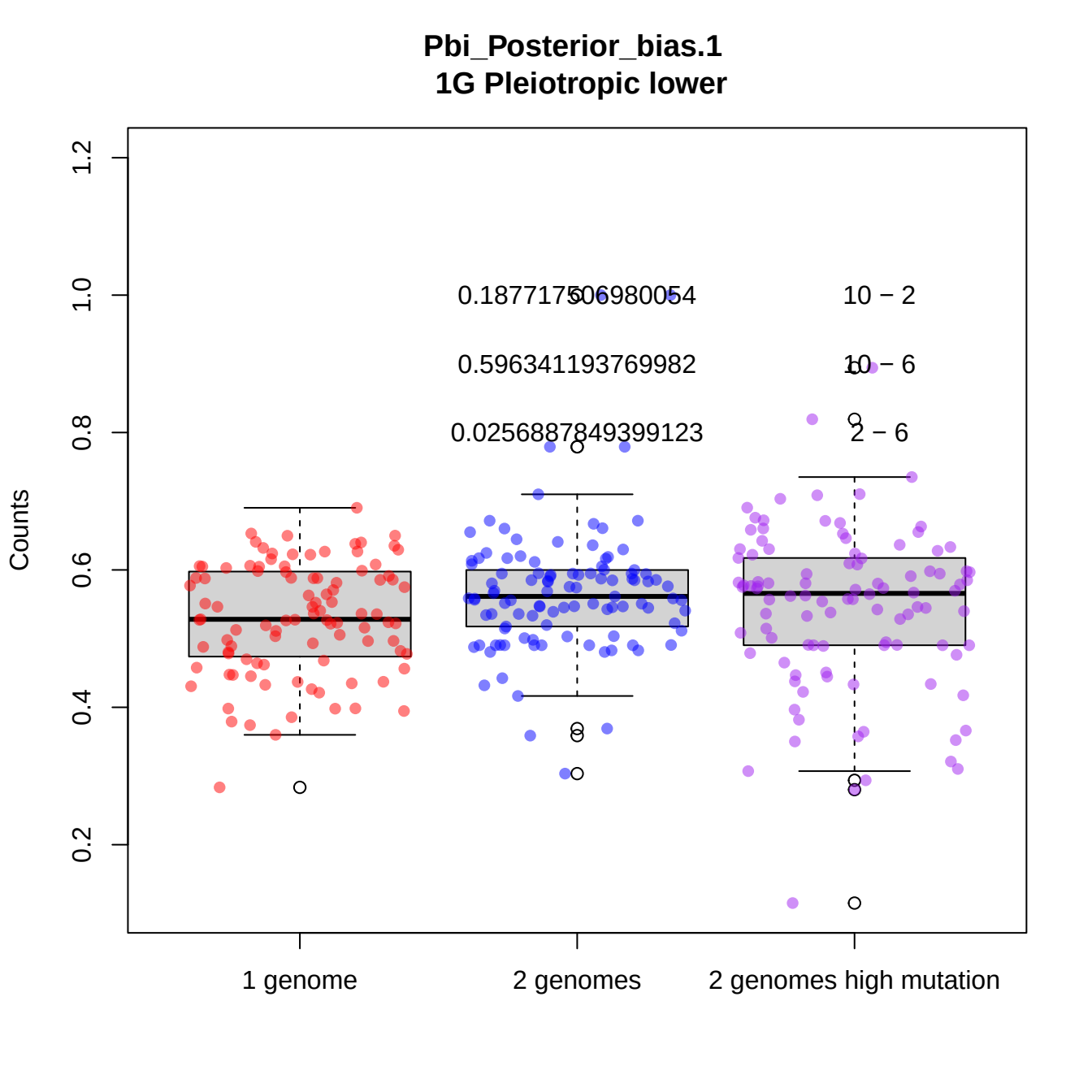

1.00
0.75
0.50
0.25
0.00
windows
30
40
50
fit
type
1
5
9

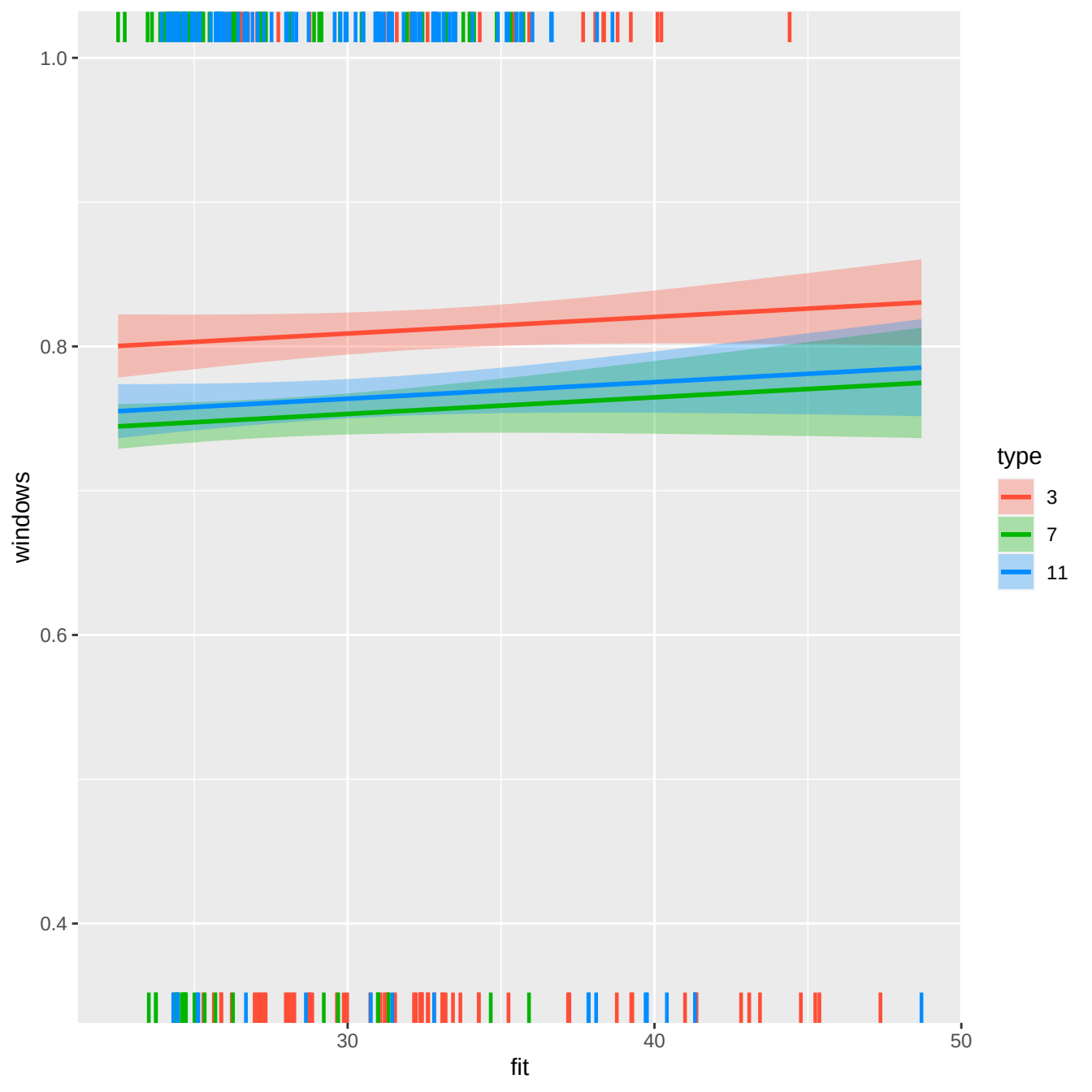

Pbi_Posterior_bias.1 1G Pleiotropic upper
1 genome
2 genomes
2 genomes high mutation
Parameter range
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

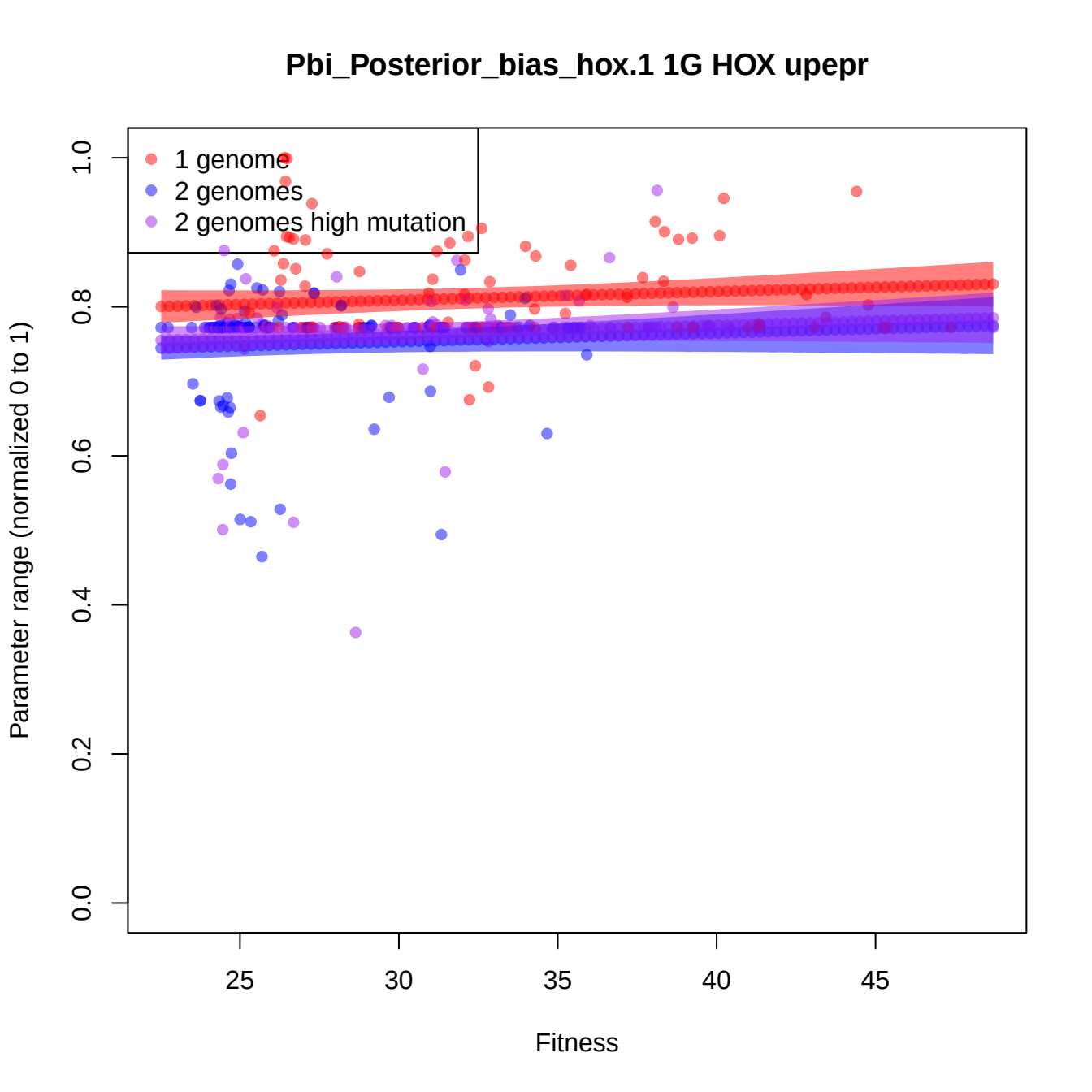

Pbi_Posterior_bias.1
1G Pleiotropic upper
1.13916450981047e−10
3.11115039914587e−10
1 − 5
1 − 9
5 − 9
Counts
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1 genome
2 genomes
2 genomes high mutation

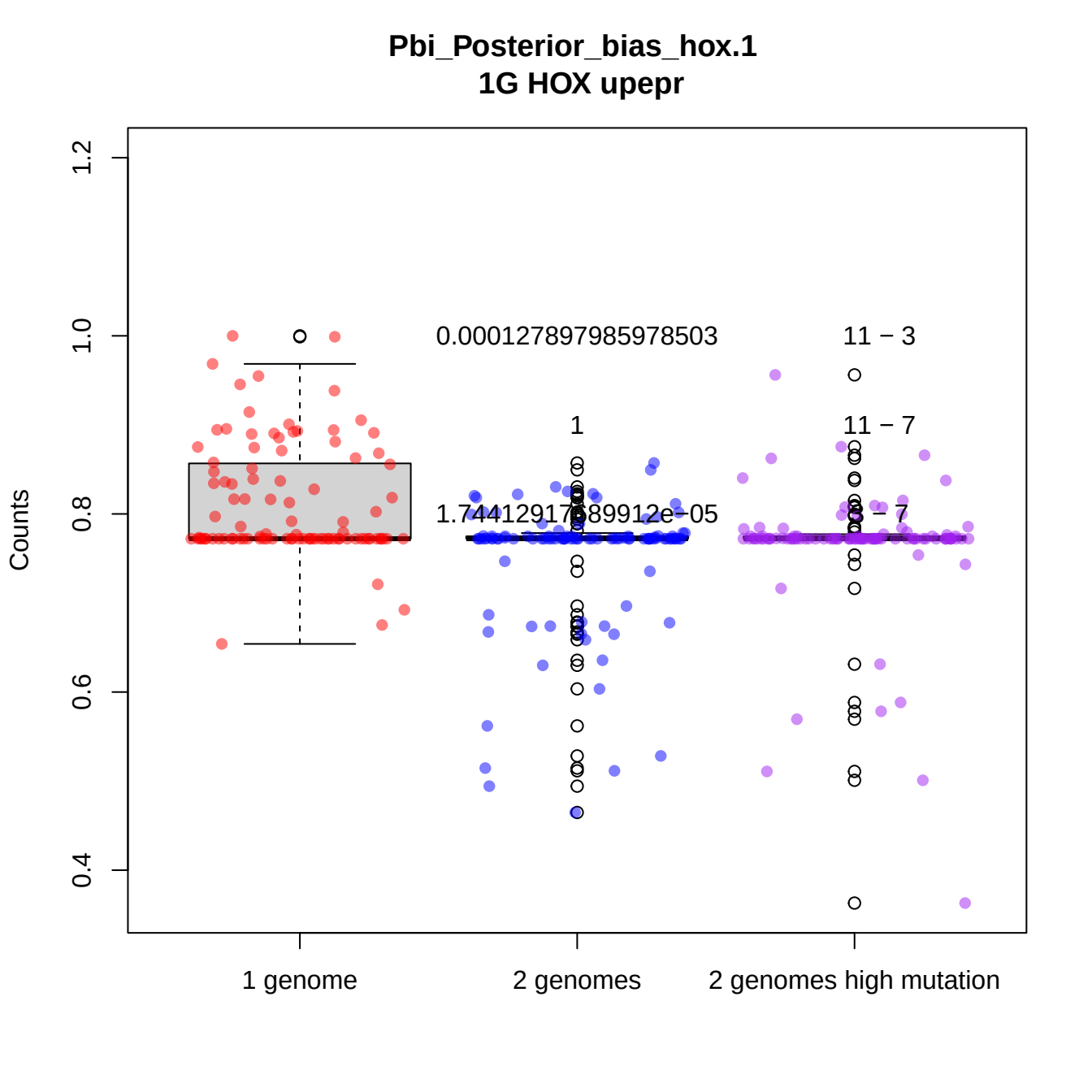

1.00
0.75
0.50
0.25
windows
30
40
50
fit
type
2
6
10

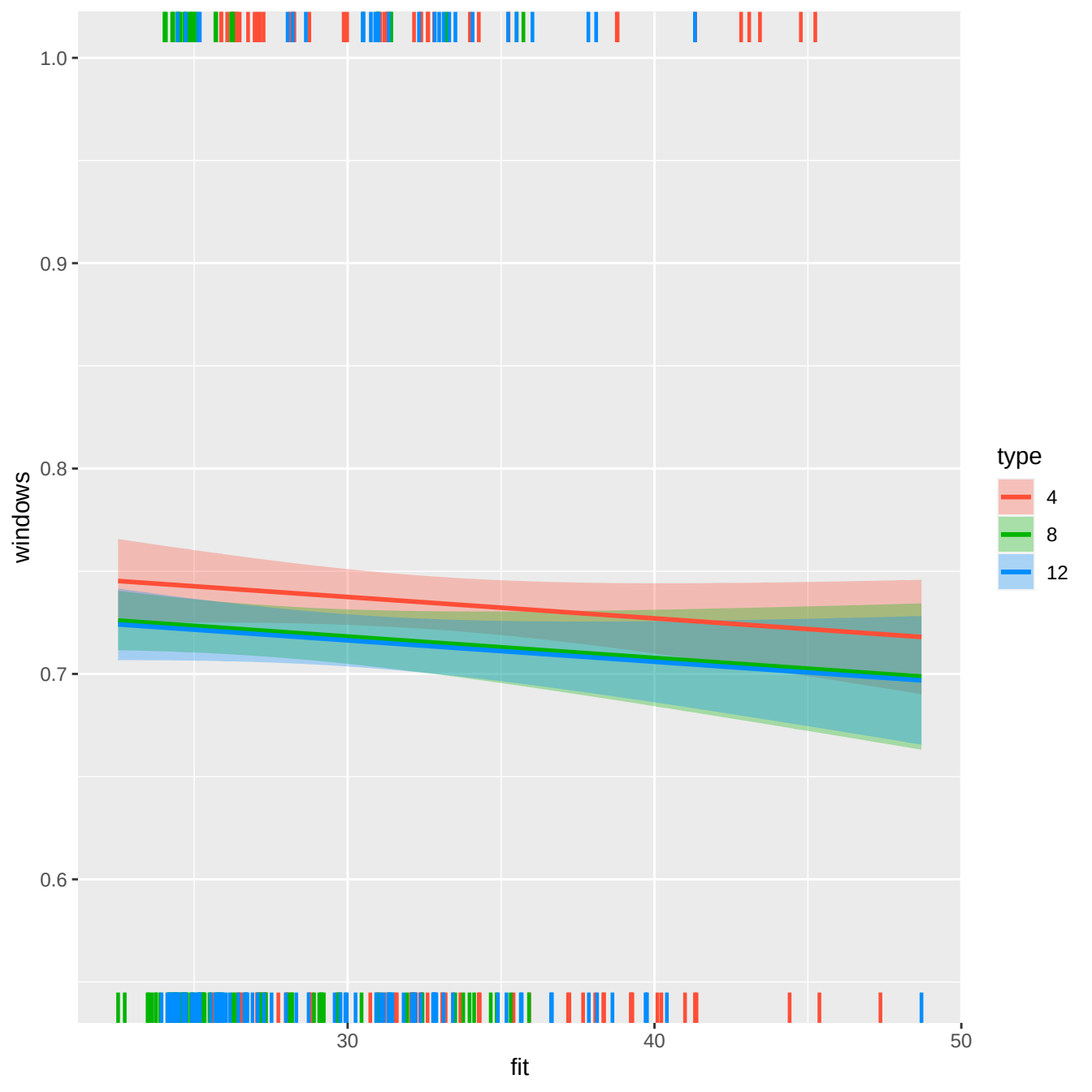

Pbi_Posterior_bias.1 1G Pleiotropic lower
1 genome
2 genomes
2 genomes high mutation
Parameter range
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

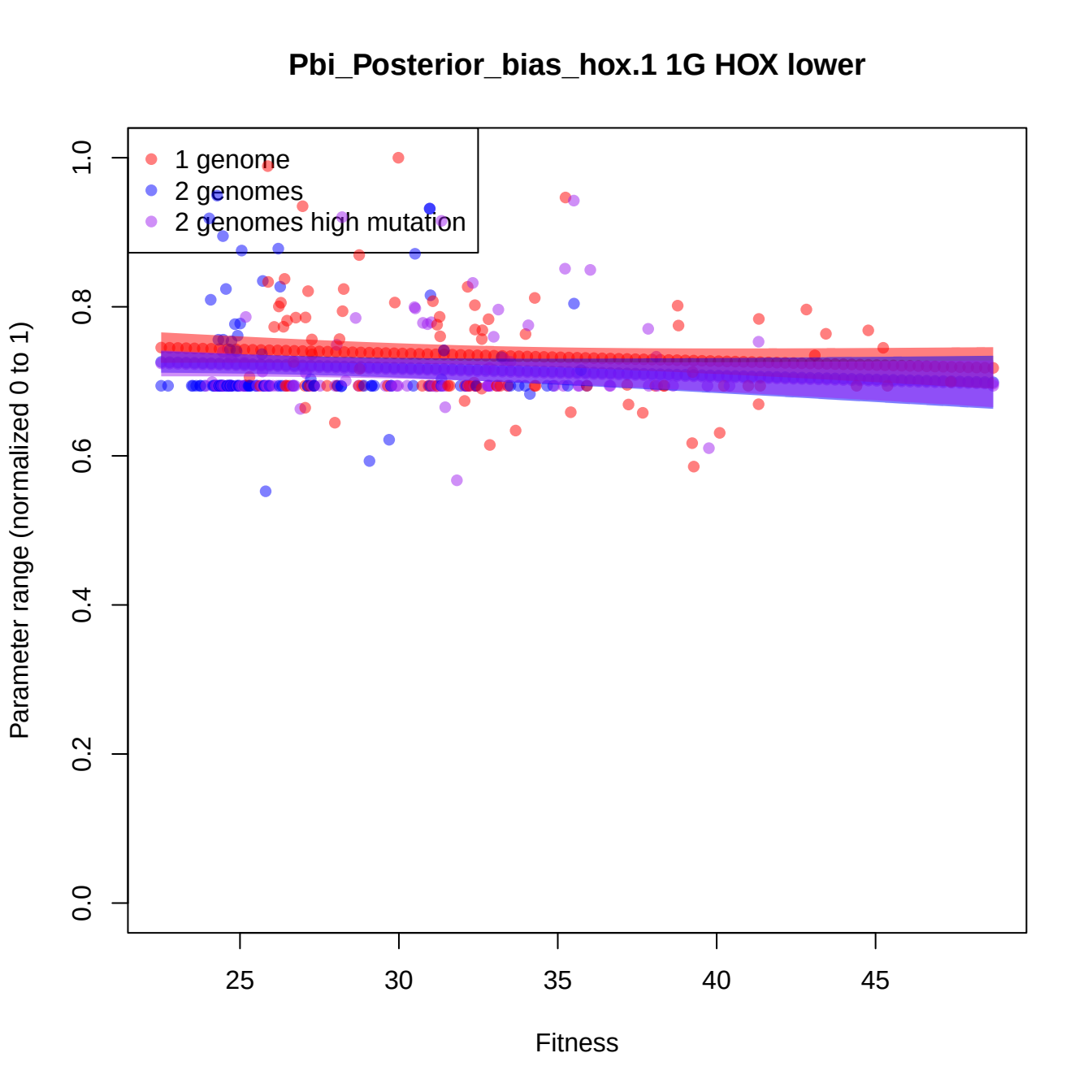

Pbi_Posterior_bias.1
1G Pleiotropic lower
Counts
1.2
1.0
0.8
0.6
0.4
0.2
0.18771750698005 4
0.596341193769982
0.0256887849399123
10 − 2
10 − 6
2 − 6
1 genome
2 genomes
2 genomes high mutation

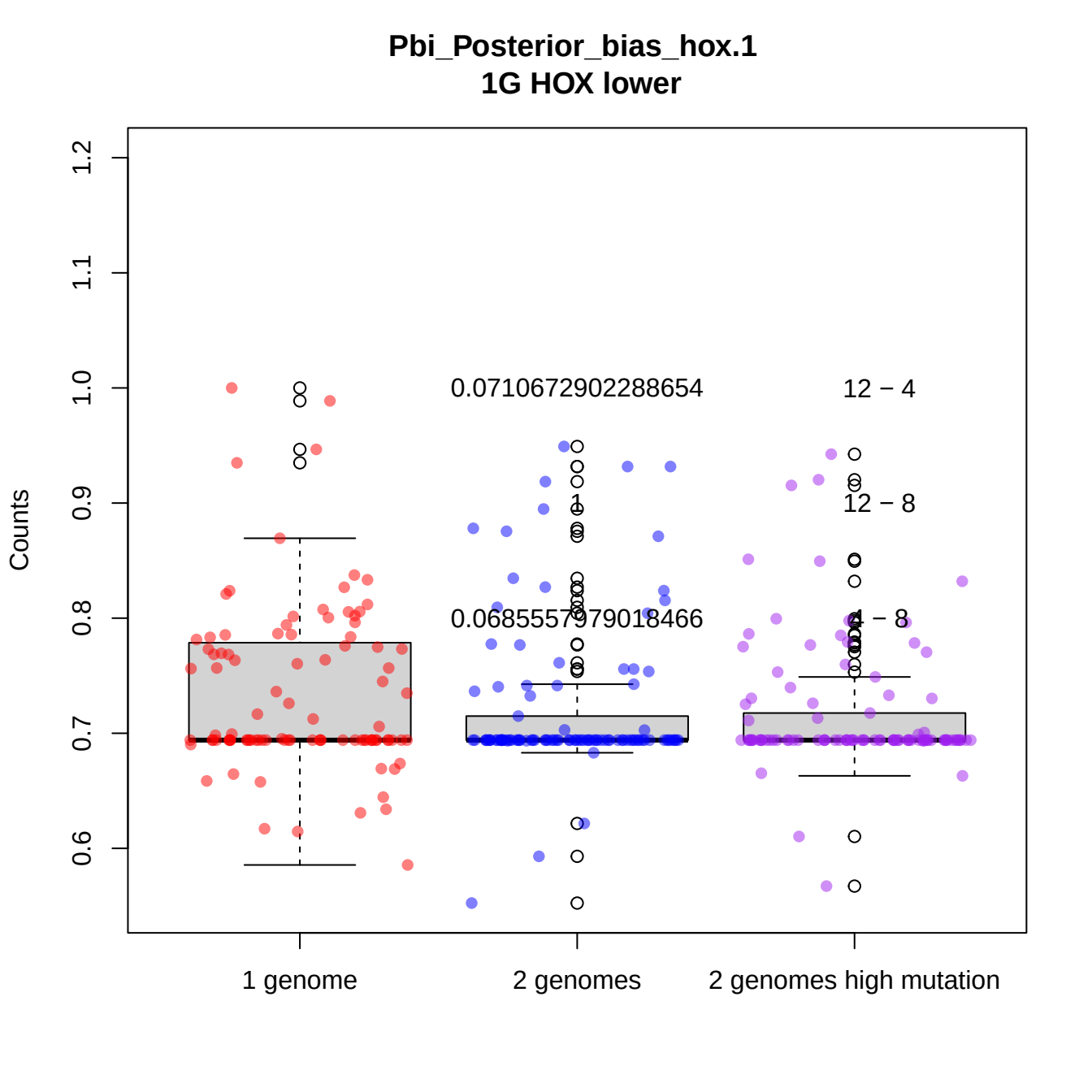

1.0
0.8
0.6
0.4
windows
30
40
50
fit
type
3
7
11

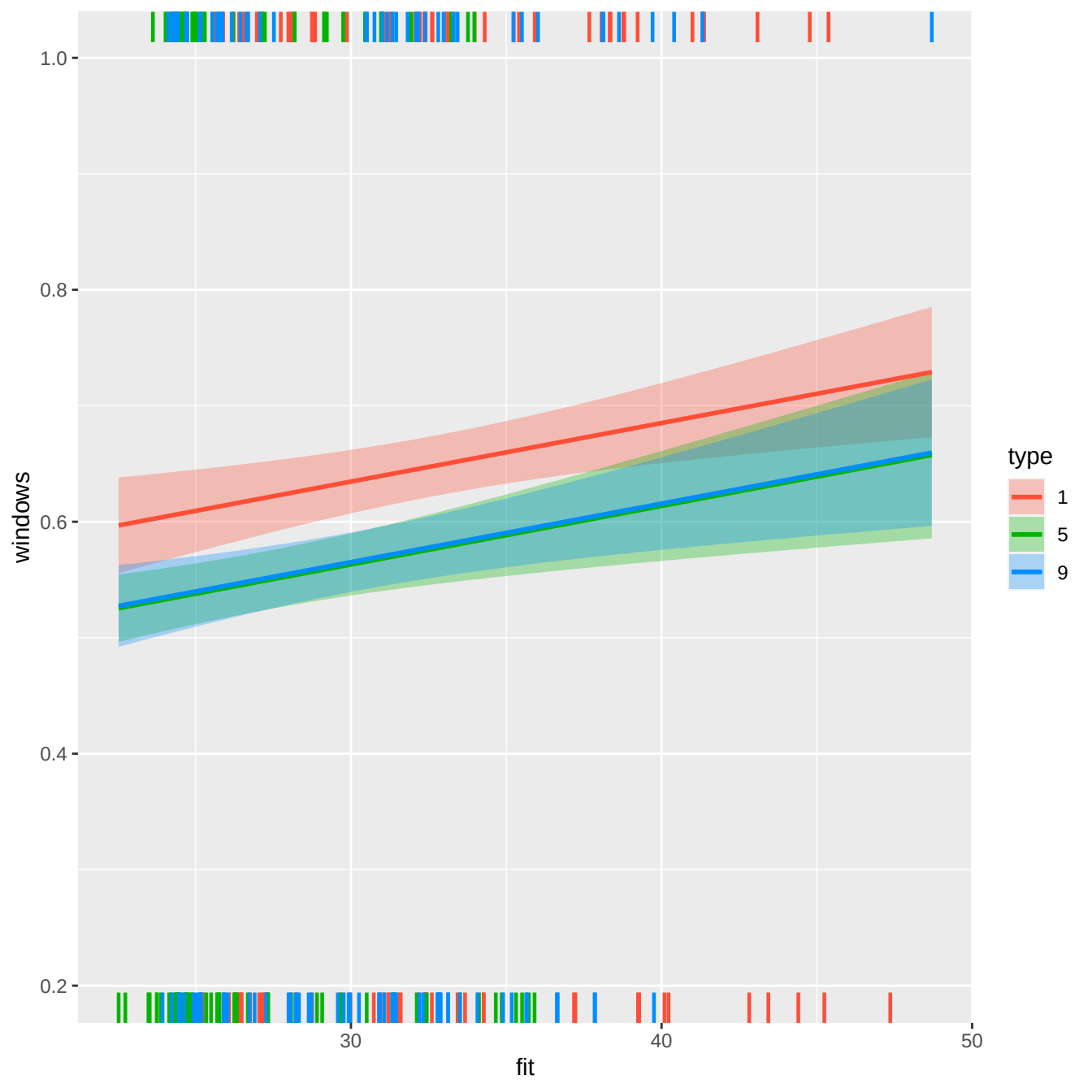

Pbi_Posterior_bias_hox.1 1G HOX upepr
1 genome
2 genomes
2 genomes high mutation
Parameter range (normalized 0 to 1)
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

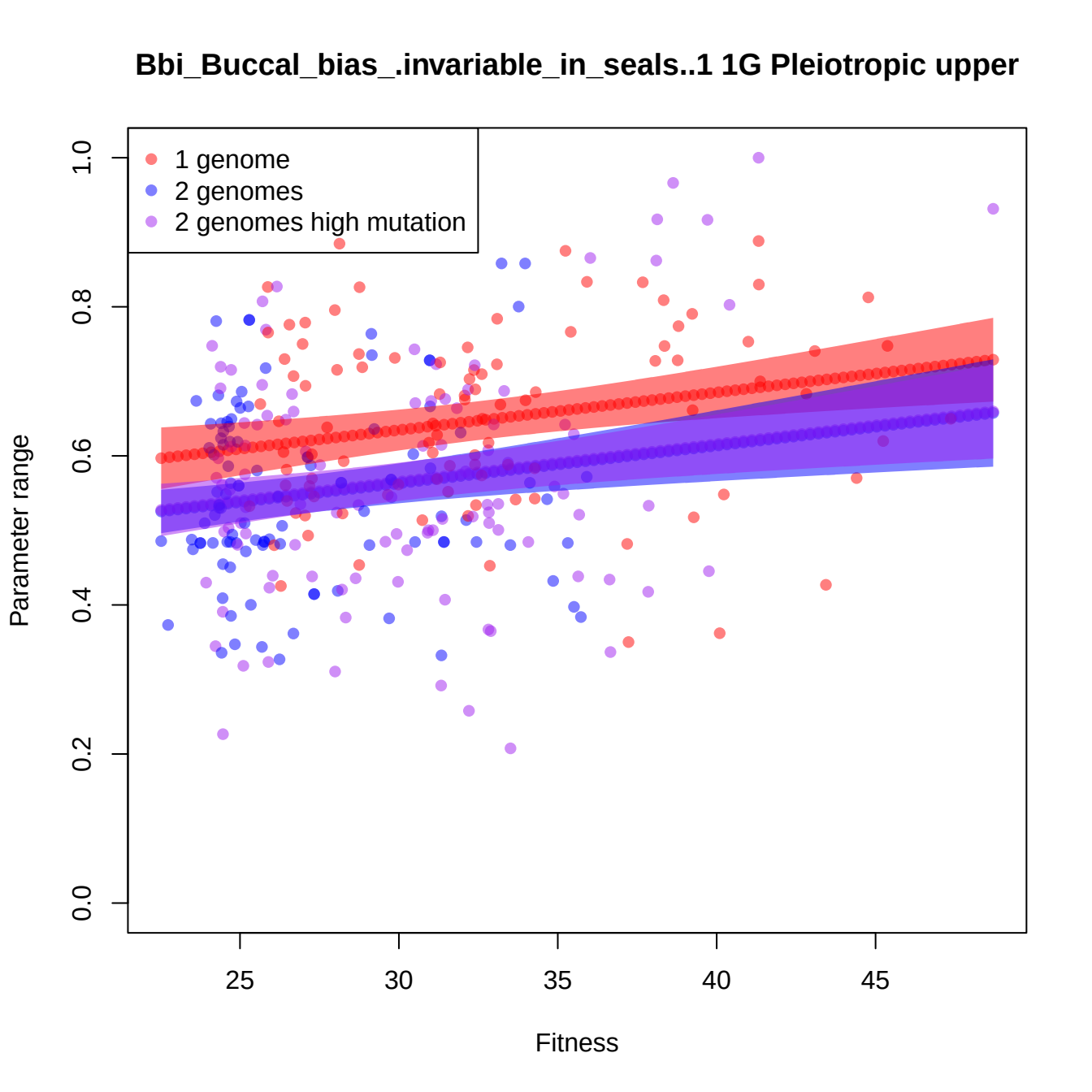

Pbi_Posterior_bias_hox.1
1G HOX upepr
Counts
1.2
1.0
0.8
0.6
0.4
0.000127897985978503
1
1.74412917689912e−05
11 − 3
11 − 7
− 7
1 genome
2 genomes
2 genomes high mutation

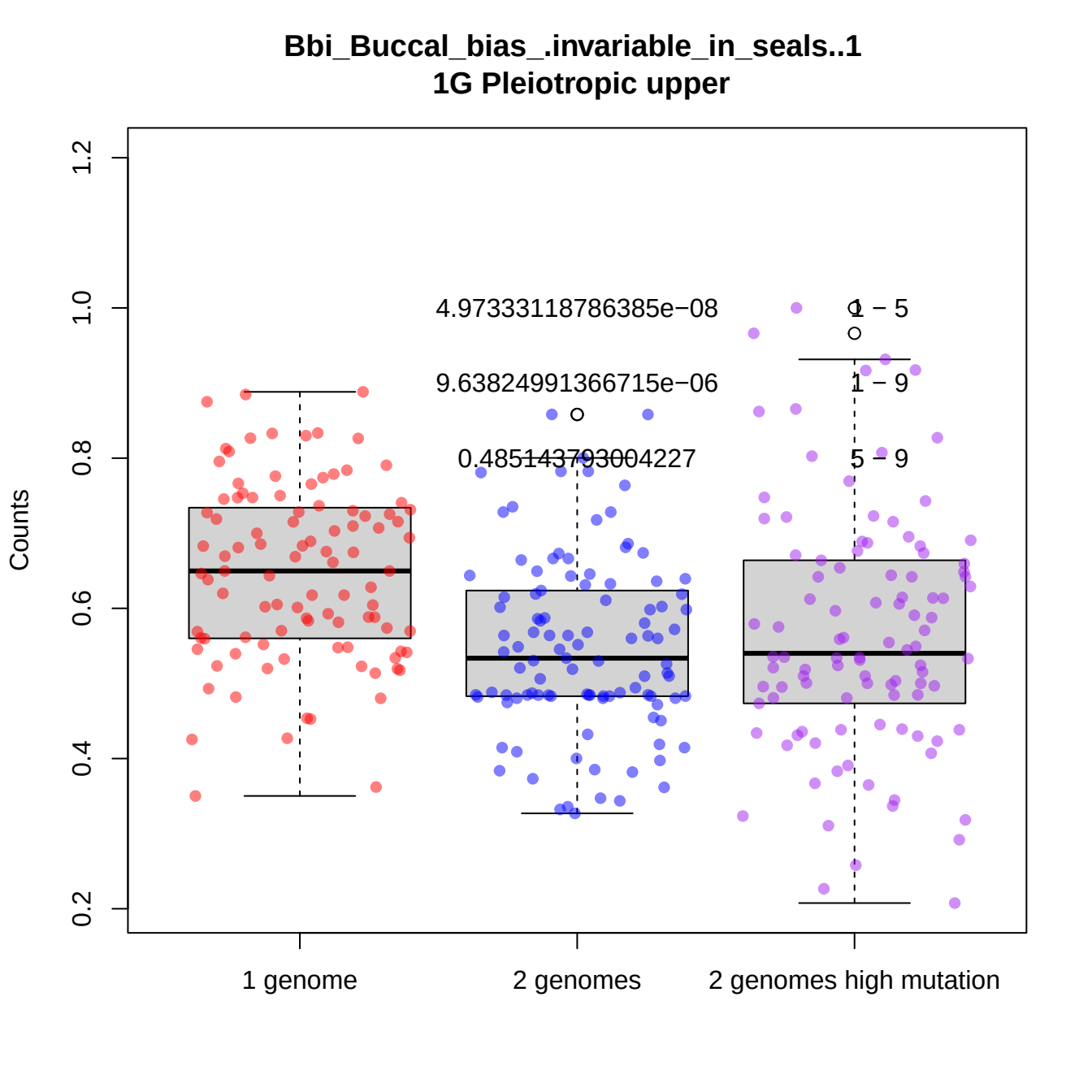

1.0
0.9
0.8
0.7
0.6
windows
30
40
50
fit
type
4
8
12

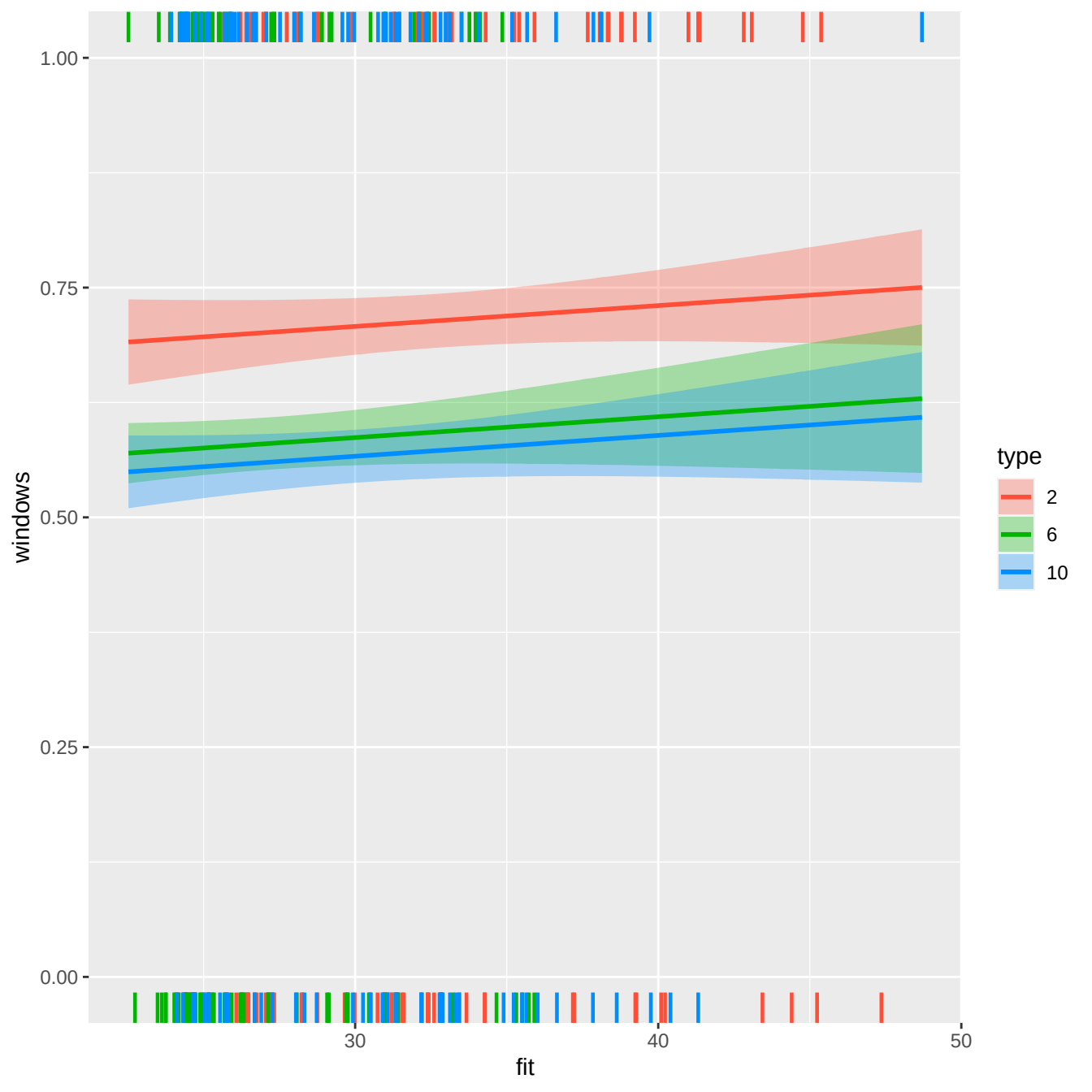

Pbi_Posterior_bias_hox.1 1G HOX lower
1 genome
2 genomes
2 genomes high mutation
Parameter range (normalized 0 to 1)
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

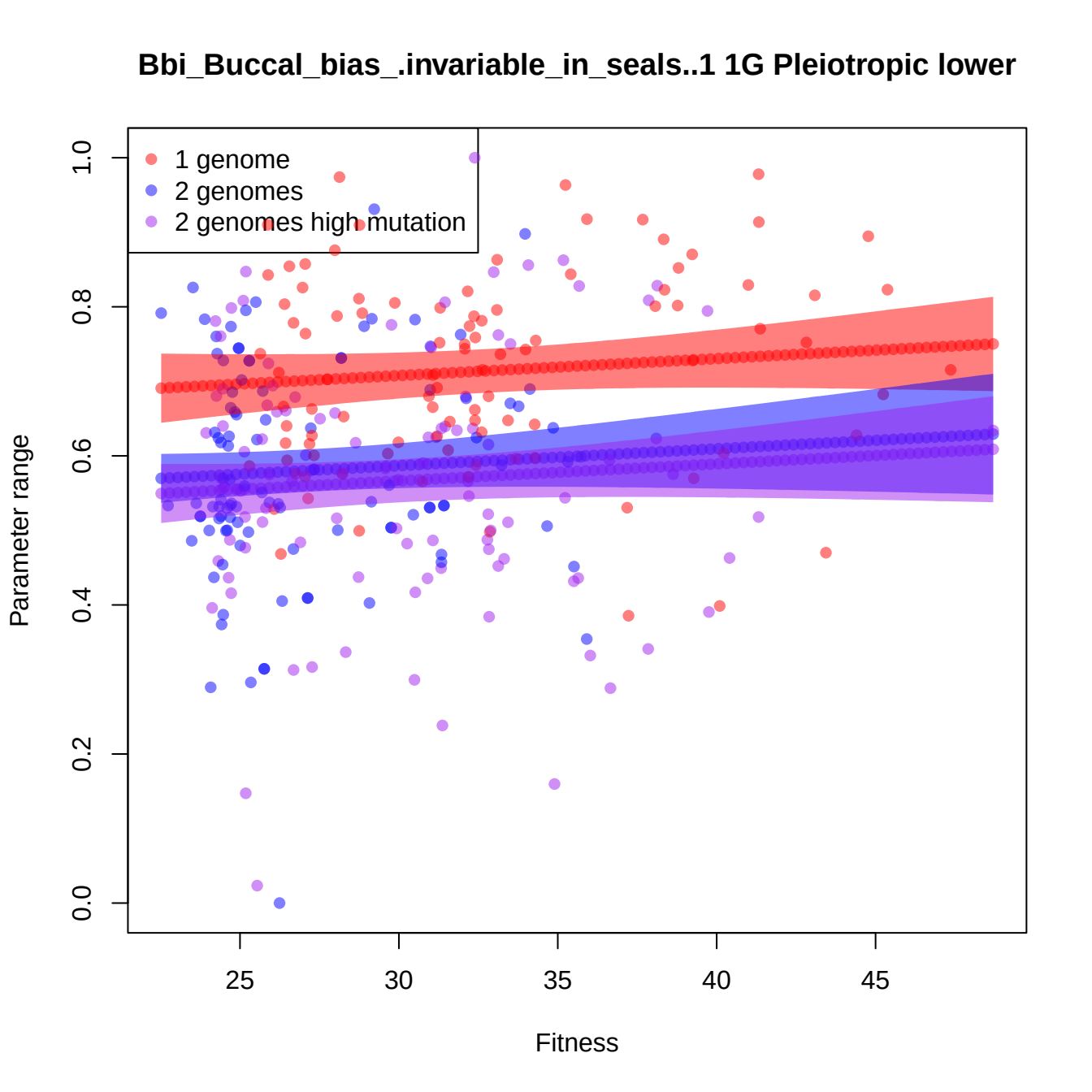

Pbi_Posterior_bias_hox.1
1G HOX lower
Counts
1.2
1.1
1.0
0.9
0.8
0.7
0.6
0.0710672902288654
0.0685557979018466
12 − 4
12 − 8
1
1 genome
2 genomes
2 genomes high mutation

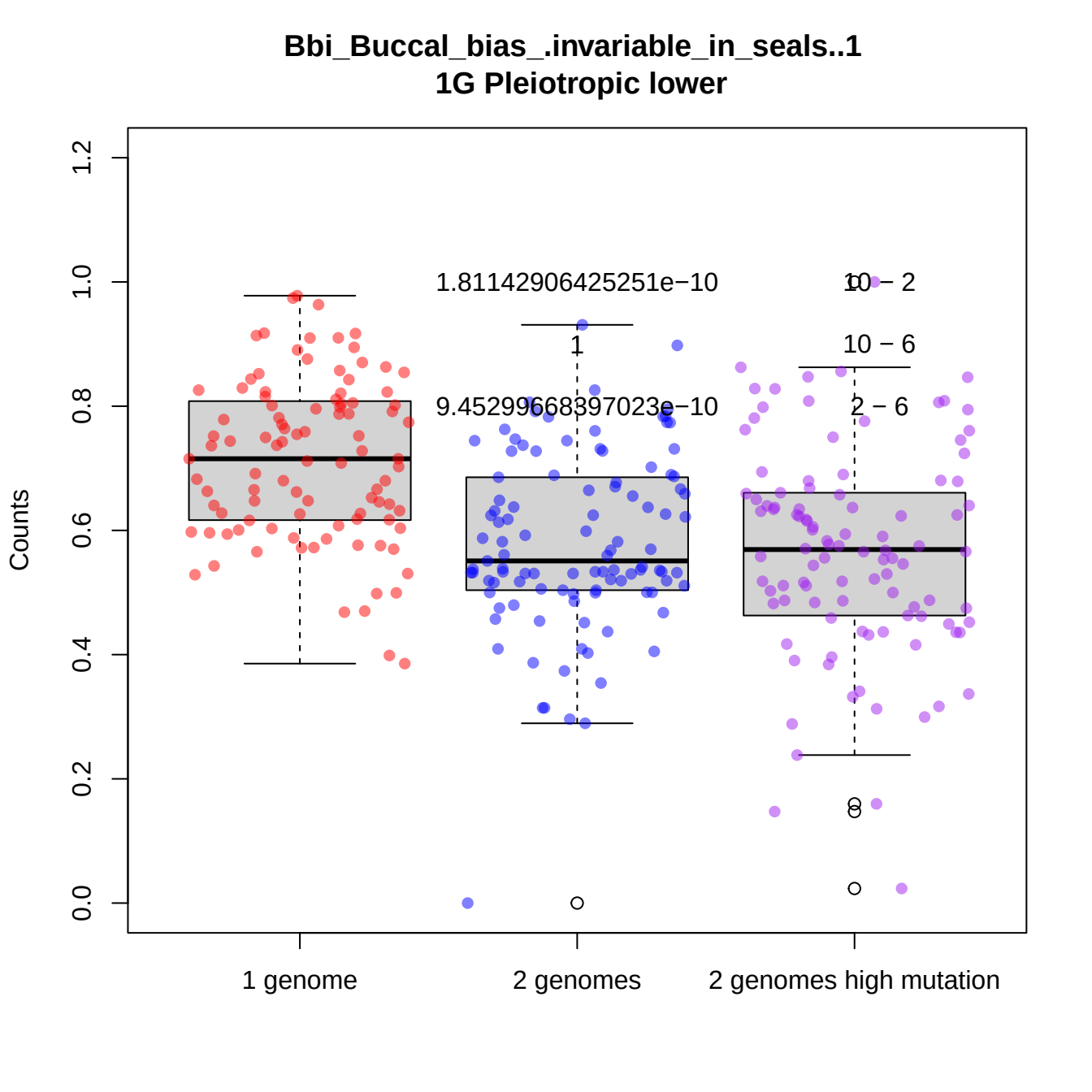

1.0
0.8
0.6
0.4
0.2
windows
30
40
50
fit
type
1
5
9

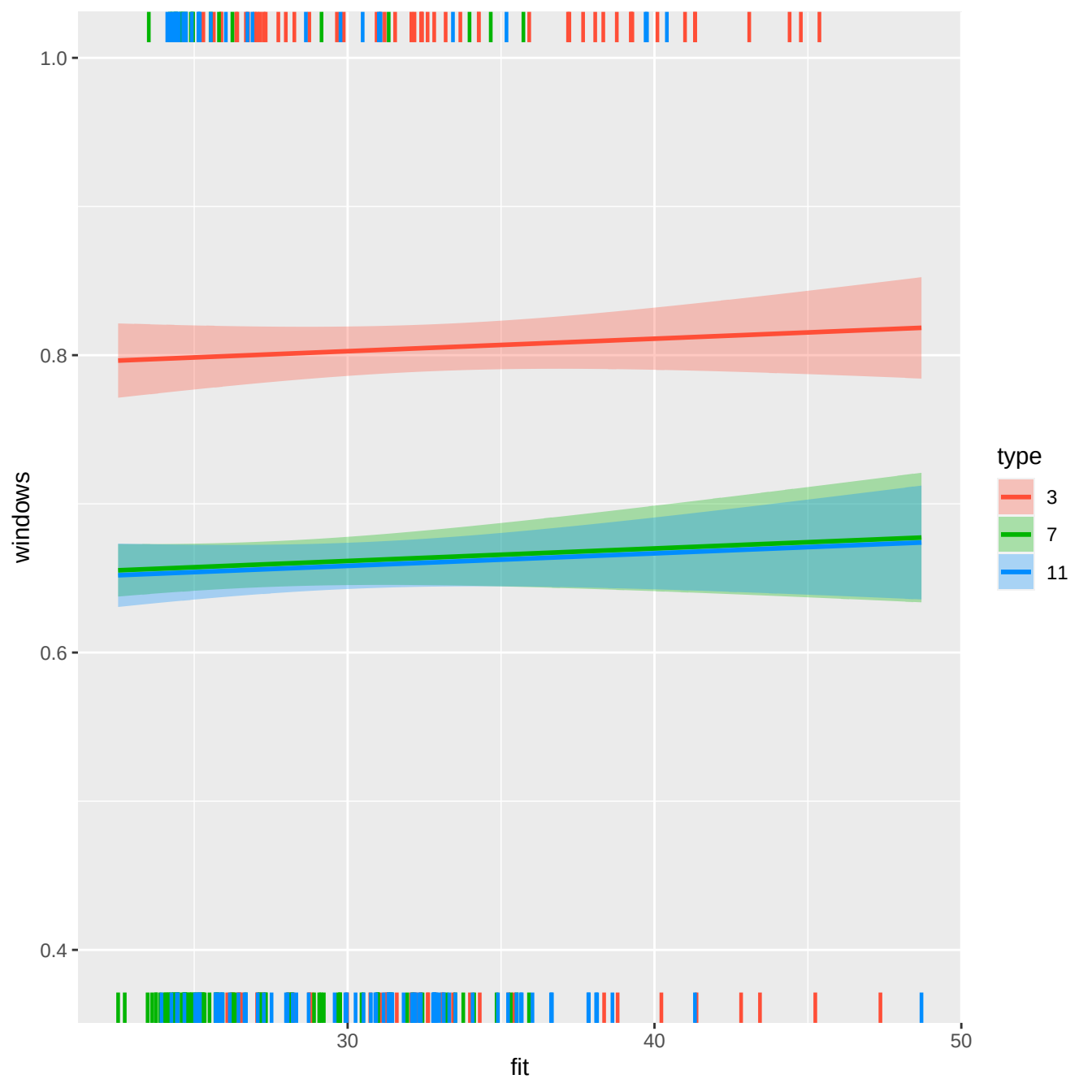

Bbi_Buccal_bias_.invariable_in_seals..1 1G Pleiotropic upper
1 genome
2 genomes
2 genomes high mutation
Parameter range
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

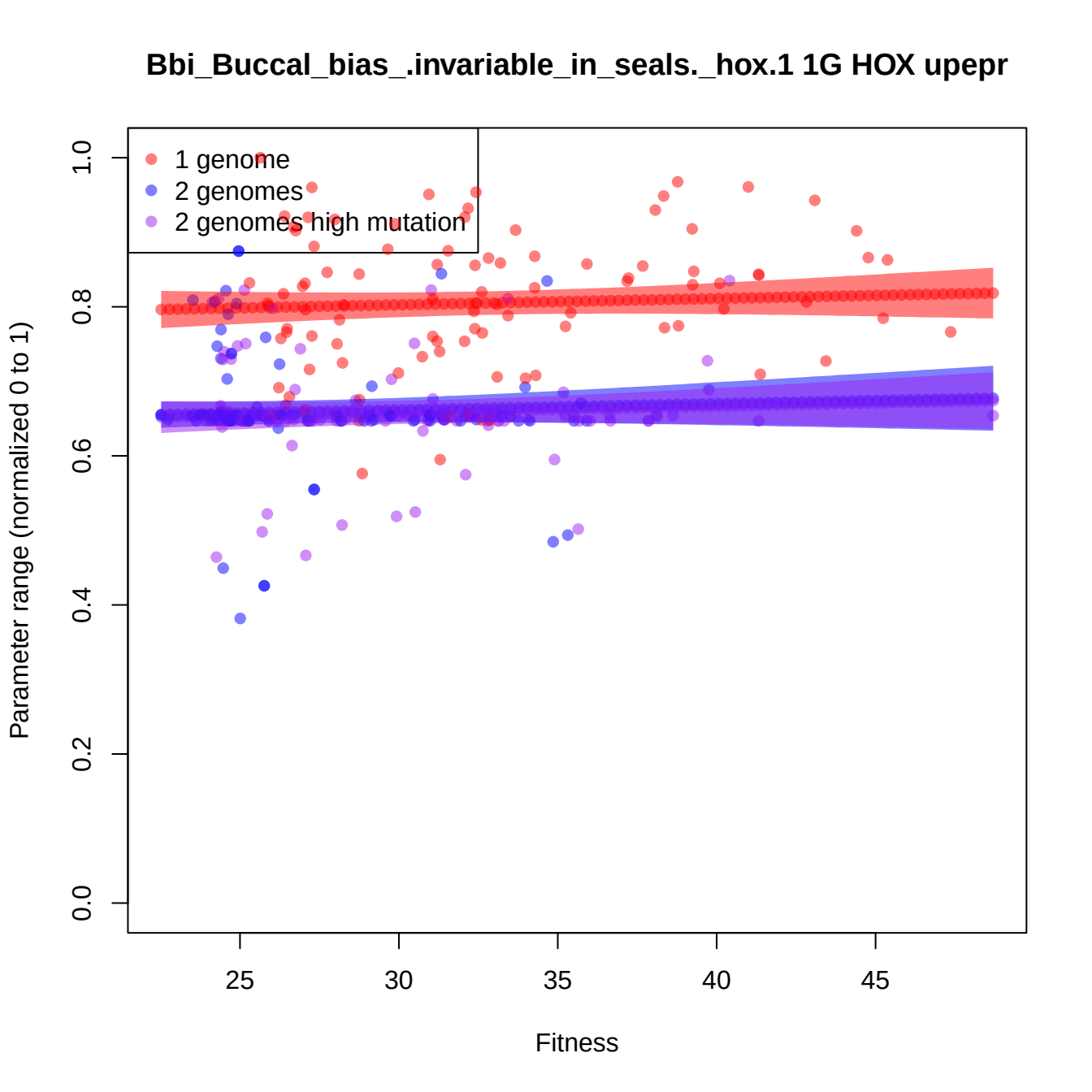

Bbi_Buccal_bias_.invariable_in_seals..1
1G Pleiotropic upper
Counts
1.2
1.0
0.8
0.6
0.4
0.2
4.97333118786385e−08
9.63824991366715e−06
0.485143793004227
1 − 5
1 − 9
5 − 9
1 genome
2 genomes
2 genomes high mutation

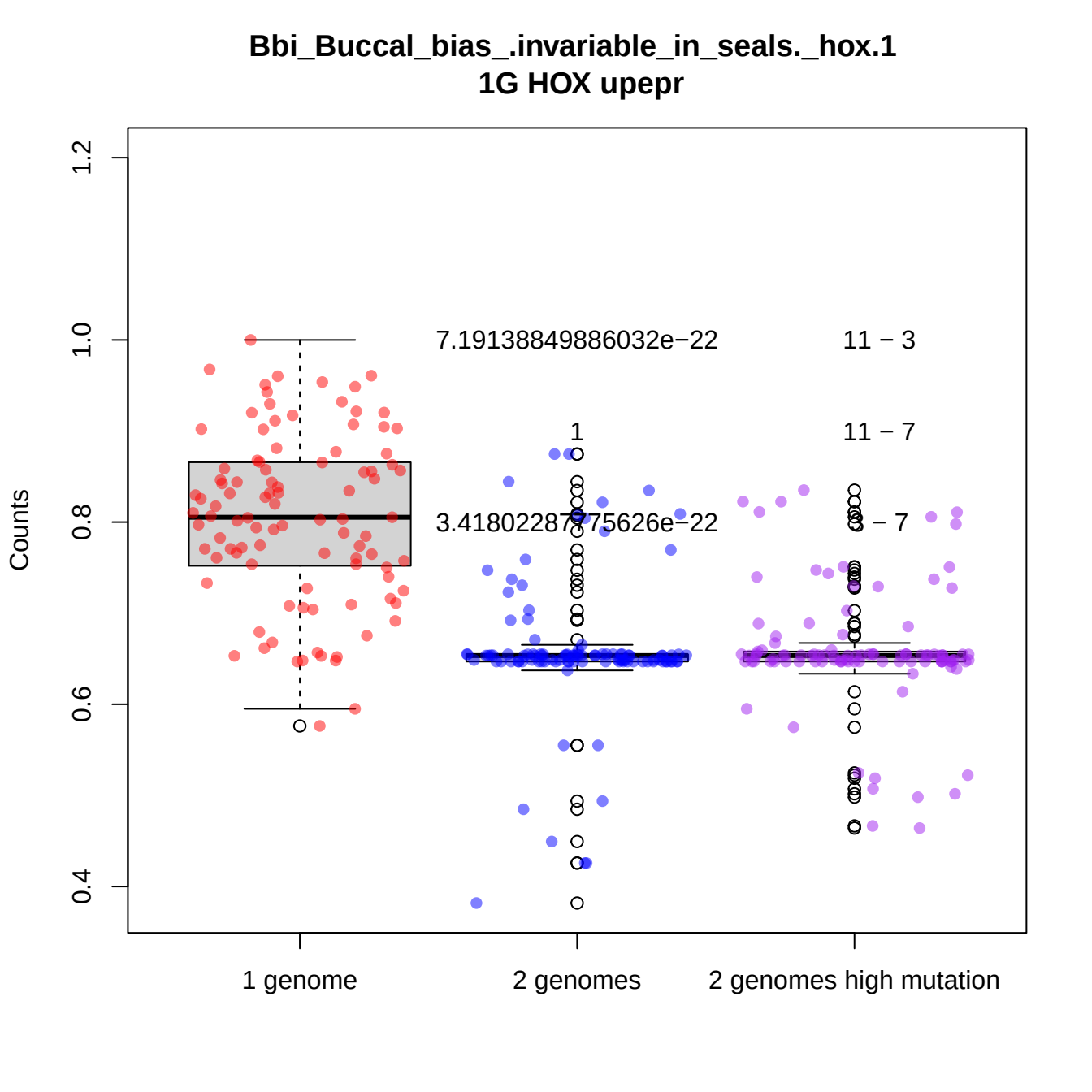

1.00
0.75
0.50
0.25
0.00
windows
30
40
50
fit
type
2
6
10

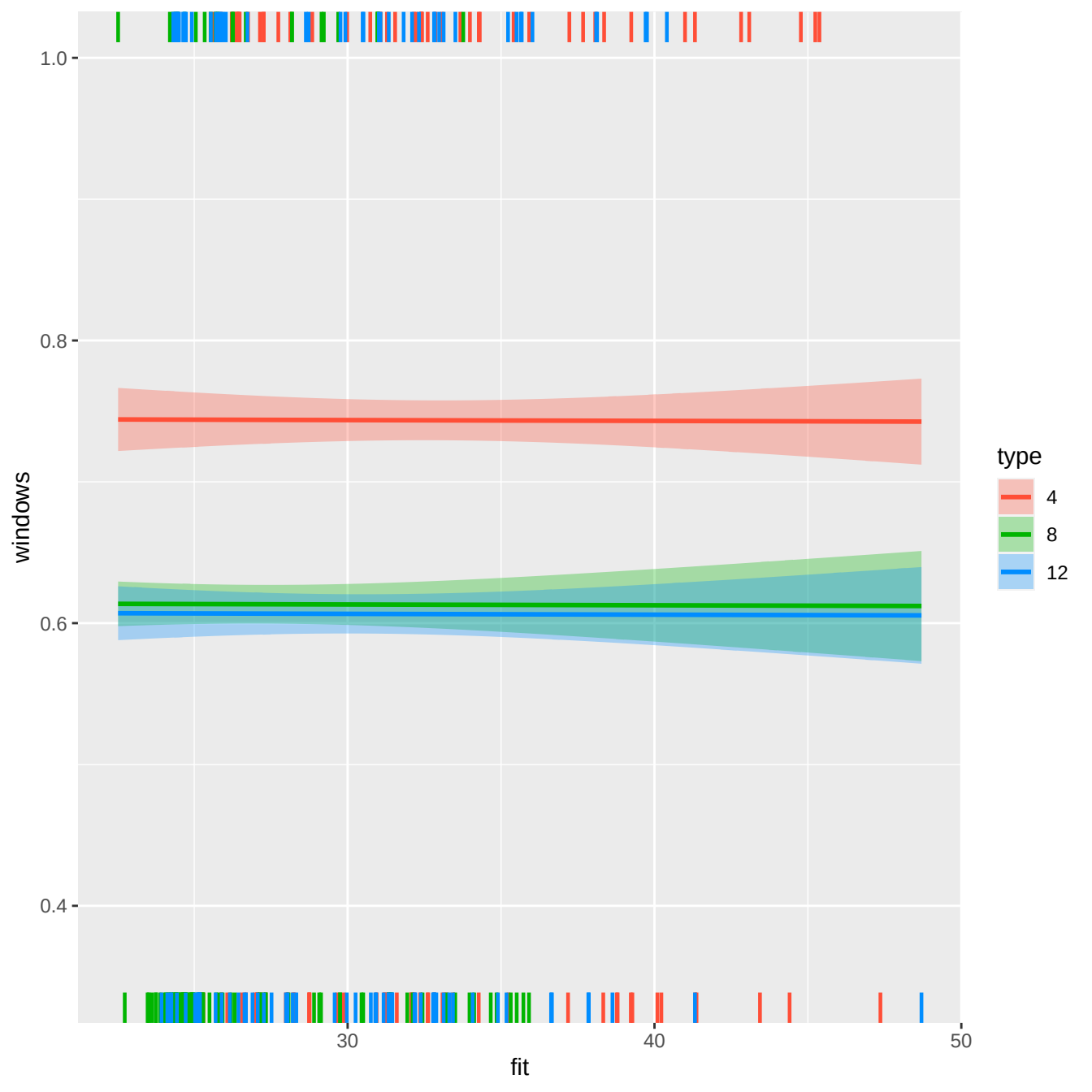

Bbi_Buccal_bias_.invariable_in_seals..1 1G Pleiotropic lower
1 genome
2 genomes
2 genomes high mutation
Parameter range
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

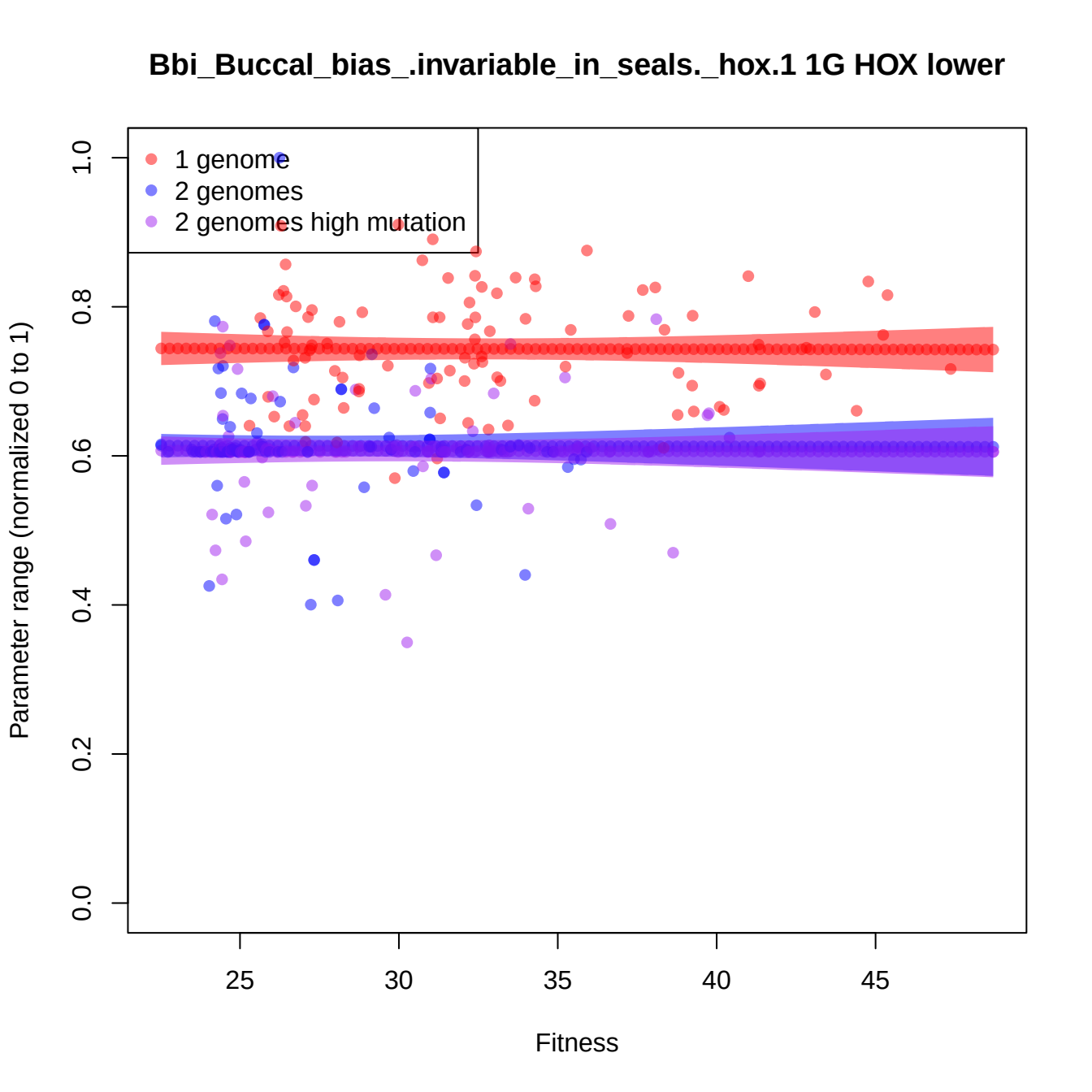

Bbi_Buccal_bias_.invariable_in_seals..1
1G Pleiotropic lower
Counts
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.81142906425251e−10
1
9.45299668397023e−10
10 − 2
10 − 6
2 − 6
1 genome
2 genomes
2 genomes high mutation

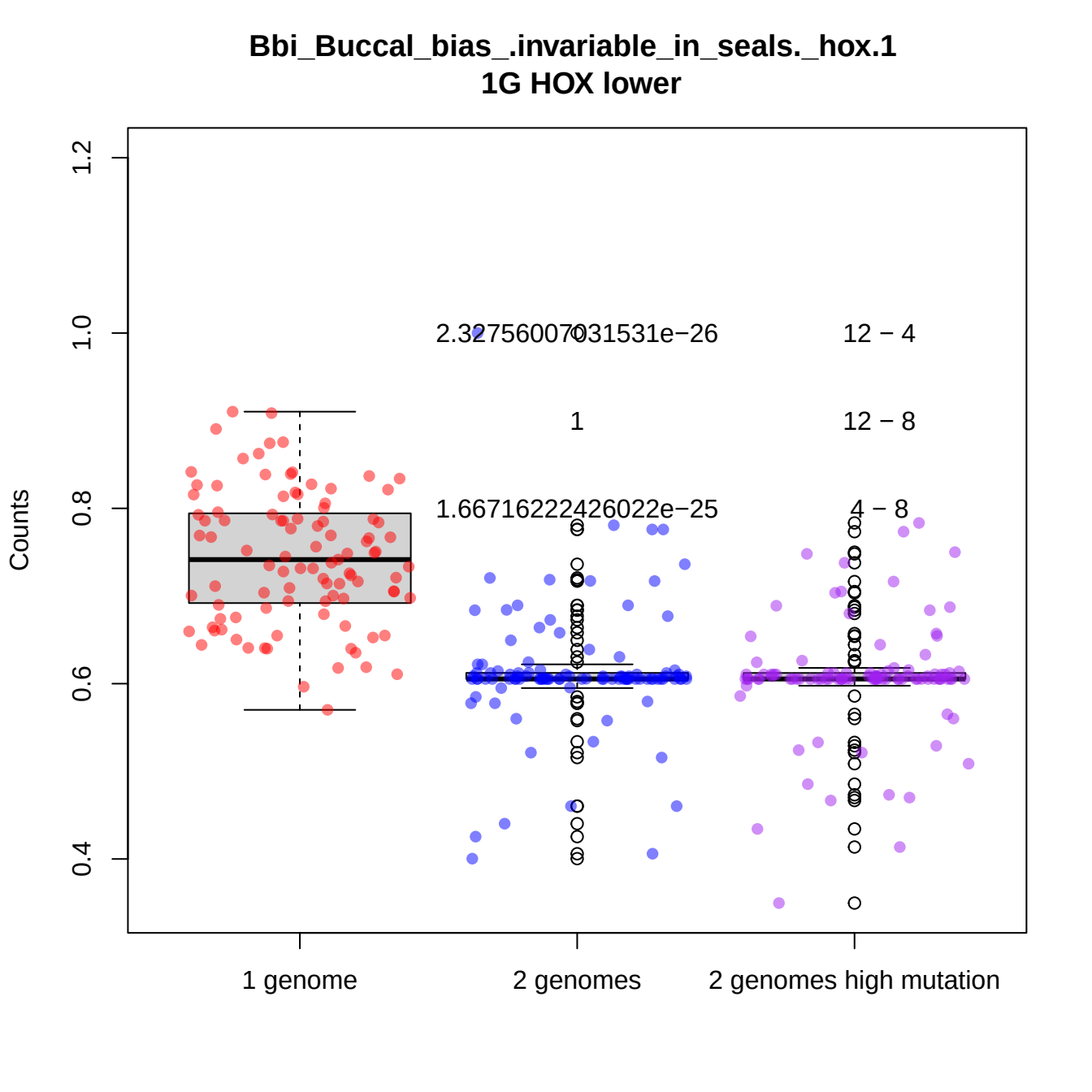

1.0
0.8
0.6
0.4
windows
30
40
50
fit
type
3
7
11

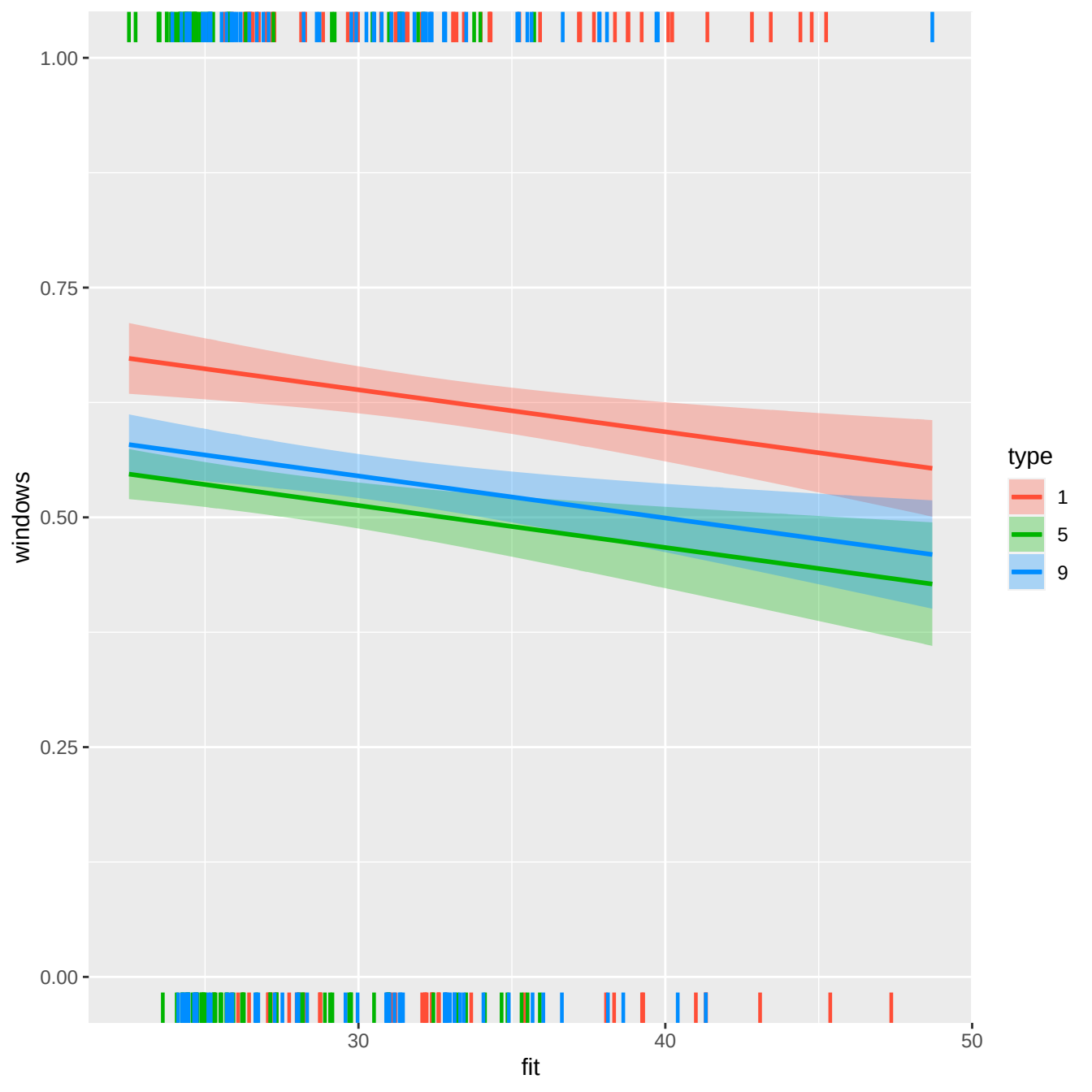

Bbi_Buccal_bias_.invariable_in_seals._hox.1 1G HOX upepr
1 genome
2 genomes
2 genomes high mutation
Parameter range (normalized 0 to 1)
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

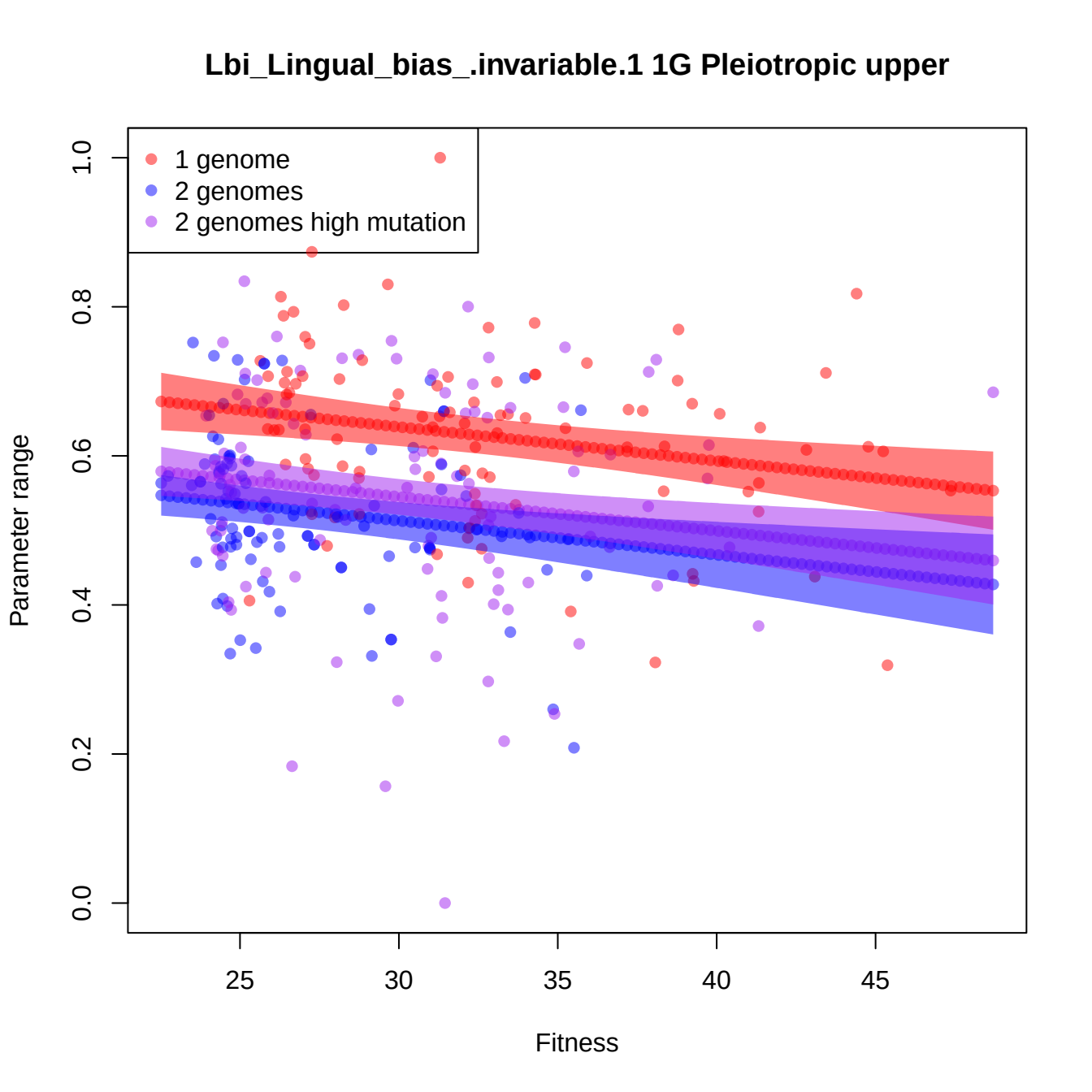

Bbi_Buccal_bias_.invariable_in_seals._hox.1
1G HOX upepr
Counts
1.2
1.0
0.8
0.6
0.4
7.19138849886032e−22
1
3.41802287975626e−22
11 − 3
11 − 7
3 − 7
1 genome
2 genomes
2 genomes high mutation

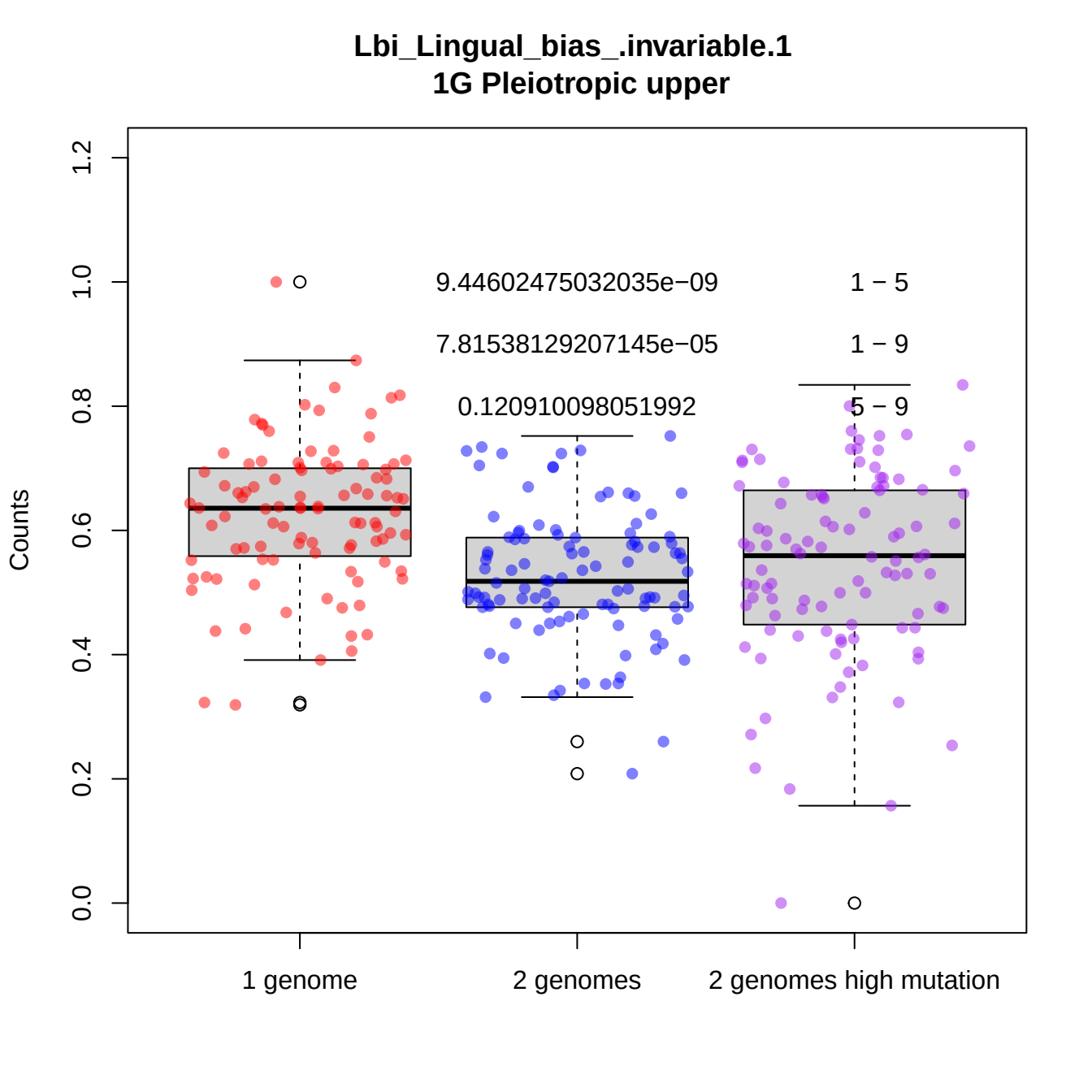

1.0
0.8
0.6
0.4
windows
30
40
50
fit
type
4
8
12

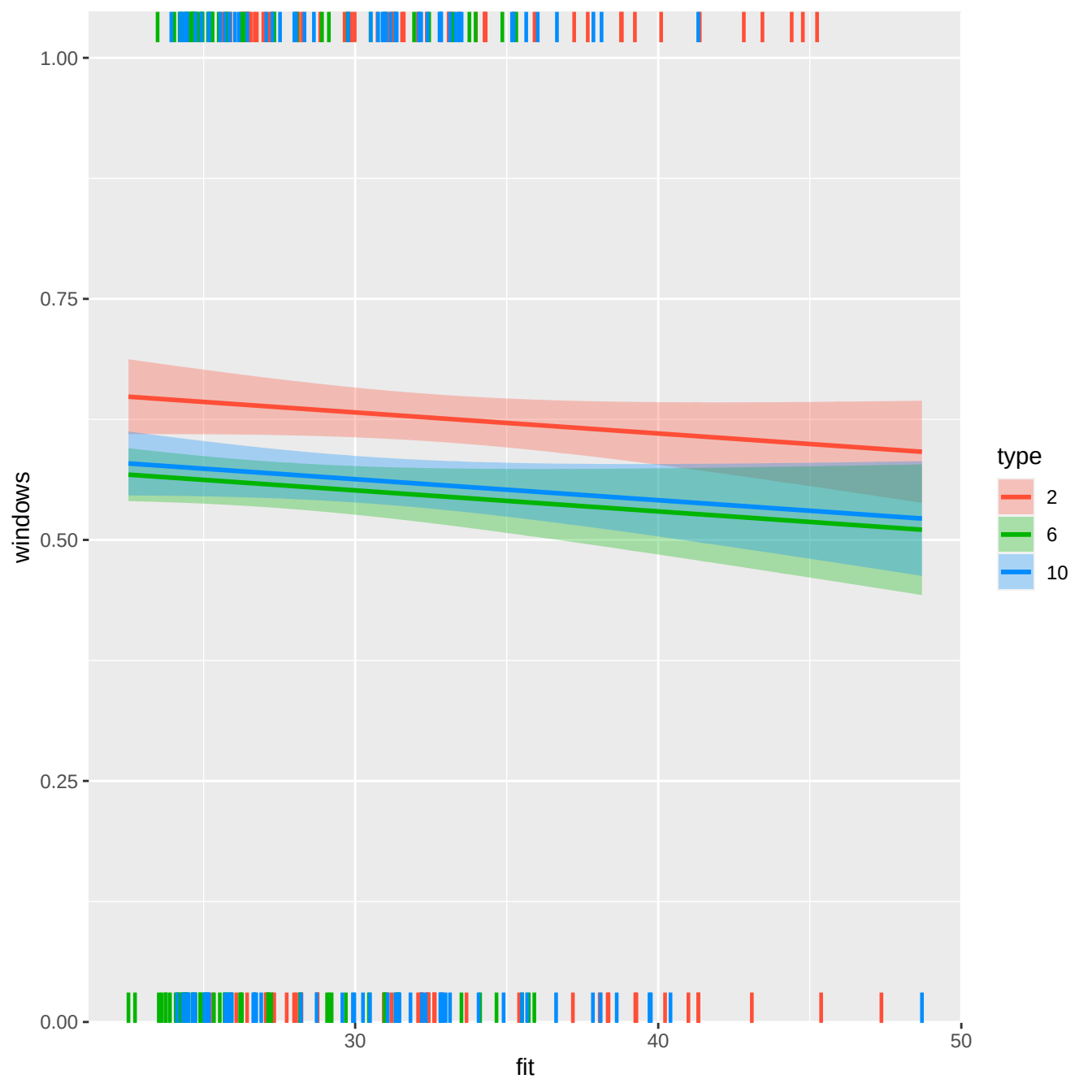

Bbi_Buccal_bias_.invariable_in_seals._hox.1 1G HOX lower
1 genome
2 genomes
2 genomes high mutation
Parameter range (normalized 0 to 1)
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

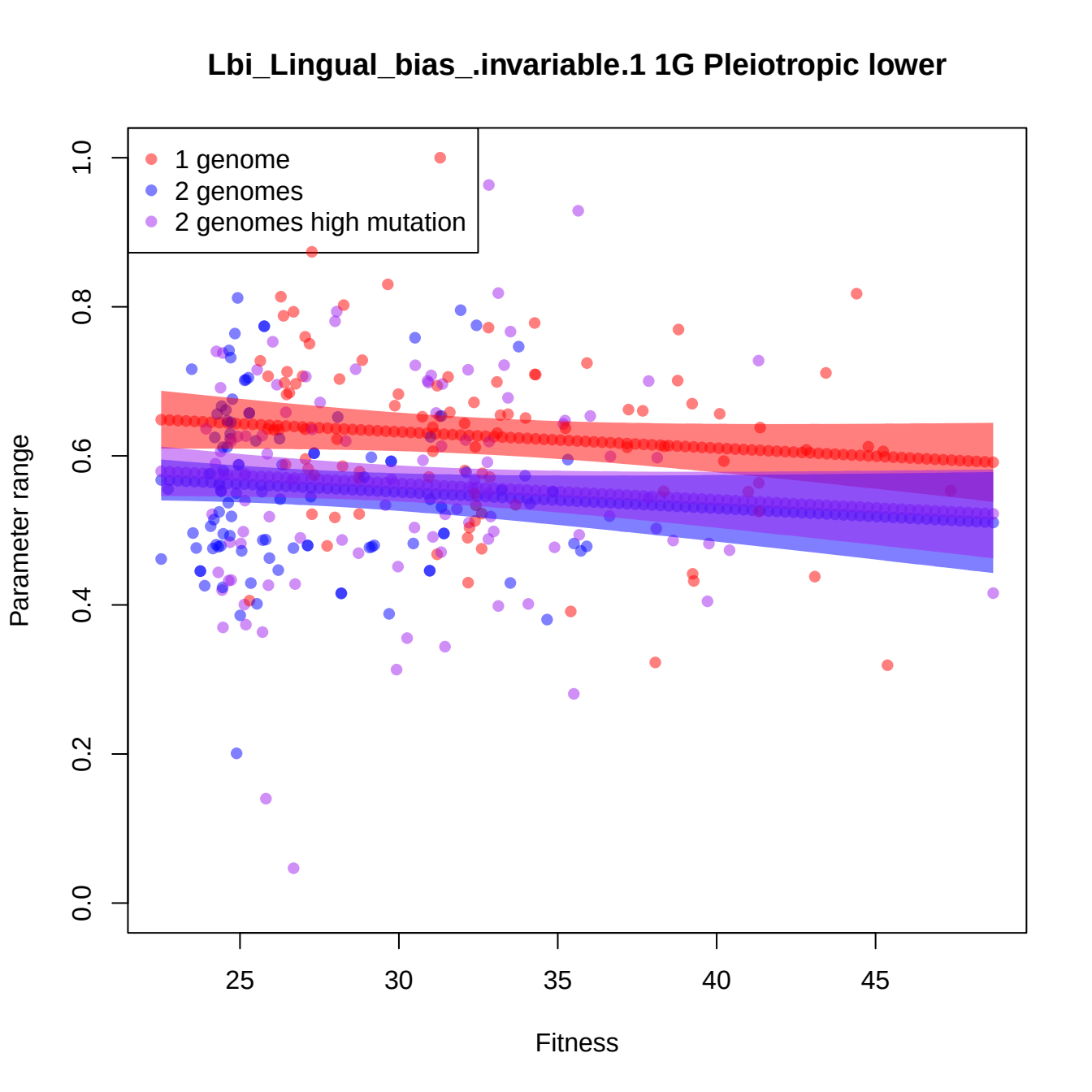

Bbi_Buccal_bias_.invariable_in_seals._hox.1
1G HOX lower
Counts
1.2
1.0
0.8
0.6
0.4
2.32756007031531e−26
1
1.66716222426022e−25
12 − 4
12 − 8
4 − 8
1 genome
2 genomes
2 genomes high mutation

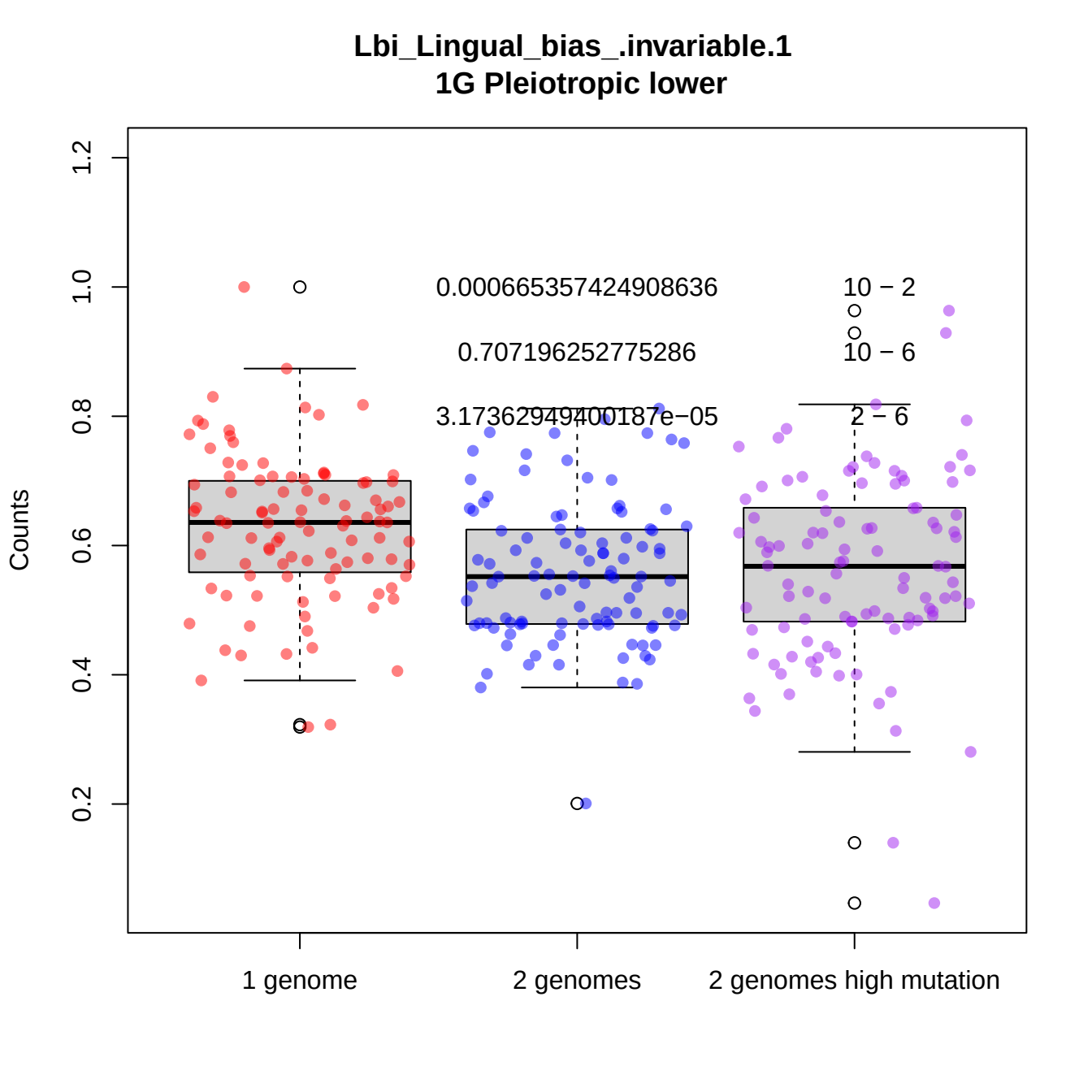

1.00
0.75
0.50
0.25
0.00
windows
30
40
50
fit
type
1
5
9

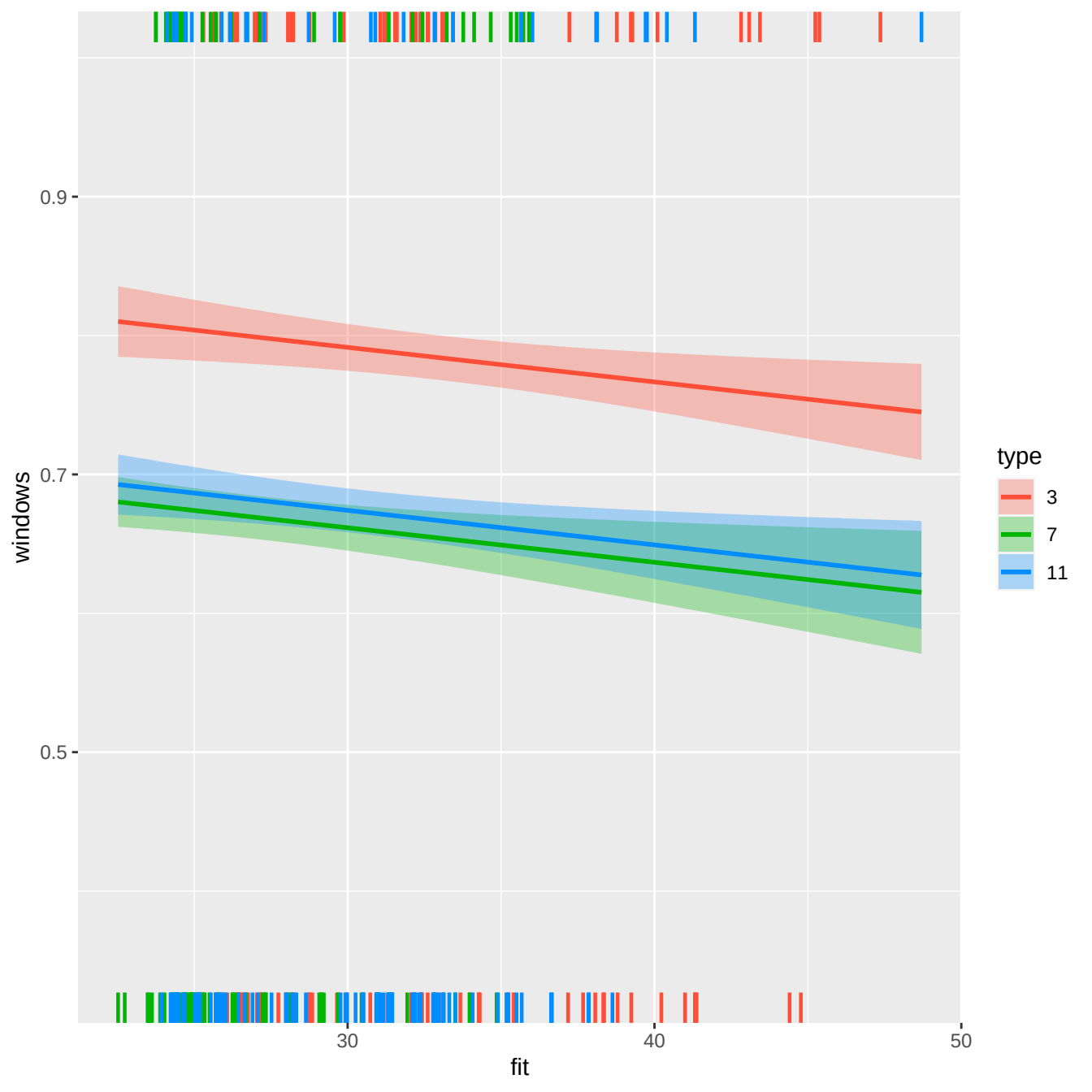

Lbi_Lingual_bias_.invariable.1 1G Pleiotropic upper
1 genome
2 genomes
2 genomes high mutation
Parameter range
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

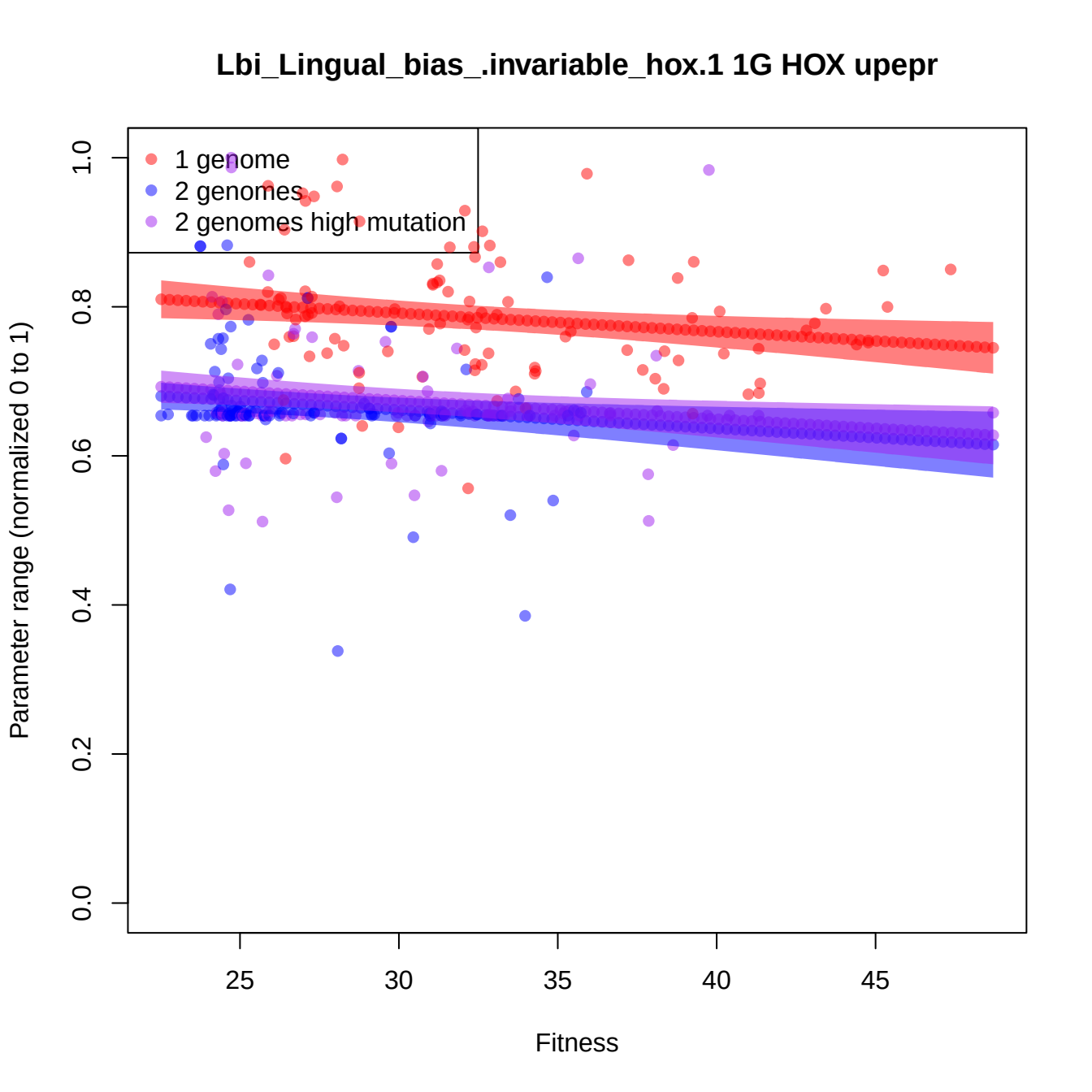

Lbi_Lingual_bias_.invariable.1
1G Pleiotropic upper
9.44602475032035e−09
7.81538129207145e−05
0.120910098051992
1 − 5
1 − 9
5 − 9
Counts
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1 genome
2 genomes
2 genomes high mutation

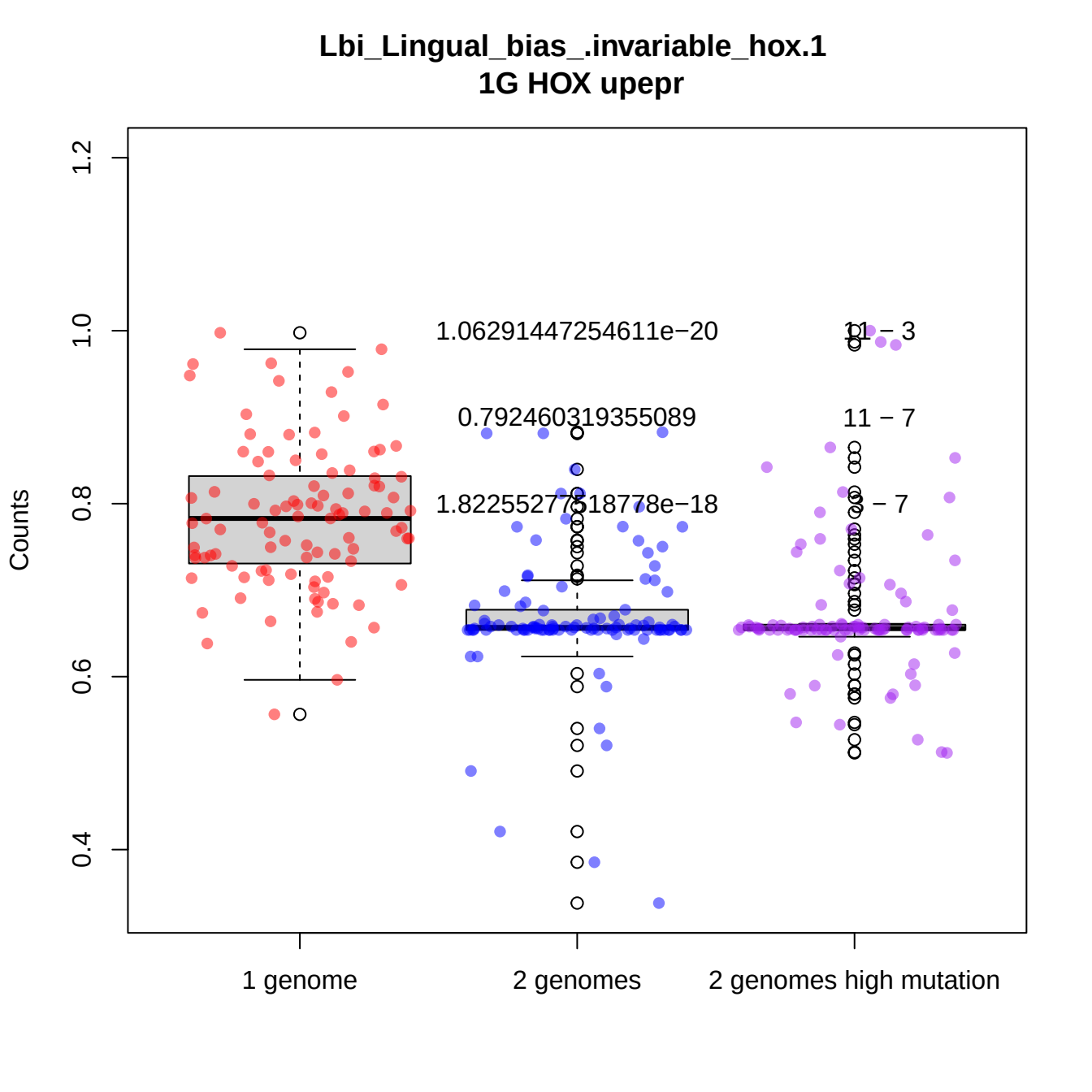

1.00
0.75
0.50
0.25
0.00
windows
30
40
50
fit
type
2
6
10

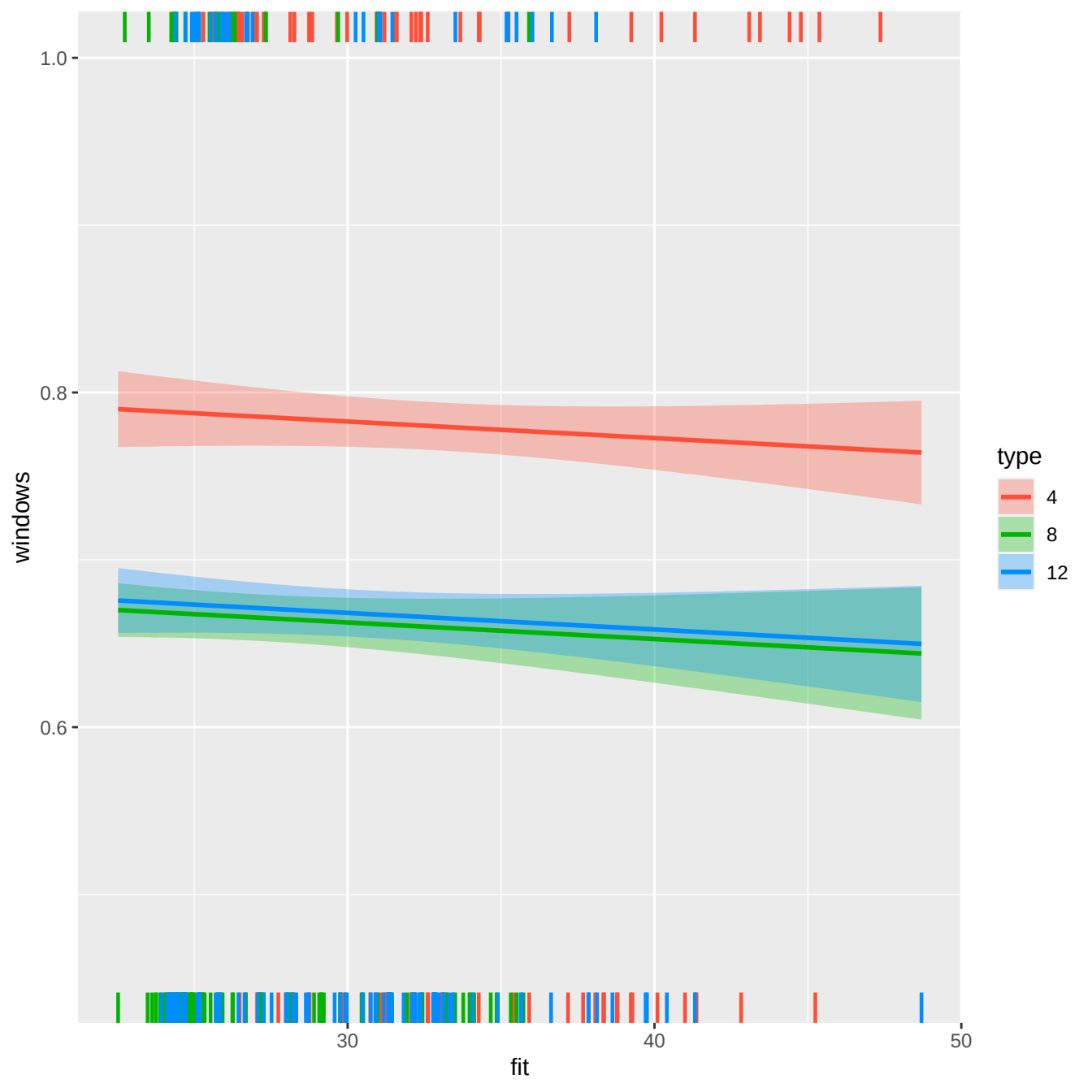

Lbi_Lingual_bias_.invariable.1 1G Pleiotropic lower
1 genome
2 genomes
2 genomes high mutation
Parameter range
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

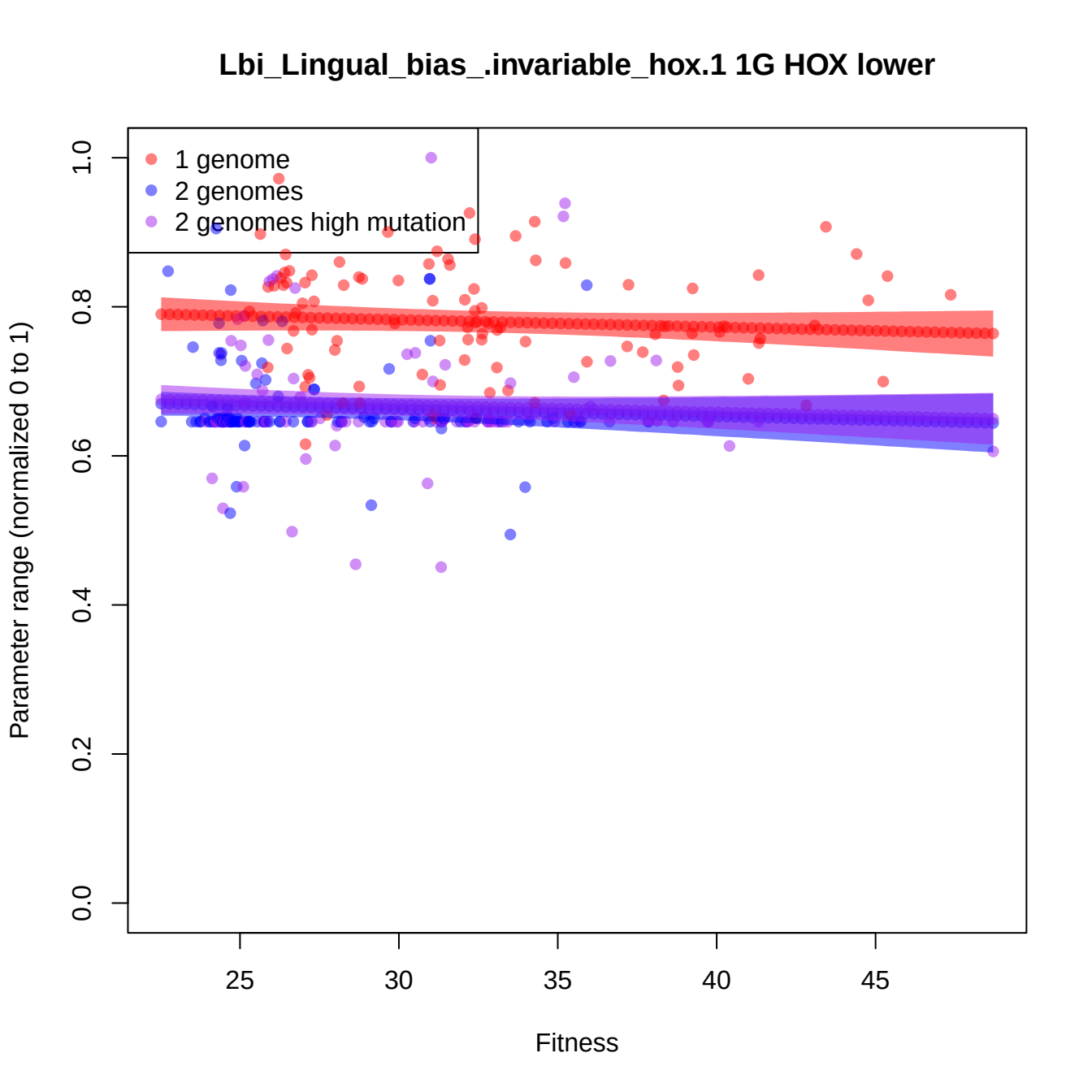

Lbi_Lingual_bias_.invariable.1
1G Pleiotropic lower
Counts
1.2
1.0
0.8
0.6
0.4
0.2
0.000665357424908636
0.707196252775286
3.17362949400187e−05
10 − 2
10 − 6
2 − 6
1 genome
2 genomes
2 genomes high mutation

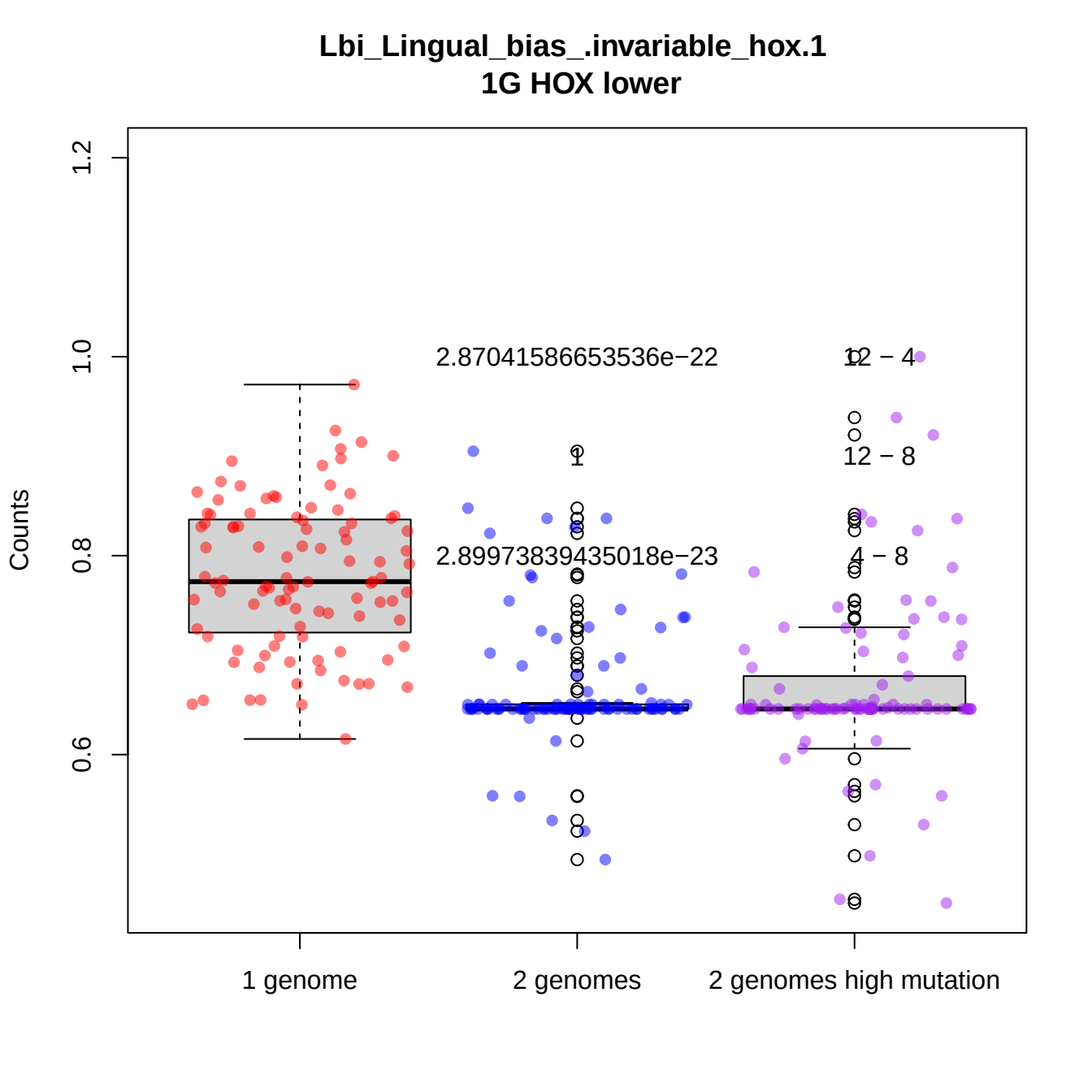

windows
0.9
0.7
0.5
30
40
50
fit
type
3
7
11


Lbi_Lingual_bias_.invariable_hox.1 1G HOX upepr
1 genome
2 genomes
2 genomes high mutation
Parameter range (normalized 0 to 1)
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness


Lbi_Lingual_bias_.invariable_hox.1
1G HOX upepr
Counts
1.2
1.0
0.8
0.6
0.4
1.06291447254611e−20
0.792460319355089
1.82255277518778e−18
11 − 3
11 − 7
11 − 7
1 genome
2 genomes
2 genomes high mutation


1.0
0.8
0.6
windows
30
40
50
fit
type
4
8
12


Lbi_Lingual_bias_.invariable_hox.1 1G HOX lower
1 genome
2 genomes
2 genomes high mutation
Parameter range (normalized 0 to 1)
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness


Lbi_Lingual_bias_.invariable_hox.1
1G HOX lower
Counts
1.2
1.0
0.8
0.6
2.87041586653536e−22
2.89973839435018e−23
1 genome
2 genomes
2 genomes high mutation